SATA 1D: IRFs at 5.2 and 10 km Spacing

Along-Track Ramps (Normal and Large Heights), Random Heights, Piecewise Ramps

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Companion to *SATA 1D: Three Open Points*. SATA 1D, $N_{rx} = 4$, PRF = 2000 Hz, DPCA, $b_{xt} = 63.7, 27.0, 4.1, 1.7$ m. Normal heights $h_j = j \cdot 200 \tan\alpha$, large heights $h_j = j \cdot 2600 \tan\alpha$; random heights and piecewise ramps keep the heights of the 200-m scenes, $N = 5$.
How to read the pages. Top: SATA whole band and per sub-band. Bottom: no SATA. Green: the reference, now the flat-terrain reconstruction (the same reconstruction with $b_{xt} = 0$, i.e. without topographic phase errors). Thin black: $|s - s_{flat}|$, the topographic error, whose peak in the grey windows ($x_j/v \pm m\,0.81$ s, $m = 1, 2$) is the topographic ambiguity. Floor: $|s_{flat}|$ in the same windows. All levels in dB relative to the peak of s_{flat}. Right: target geometry in 2D and 3D.

1 Configuration E: along-track ramps, targets 5.2 km apart, normal heights

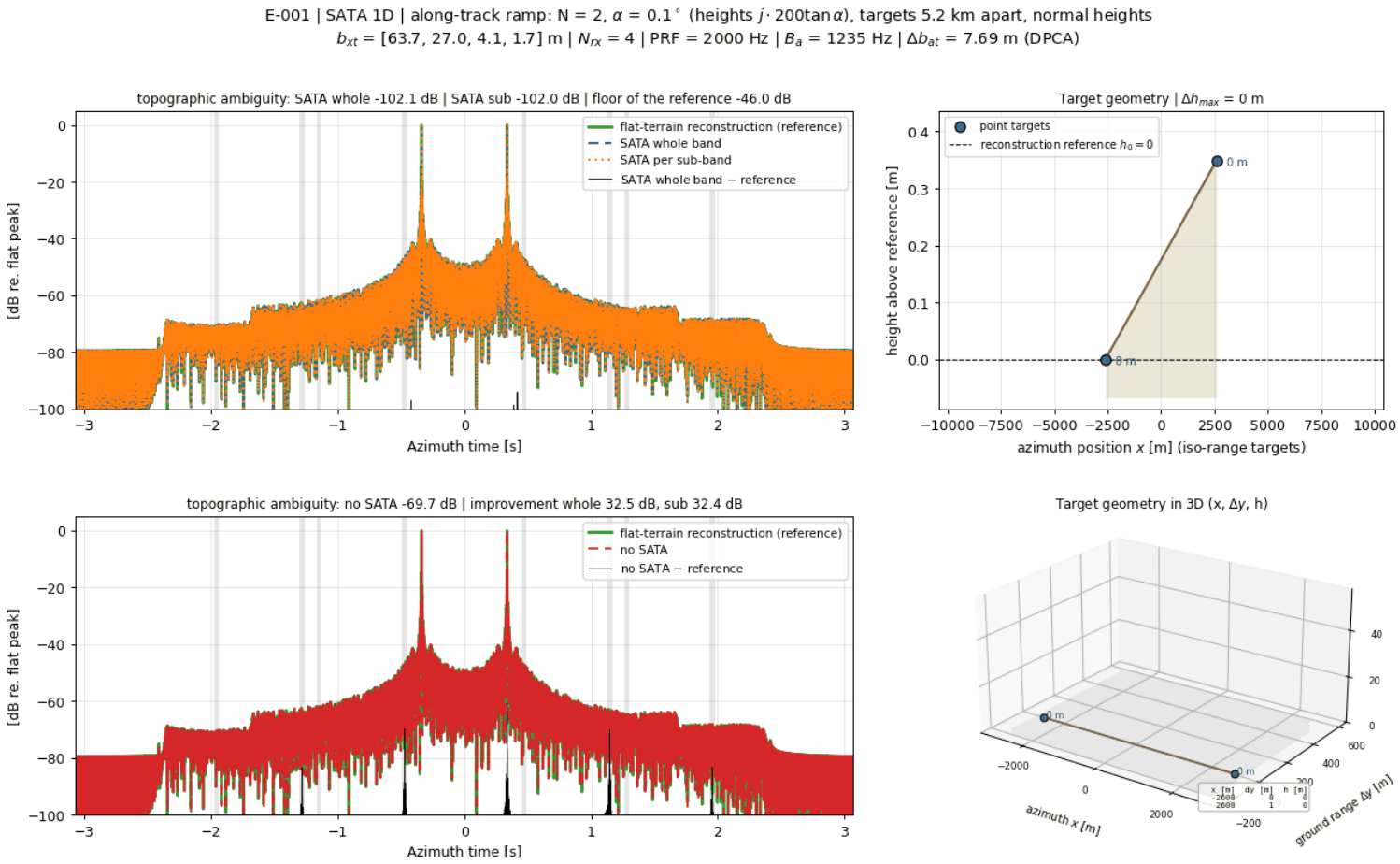


Figure 1: E-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −69.7, SATA whole band −102.1, per sub-band −102.0 dB (improvement 32.5 dB); floor −46.0 dB.

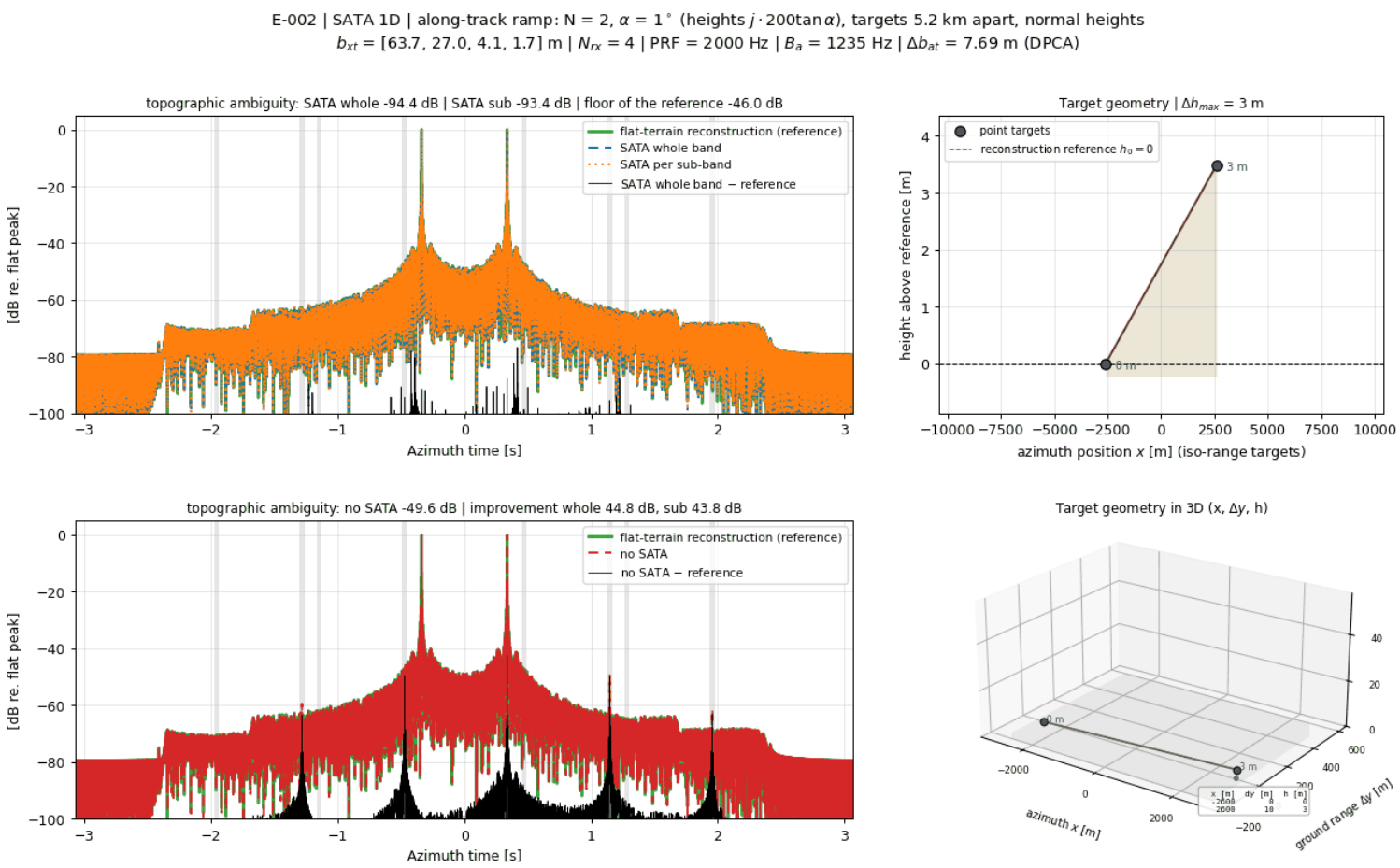


Figure 2: E-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −49.6, SATA whole band −94.4, per sub-band −93.4 dB (improvement 44.8 dB); floor −46.0 dB.

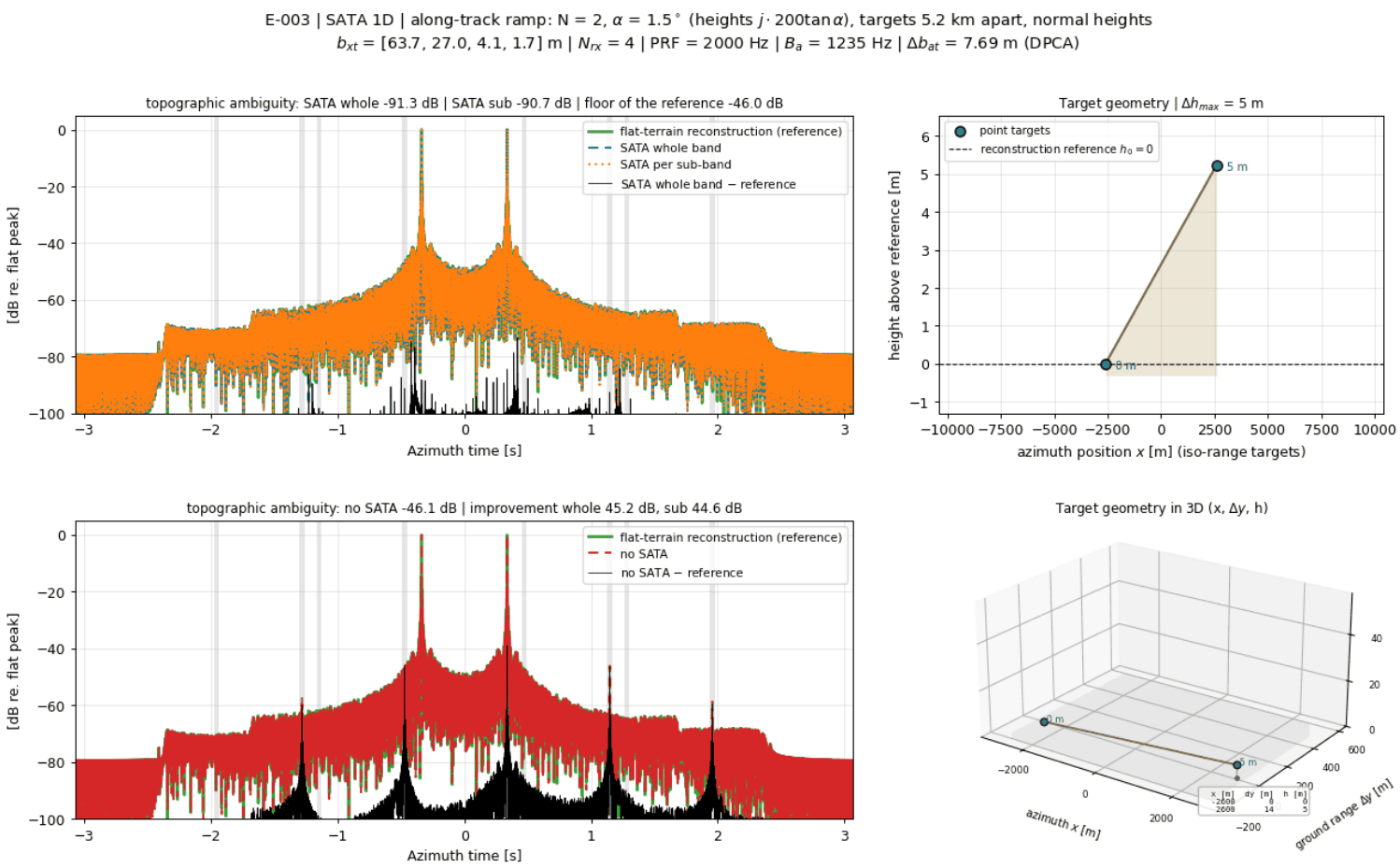


Figure 3: E-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -46.1, SATA whole band -91.3, per sub-band -90.7 dB (improvement 45.2 dB); floor -46.0 dB.

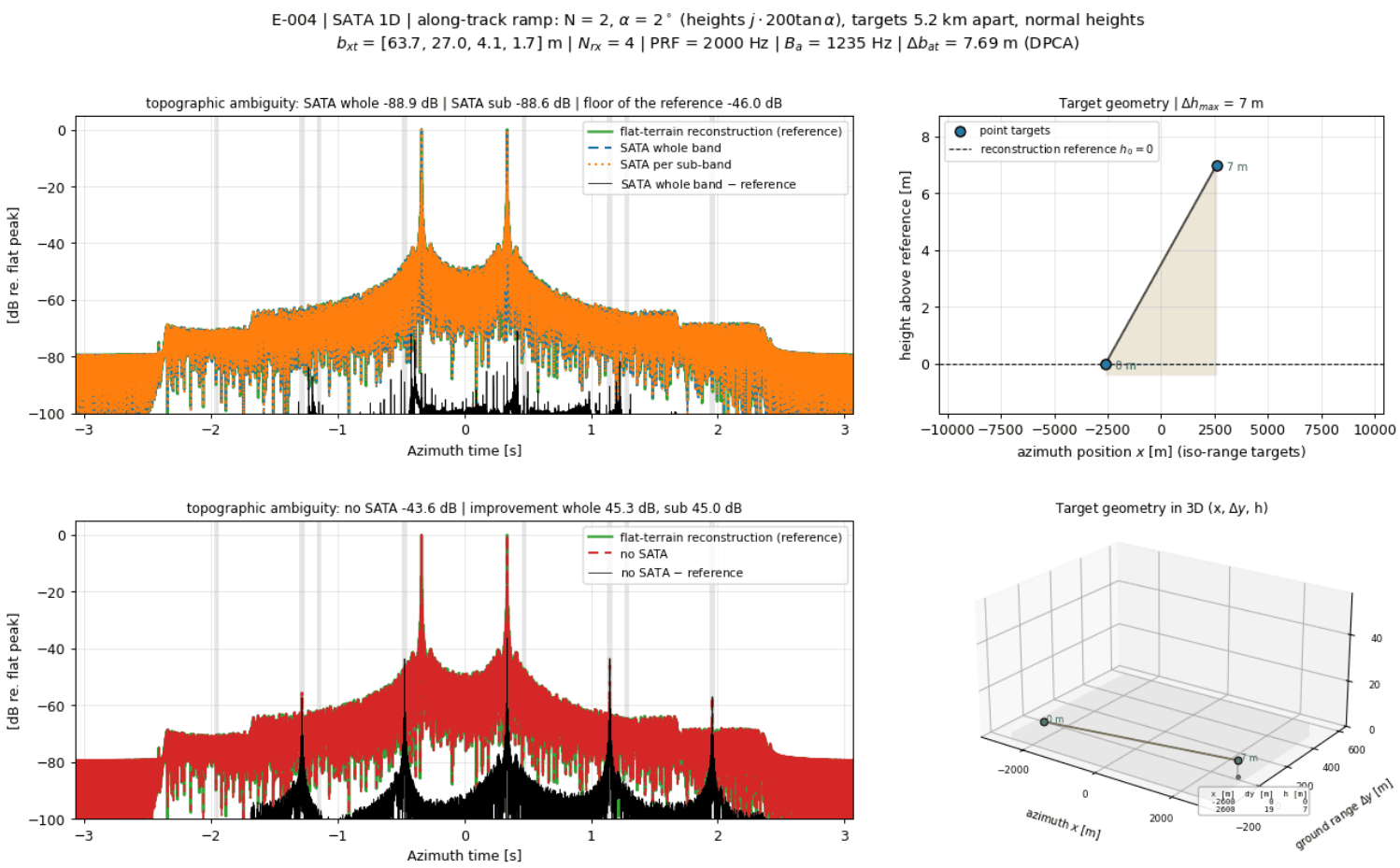


Figure 4: E-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −43.6, SATA whole band −88.9, per sub-band −88.6 dB (improvement 45.3 dB); floor −46.0 dB.

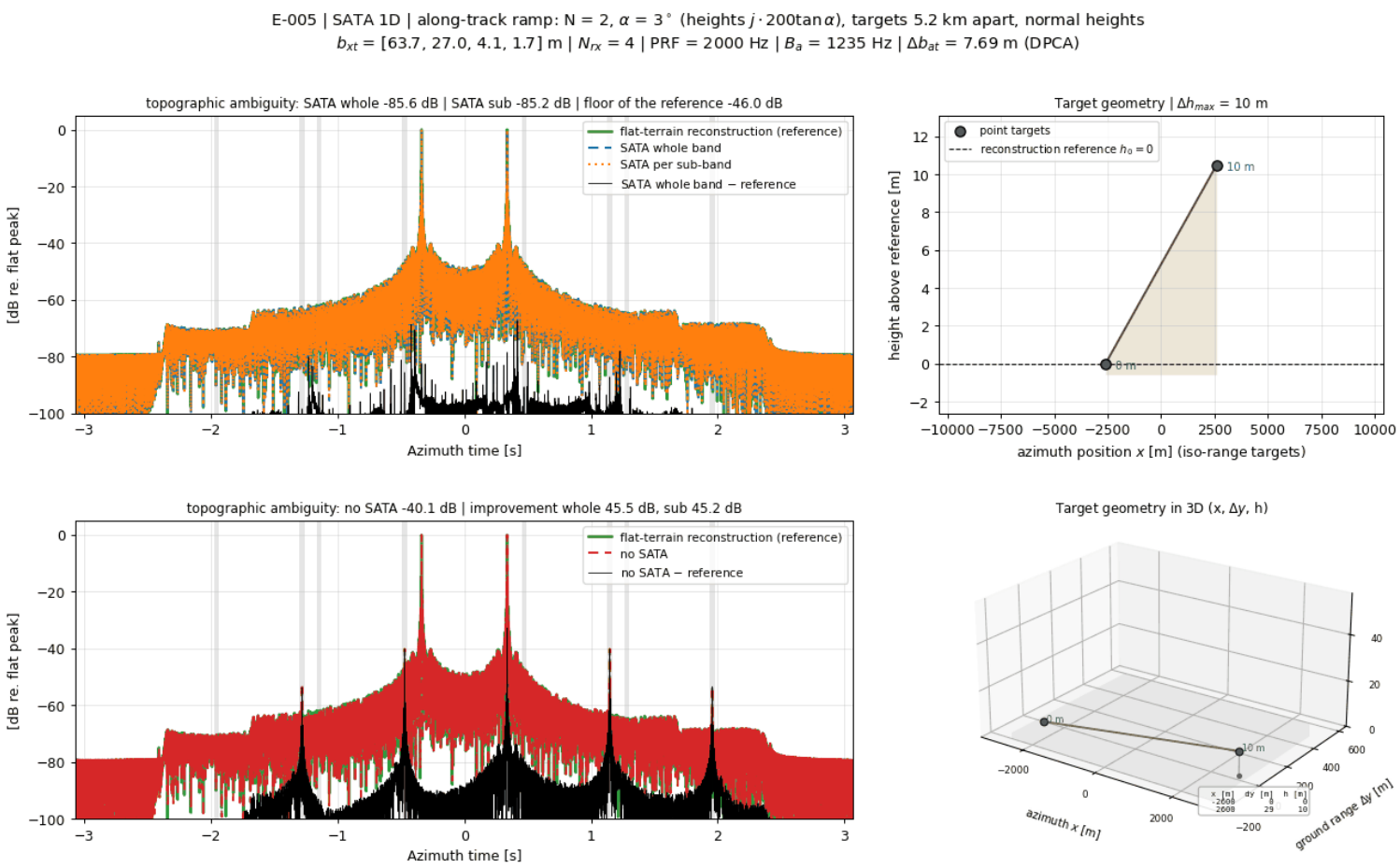


Figure 5: E-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −40.1, SATA whole band −85.6, per sub-band −85.2 dB (improvement 45.5 dB); floor −46.0 dB.

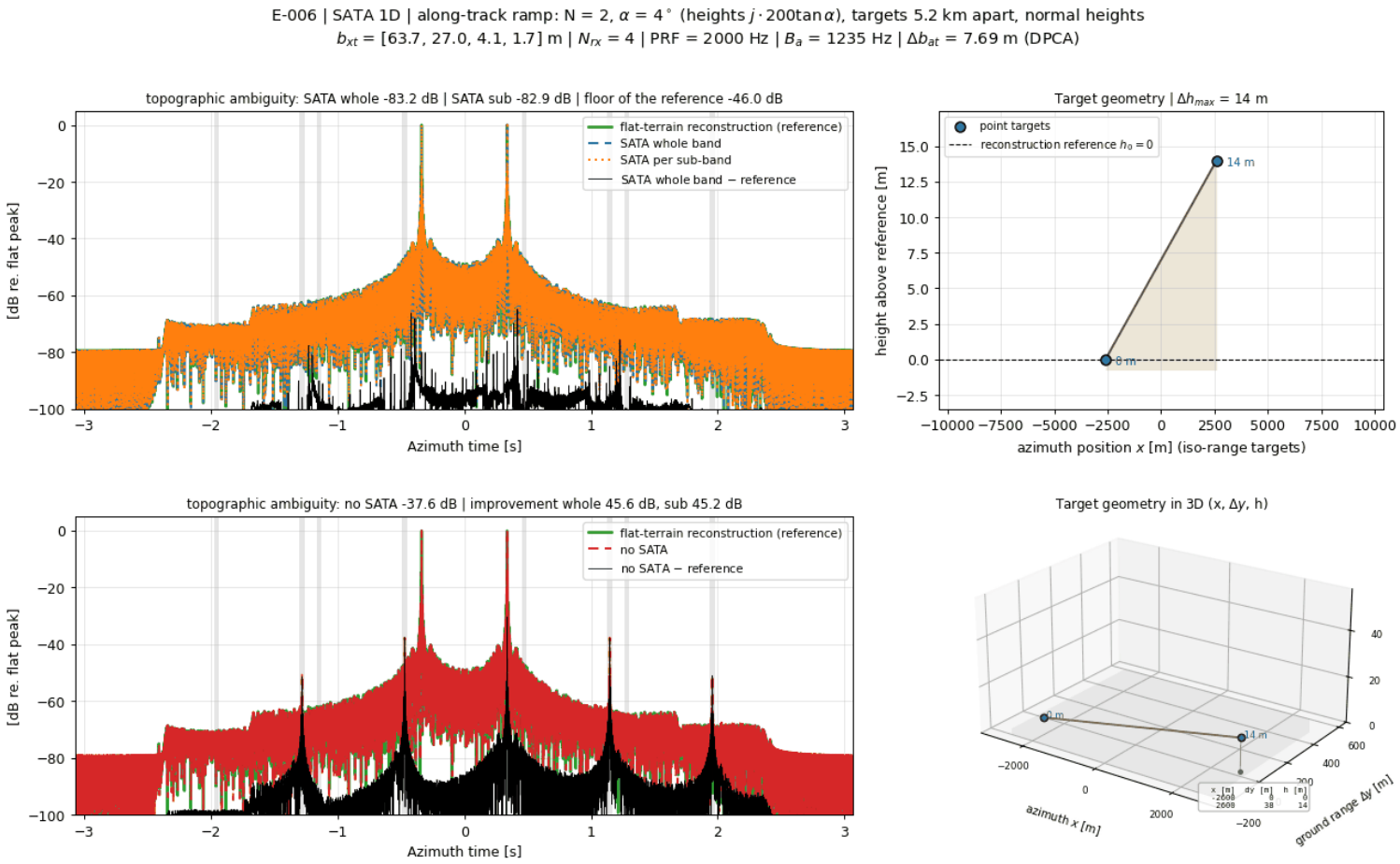


Figure 6: E-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −37.6, SATA whole band −83.2, per sub-band −82.9 dB (improvement 45.6 dB); floor −46.0 dB.

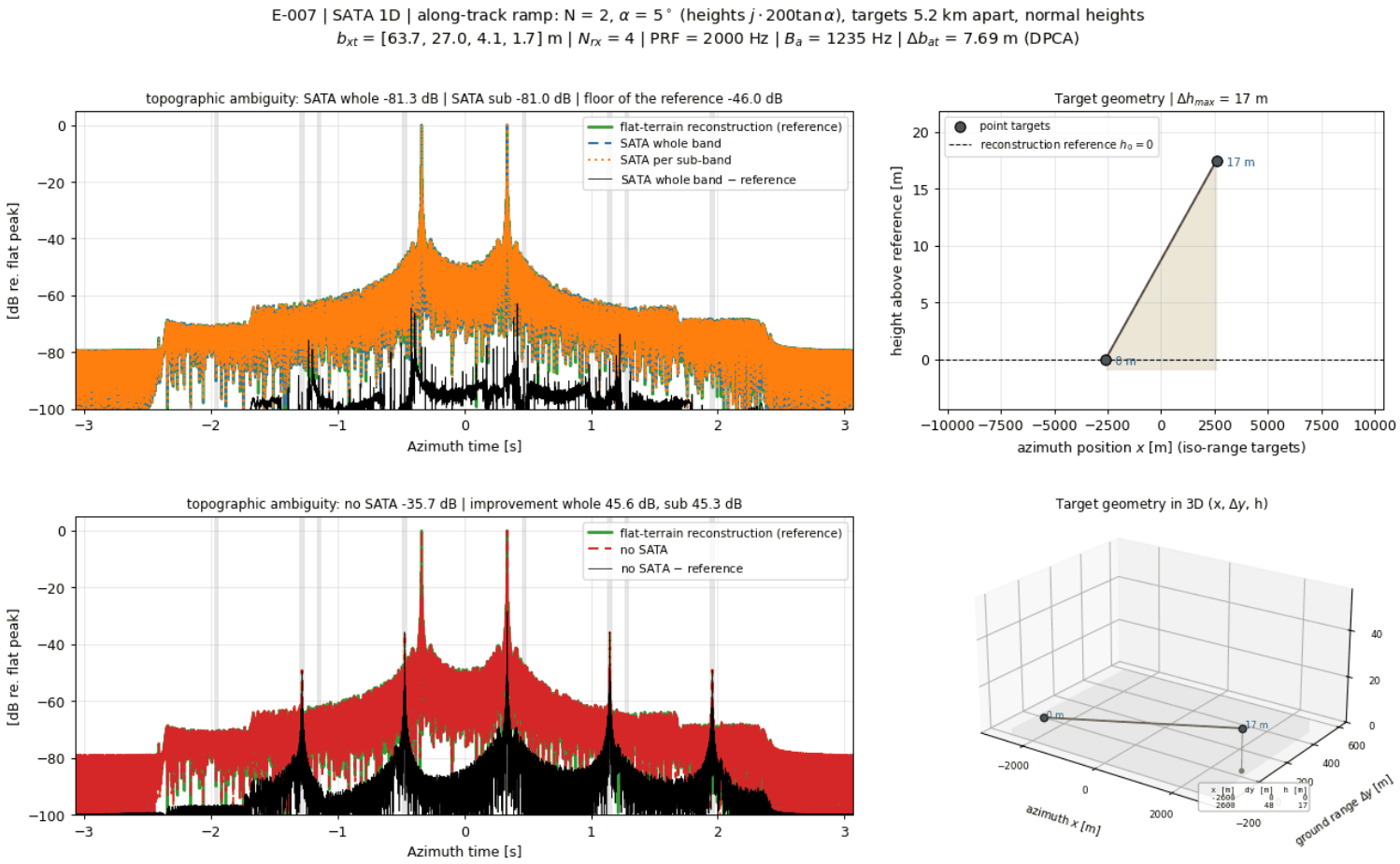


Figure 7: E-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −35.7, SATA whole band −81.3, per sub-band −81.0 dB (improvement 45.6 dB); floor −46.0 dB.

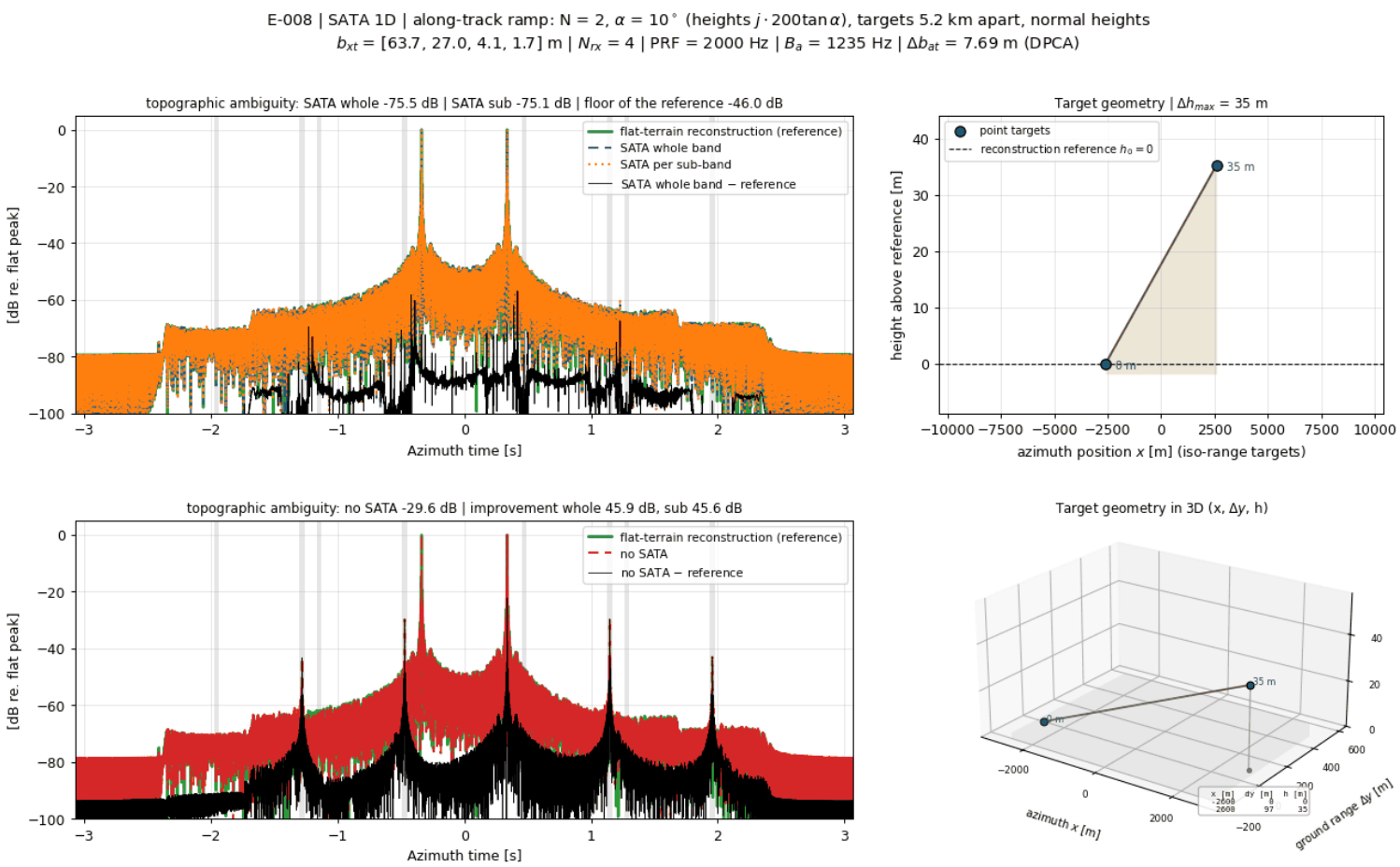


Figure 8: E-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −29.6, SATA whole band −75.5, per sub-band −75.1 dB (improvement 45.9 dB); floor −46.0 dB.

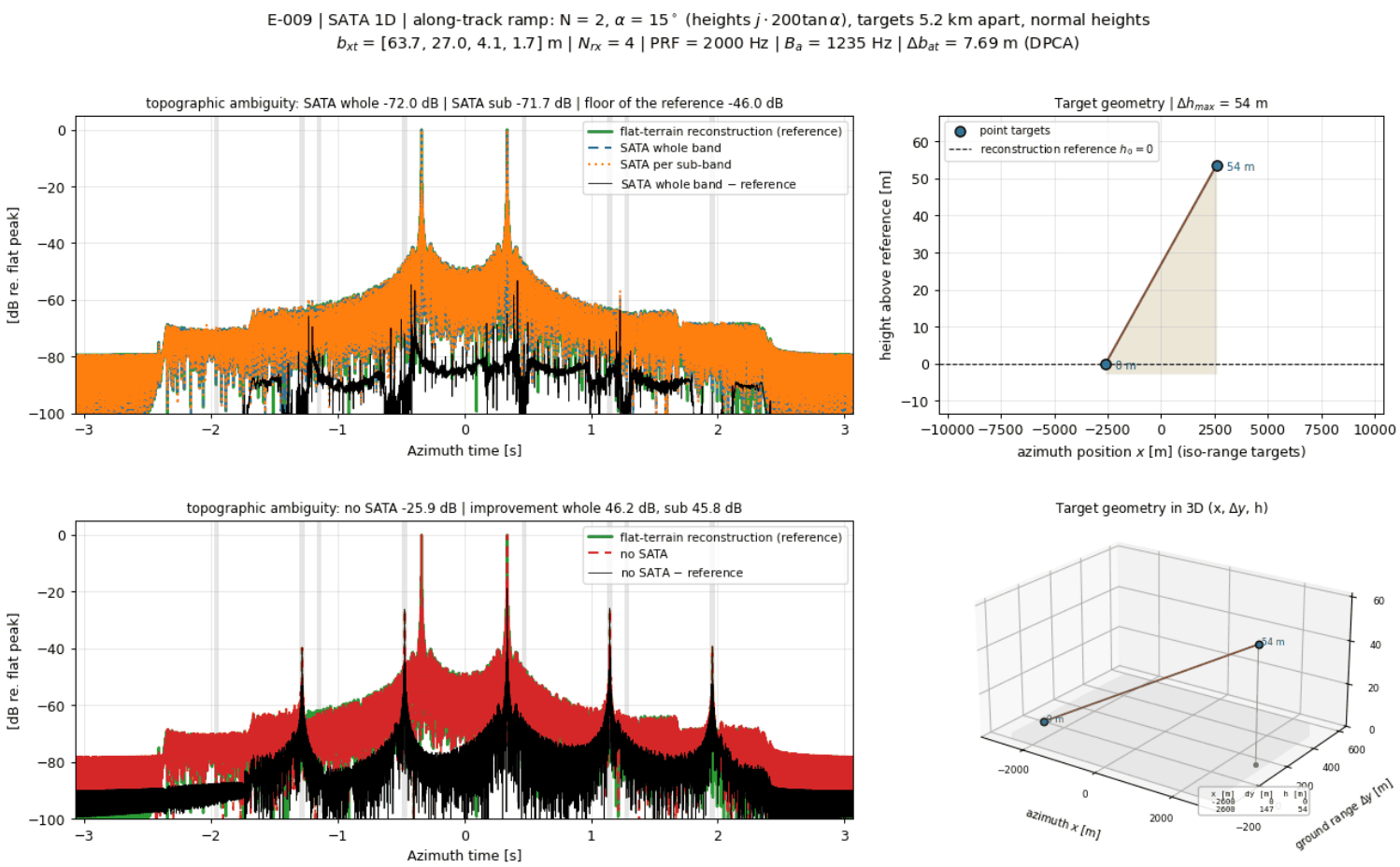


Figure 9: E-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −25.9, SATA whole band −72.0, per sub-band −71.7 dB (improvement 46.2 dB); floor −46.0 dB.

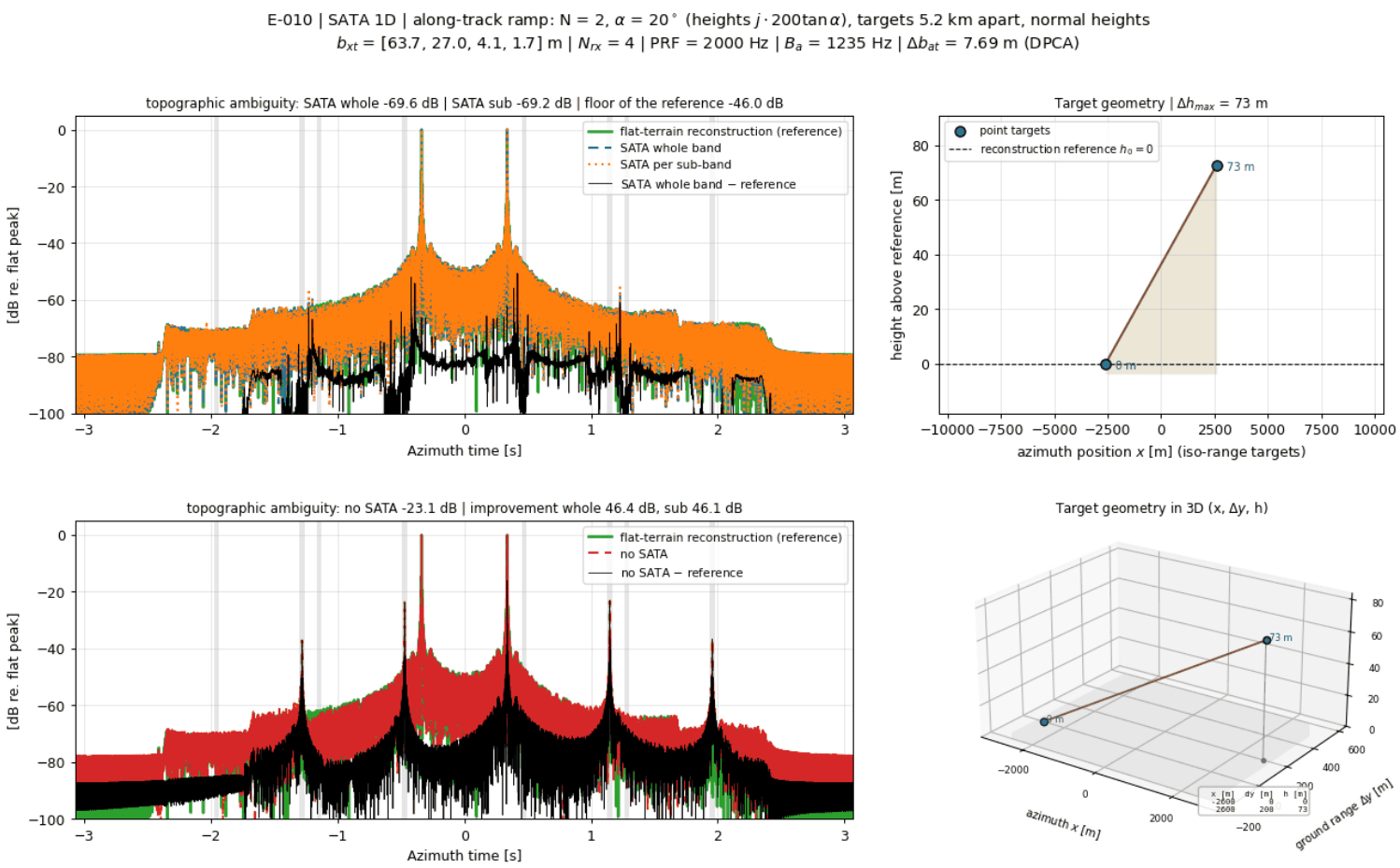


Figure 10: E-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -23.1, SATA whole band -69.6, per sub-band -69.2 dB (improvement 46.4 dB); floor -46.0 dB.

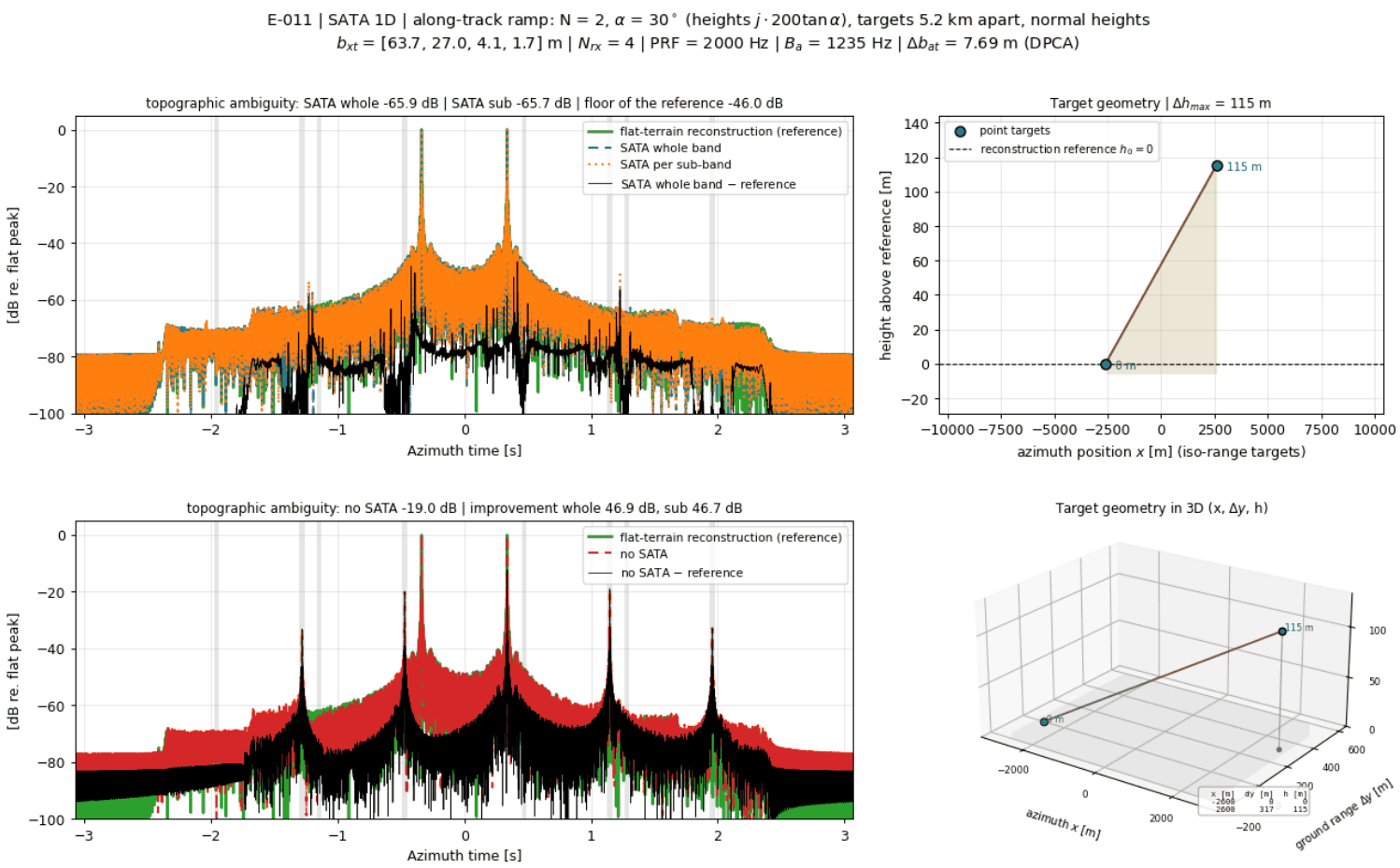


Figure 11: E-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −19.0, SATA whole band −65.9, per sub-band −65.7 dB (improvement 46.9 dB); floor −46.0 dB.

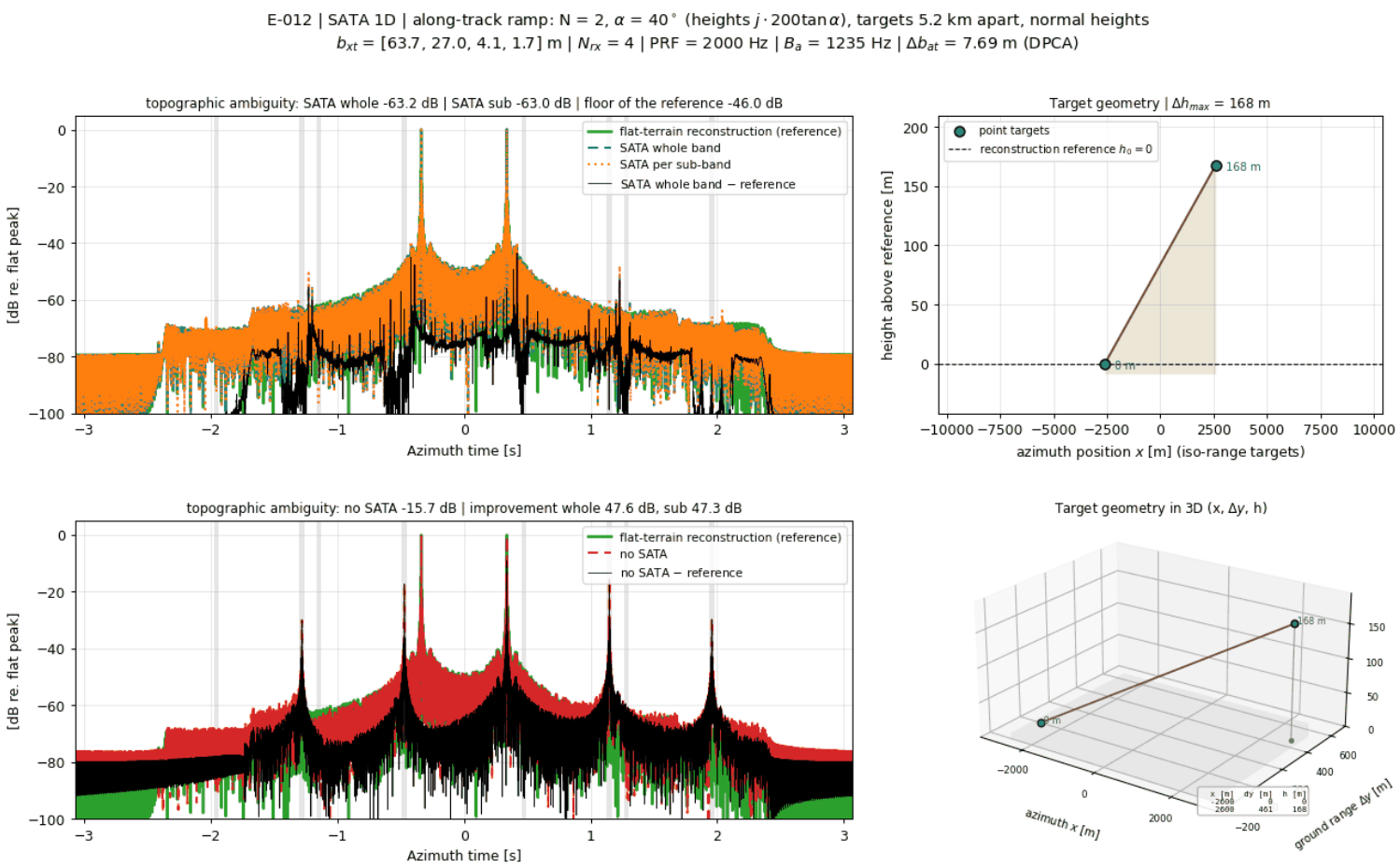


Figure 12: E-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −15.7, SATA whole band −63.2, per sub-band −63.0 dB (improvement 47.6 dB); floor −46.0 dB.

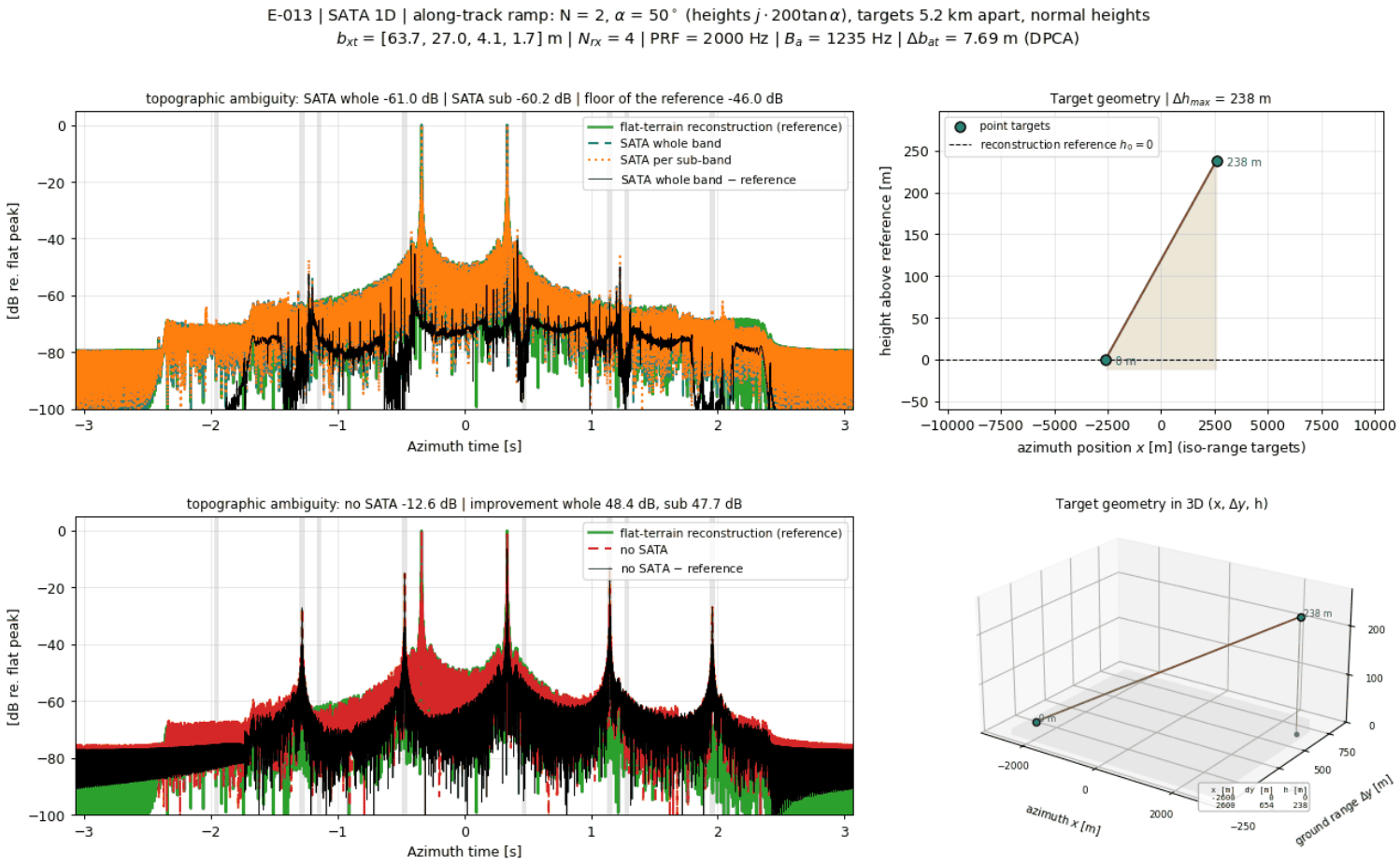


Figure 13: E-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −12.6, SATA whole band −61.0, per sub-band −60.2 dB (improvement 48.4 dB); floor −46.0 dB.

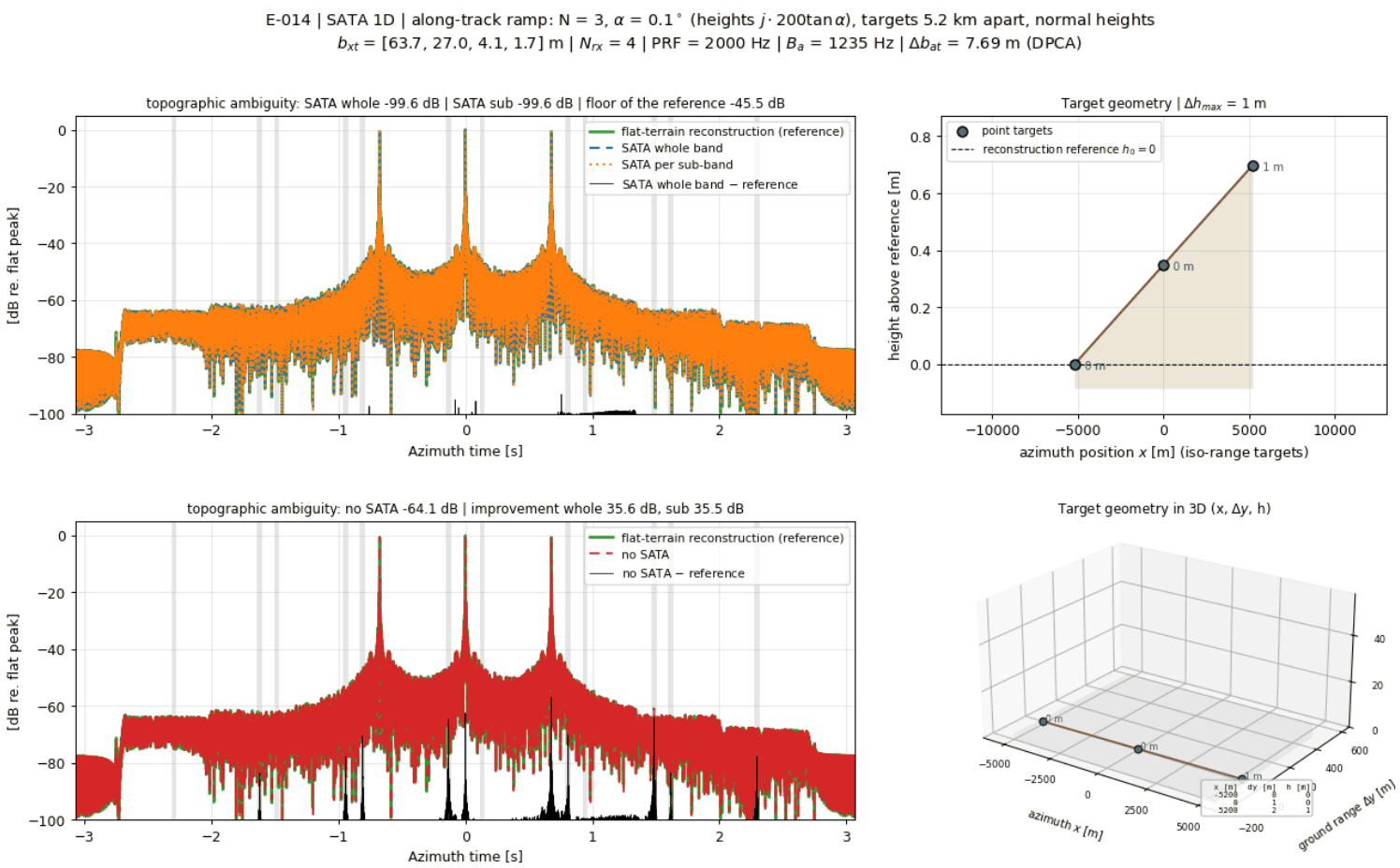


Figure 14: E-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -64.1, SATA whole band -99.6, per sub-band -99.6 dB (improvement 35.6 dB); floor -45.5 dB.

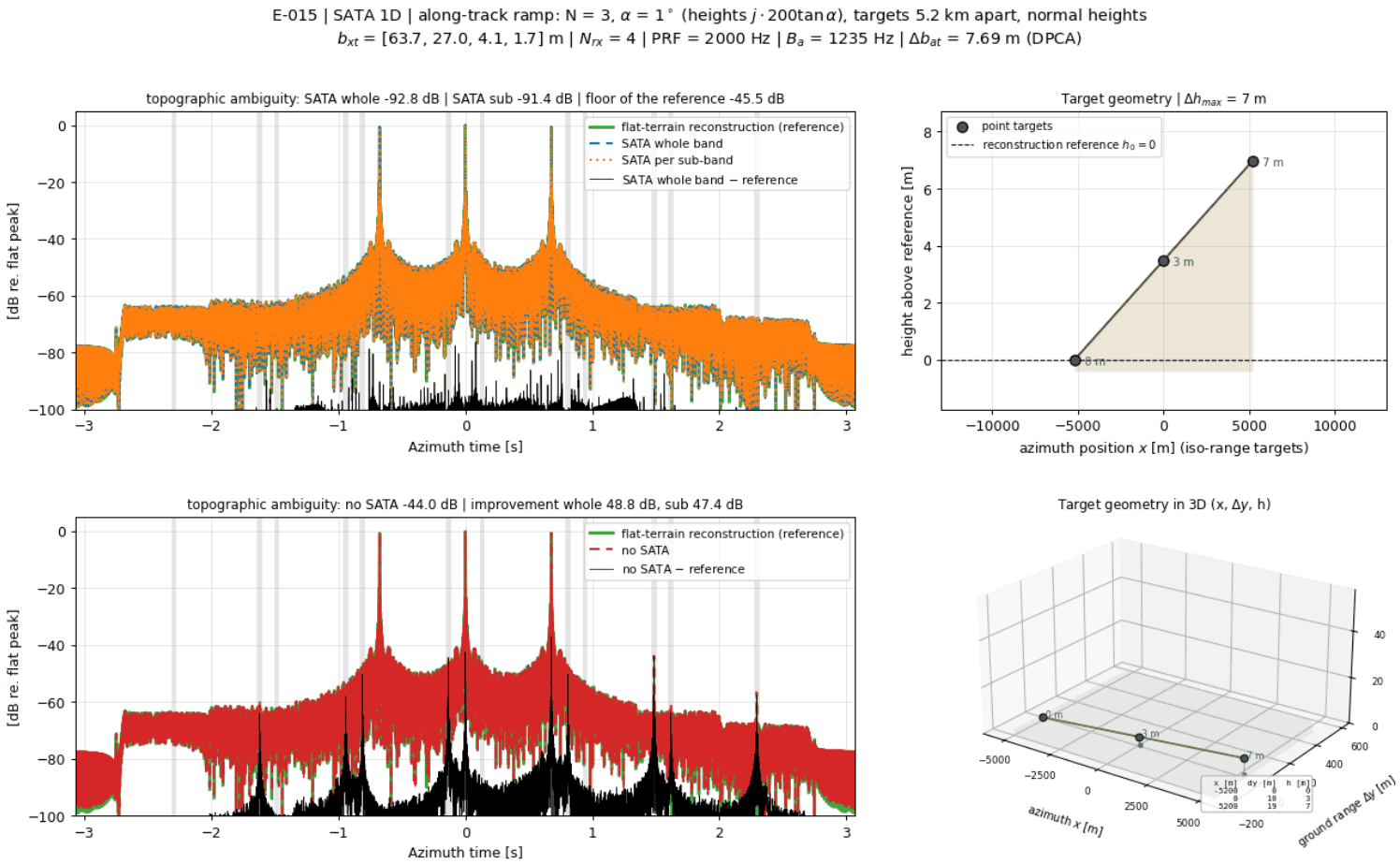


Figure 15: E-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -44.0, SATA whole band -92.8, per sub-band -91.4 dB (improvement 48.8 dB); floor -45.5 dB.

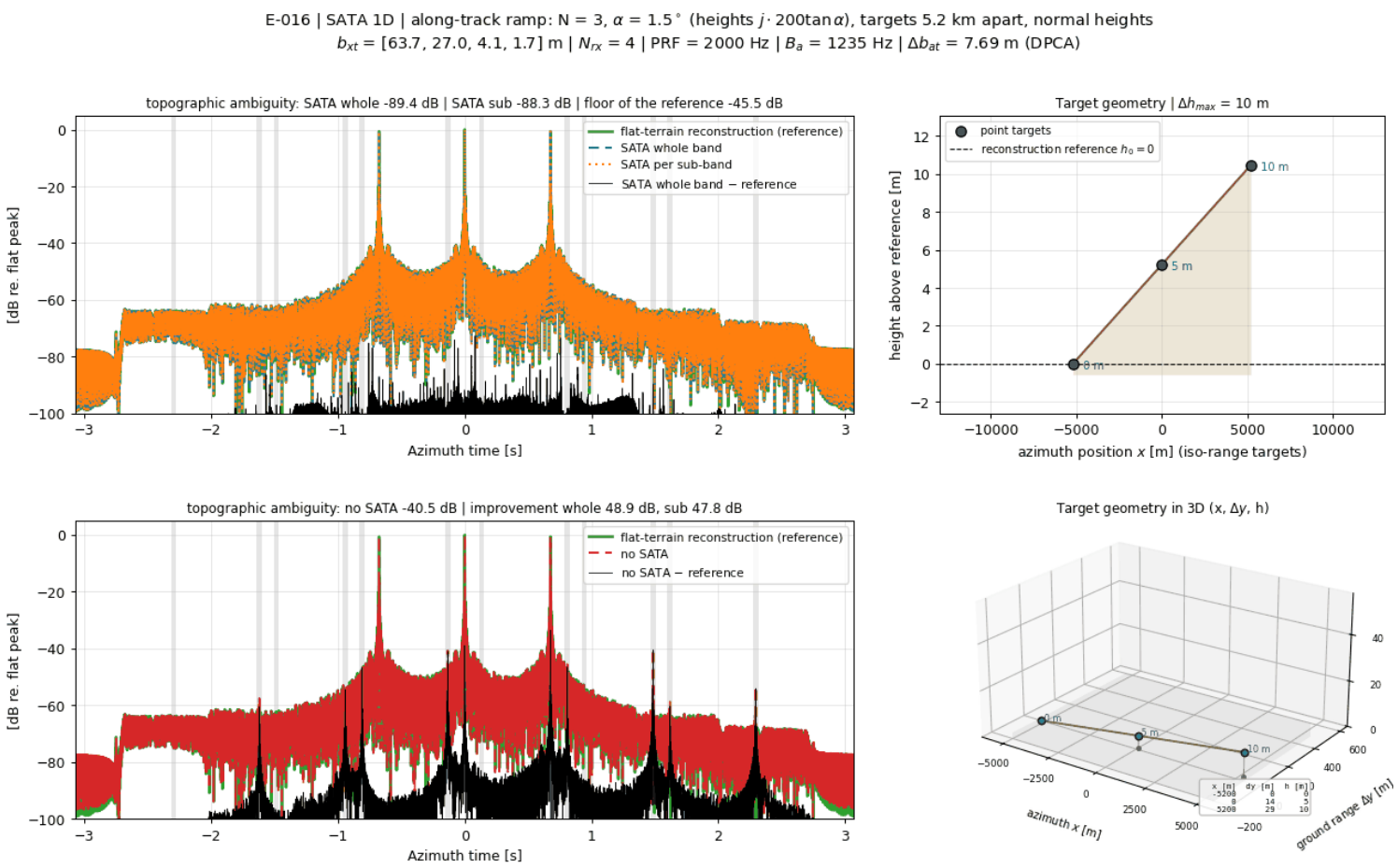


Figure 16: E-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −40.5, SATA whole band −89.4, per sub-band −88.3 dB (improvement 48.9 dB); floor −45.5 dB.

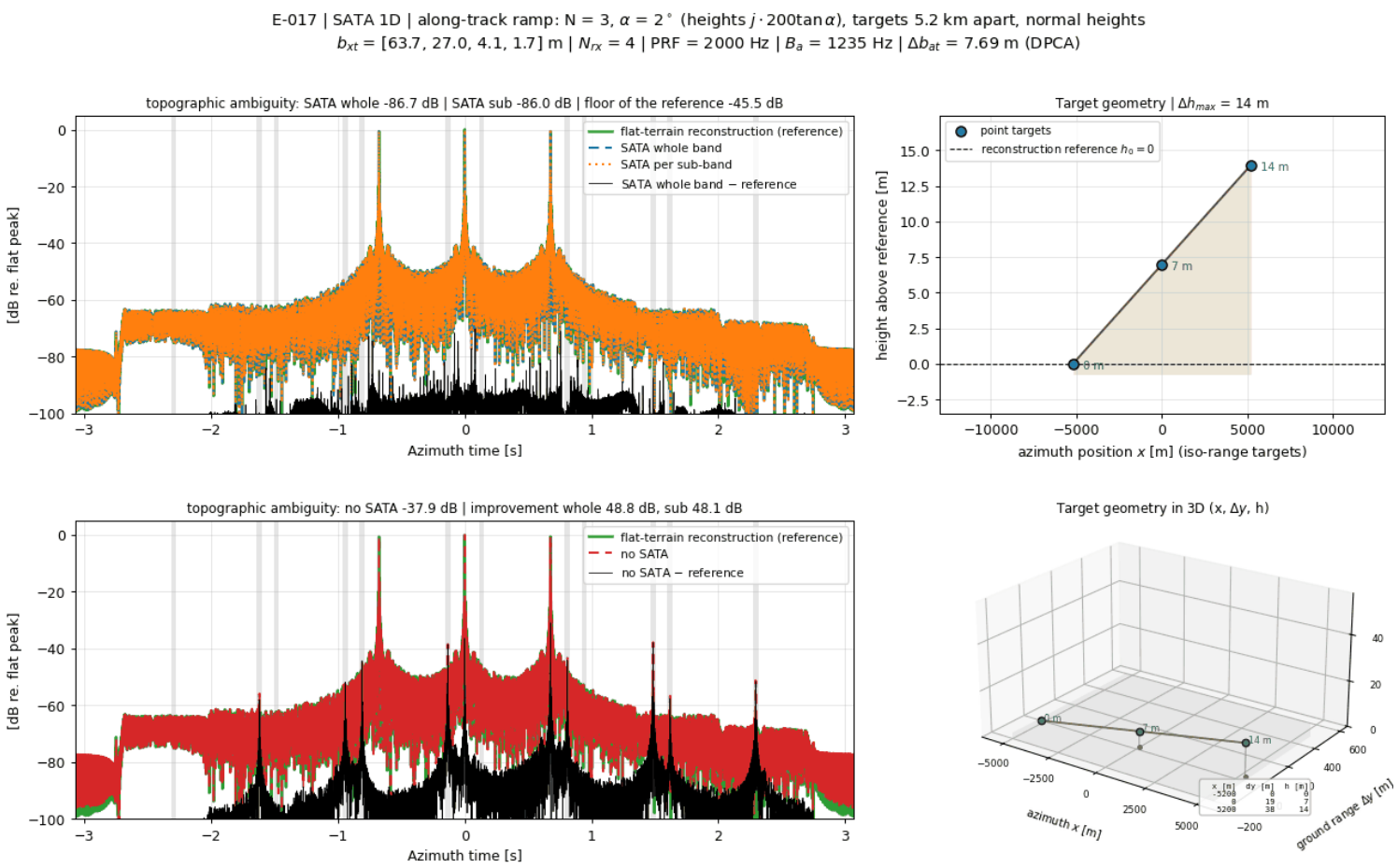


Figure 17: E-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −37.9, SATA whole band −86.7, per sub-band −86.0 dB (improvement 48.8 dB); floor −45.5 dB.

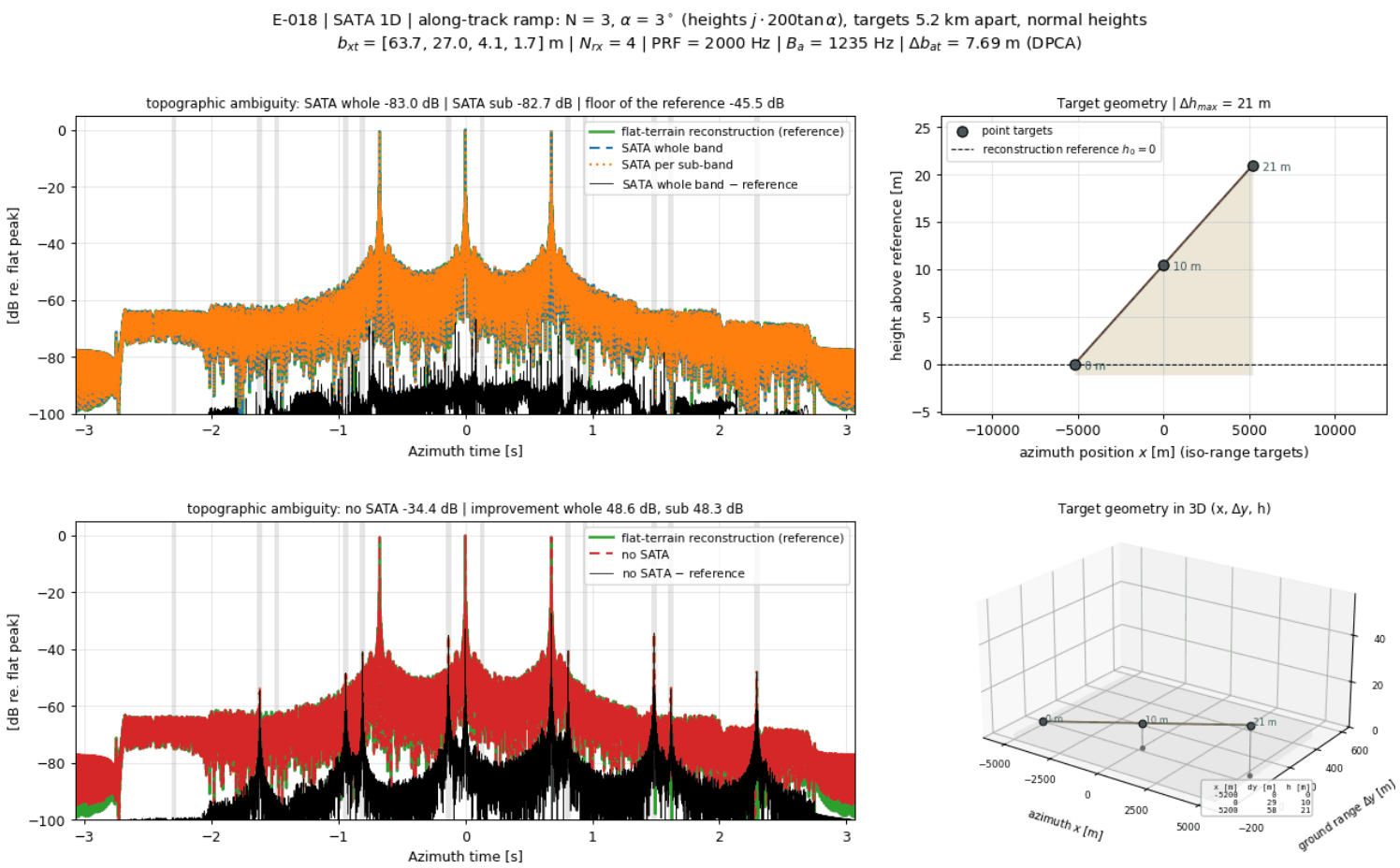


Figure 18: E-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −34.4, SATA whole band −83.0, per sub-band −82.7 dB (improvement 48.6 dB); floor −45.5 dB.

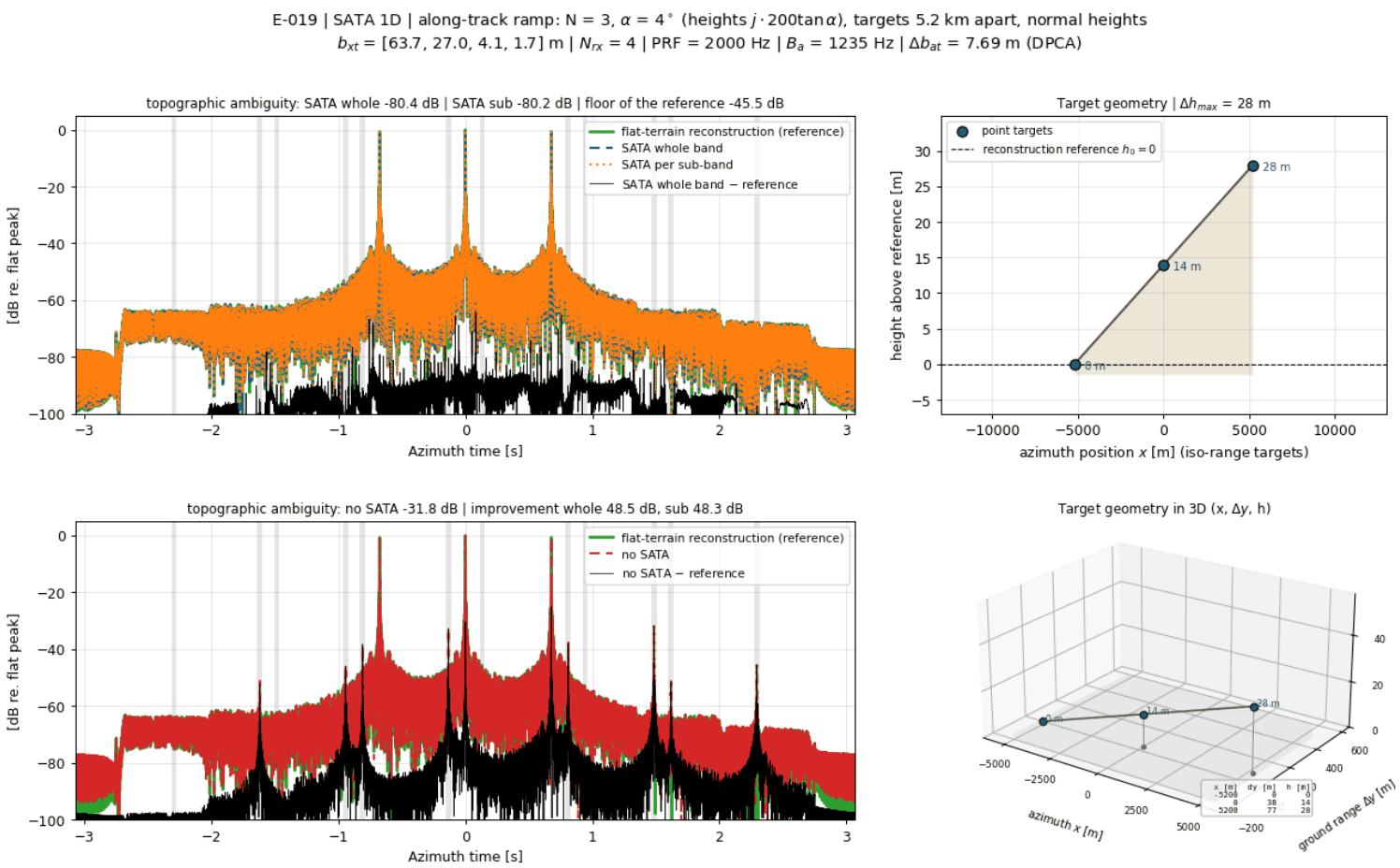


Figure 19: E-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −31.8, SATA whole band −80.4, per sub-band −80.2 dB (improvement 48.5 dB); floor −45.5 dB.

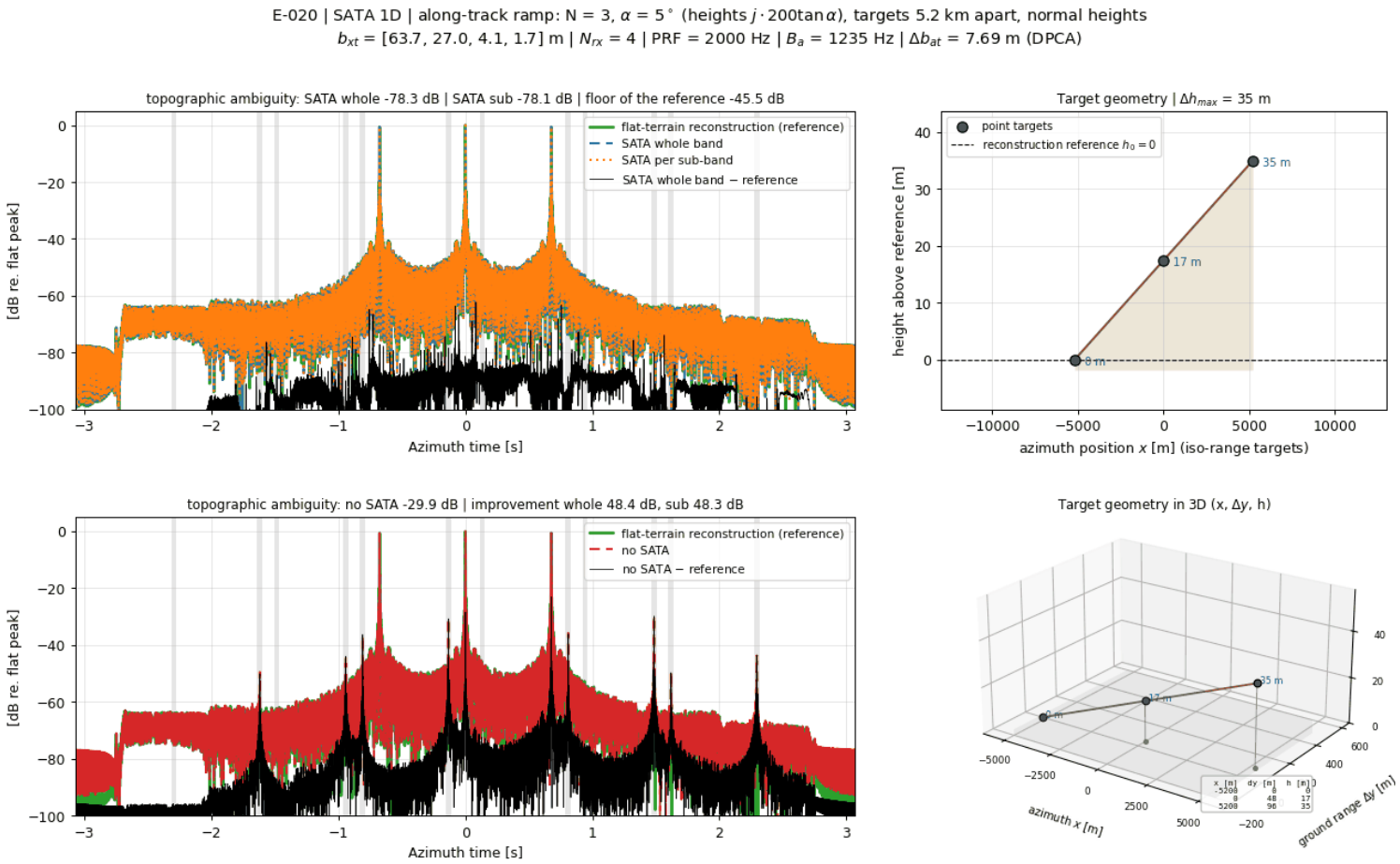


Figure 20: E-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −29.9, SATA whole band −78.3, per sub-band −78.1 dB (improvement 48.4 dB); floor −45.5 dB.

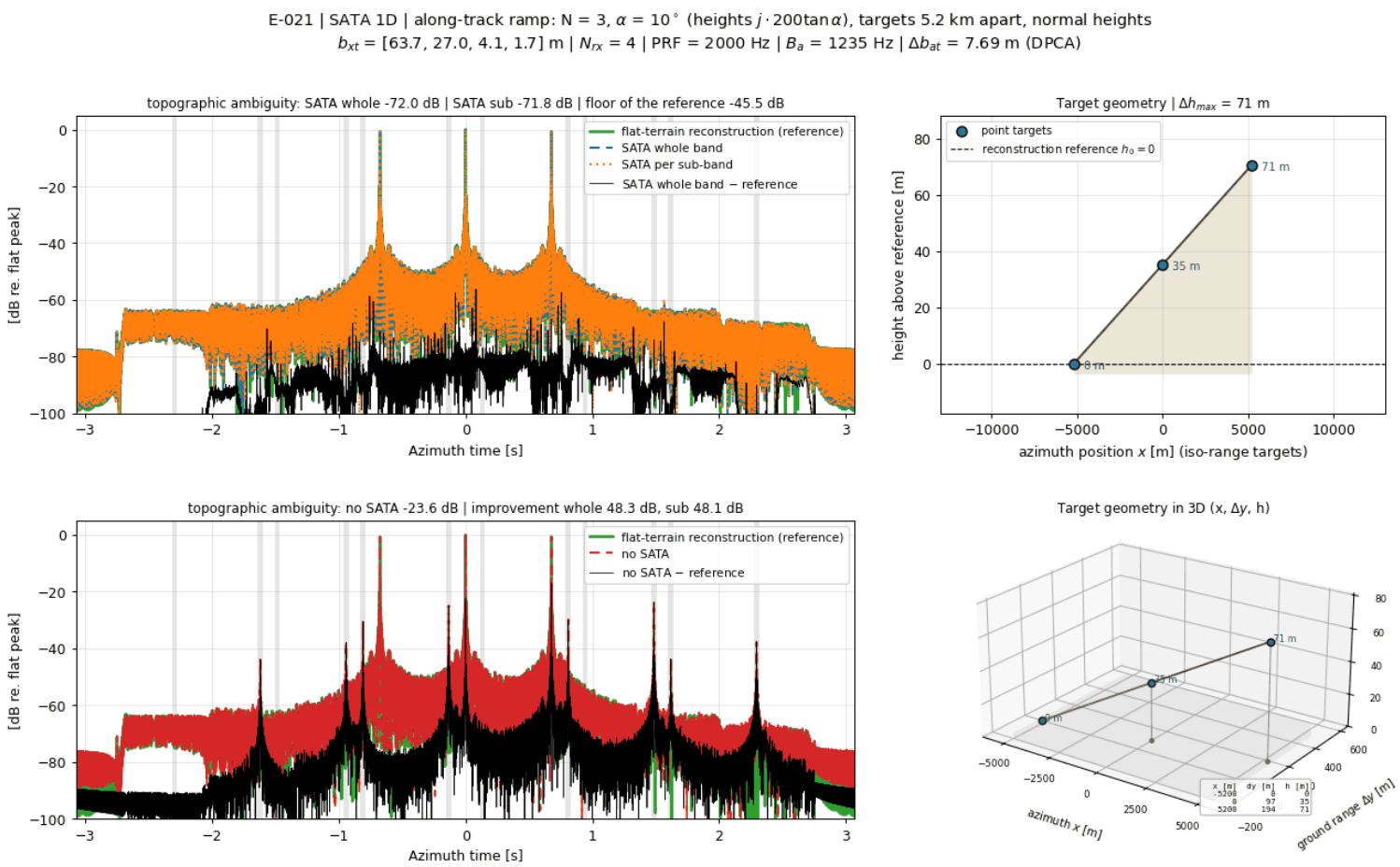


Figure 21: E-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -23.6, SATA whole band -72.0, per sub-band -71.8 dB (improvement 48.3 dB); floor -45.5 dB.

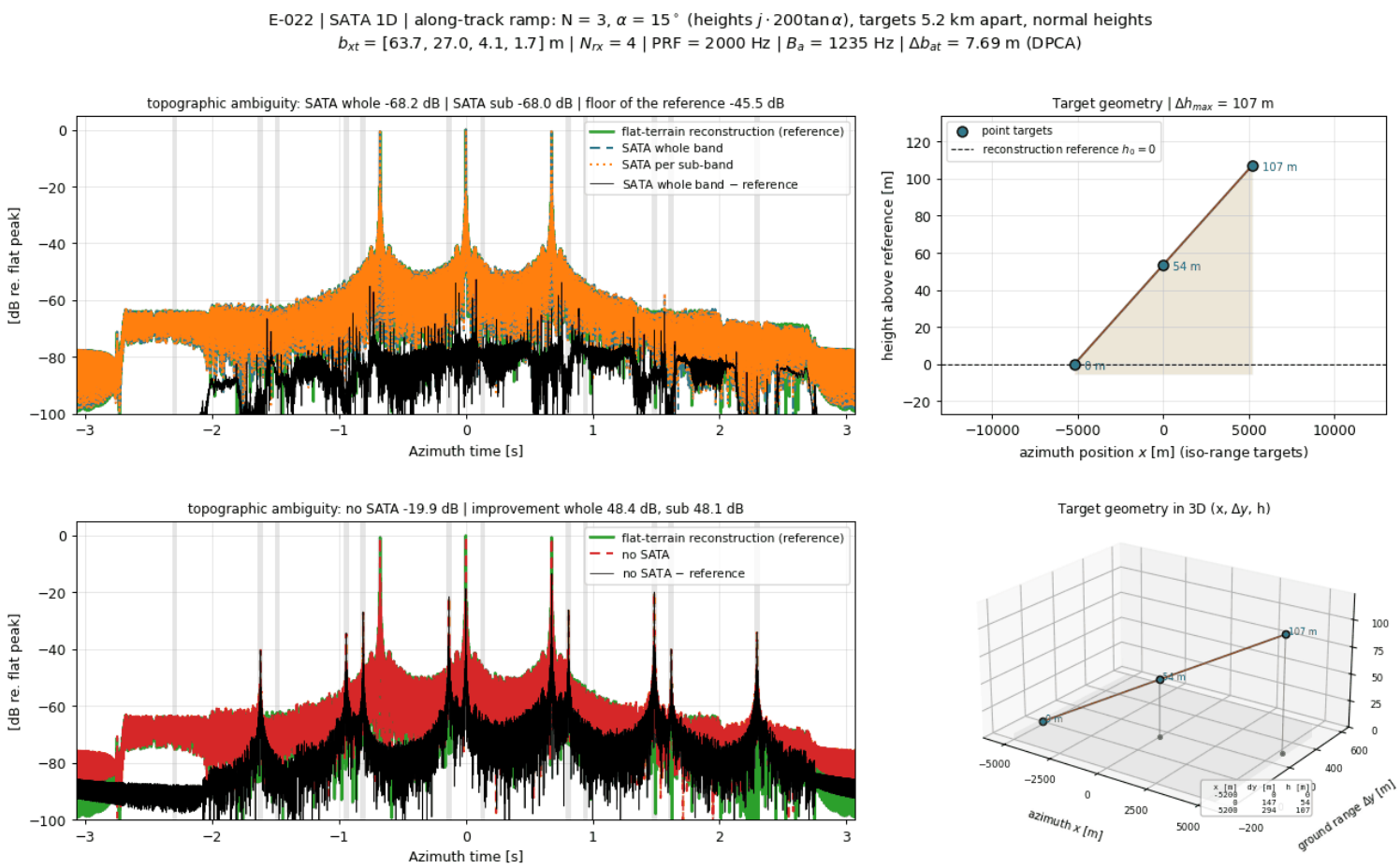


Figure 22: E-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −19.9, SATA whole band −68.2, per sub-band −68.0 dB (improvement 48.4 dB); floor −45.5 dB.

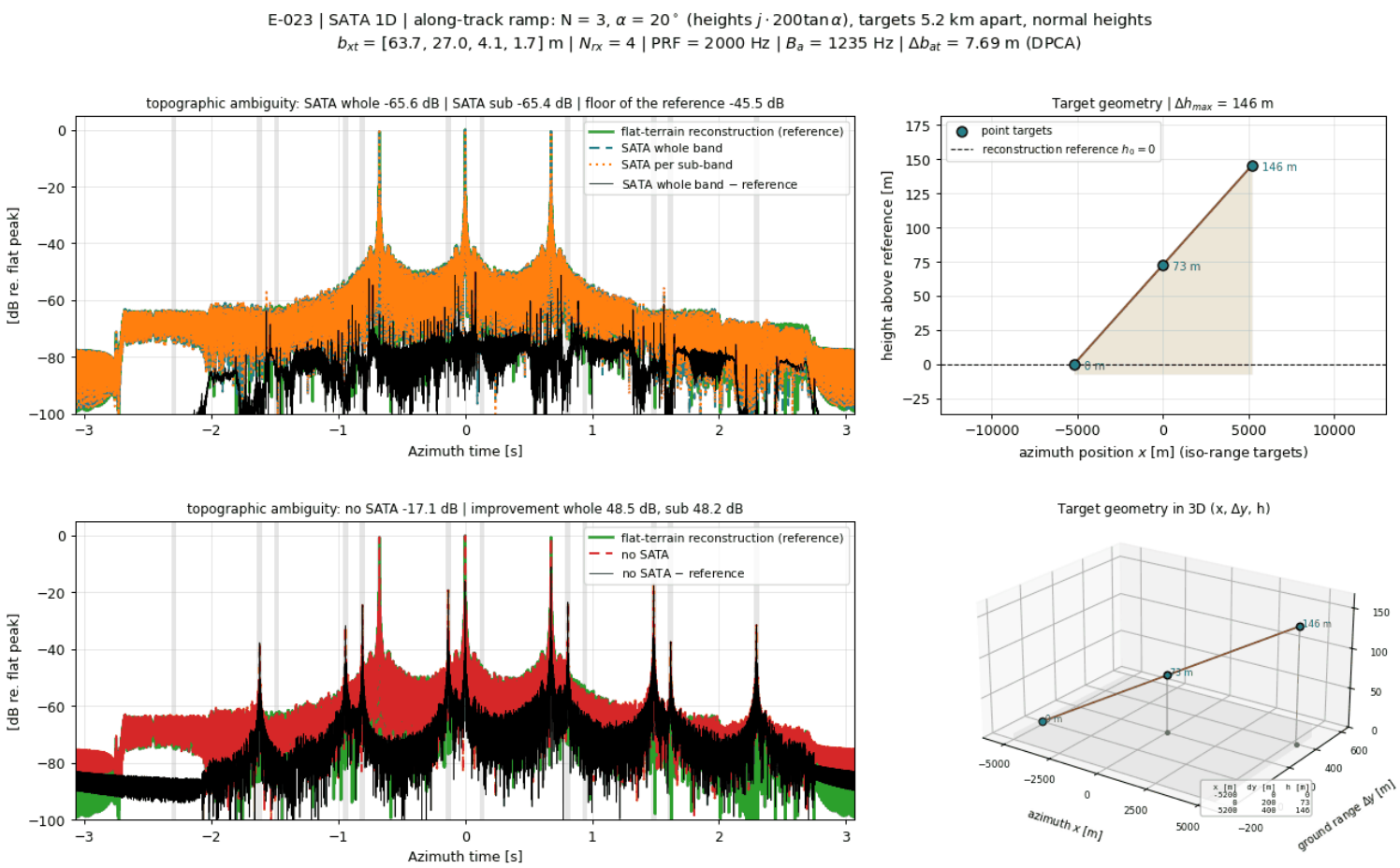


Figure 23: E-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −17.1, SATA whole band −65.6, per sub-band −65.4 dB (improvement 48.5 dB); floor −45.5 dB.

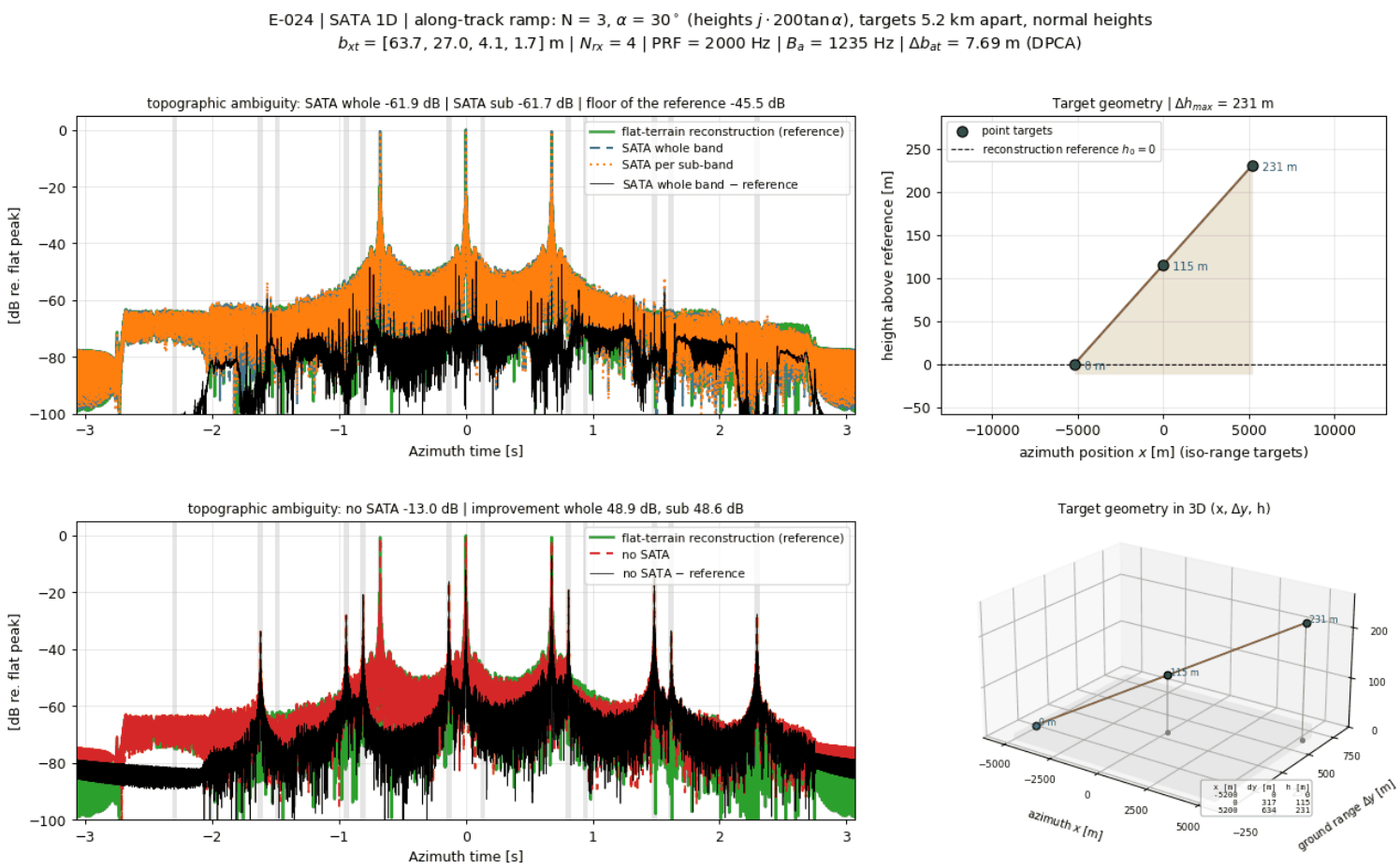


Figure 24: E-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −13.0, SATA whole band −61.9, per sub-band −61.7 dB (improvement 48.9 dB); floor −45.5 dB.

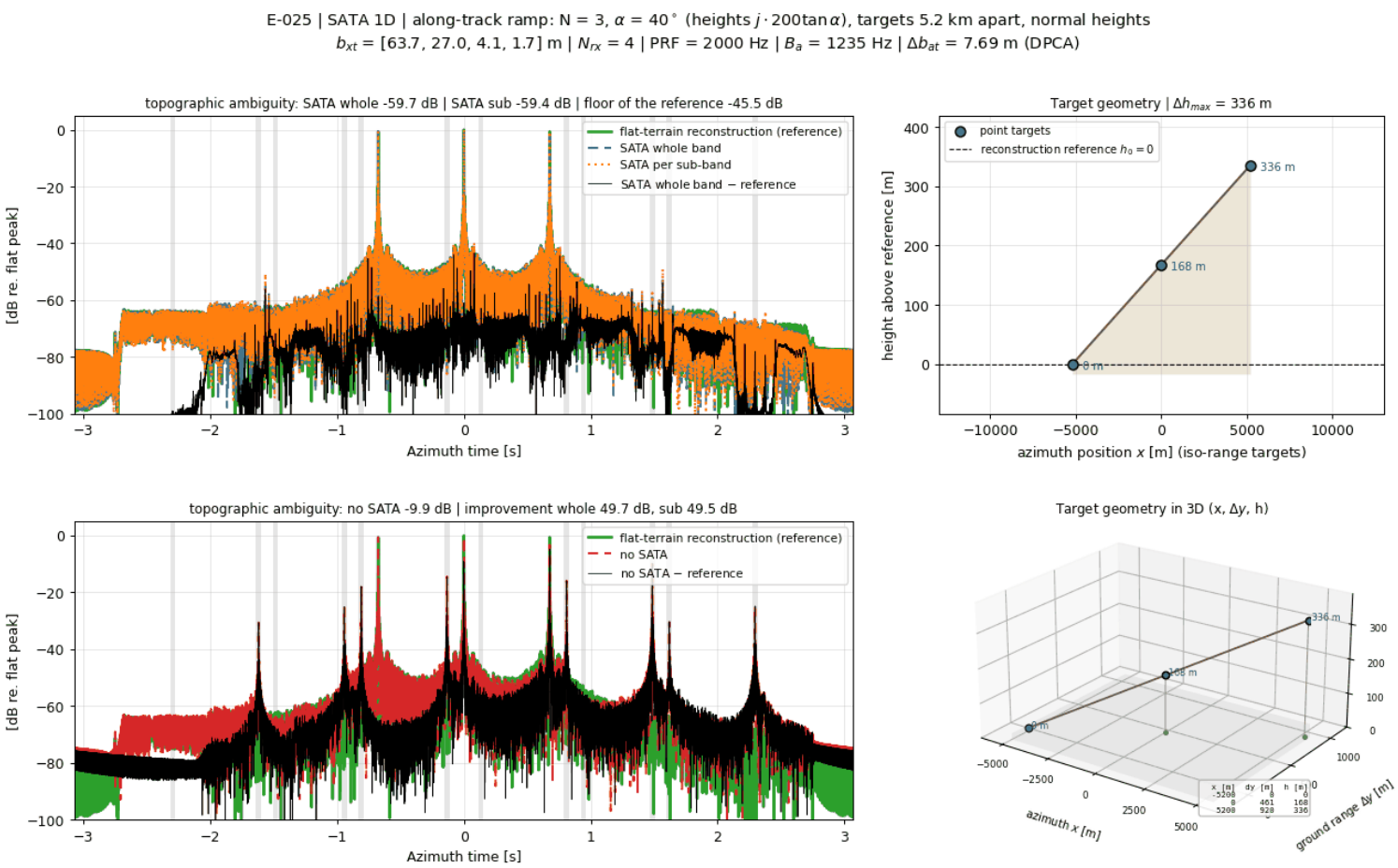


Figure 25: E-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −9.9, SATA whole band −59.7, per sub-band −59.4 dB (improvement 49.7 dB); floor −45.5 dB.

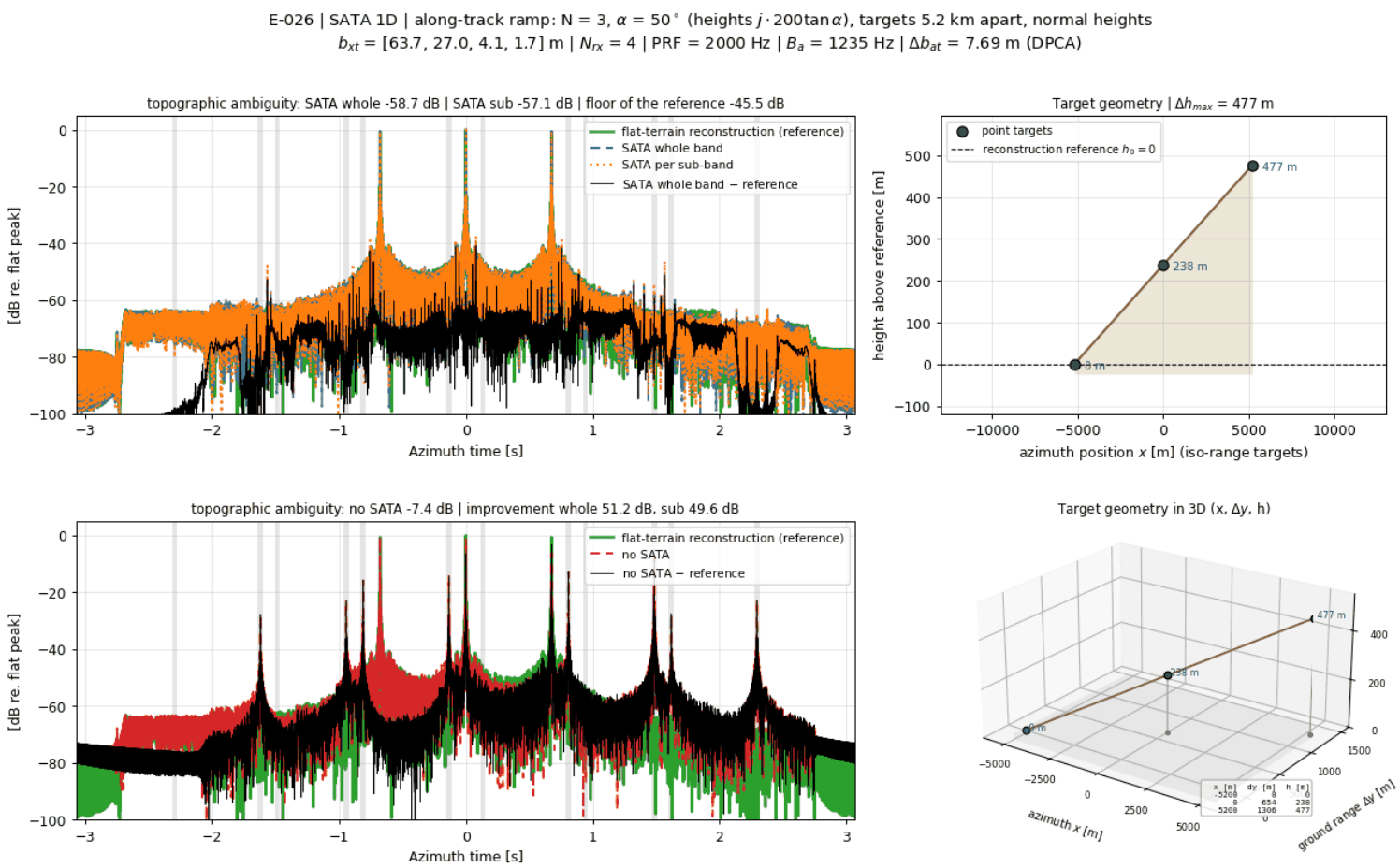


Figure 26: E-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −7.4, SATA whole band −58.7, per sub-band −57.1 dB (improvement 51.2 dB); floor −45.5 dB.

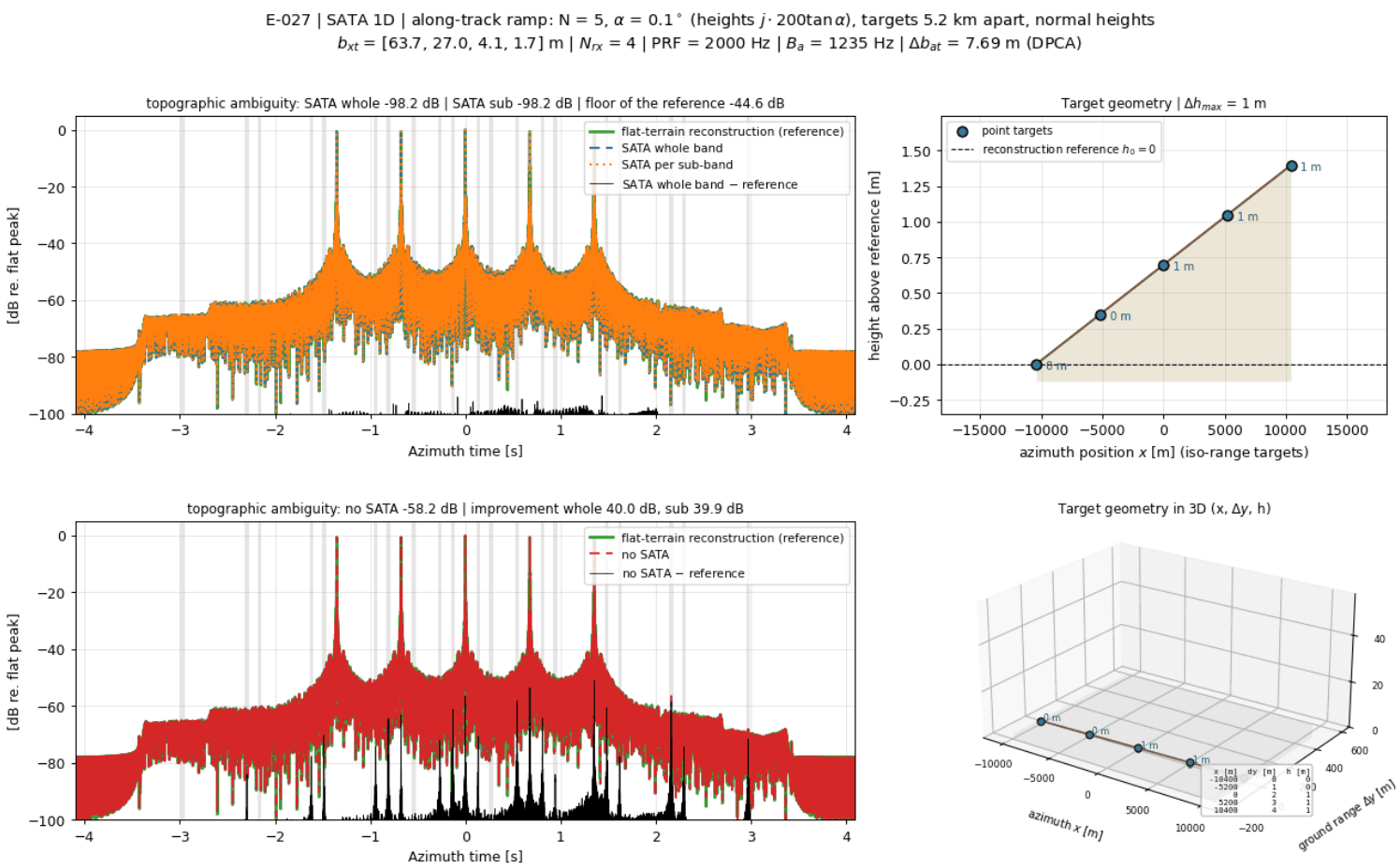


Figure 27: E-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -58.2, SATA whole band -98.2, per sub-band -98.2 dB (improvement 40.0 dB); floor -44.6 dB.

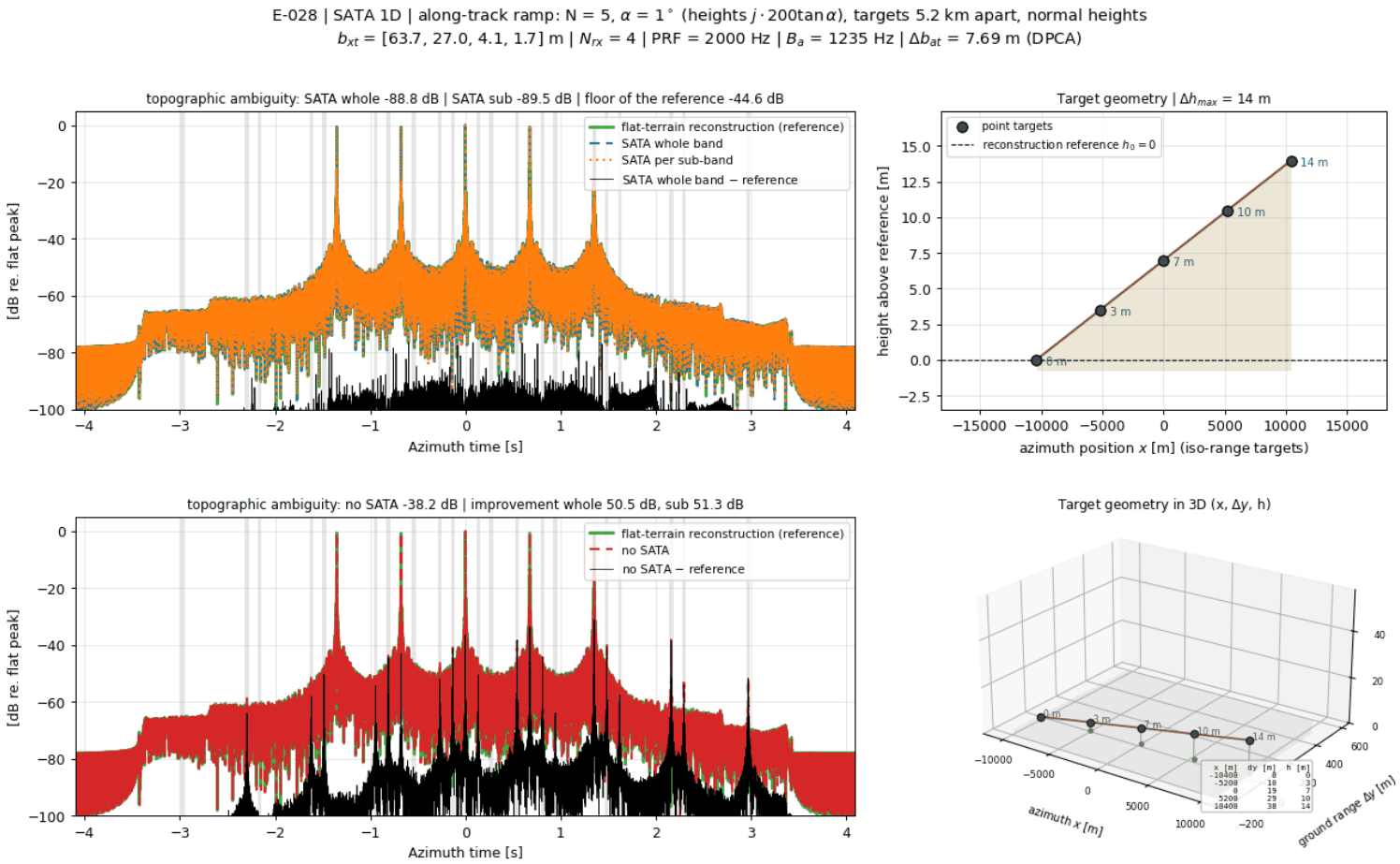


Figure 28: E-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −38.2, SATA whole band −88.8, per sub-band −89.5 dB (improvement 50.5 dB); floor −44.6 dB.

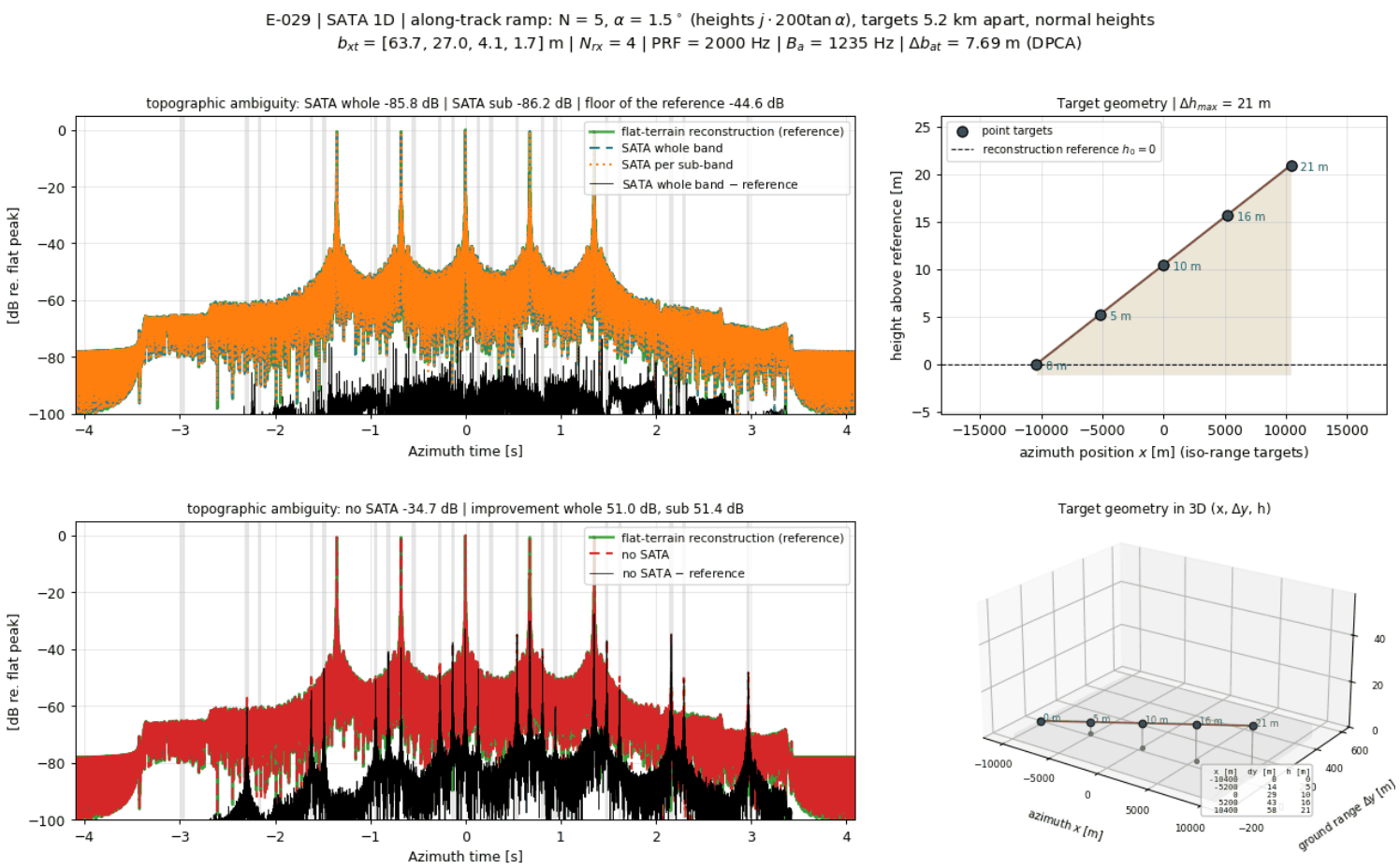


Figure 29: E-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -34.7, SATA whole band -85.8, per sub-band -86.2 dB (improvement 51.0 dB); floor -44.6 dB.

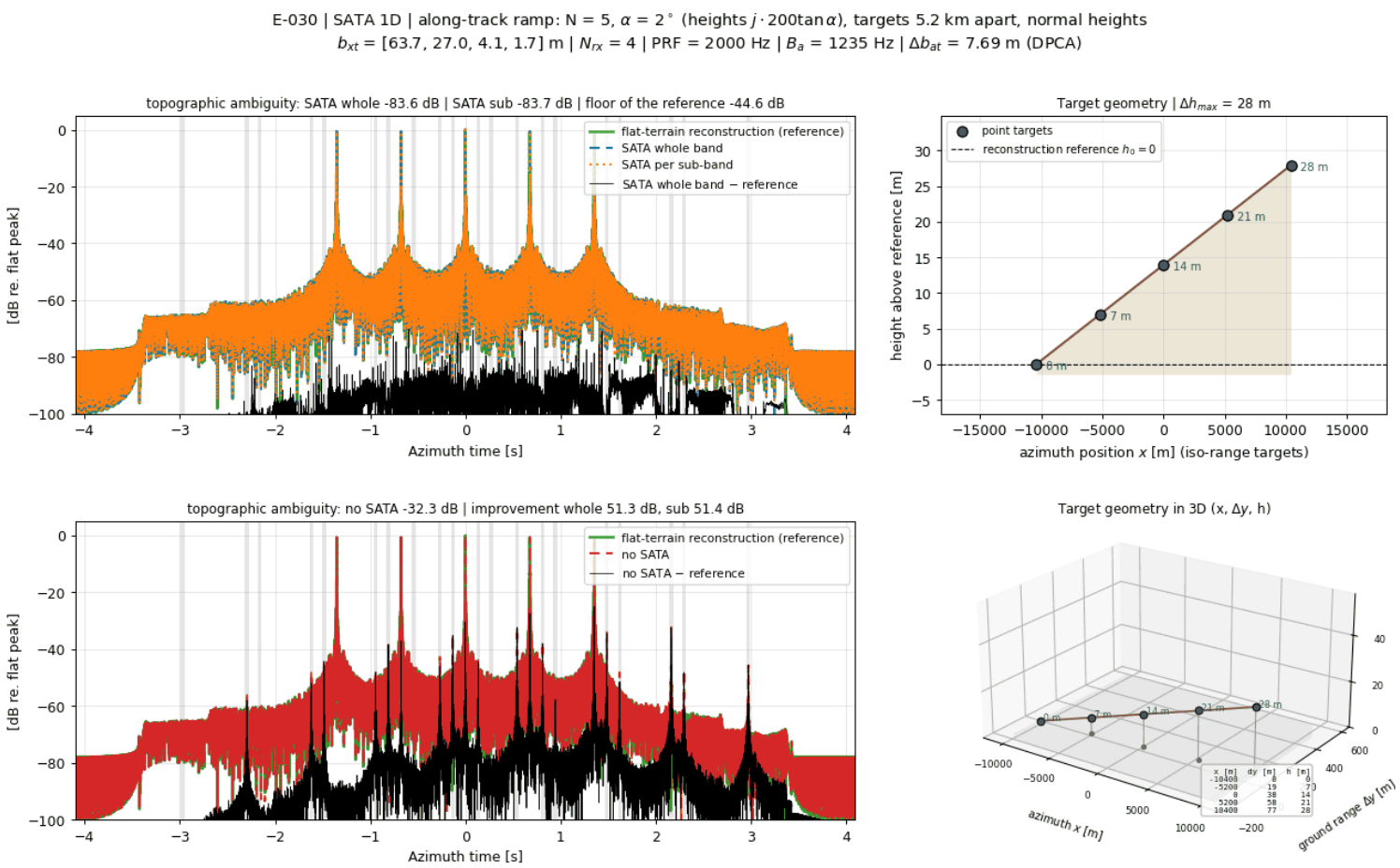


Figure 30: E-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −32.3, SATA whole band −83.6, per sub-band −83.7 dB (improvement 51.3 dB); floor −44.6 dB.

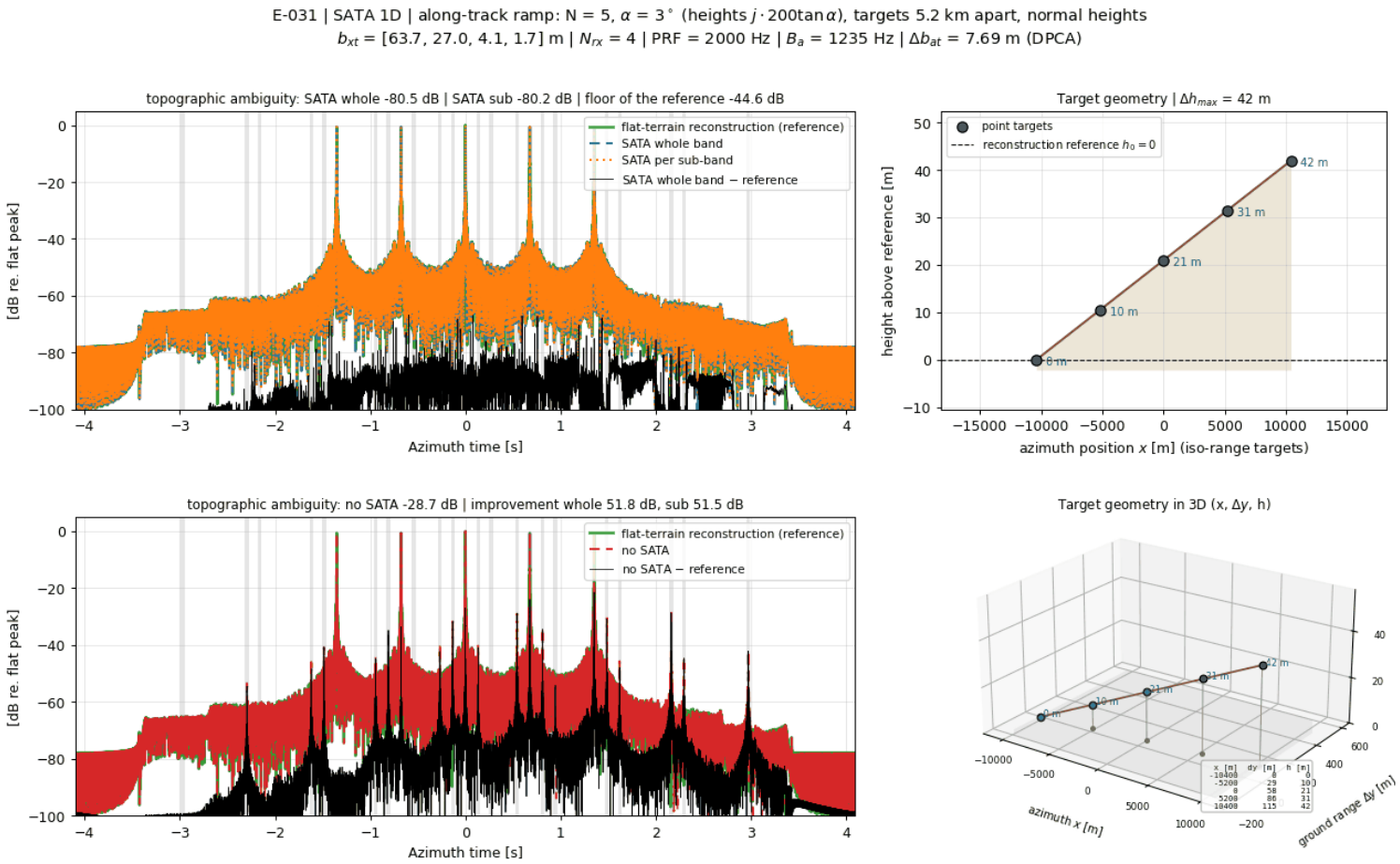


Figure 31: E-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −28.7, SATA whole band −80.5, per sub-band −80.2 dB (improvement 51.8 dB); floor −44.6 dB.

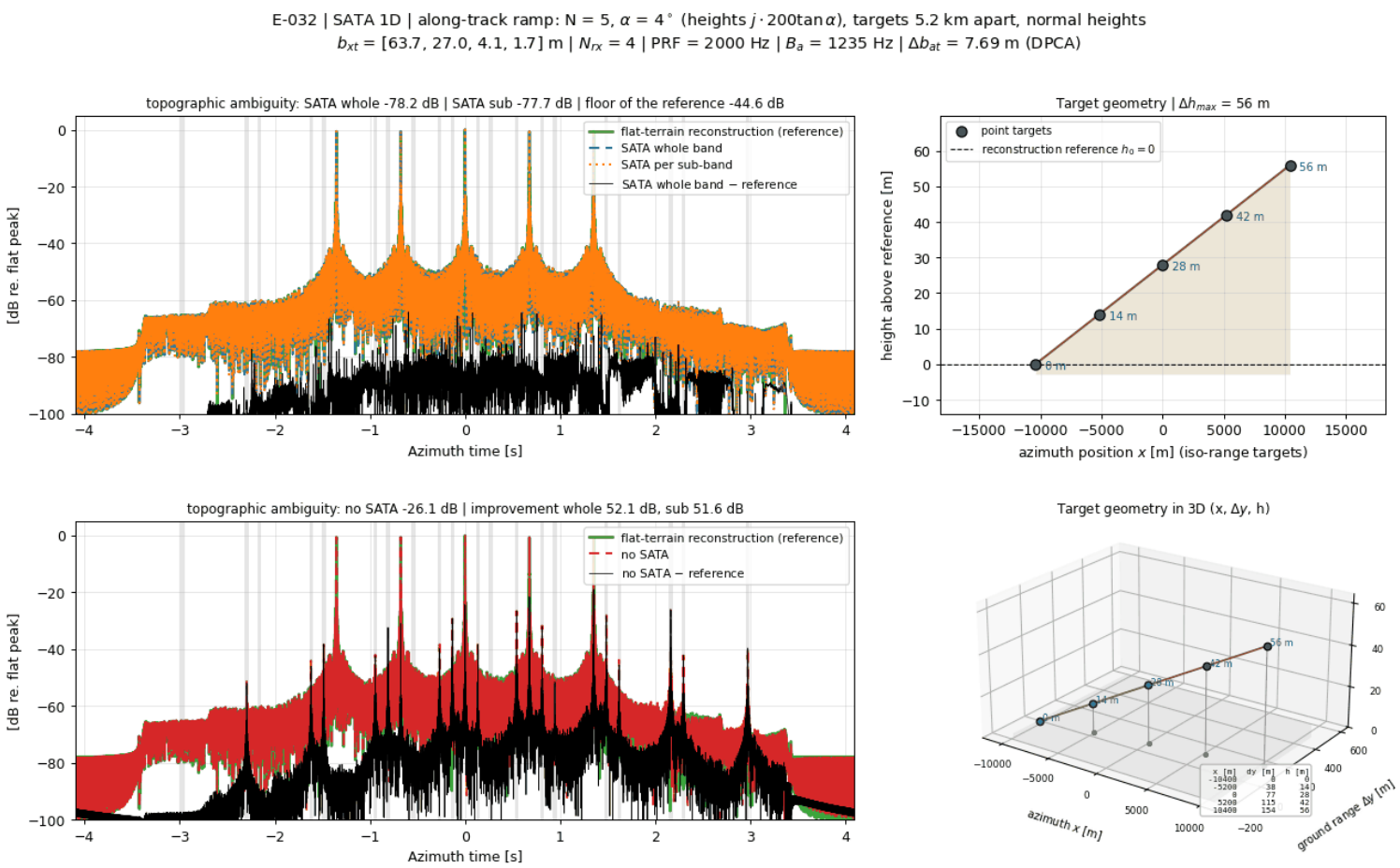


Figure 32: E-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −26.1, SATA whole band −78.2, per sub-band −77.7 dB (improvement 52.1 dB); floor −44.6 dB.

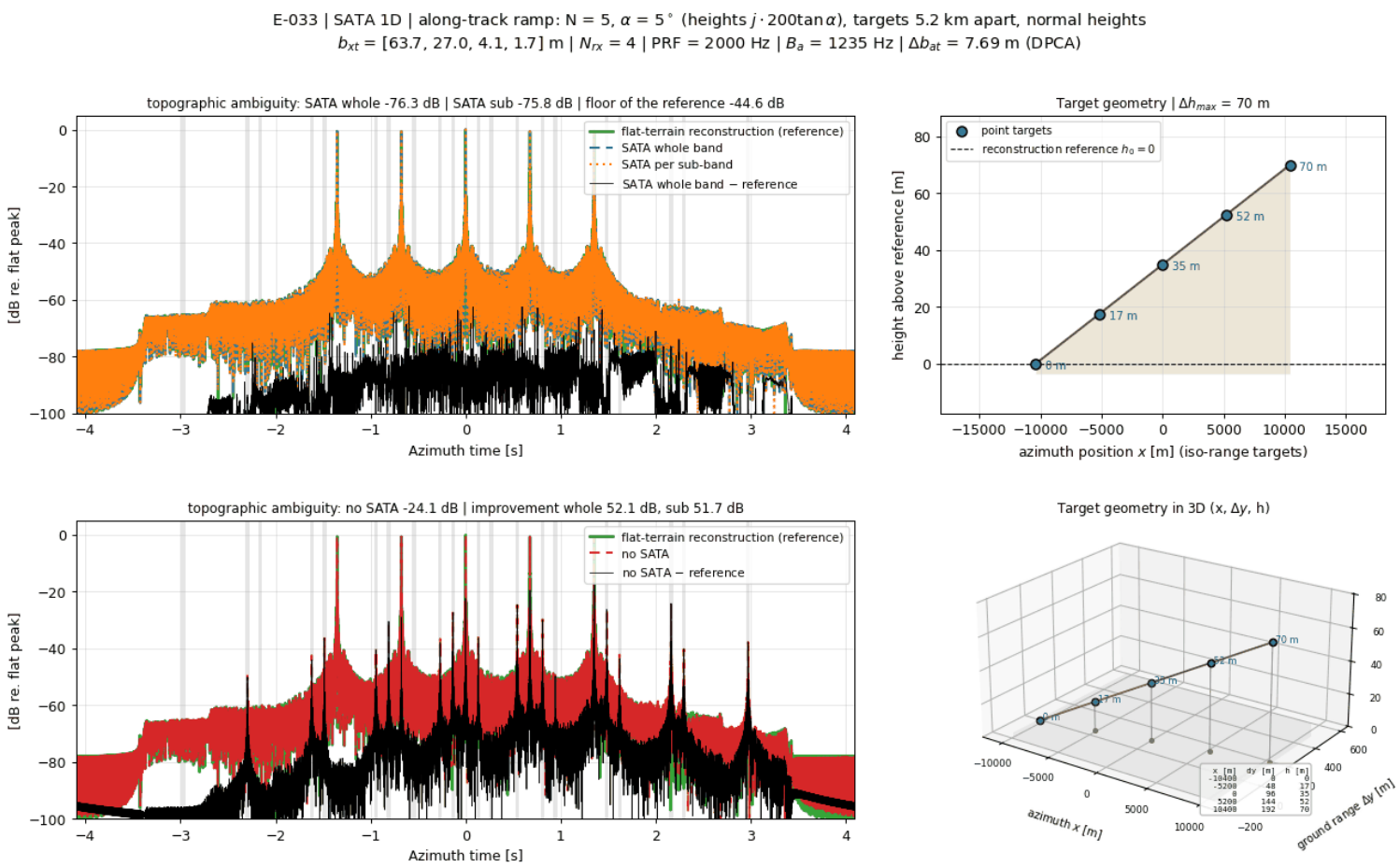


Figure 33: E-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −24.1, SATA whole band −76.3, per sub-band −75.8 dB (improvement 52.1 dB); floor −44.6 dB.

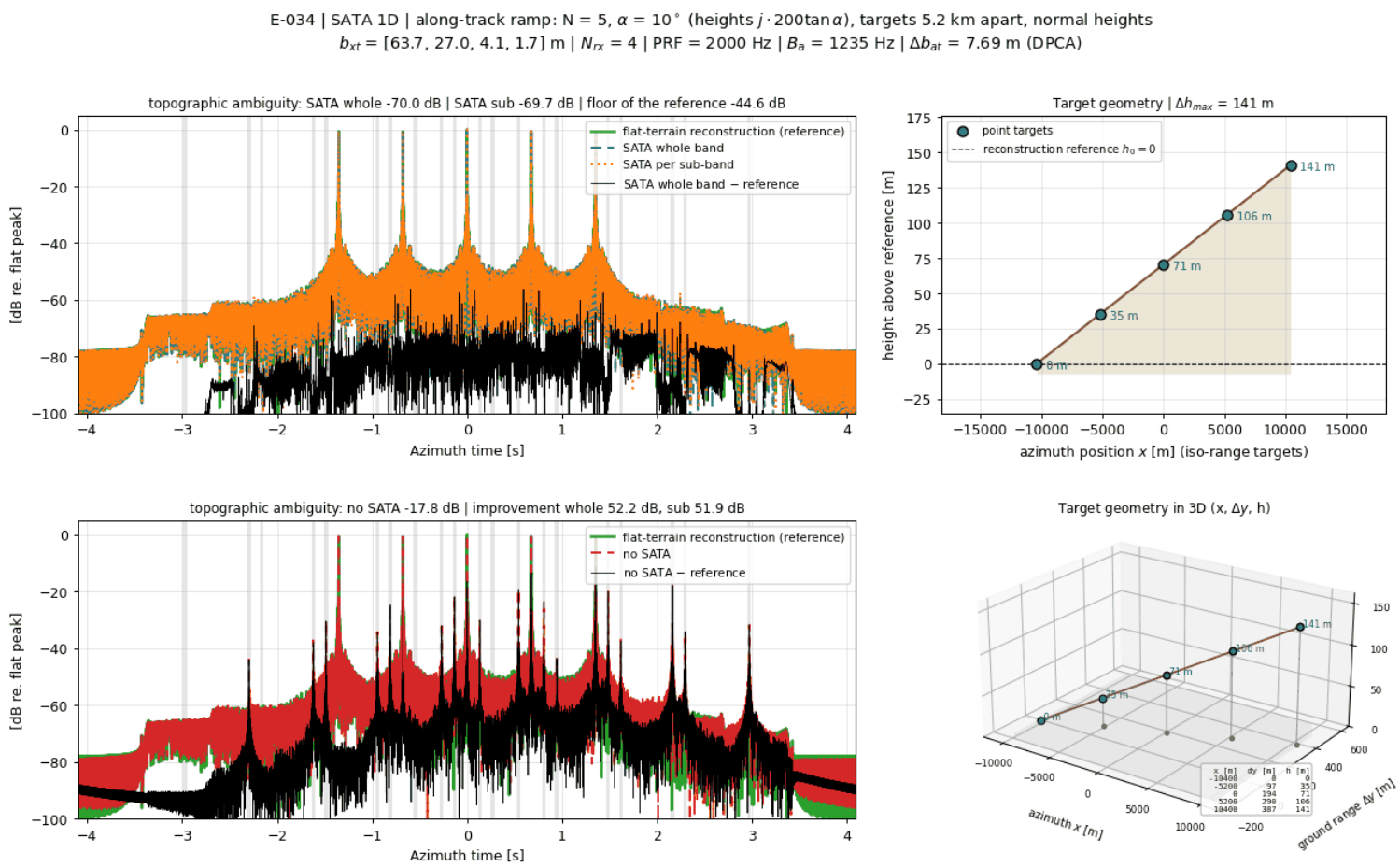


Figure 34: E-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −17.8, SATA whole band −70.0, per sub-band −69.7 dB (improvement 52.2 dB); floor −44.6 dB.

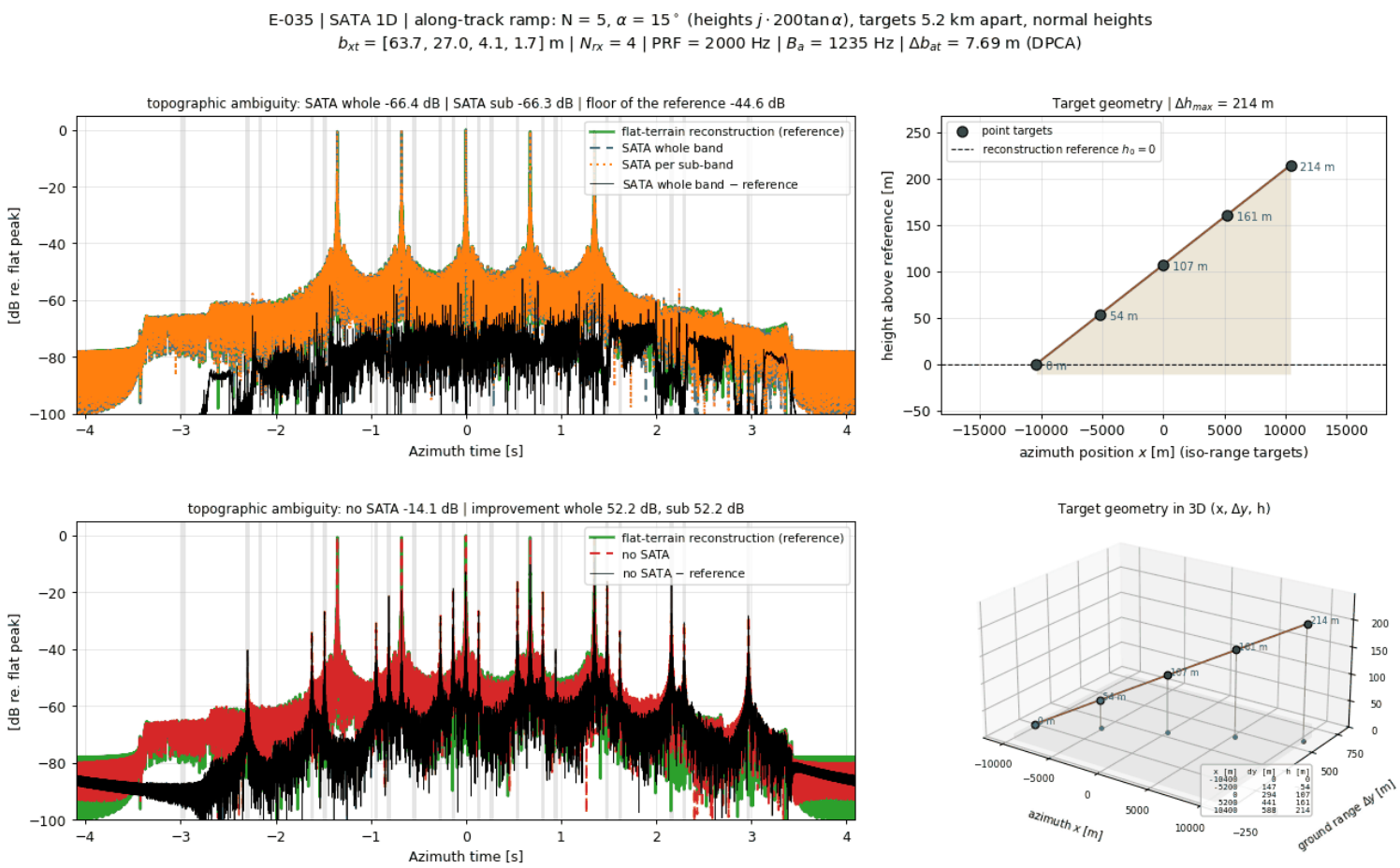


Figure 35: E-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −14.1, SATA whole band −66.4, per sub-band −66.3 dB (improvement 52.2 dB); floor −44.6 dB.

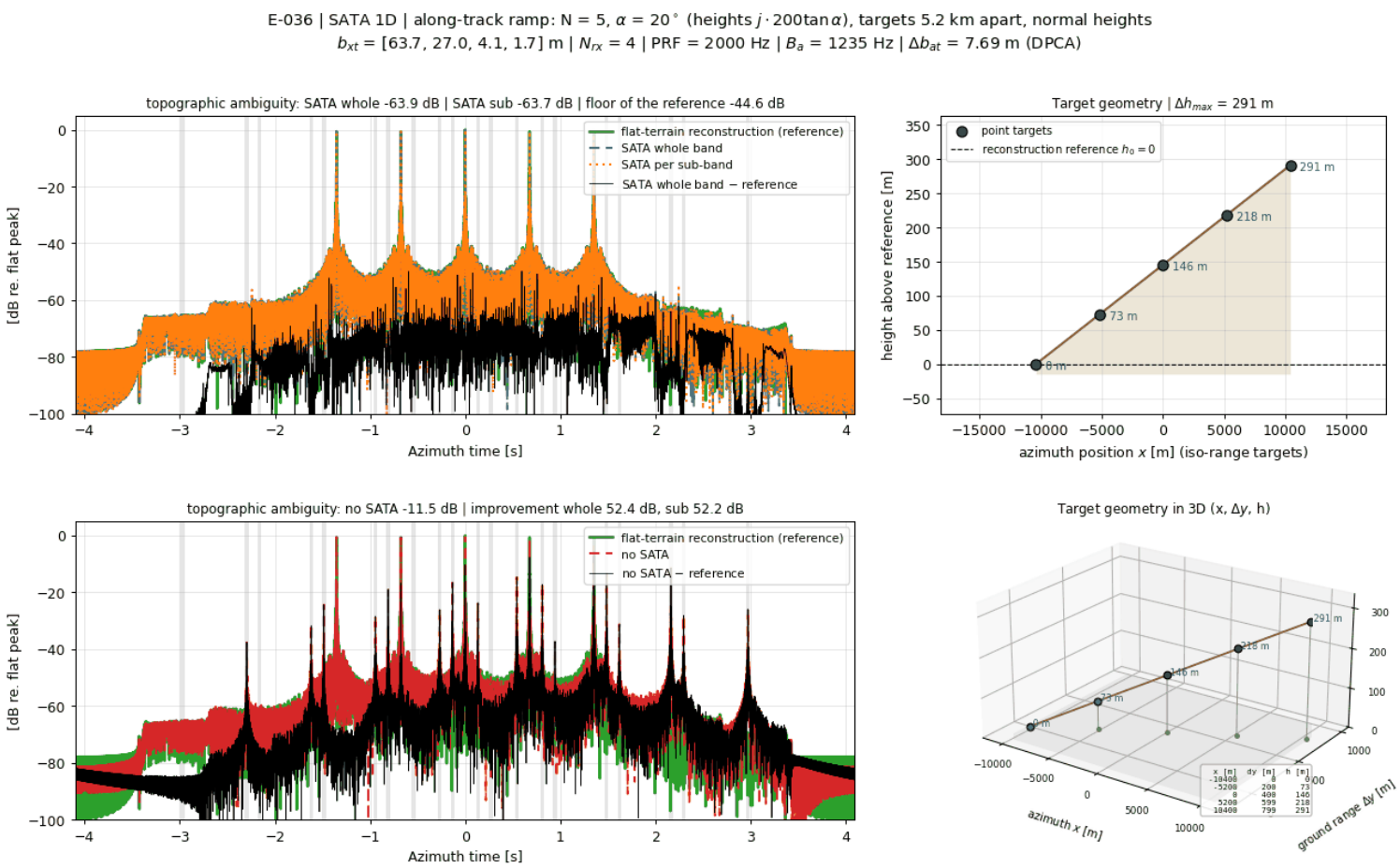


Figure 36: E-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -11.5, SATA whole band -63.9, per sub-band -63.7 dB (improvement 52.4 dB); floor -44.6 dB.

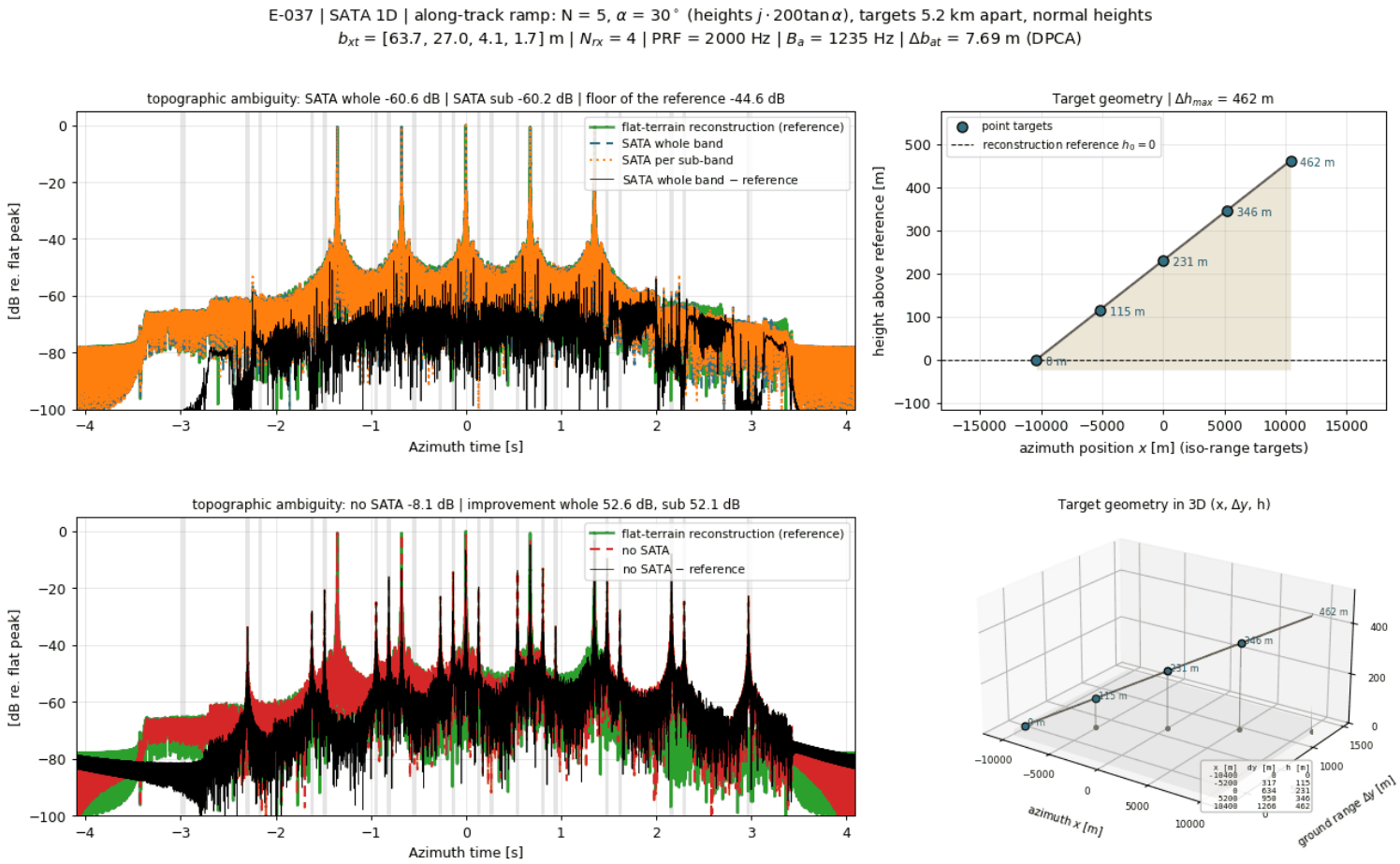


Figure 37: E-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −8.1, SATA whole band −60.6, per sub-band −60.2 dB (improvement 52.6 dB); floor −44.6 dB.

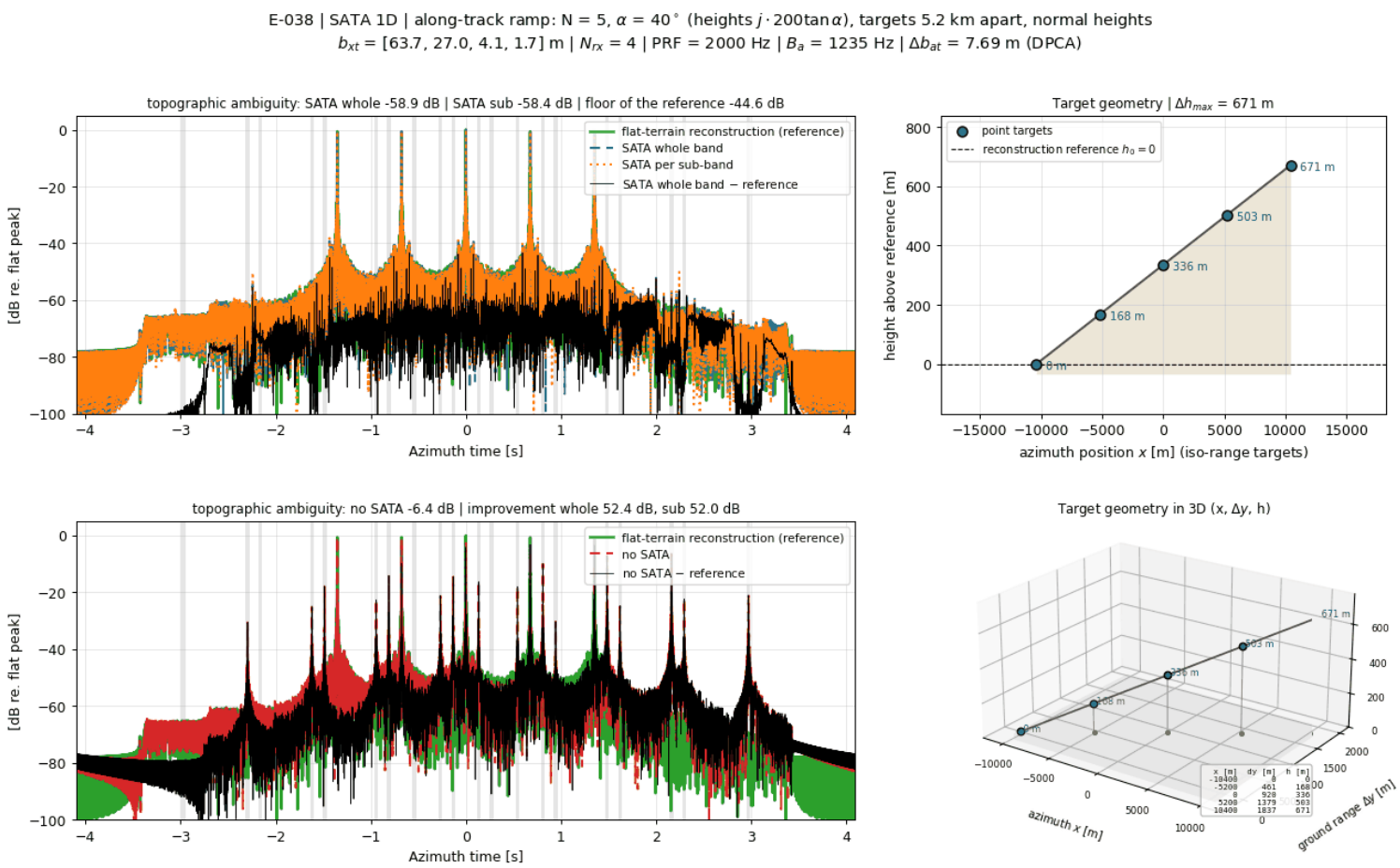


Figure 38: E-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −6.4, SATA whole band −58.9, per sub-band −58.4 dB (improvement 52.4 dB); floor −44.6 dB.

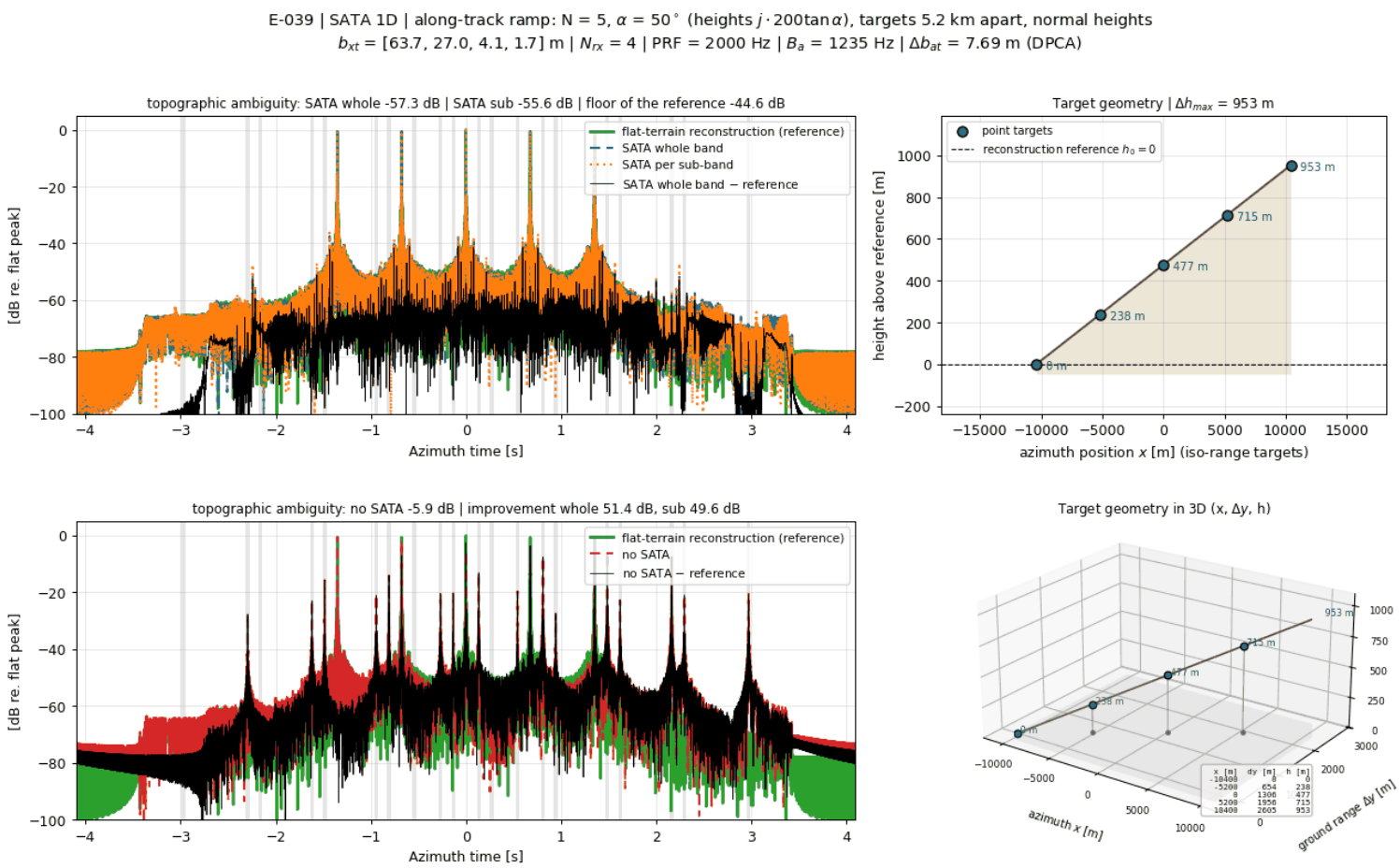


Figure 39: E-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -5.9, SATA whole band -57.3, per sub-band -55.6 dB (improvement 51.4 dB); floor -44.6 dB.

2 Configuration F: along-track ramps, targets 5.2 km apart, large heights

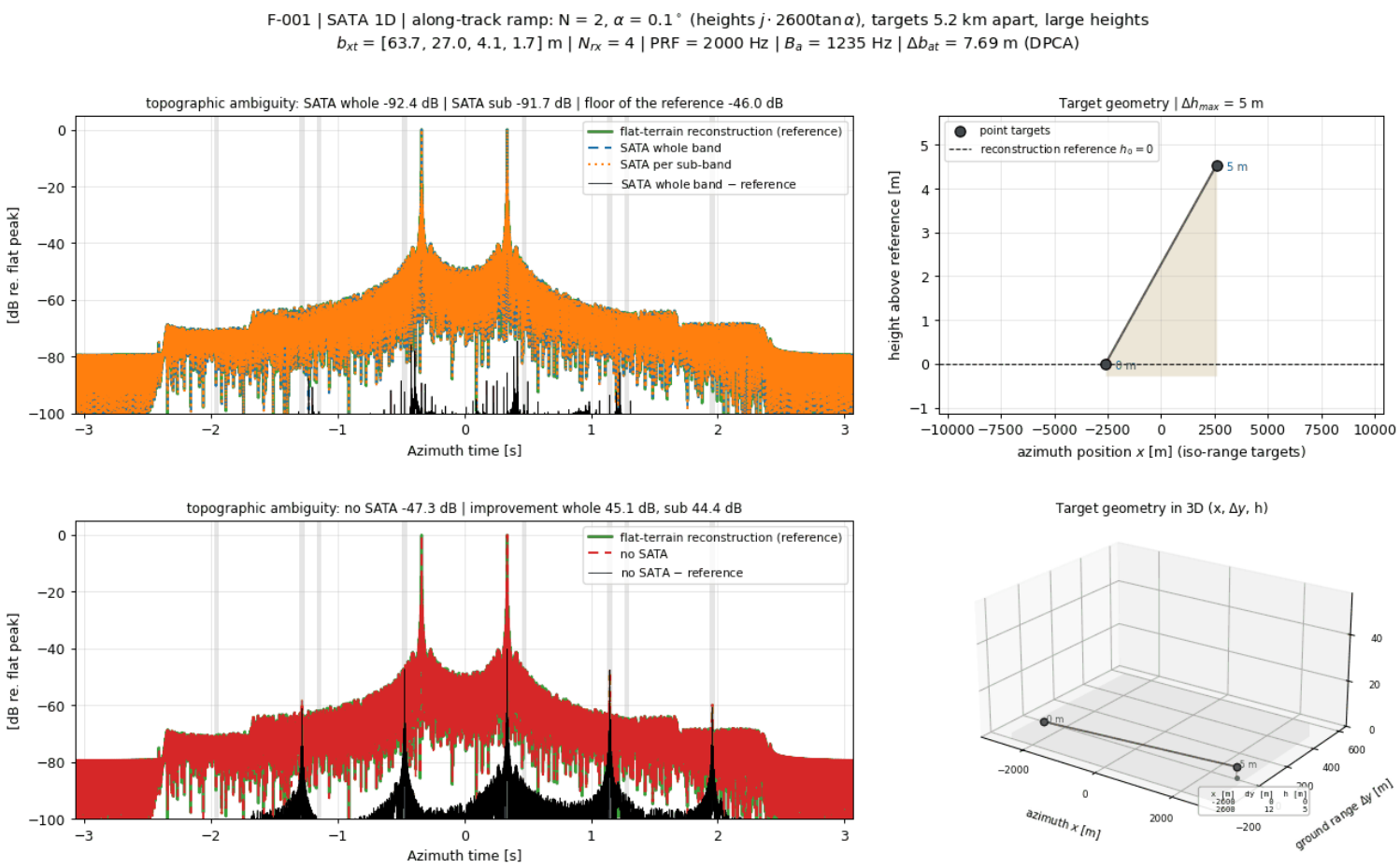


Figure 40: F-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −47.3, SATA whole band −92.4, per sub-band −91.7 dB (improvement 45.1 dB); floor −46.0 dB.

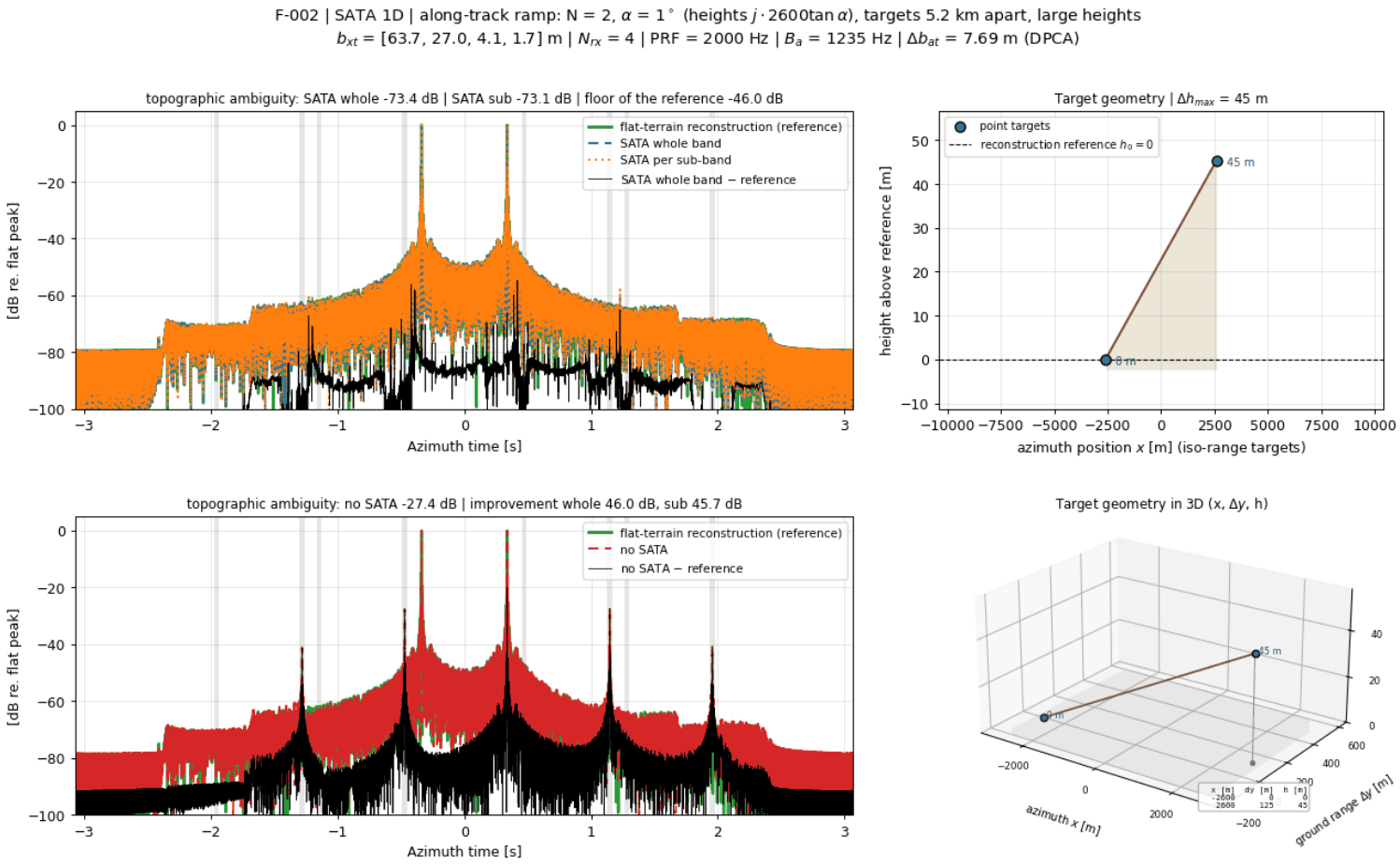


Figure 41: F-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −27.4, SATA whole band −73.4, per sub-band −73.1 dB (improvement 46.0 dB); floor −46.0 dB.

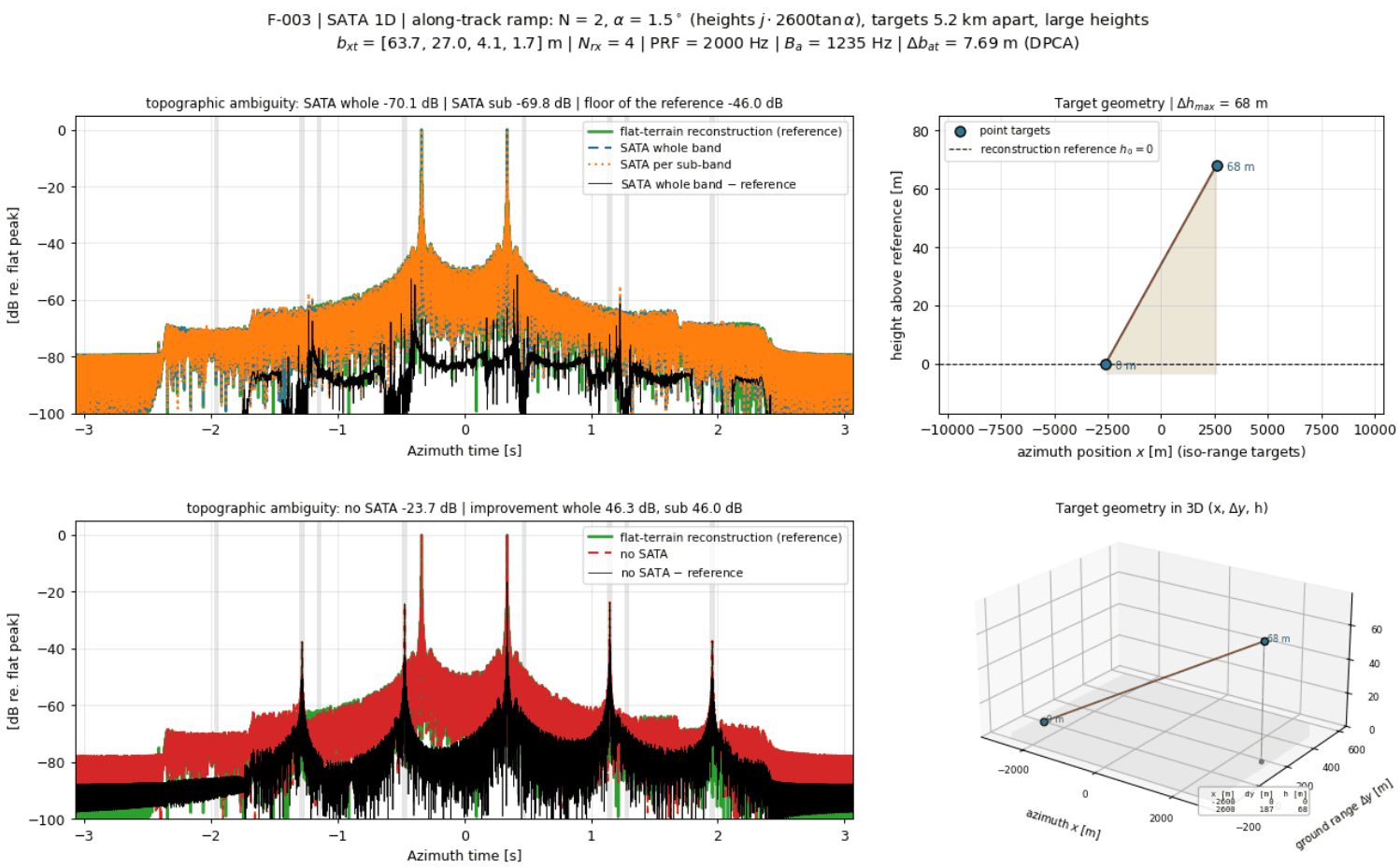


Figure 42: F-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −23.7, SATA whole band −70.1, per sub-band −69.8 dB (improvement 46.3 dB); floor −46.0 dB.

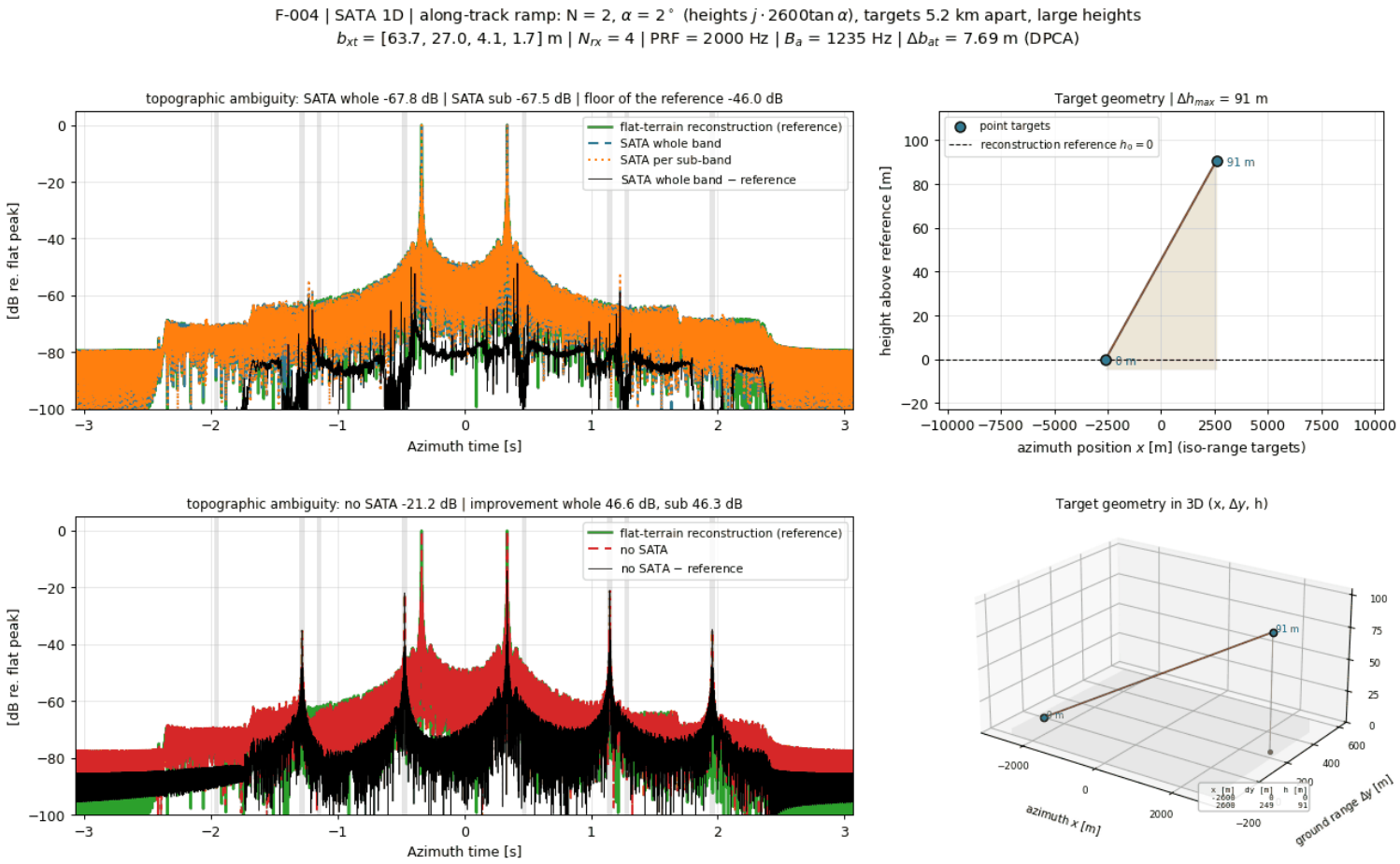


Figure 43: F-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −21.2, SATA whole band −67.8, per sub-band −67.5 dB (improvement 46.6 dB); floor −46.0 dB.

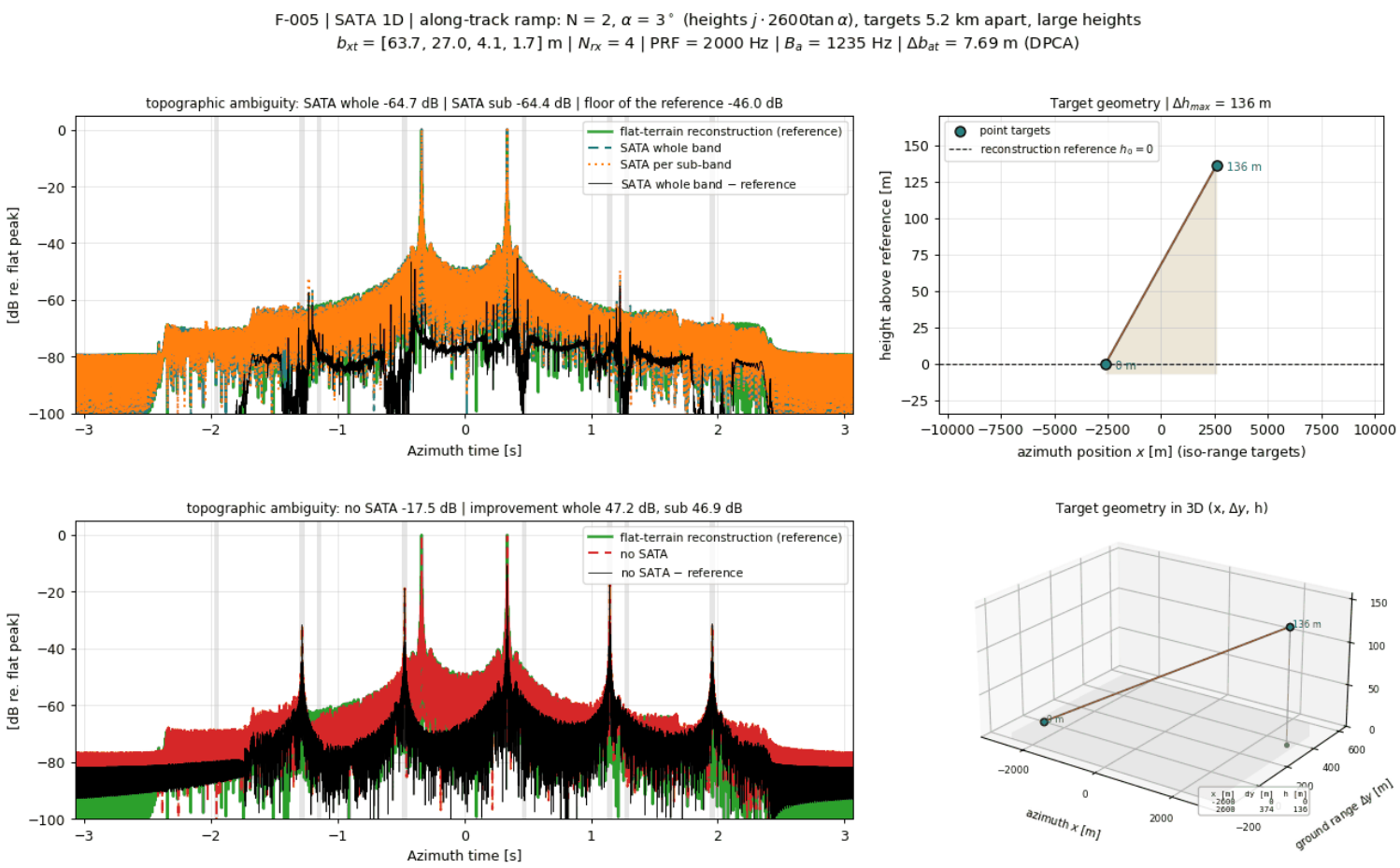


Figure 44: F-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −17.5, SATA whole band −64.7, per sub-band −64.4 dB (improvement 47.2 dB); floor −46.0 dB.

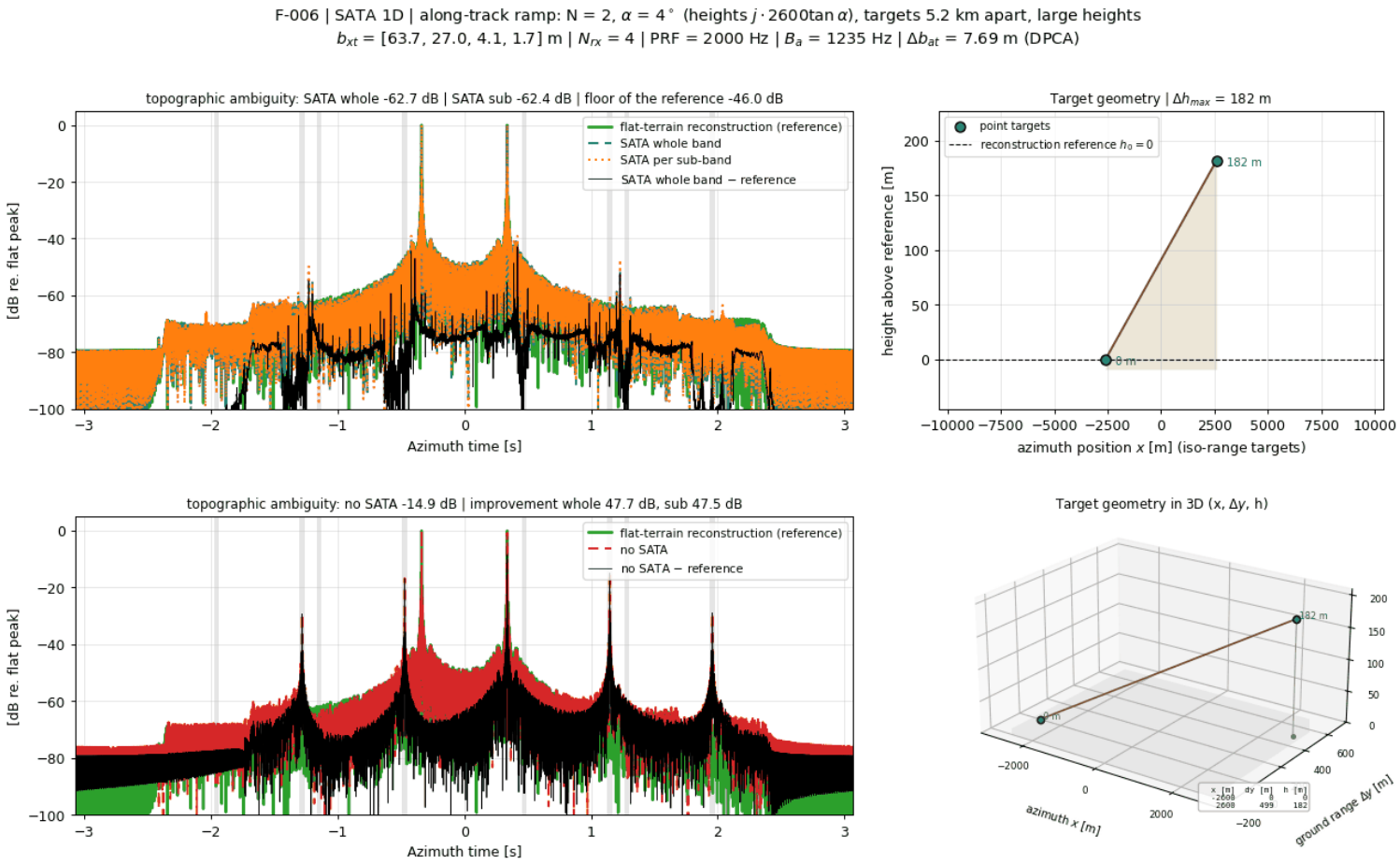


Figure 45: F-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −14.9, SATA whole band −62.7, per sub-band −62.4 dB (improvement 47.7 dB); floor −46.0 dB.

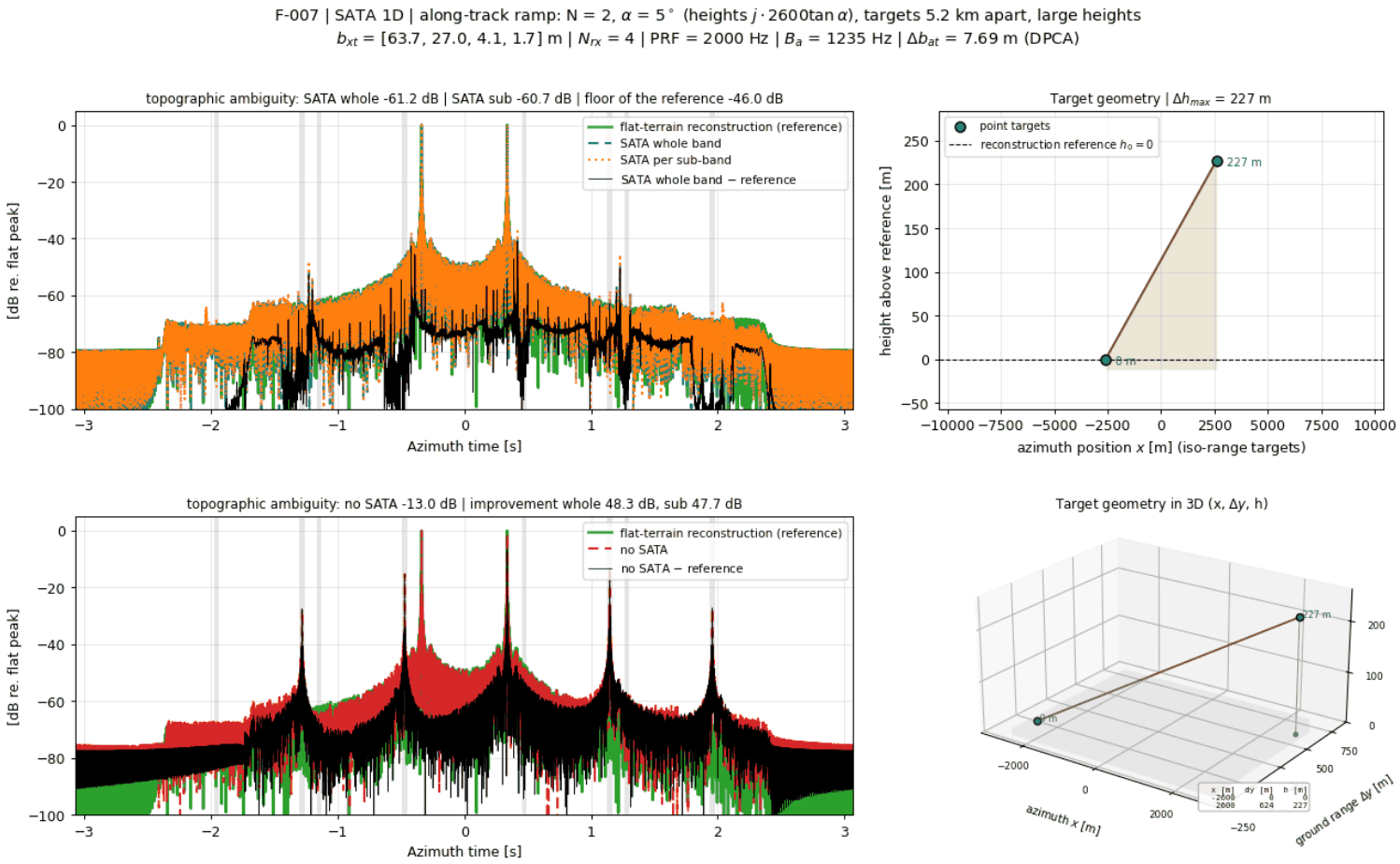


Figure 46: F-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −13.0, SATA whole band −61.2, per sub-band −60.7 dB (improvement 48.3 dB); floor −46.0 dB.

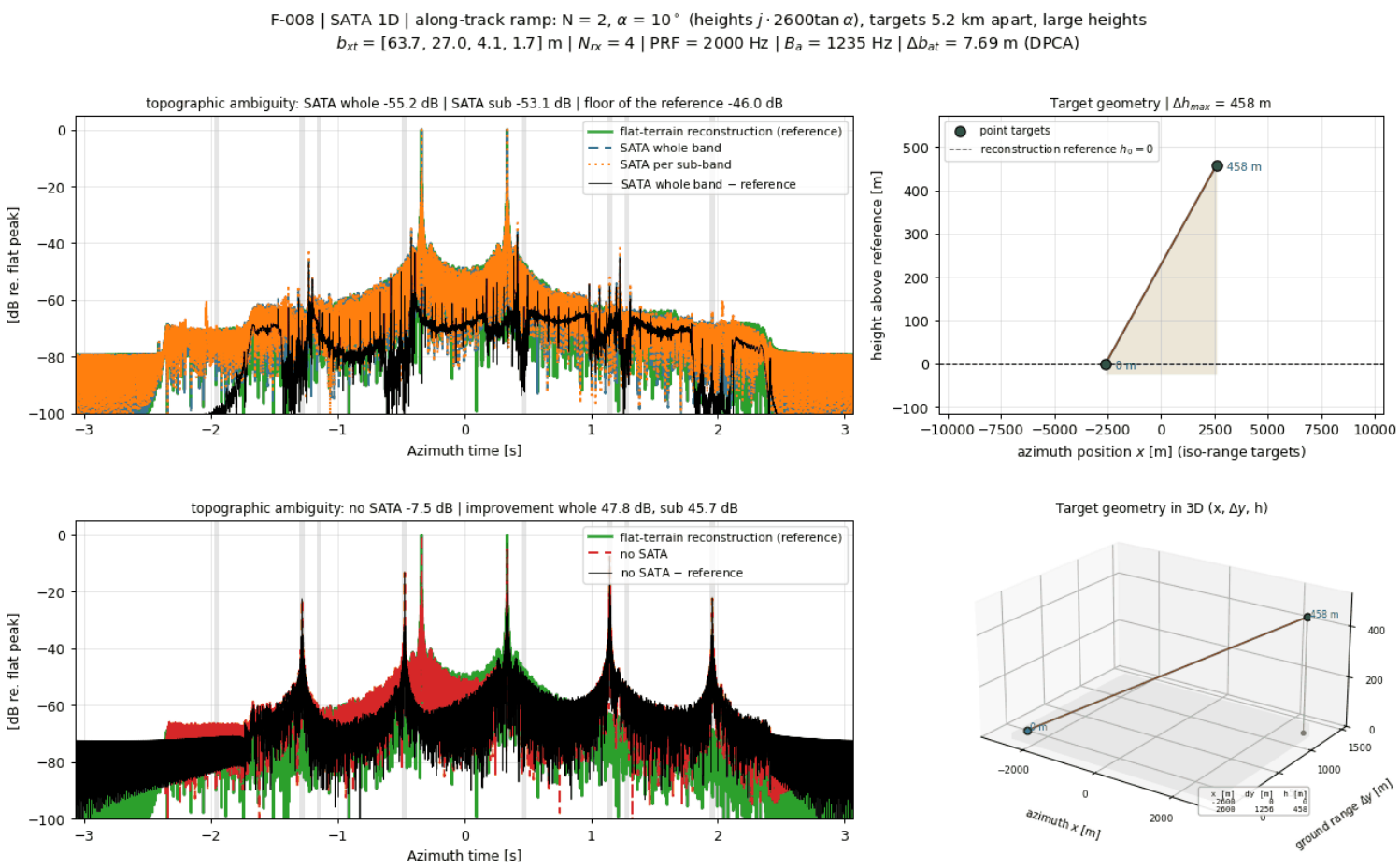


Figure 47: F-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.5, SATA whole band −55.2, per sub-band −53.1 dB (improvement 47.8 dB); floor −46.0 dB.

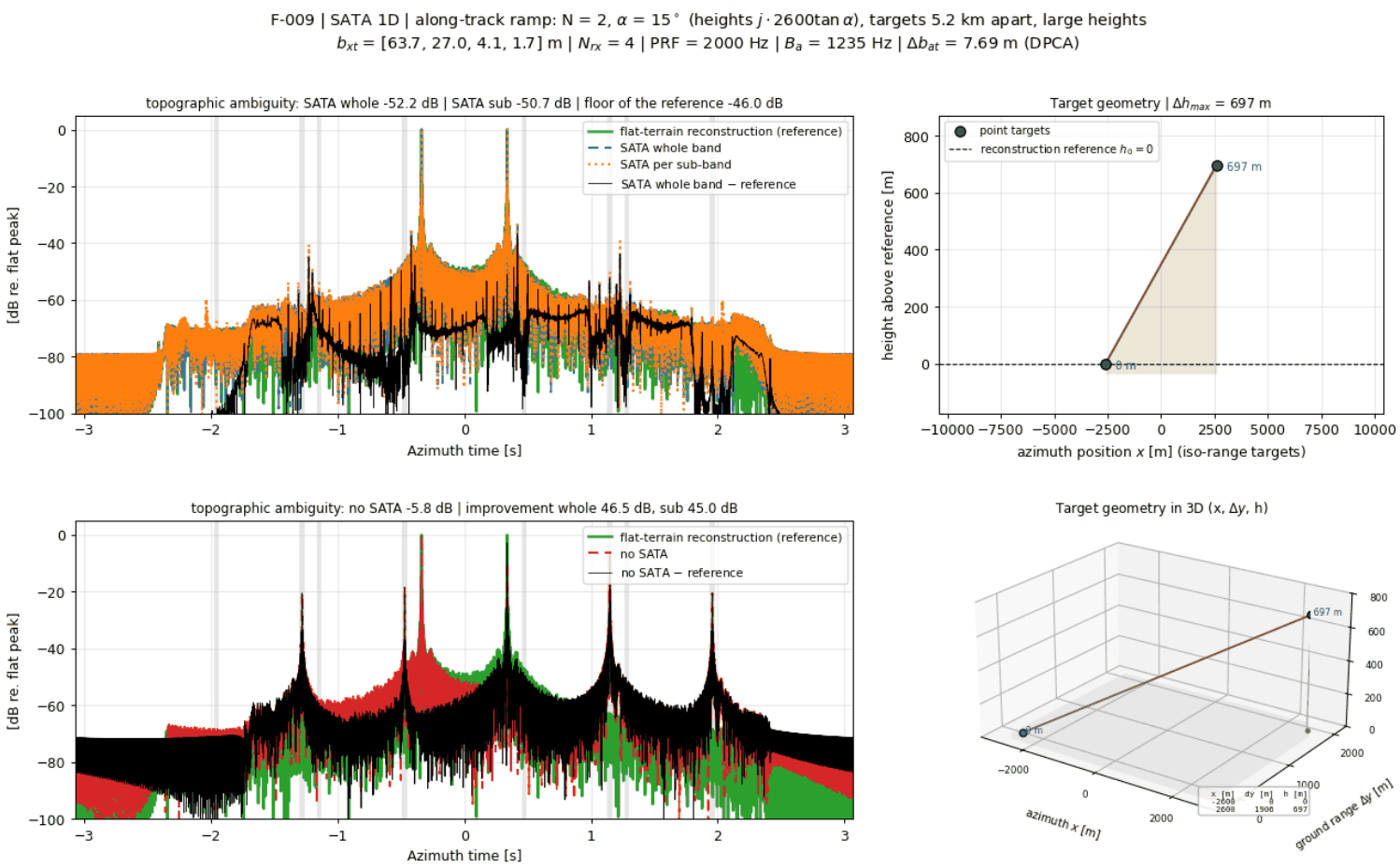


Figure 48: F-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −5.8, SATA whole band −52.2, per sub-band −50.7 dB (improvement 46.5 dB); floor −46.0 dB.

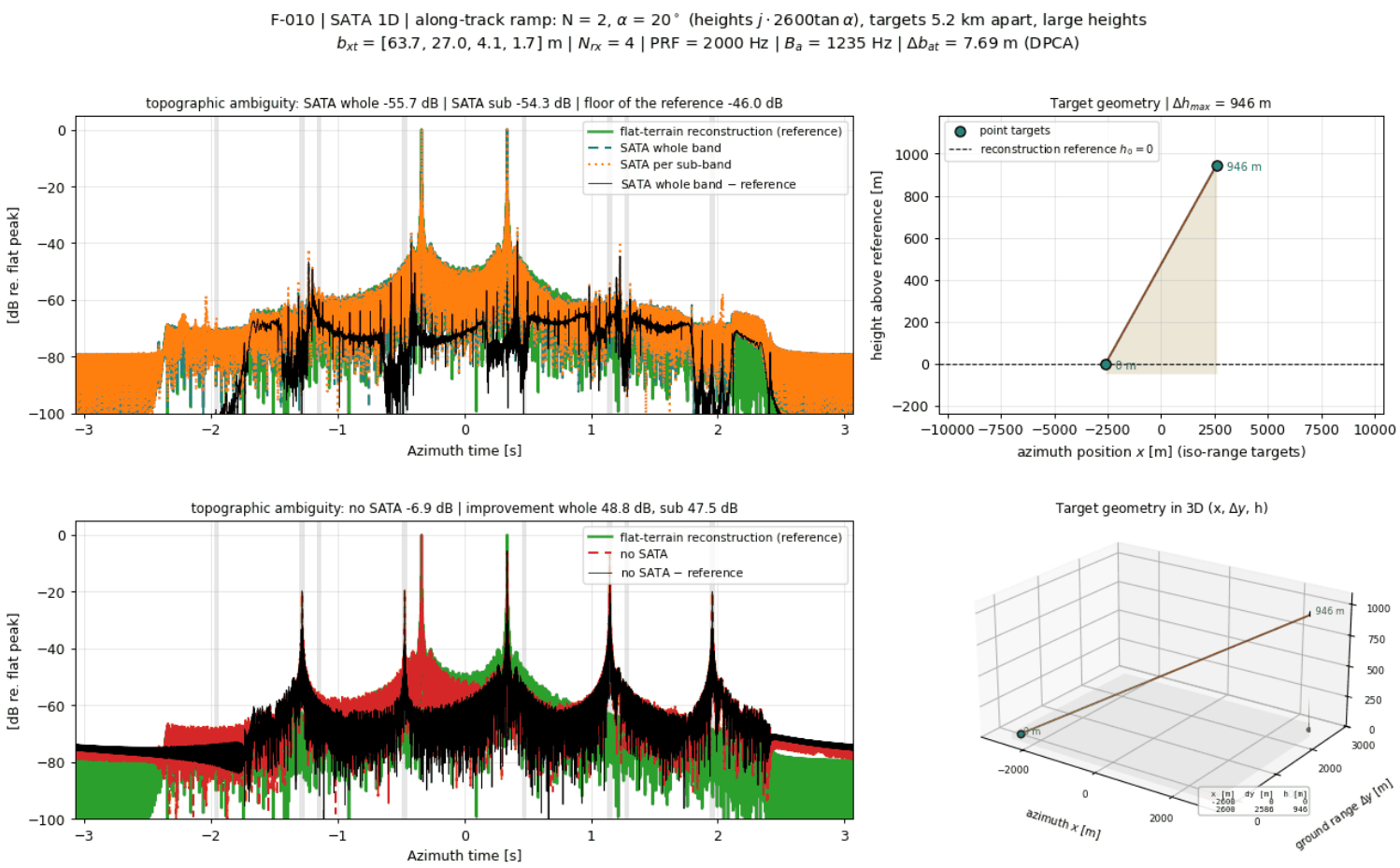


Figure 49: F-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.9, SATA whole band −55.7, per sub-band −54.3 dB (improvement 48.8 dB); floor −46.0 dB.

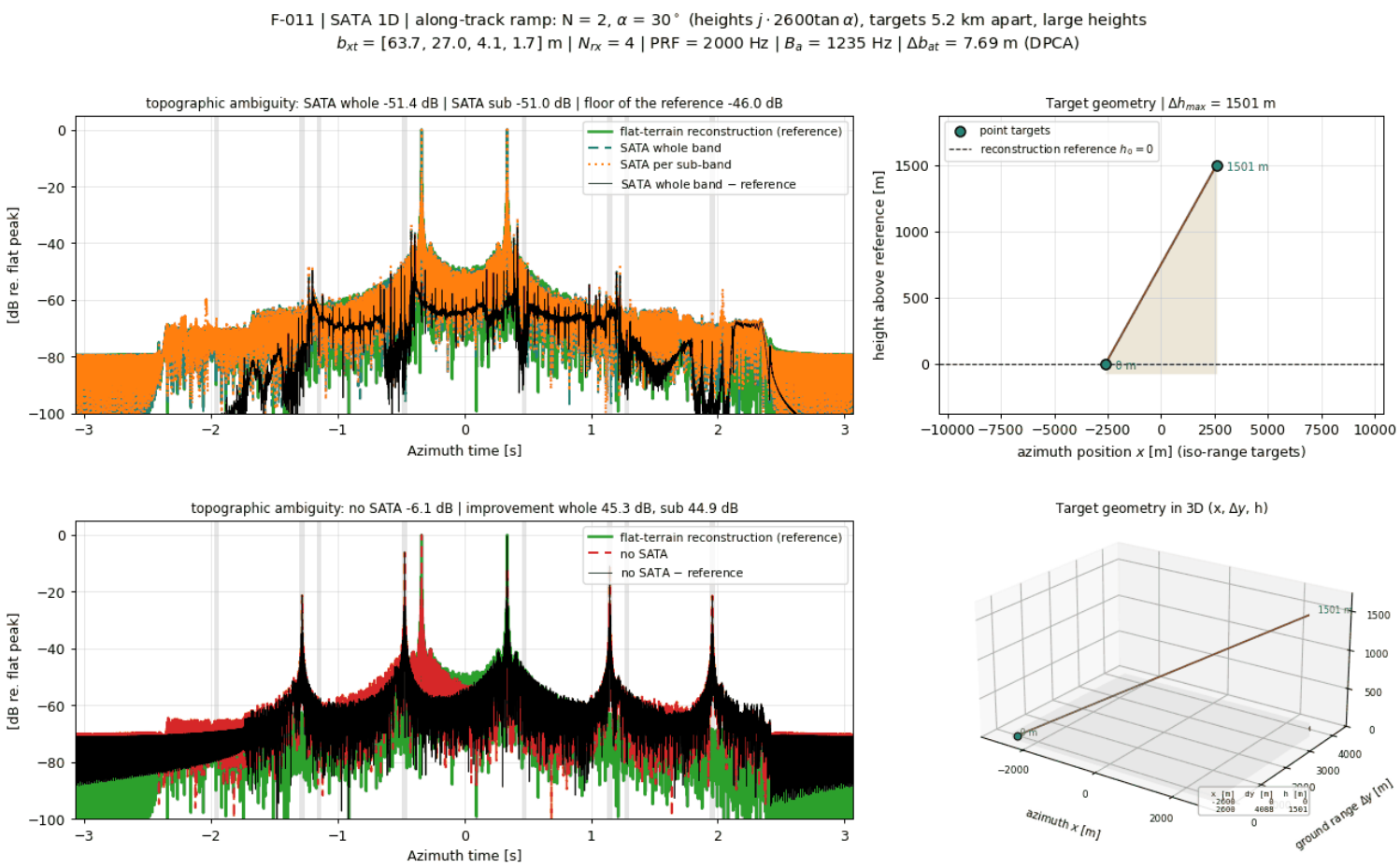


Figure 50: F-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.1, SATA whole band −51.4, per sub-band −51.0 dB (improvement 45.3 dB); floor −46.0 dB.

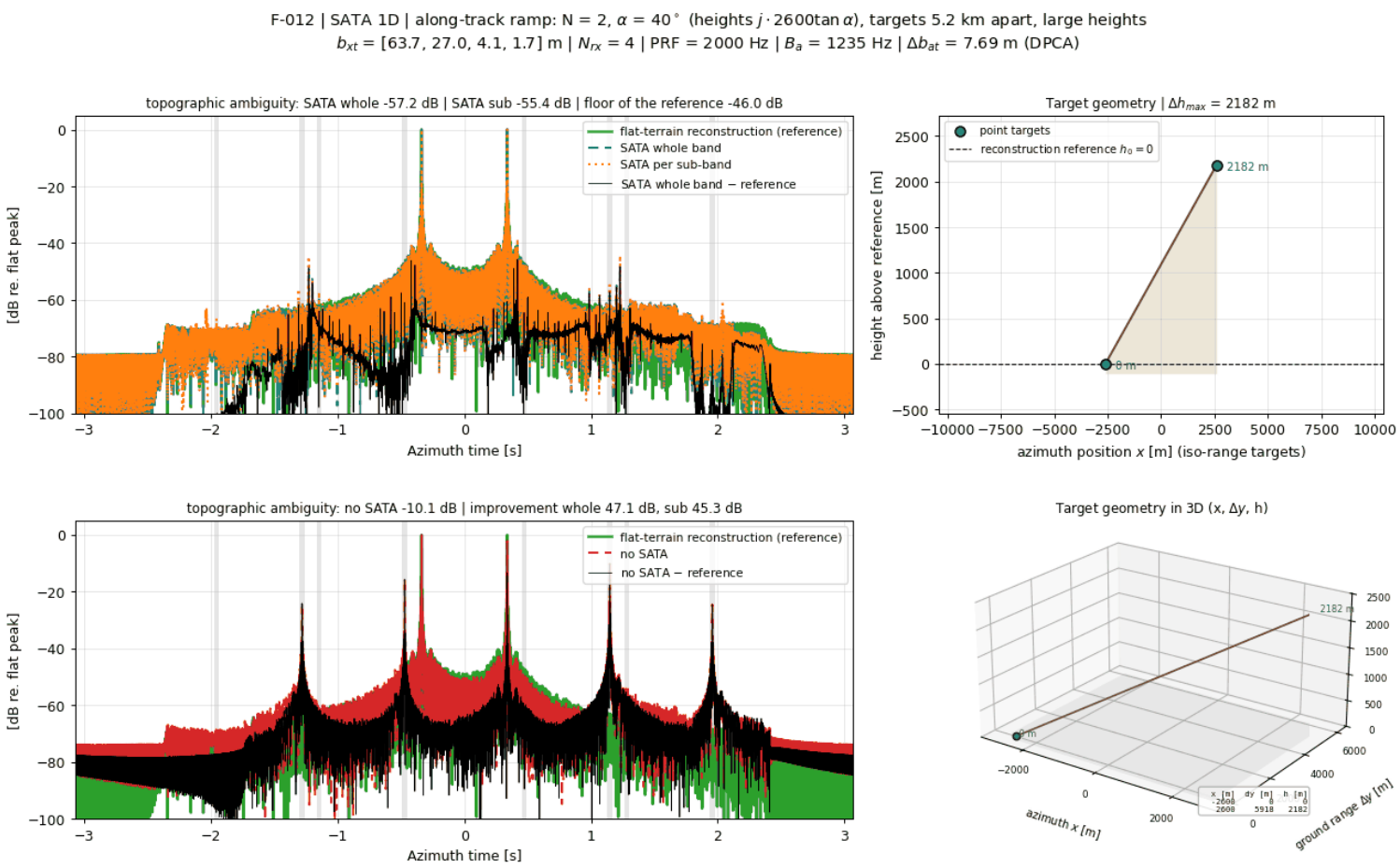


Figure 51: F-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −10.1, SATA whole band −57.2, per sub-band −55.4 dB (improvement 47.1 dB); floor −46.0 dB.

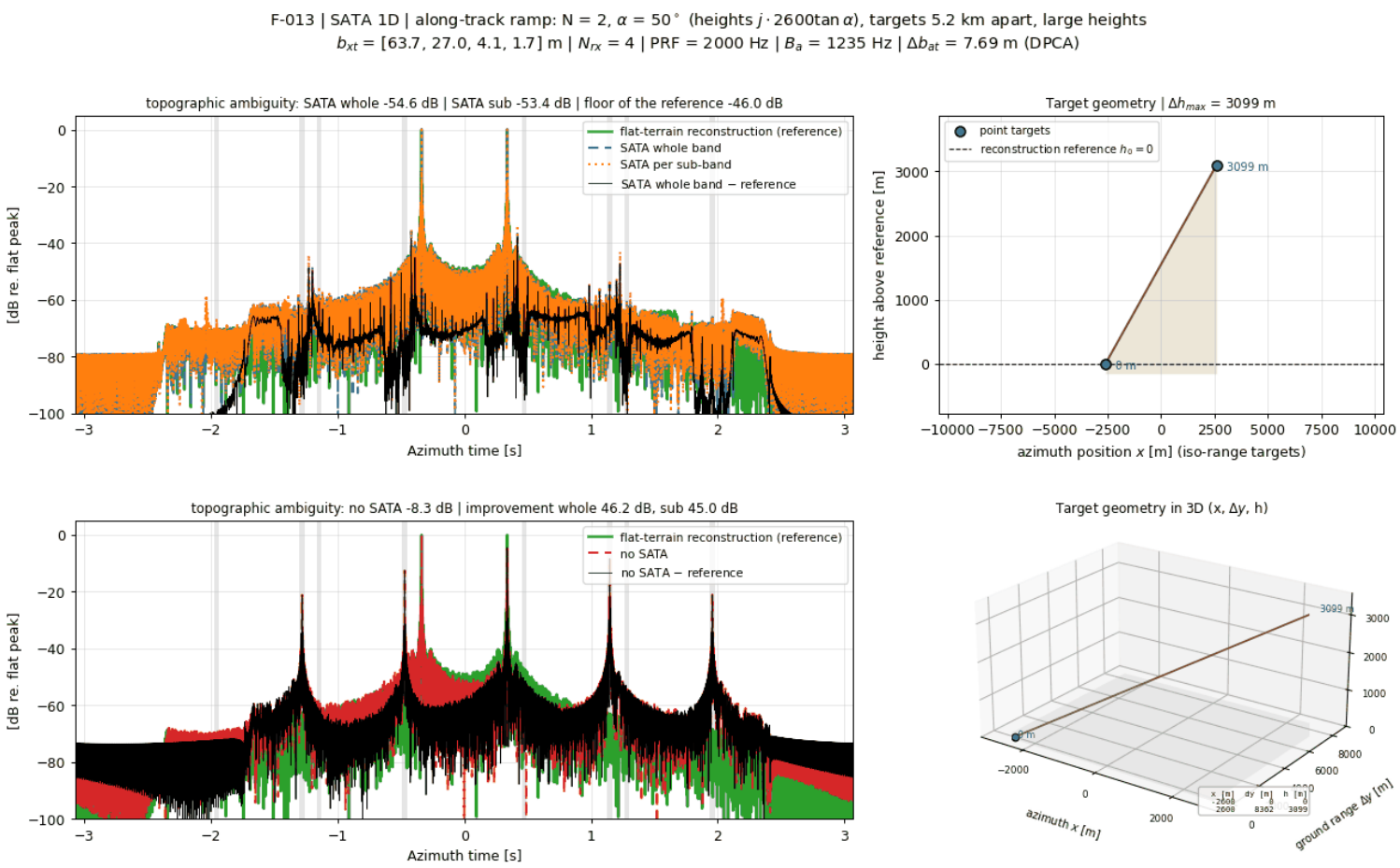


Figure 52: F-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −8.3, SATA whole band −54.6, per sub-band −53.4 dB (improvement 46.2 dB); floor −46.0 dB.

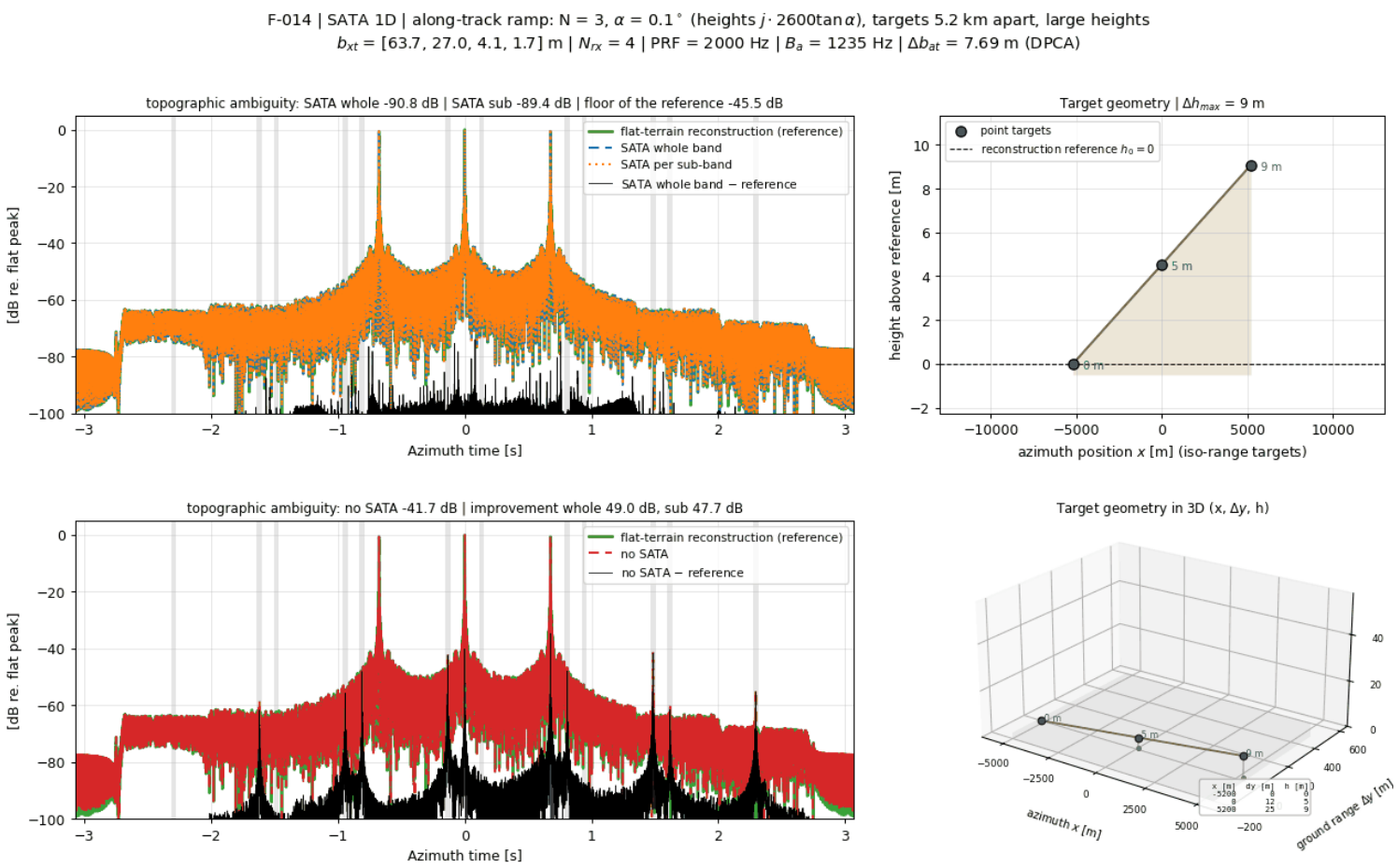


Figure 53: F-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −41.7, SATA whole band −90.8, per sub-band −89.4 dB (improvement 49.0 dB); floor −45.5 dB.

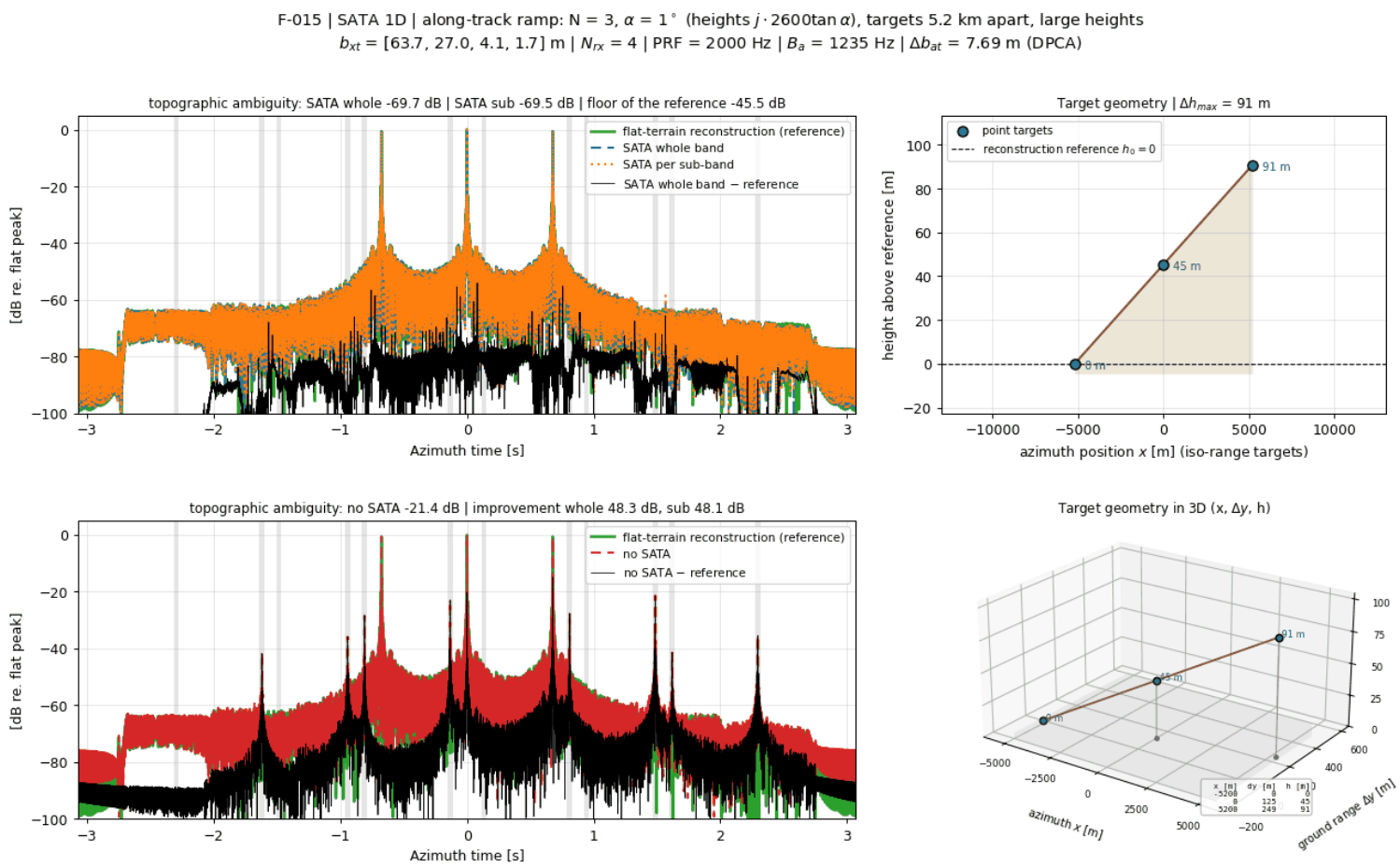


Figure 54: F-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −21.4, SATA whole band −69.7, per sub-band −69.5 dB (improvement 48.3 dB); floor −45.5 dB.

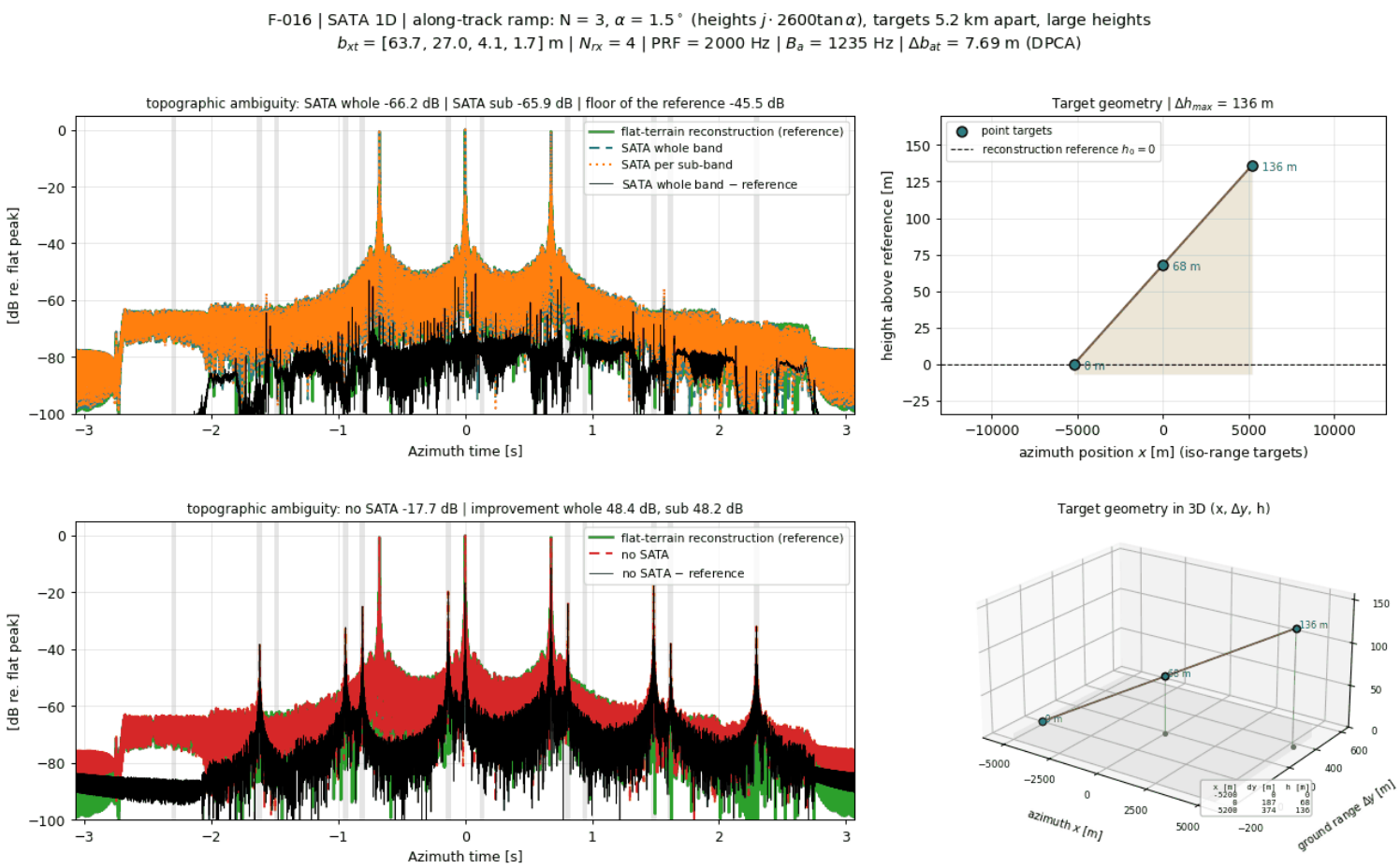


Figure 55: F-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −17.7, SATA whole band −66.2, per sub-band −65.9 dB (improvement 48.4 dB); floor −45.5 dB.

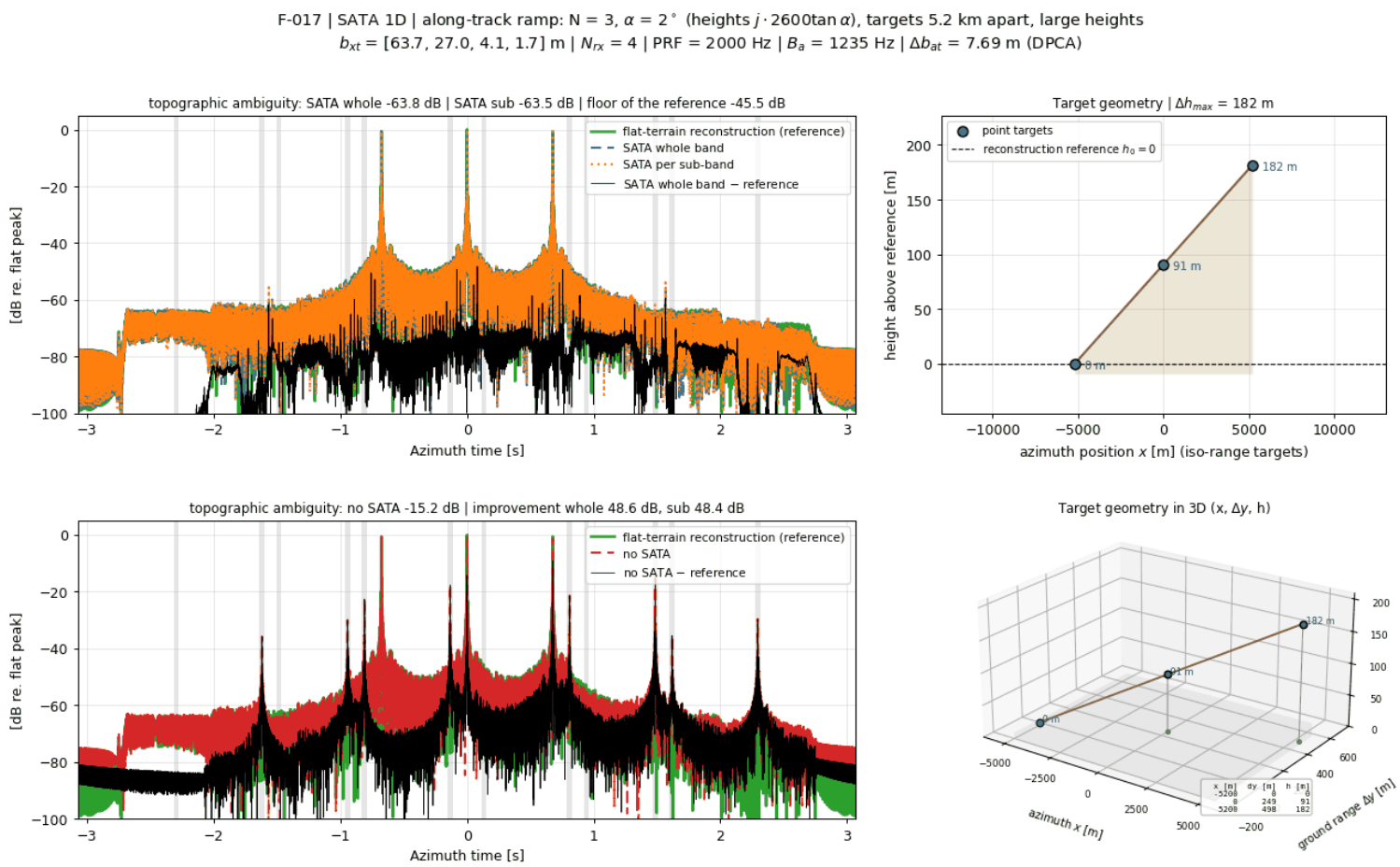


Figure 56: F-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −15.2, SATA whole band −63.8, per sub-band −63.5 dB (improvement 48.6 dB); floor −45.5 dB.

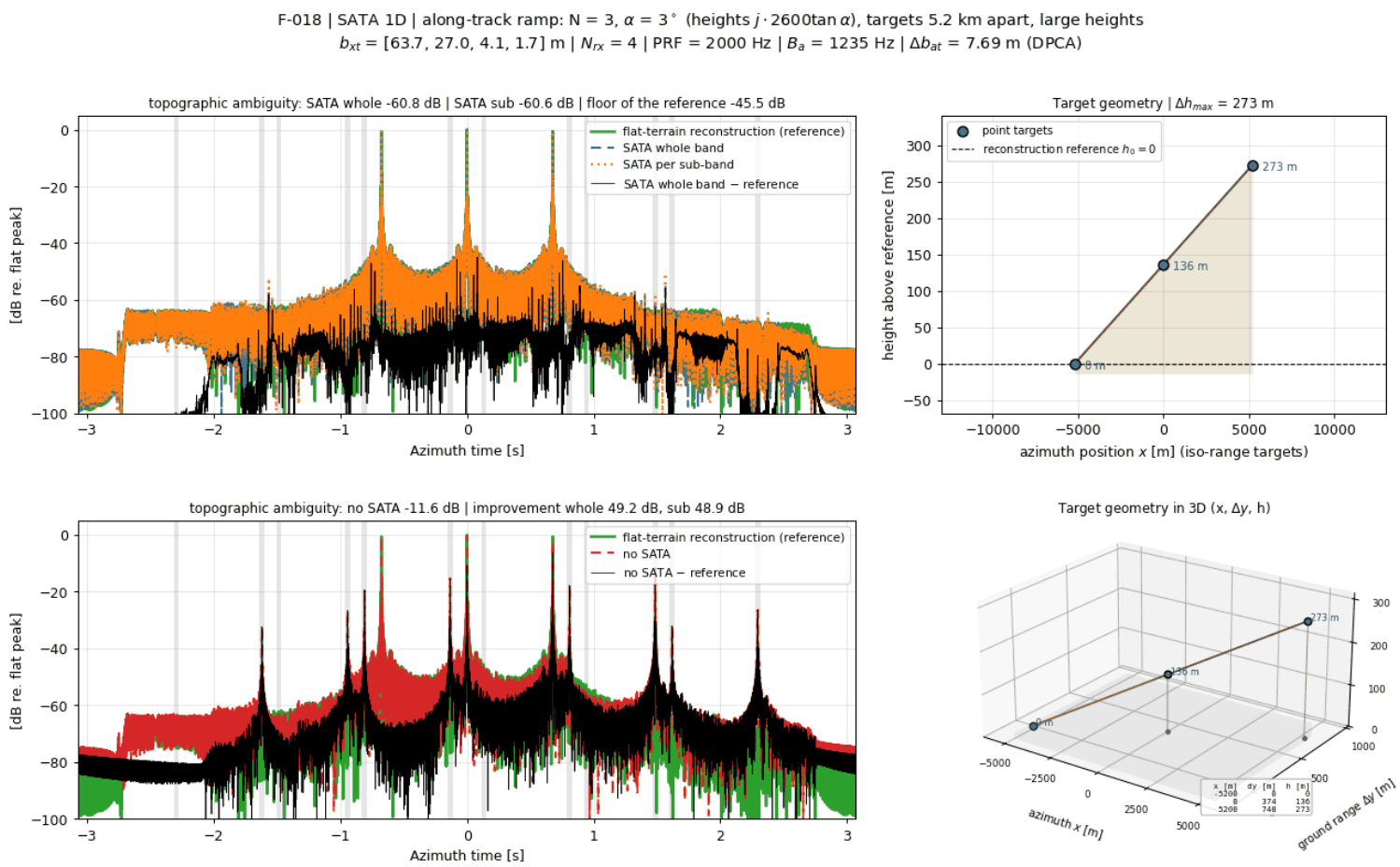


Figure 57: F-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −11.6, SATA whole band −60.8, per sub-band −60.6 dB (improvement 49.2 dB); floor −45.5 dB.

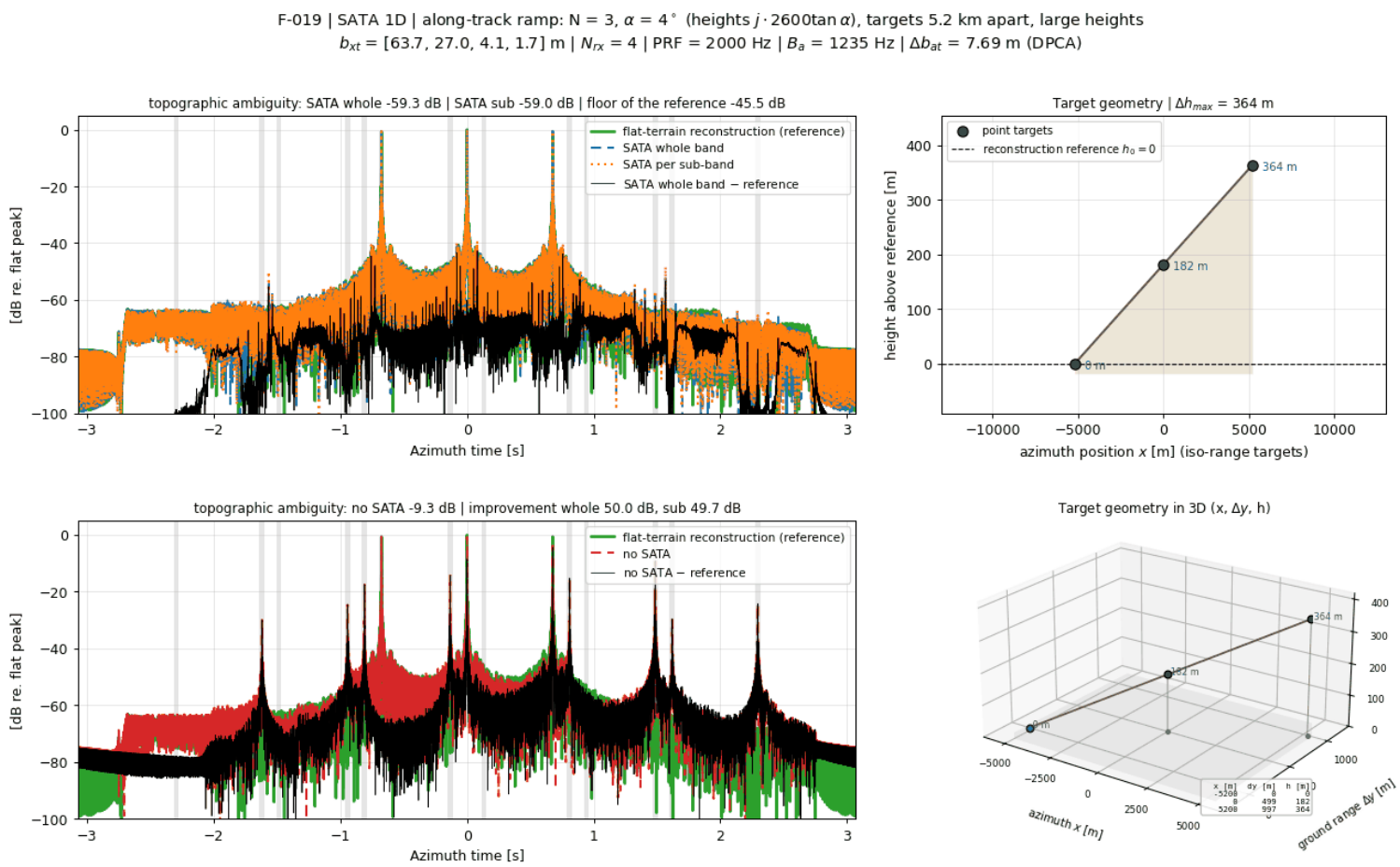


Figure 58: F-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −9.3, SATA whole band −59.3, per sub-band −59.0 dB (improvement 50.0 dB); floor −45.5 dB.

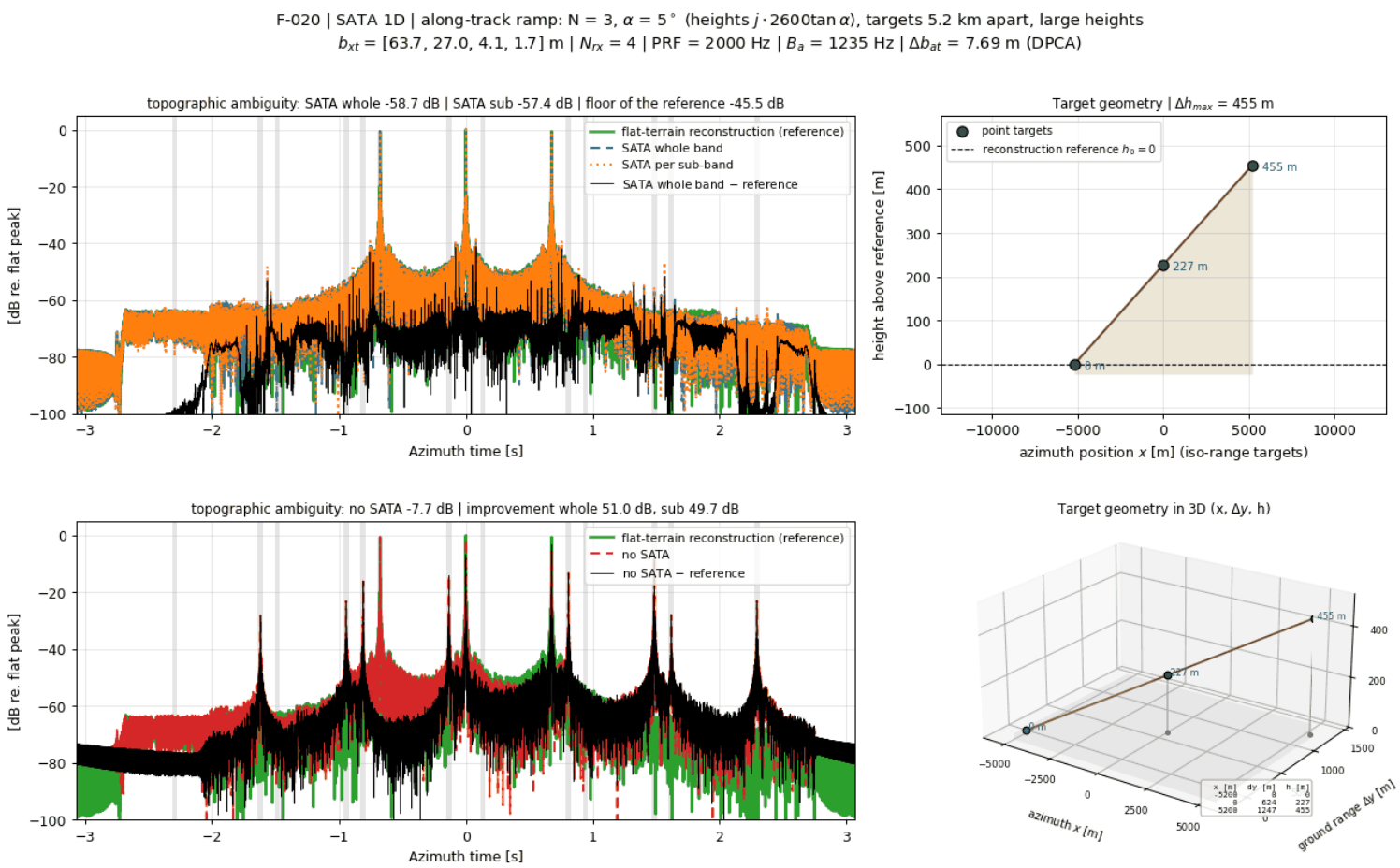


Figure 59: F-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.7, SATA whole band −58.7, per sub-band −57.4 dB (improvement 51.0 dB); floor −45.5 dB.

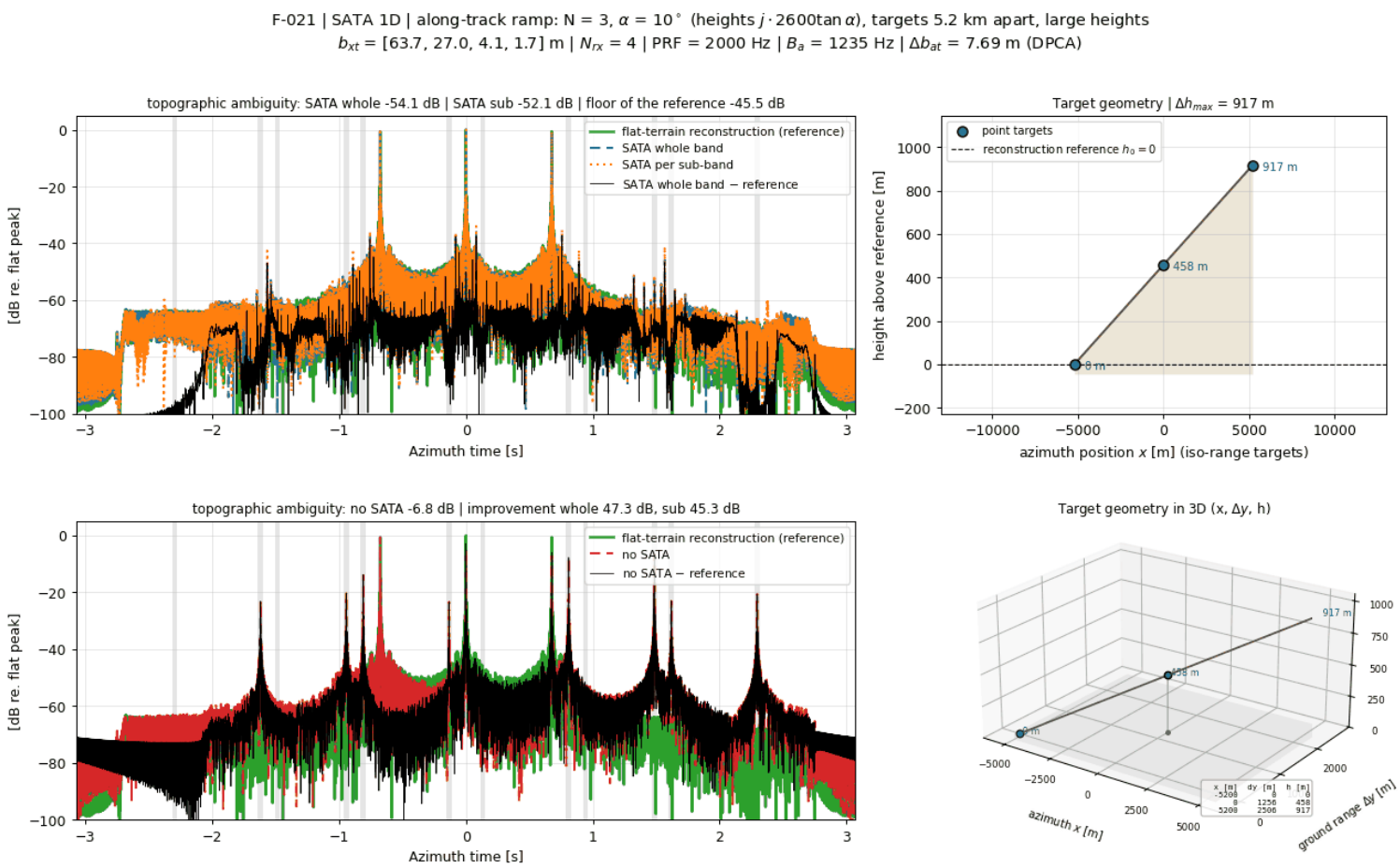


Figure 60: F-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.8, SATA whole band −54.1, per sub-band −52.1 dB (improvement 47.3 dB); floor −45.5 dB.

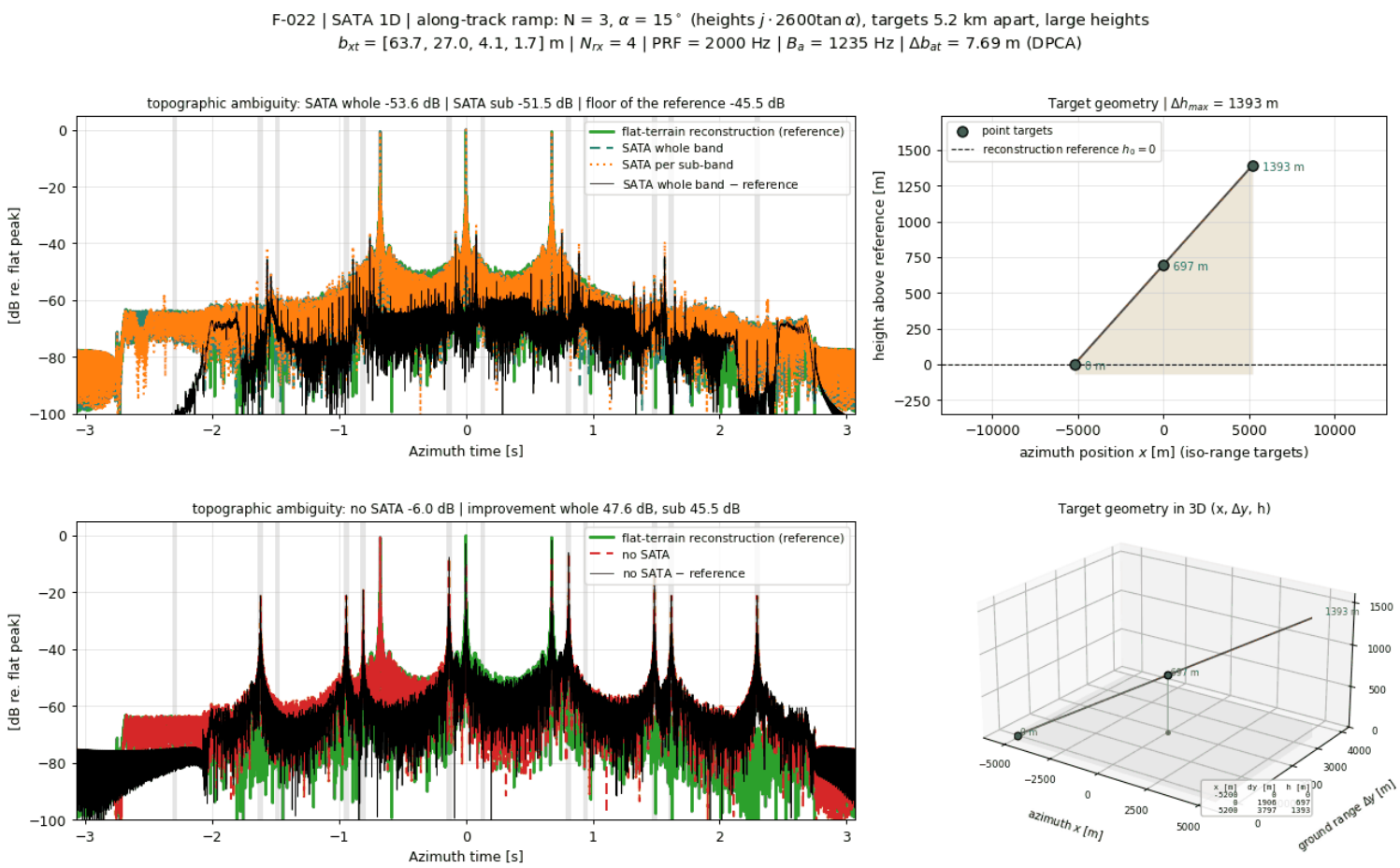


Figure 61: F-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.0, SATA whole band −53.6, per sub-band −51.5 dB (improvement 47.6 dB); floor −45.5 dB.

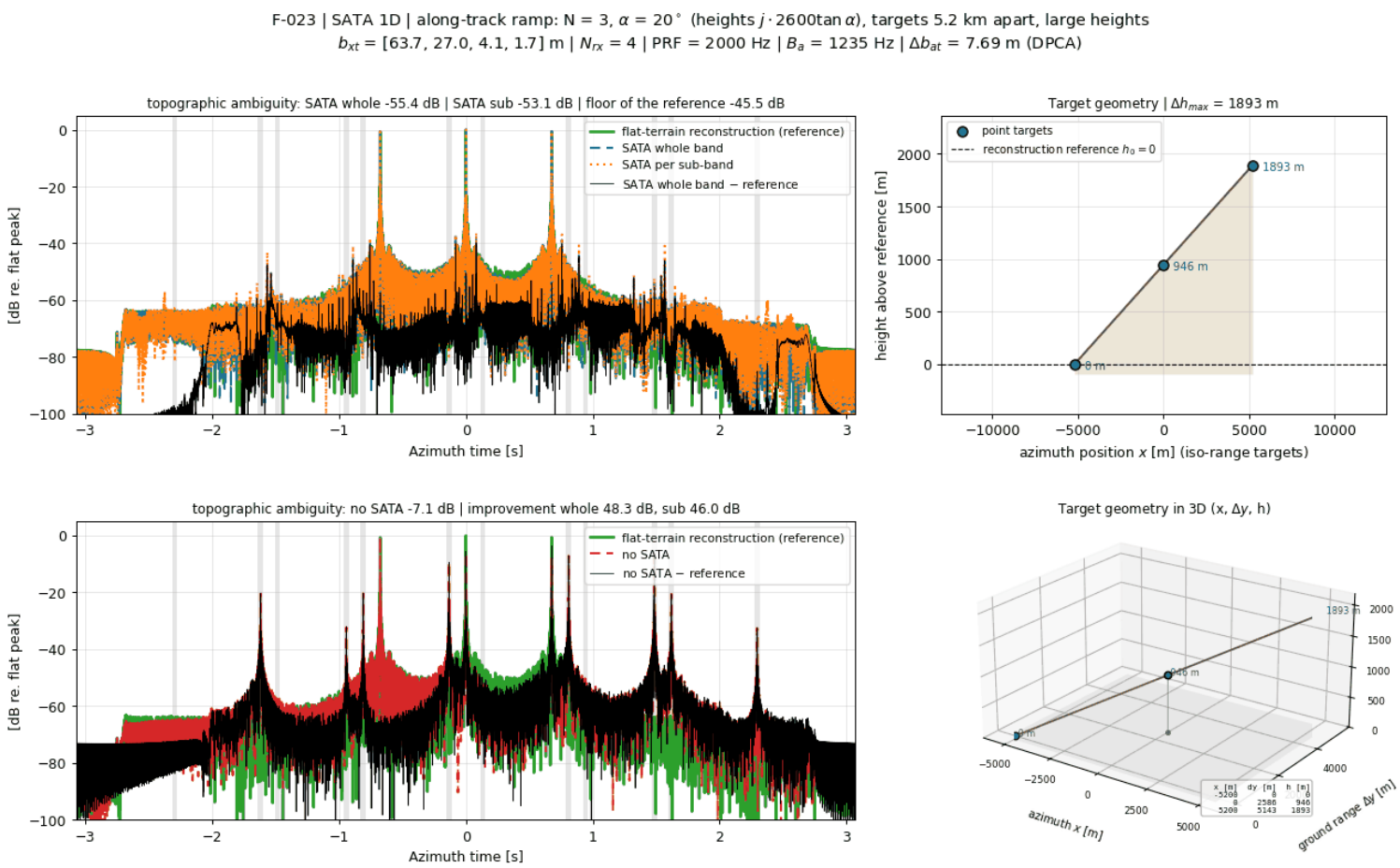


Figure 62: F-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.1, SATA whole band −55.4, per sub-band −53.1 dB (improvement 48.3 dB); floor −45.5 dB.

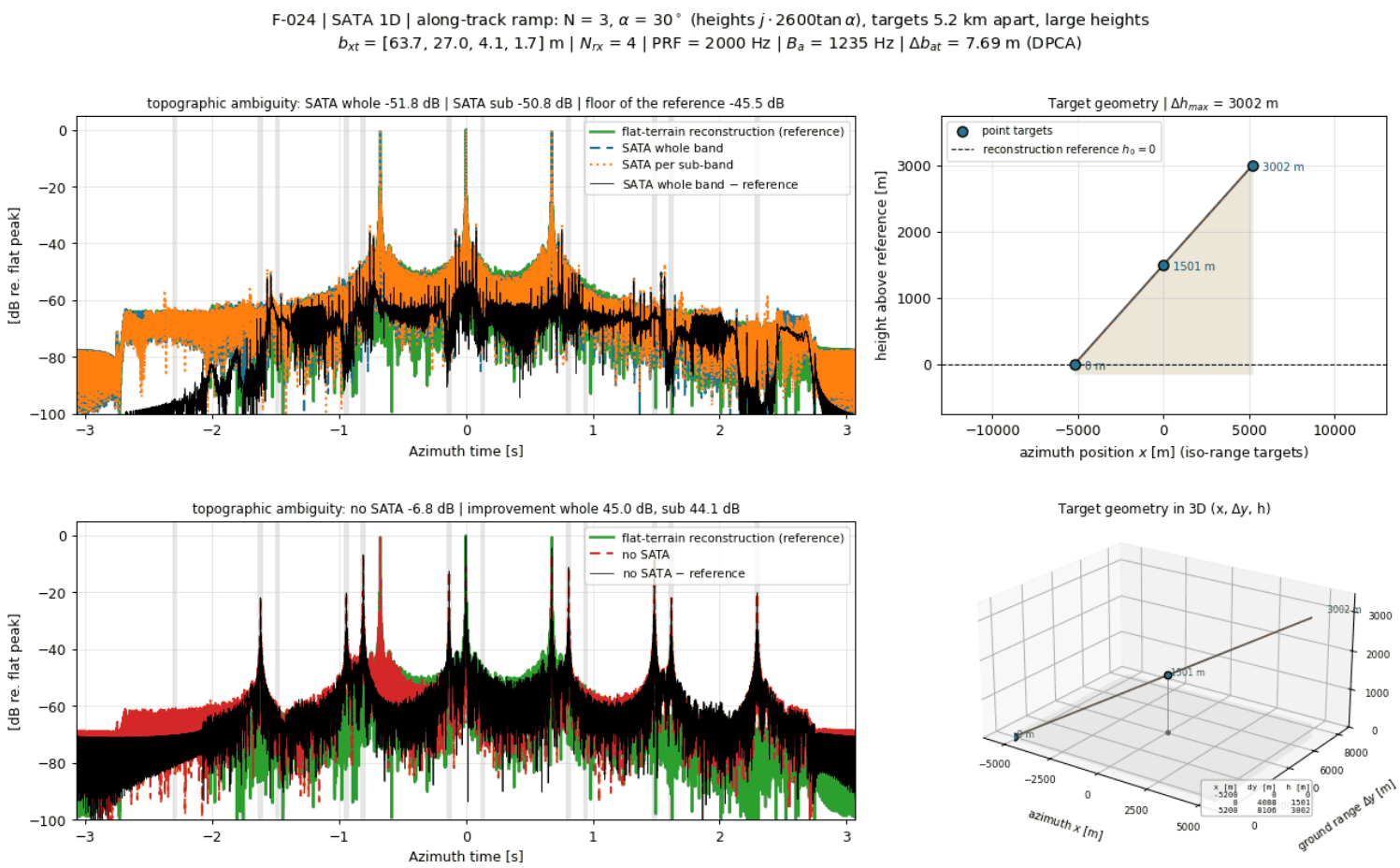


Figure 63: F-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.8, SATA whole band −51.8, per sub-band −50.8 dB (improvement 45.0 dB); floor −45.5 dB.

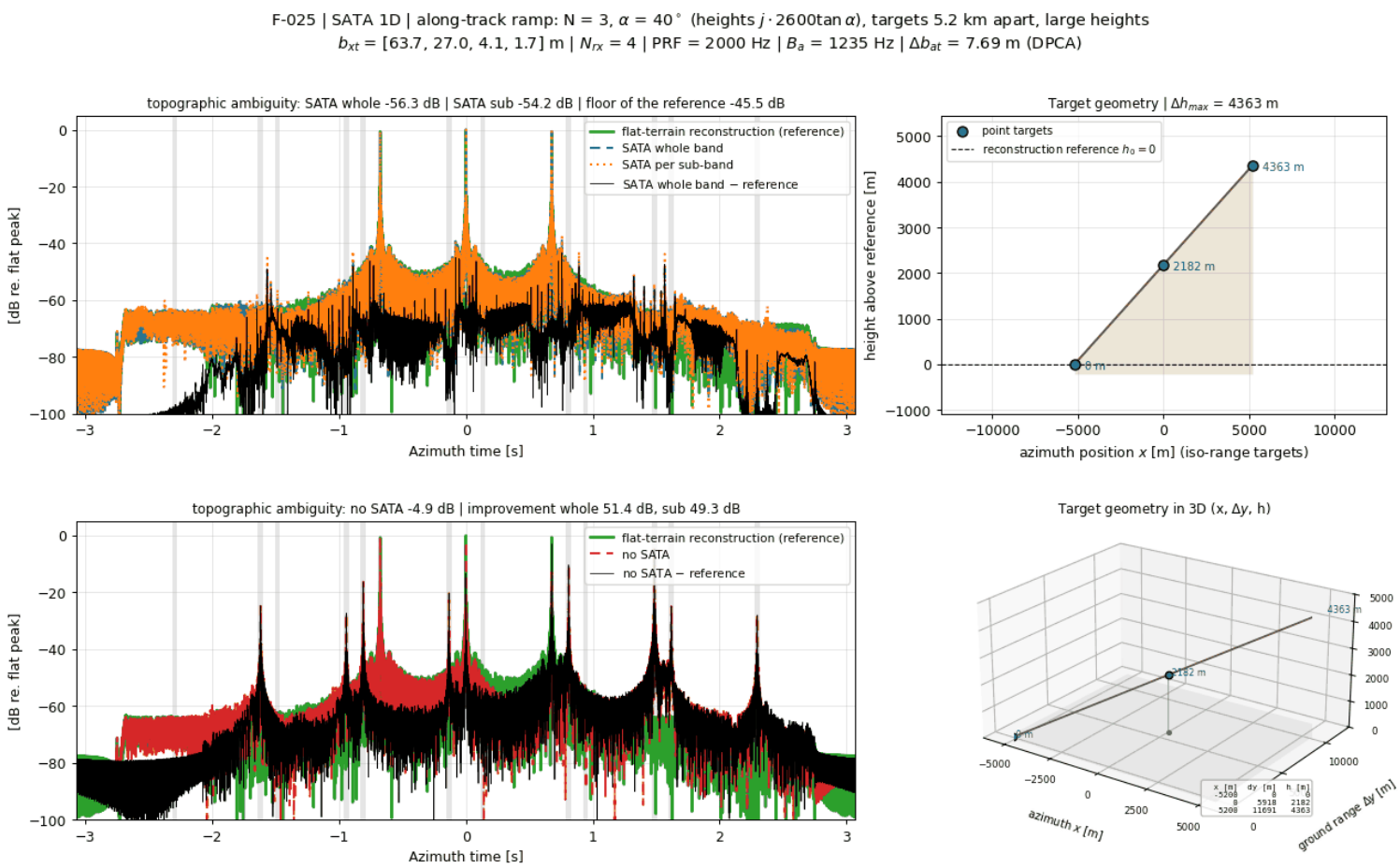


Figure 64: F-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −4.9, SATA whole band −56.3, per sub-band −54.2 dB (improvement 51.4 dB); floor −45.5 dB.

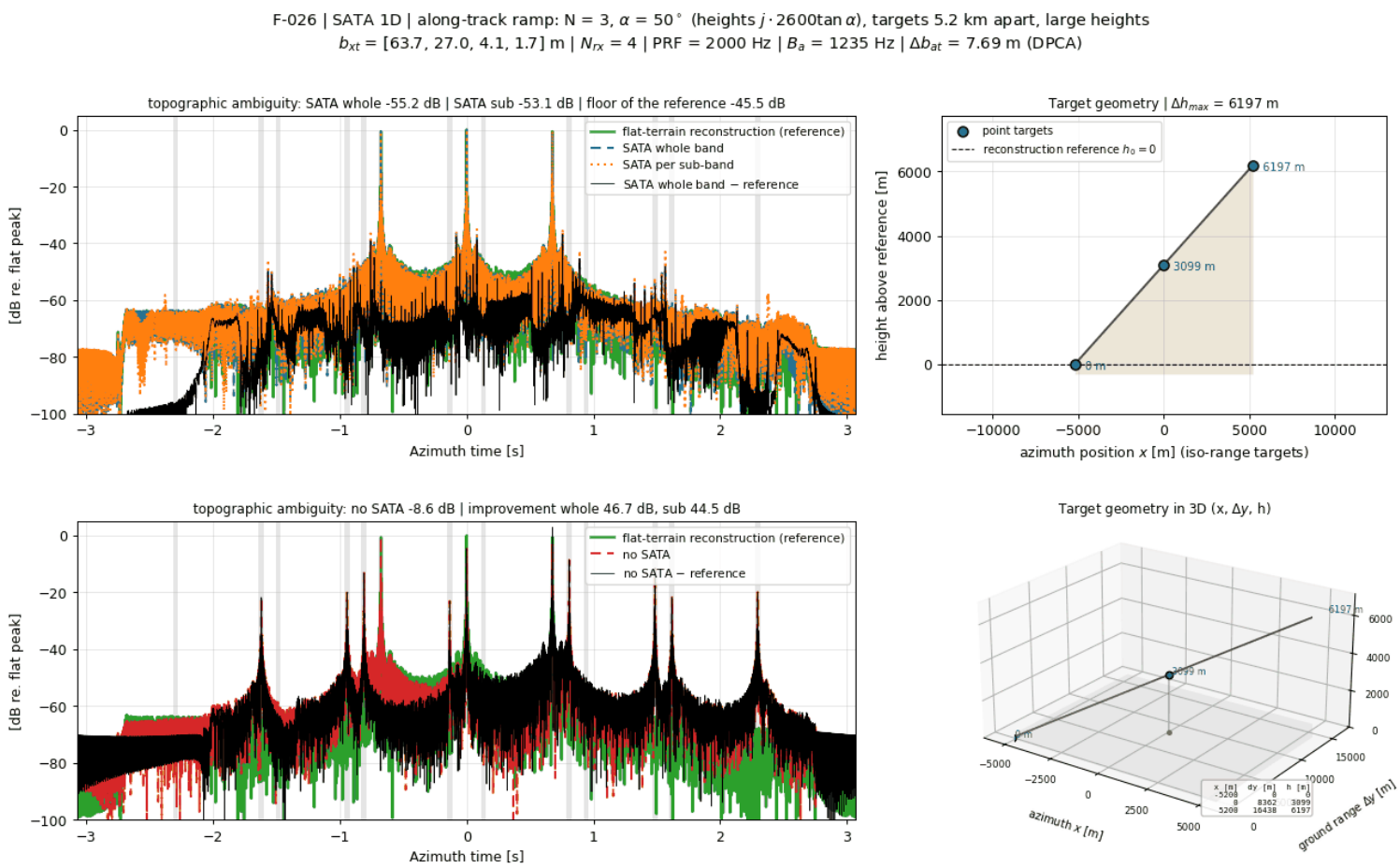


Figure 65: F-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −8.6, SATA whole band −55.2, per sub-band −53.1 dB (improvement 46.7 dB); floor −45.5 dB.

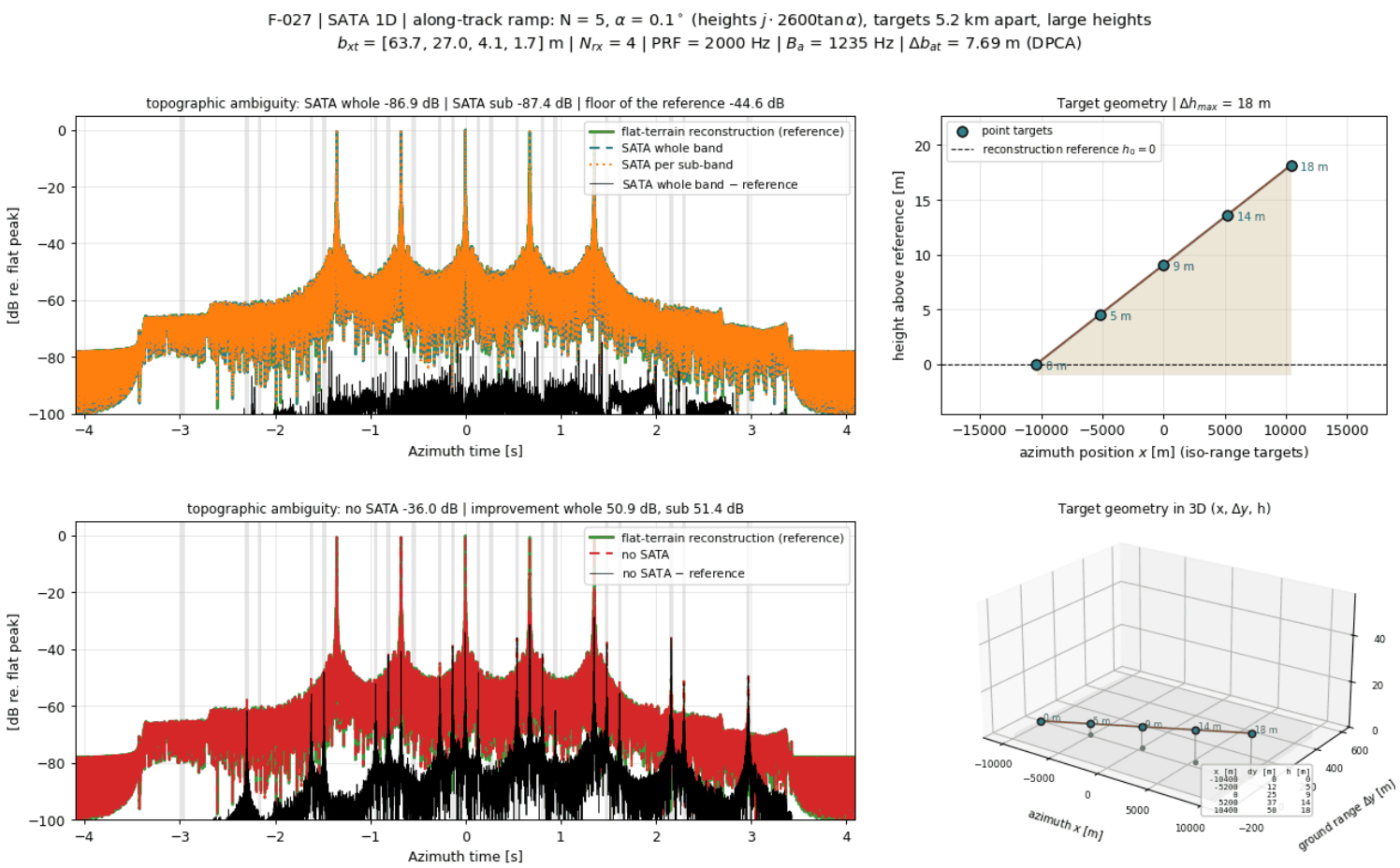


Figure 66: F-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −36.0, SATA whole band −86.9, per sub-band −87.4 dB (improvement 50.9 dB); floor −44.6 dB.

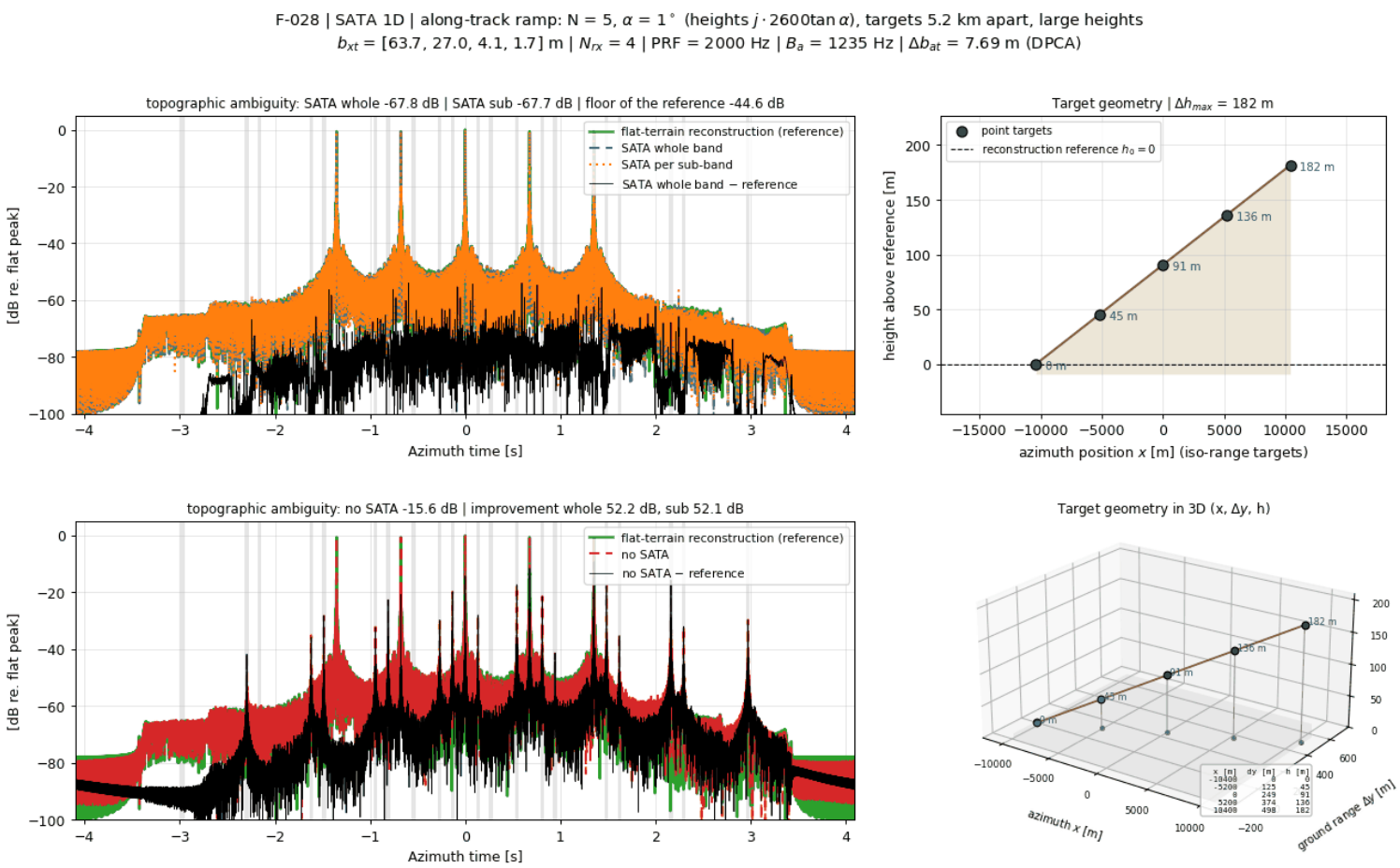


Figure 67: F-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −15.6, SATA whole band −67.8, per sub-band −67.7 dB (improvement 52.2 dB); floor −44.6 dB.

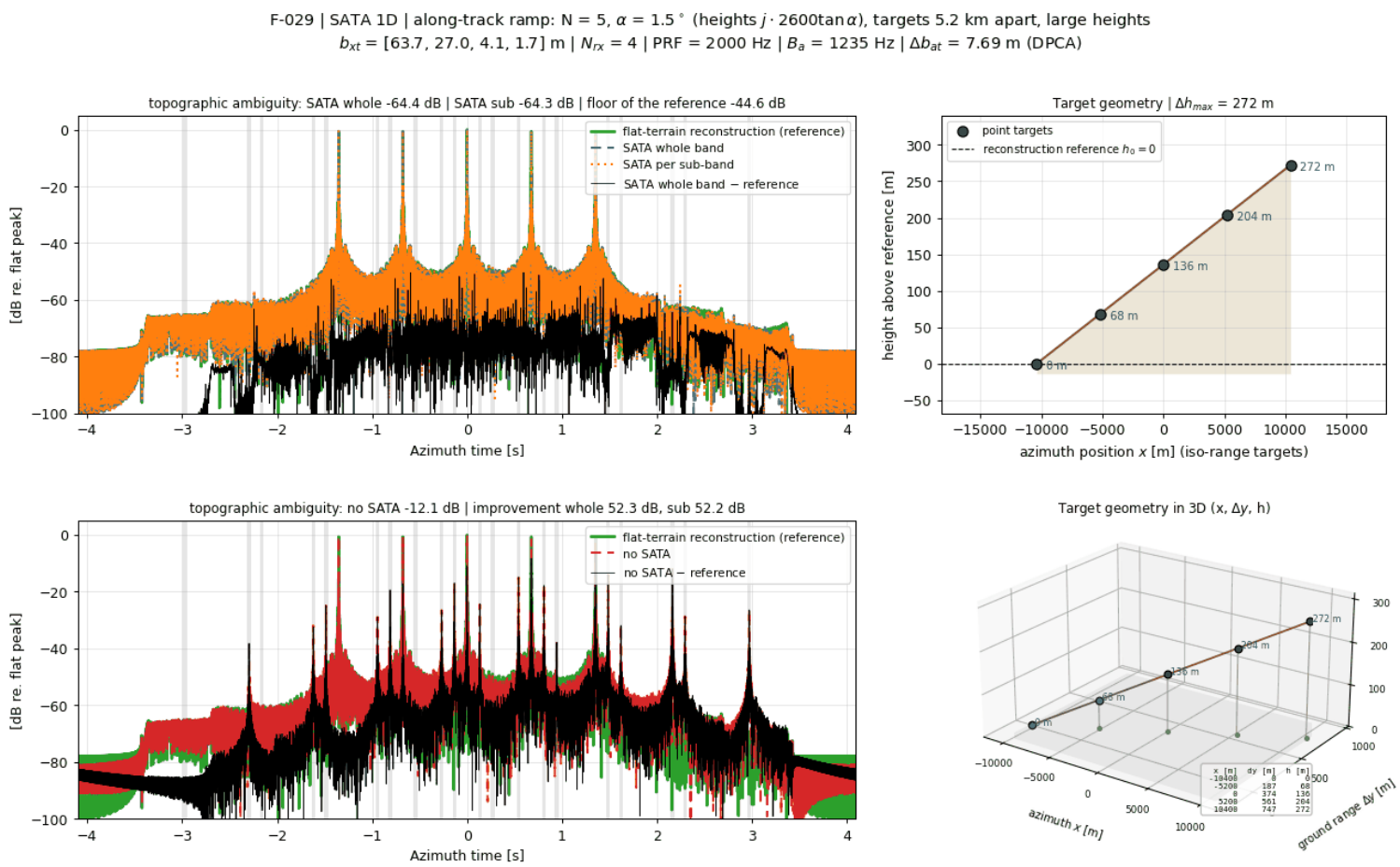


Figure 68: F-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −12.1, SATA whole band −64.4, per sub-band −64.3 dB (improvement 52.3 dB); floor −44.6 dB.

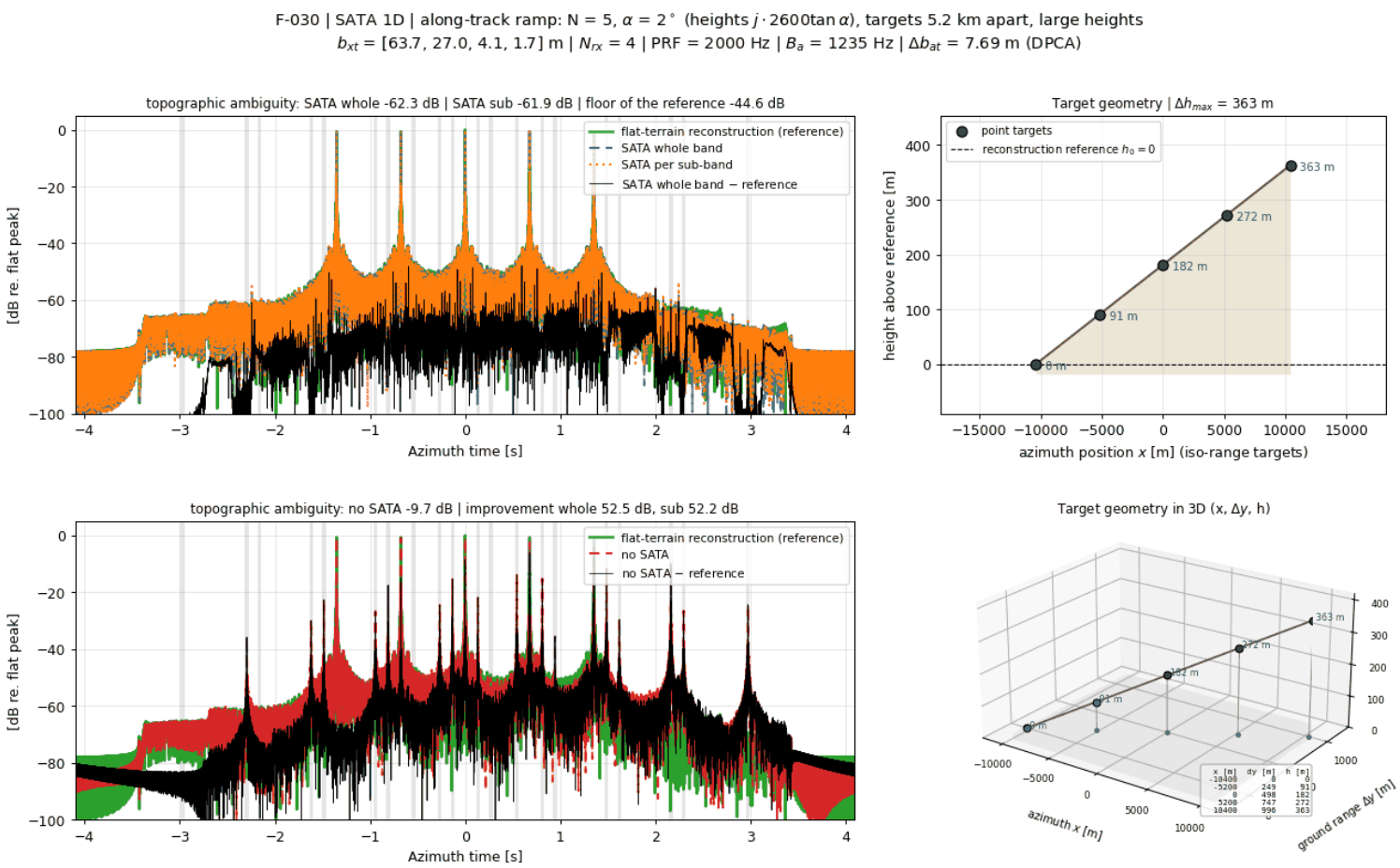


Figure 69: F-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan \alpha$). Topographic ambiguity: no SATA −9.7, SATA whole band −62.3, per sub-band −61.9 dB (improvement 52.5 dB); floor −44.6 dB.

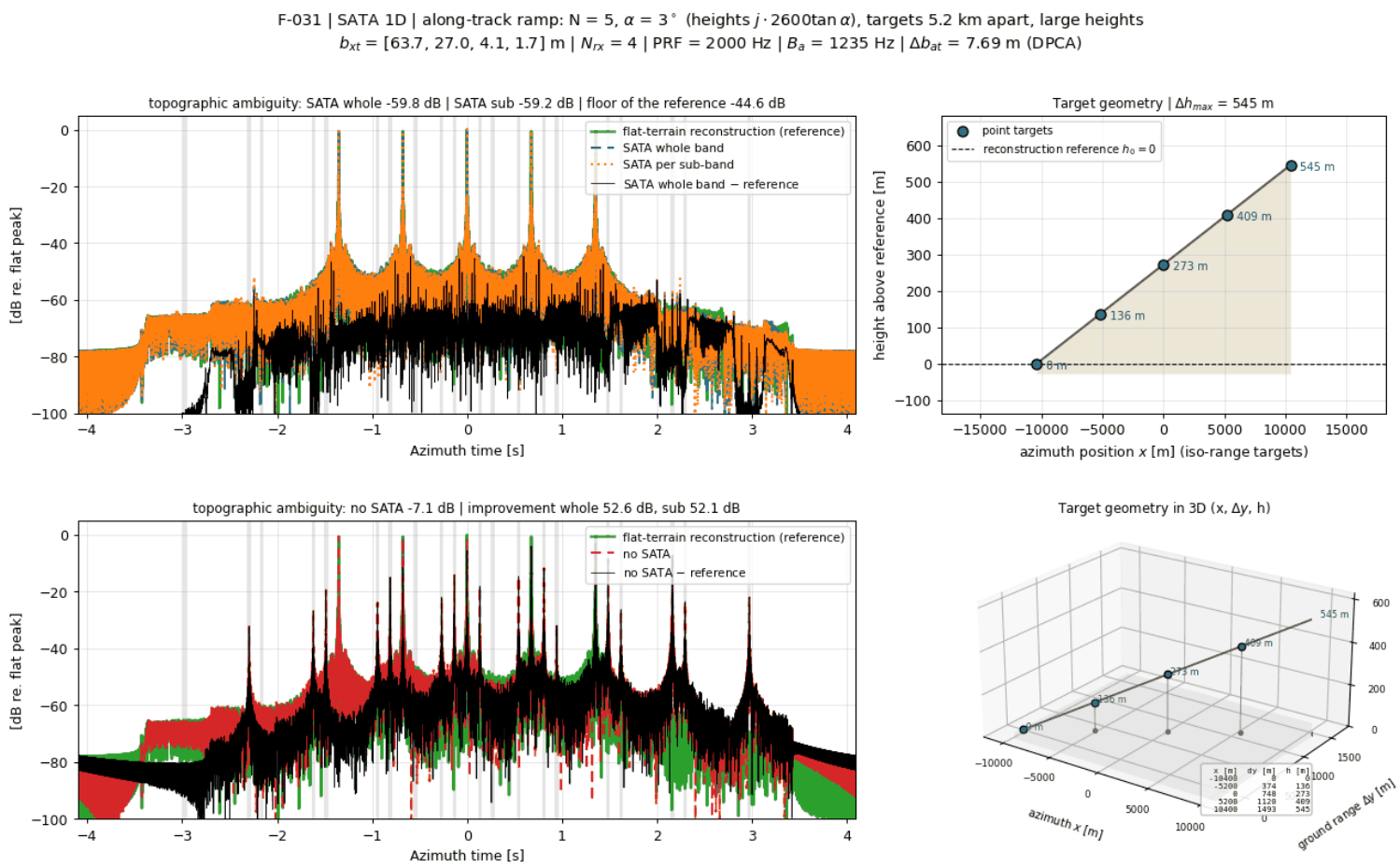


Figure 70: F-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA -7.1, SATA whole band -59.8, per sub-band -59.2 dB (improvement 52.6 dB); floor -44.6 dB.

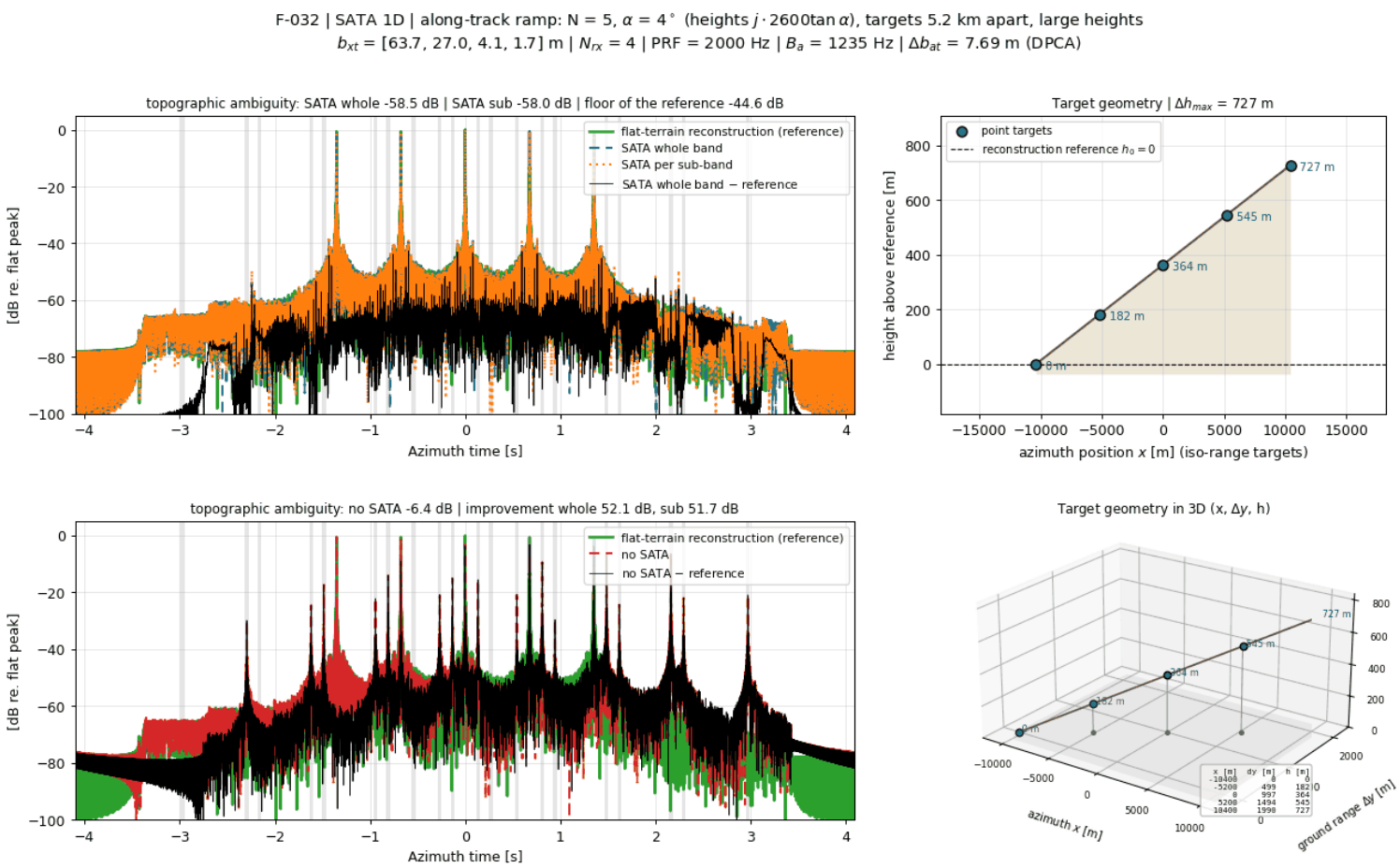


Figure 71: F-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.4, SATA whole band −58.5, per sub-band −58.0 dB (improvement 52.1 dB); floor −44.6 dB.

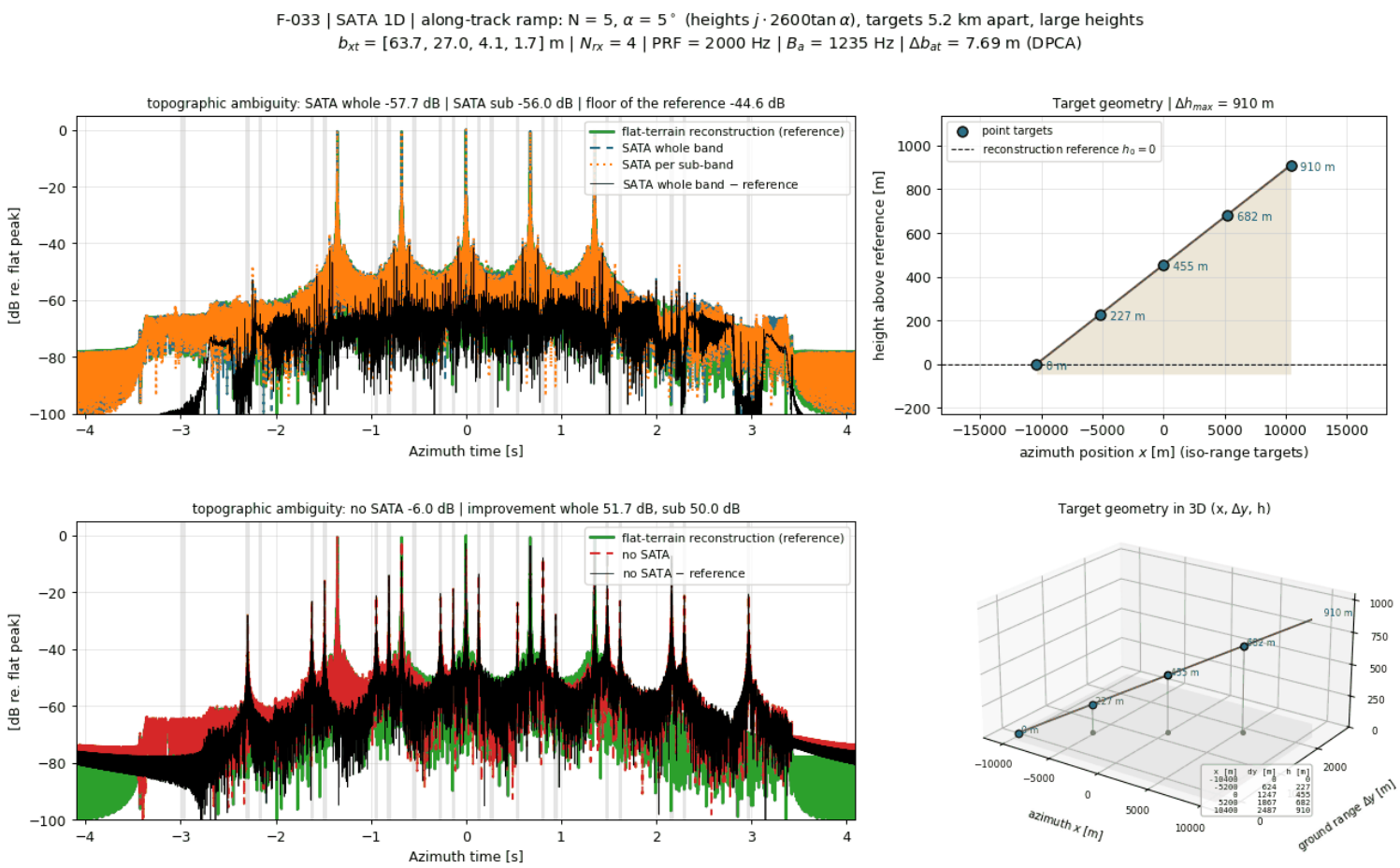


Figure 72: F-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.0, SATA whole band −57.7, per sub-band −56.0 dB (improvement 51.7 dB); floor −44.6 dB.

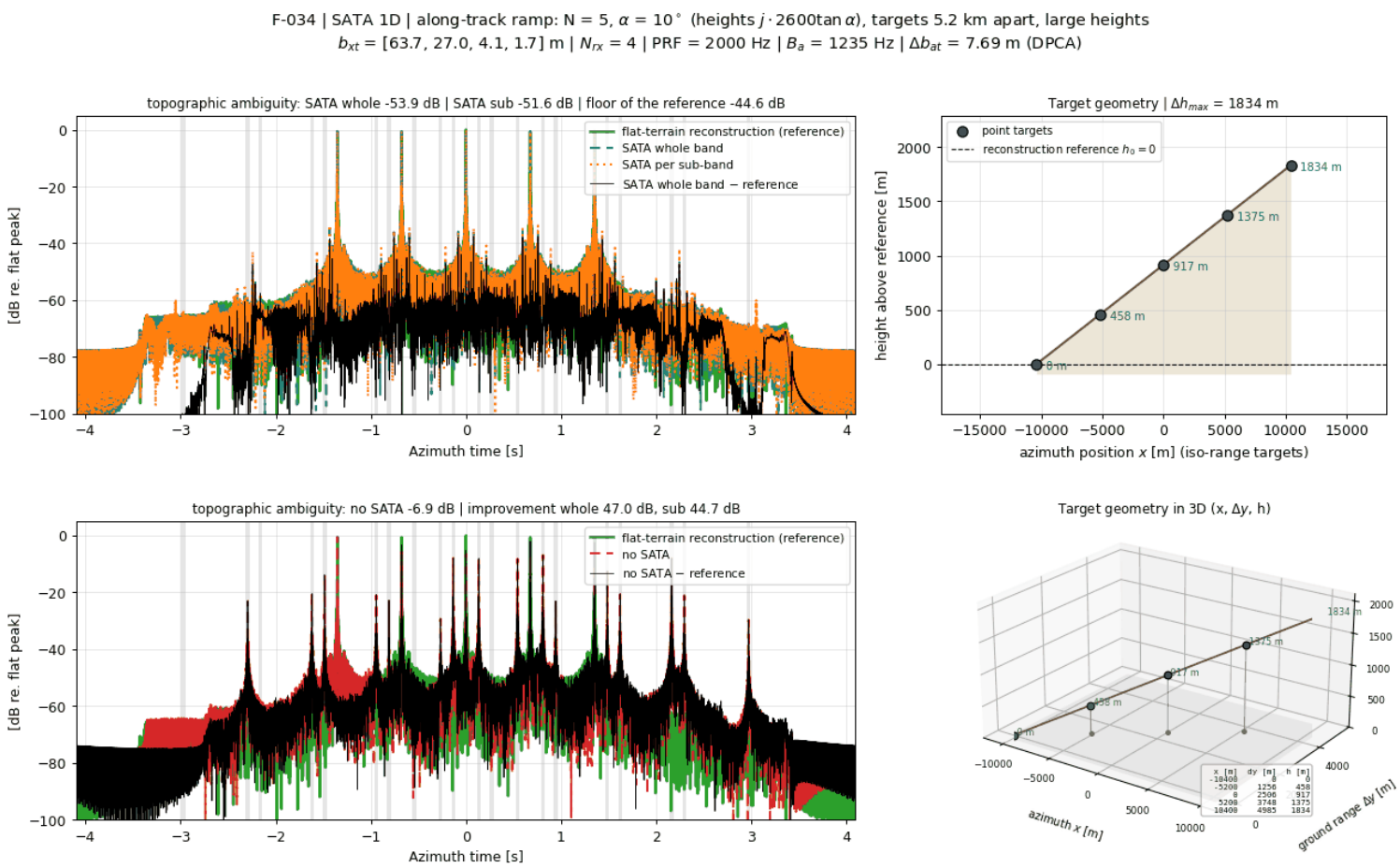


Figure 73: F-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.9, SATA whole band −53.9, per sub-band −51.6 dB (improvement 47.0 dB); floor −44.6 dB.

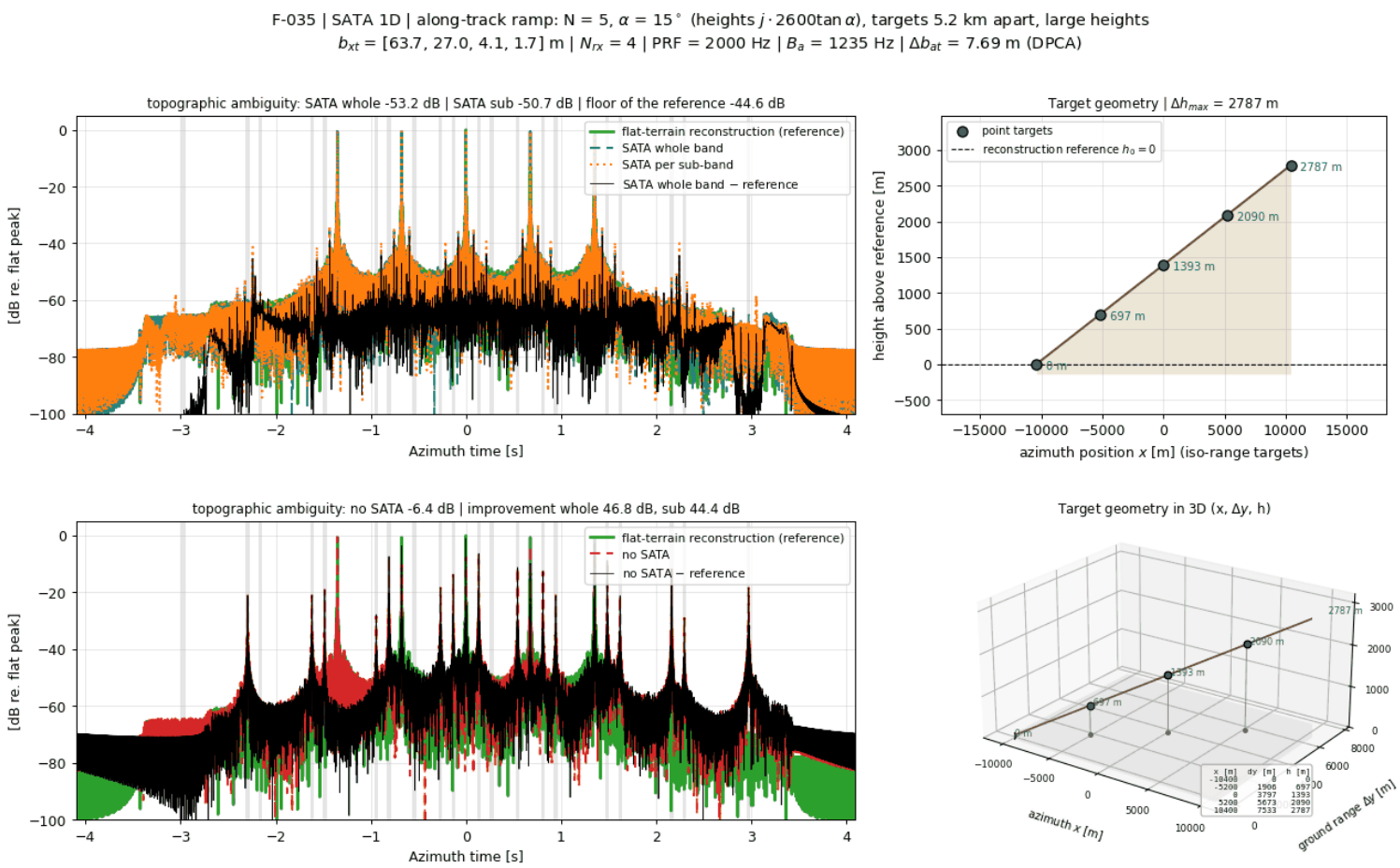


Figure 74: F-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.4, SATA whole band −53.2, per sub-band −50.7 dB (improvement 46.8 dB); floor −44.6 dB.

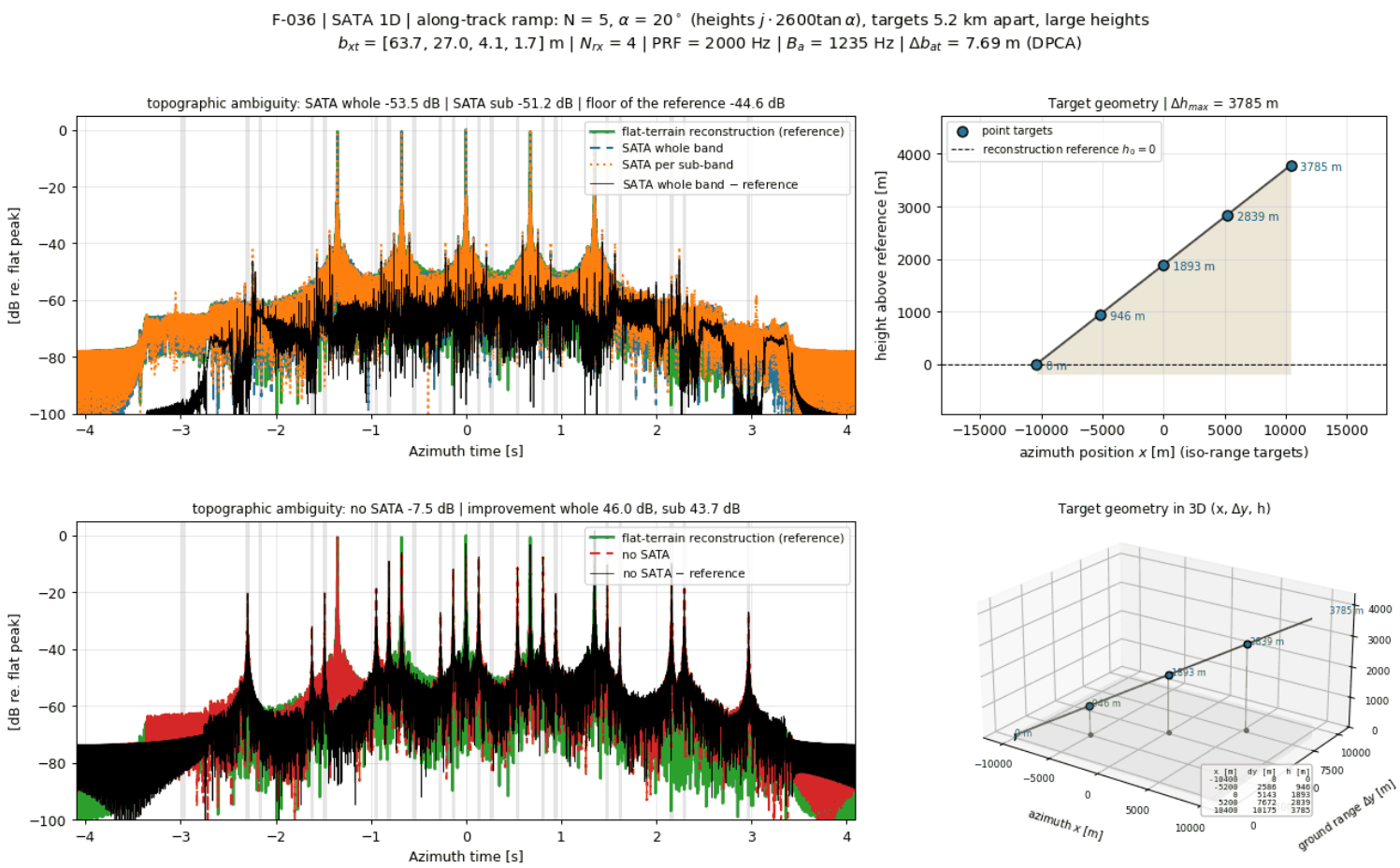


Figure 75: F-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA -7.5, SATA whole band -53.5, per sub-band -51.2 dB (improvement 46.0 dB); floor -44.6 dB.

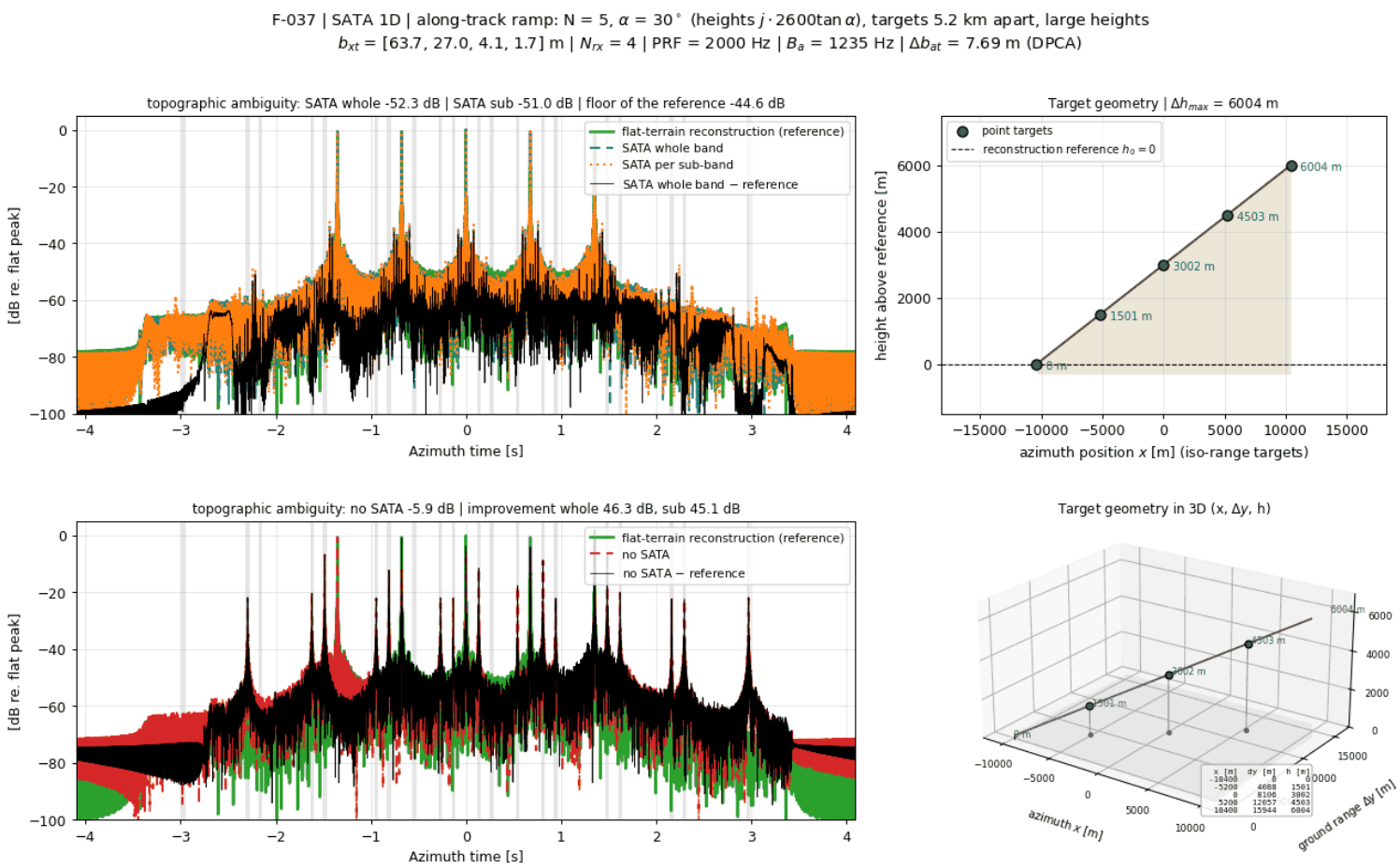


Figure 76: F-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −5.9, SATA whole band −52.3, per sub-band −51.0 dB (improvement 46.3 dB); floor −44.6 dB.

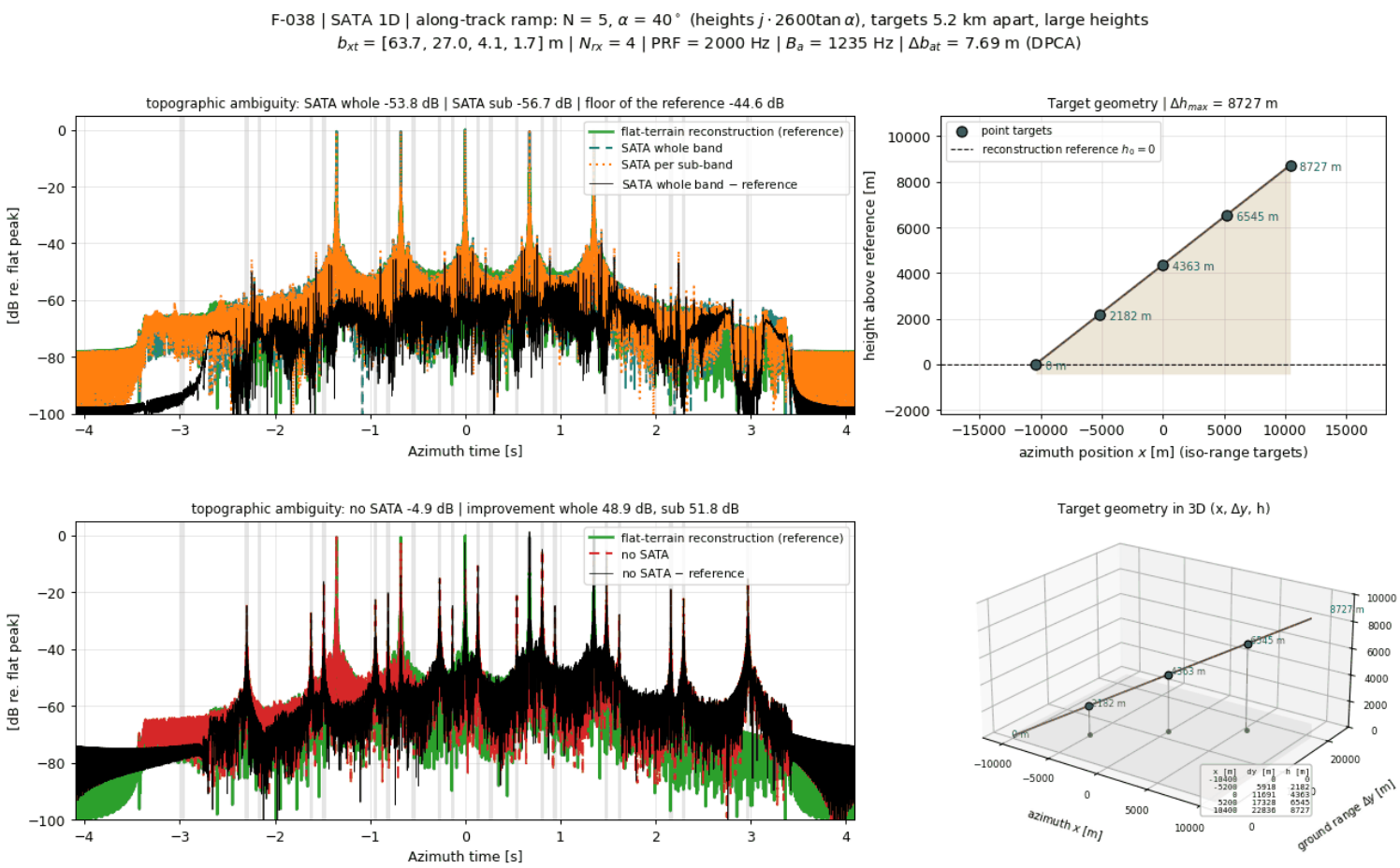


Figure 77: F-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −4.9, SATA whole band −53.8, per sub-band −56.7 dB (improvement 48.9 dB); floor −44.6 dB.

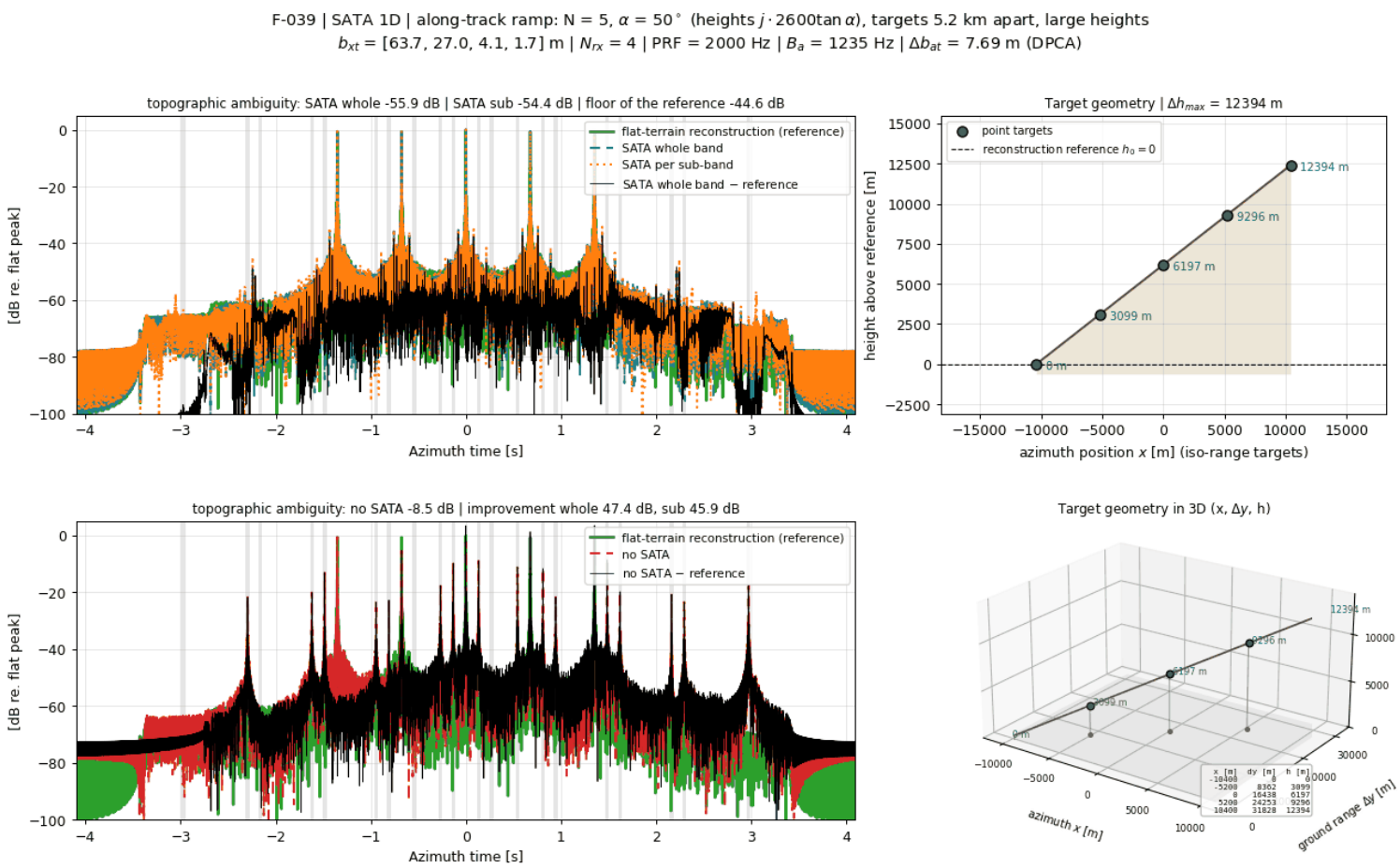


Figure 78: F-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −8.5, SATA whole band −55.9, per sub-band −54.4 dB (improvement 47.4 dB); floor −44.6 dB.

3 Configuration G: along-track ramps, targets 10 km apart, normal heights

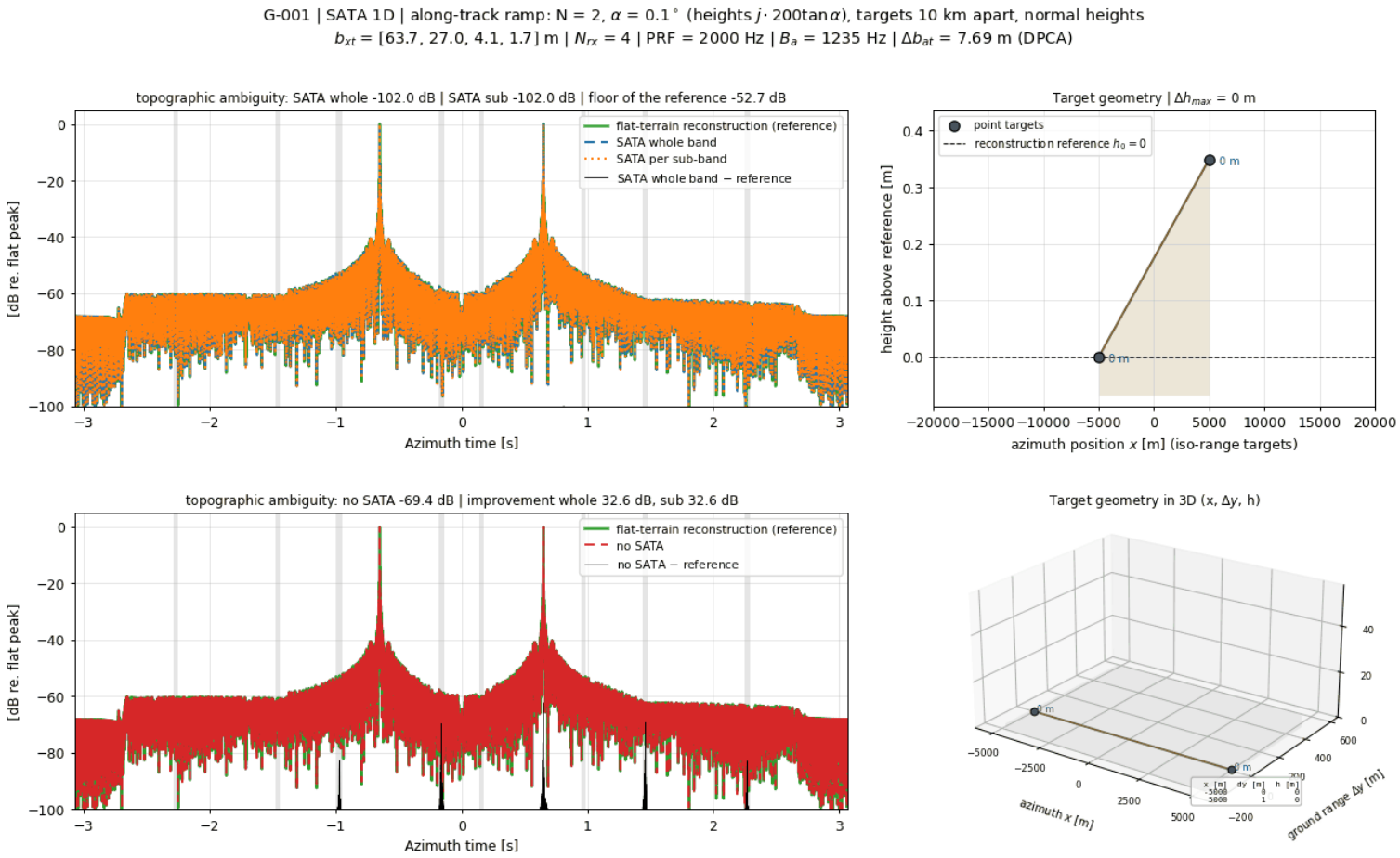


Figure 79: G-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −69.4, SATA whole band −102.0, per sub-band −102.0 dB (improvement 32.6 dB); floor −52.7 dB.

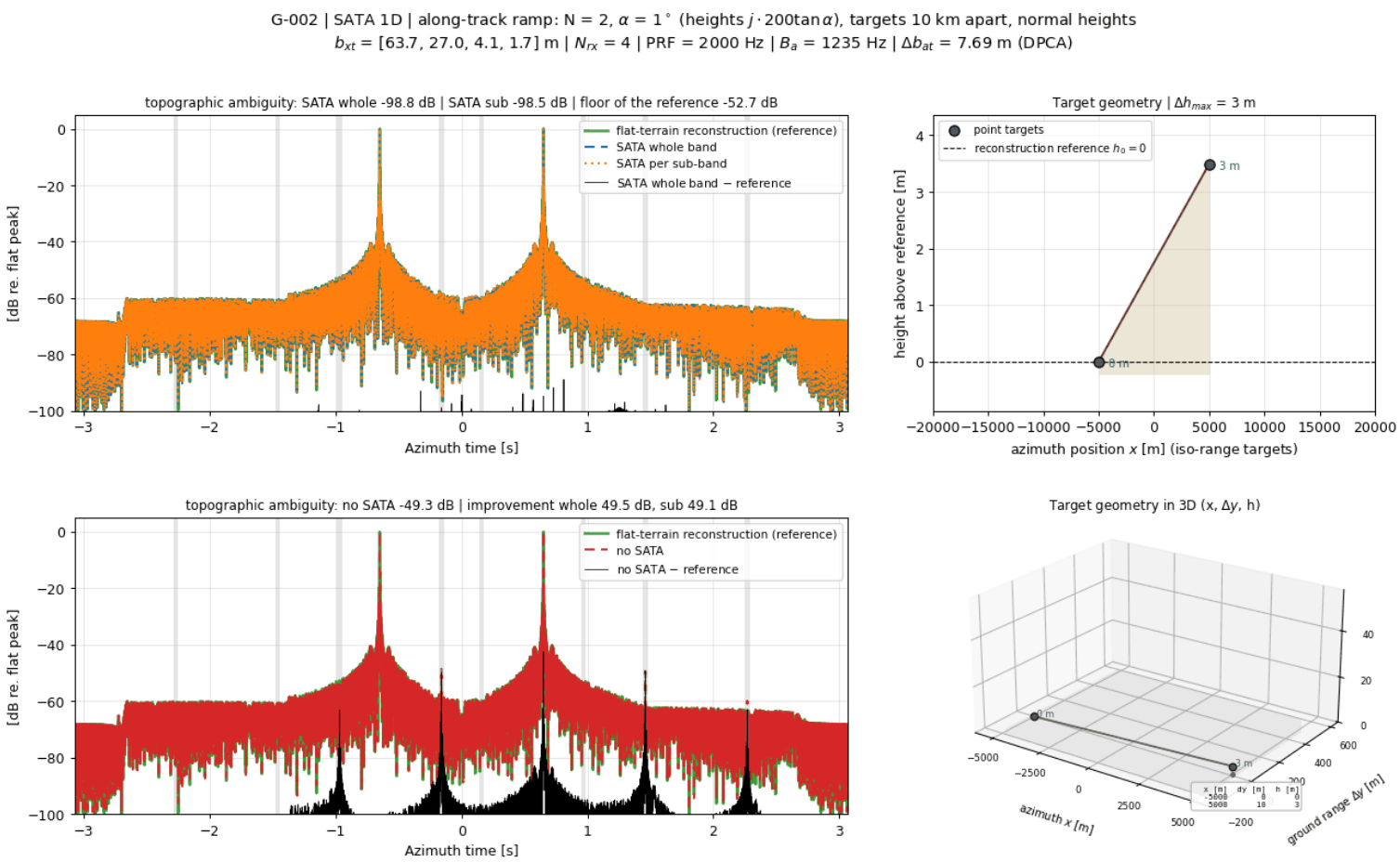


Figure 80: G-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −49.3, SATA whole band −98.8, per sub-band −98.5 dB (improvement 49.5 dB); floor −52.7 dB.

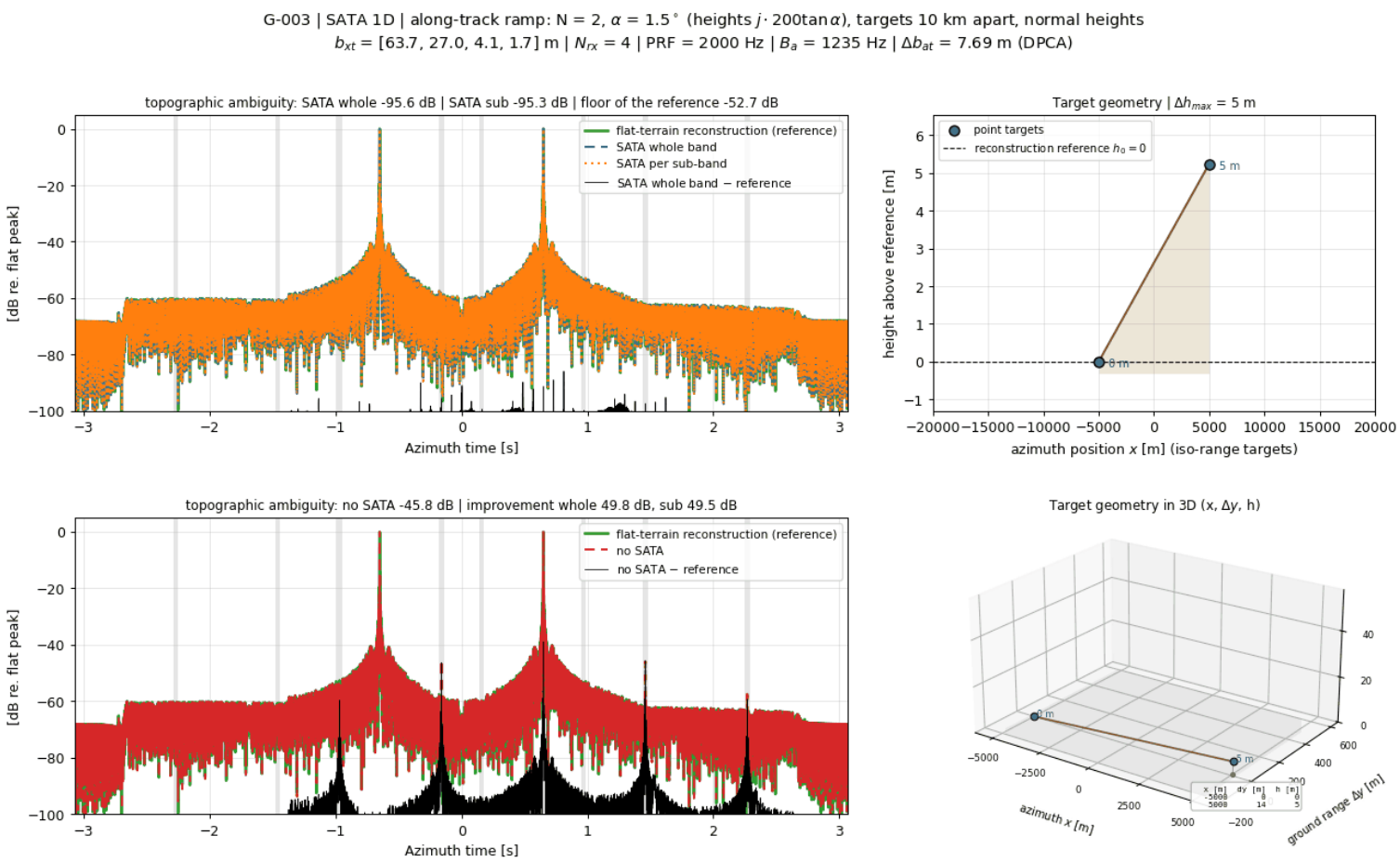


Figure 81: G-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -45.8, SATA whole band -95.6, per sub-band -95.3 dB (improvement 49.8 dB); floor -52.7 dB.

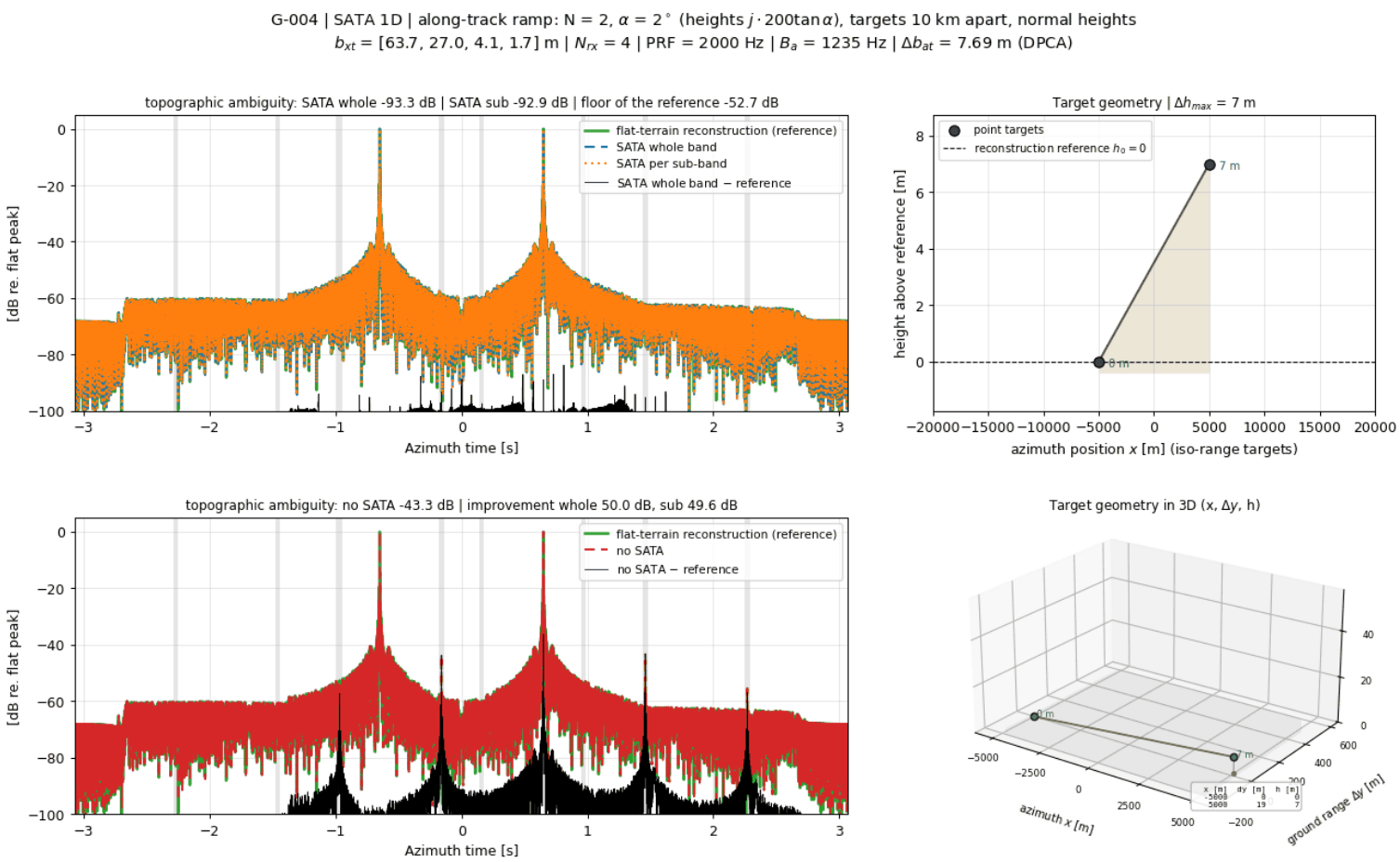


Figure 82: G-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −43.3, SATA whole band −93.3, per sub-band −92.9 dB (improvement 50.0 dB); floor −52.7 dB.

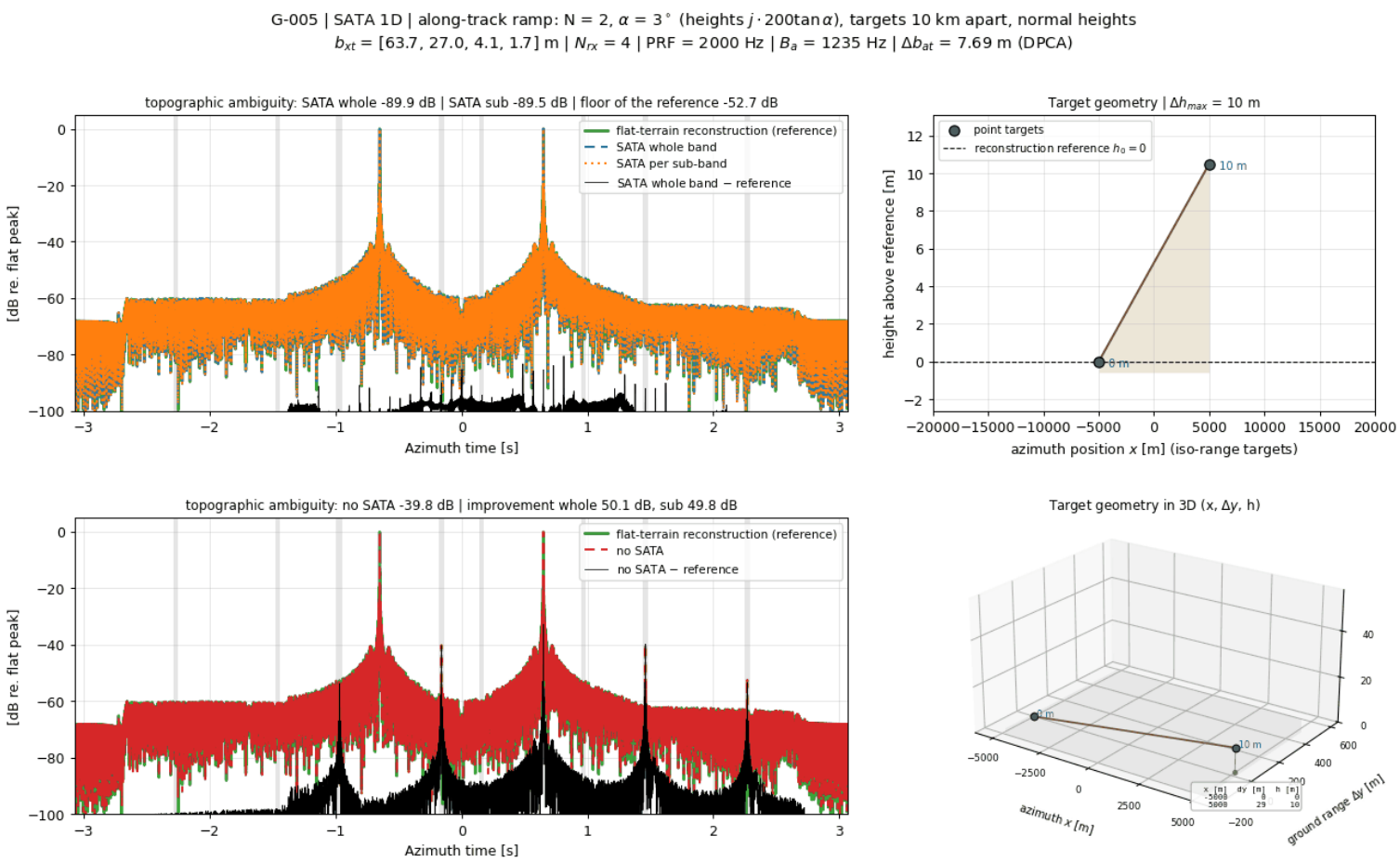


Figure 83: G-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −39.8, SATA whole band −89.9, per sub-band −89.5 dB (improvement 50.1 dB); floor −52.7 dB.

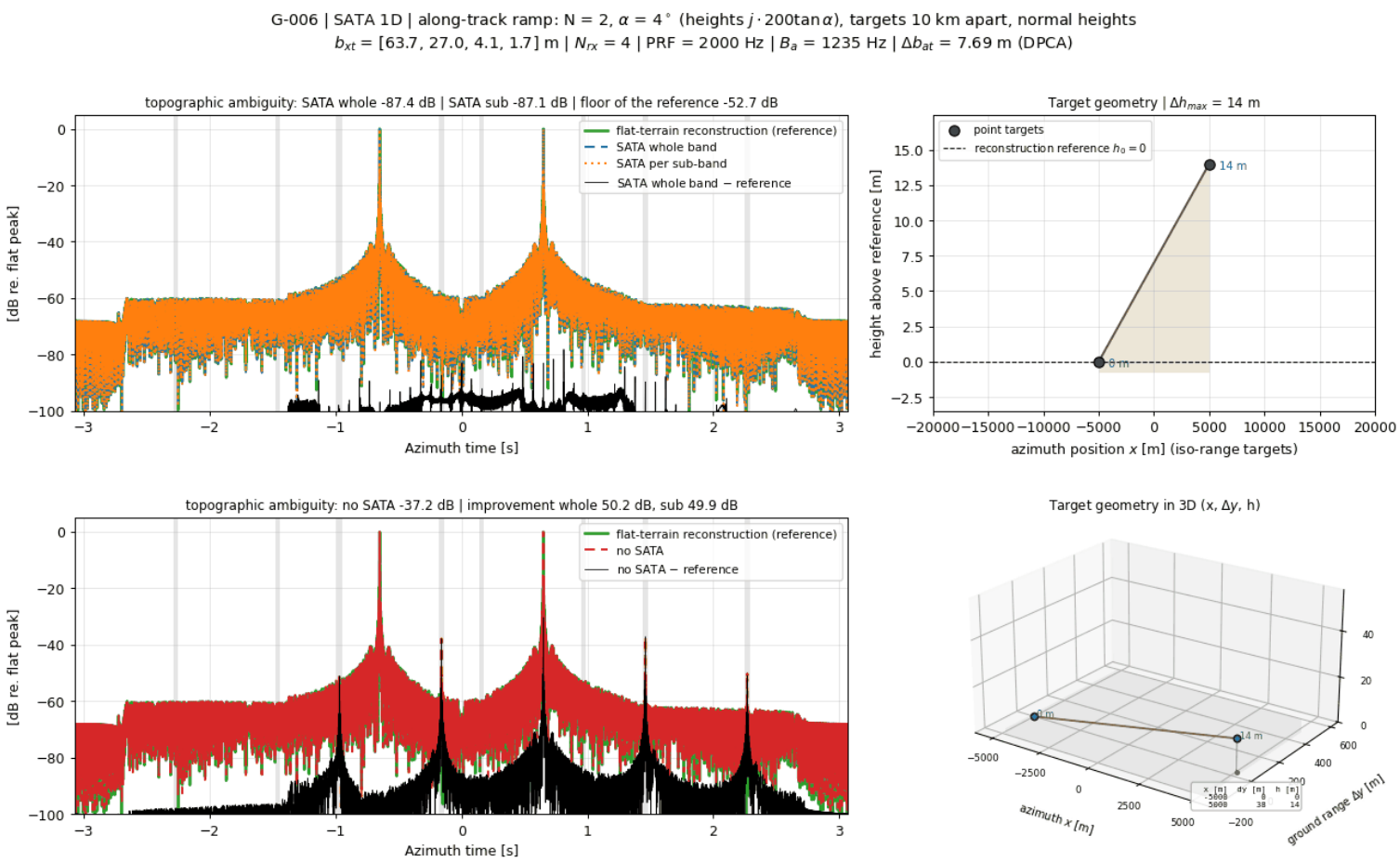


Figure 84: G-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −37.2, SATA whole band −87.4, per sub-band −87.1 dB (improvement 50.2 dB); floor −52.7 dB.

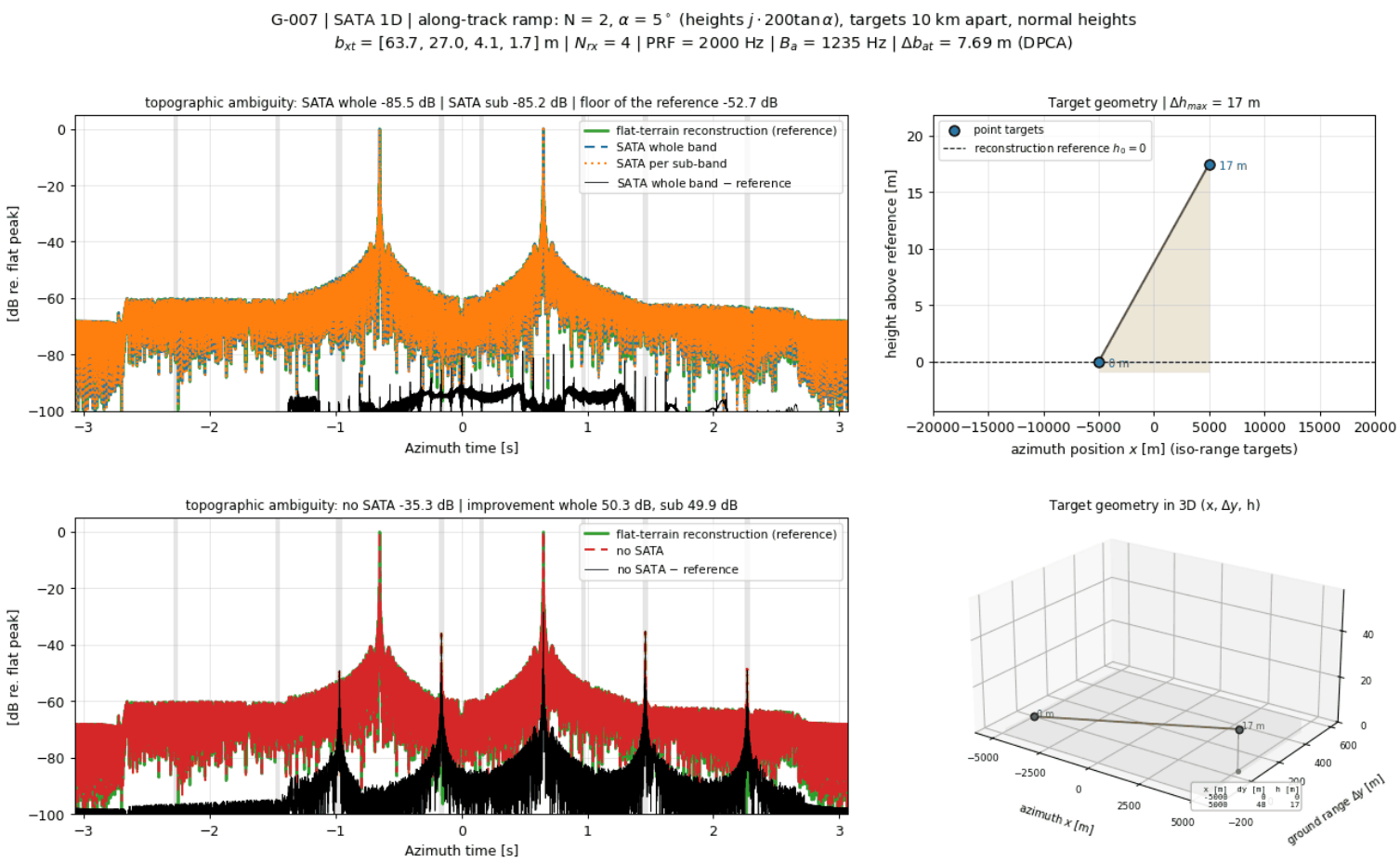


Figure 85: G-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −35.3, SATA whole band −85.5, per sub-band −85.2 dB (improvement 50.3 dB); floor −52.7 dB.

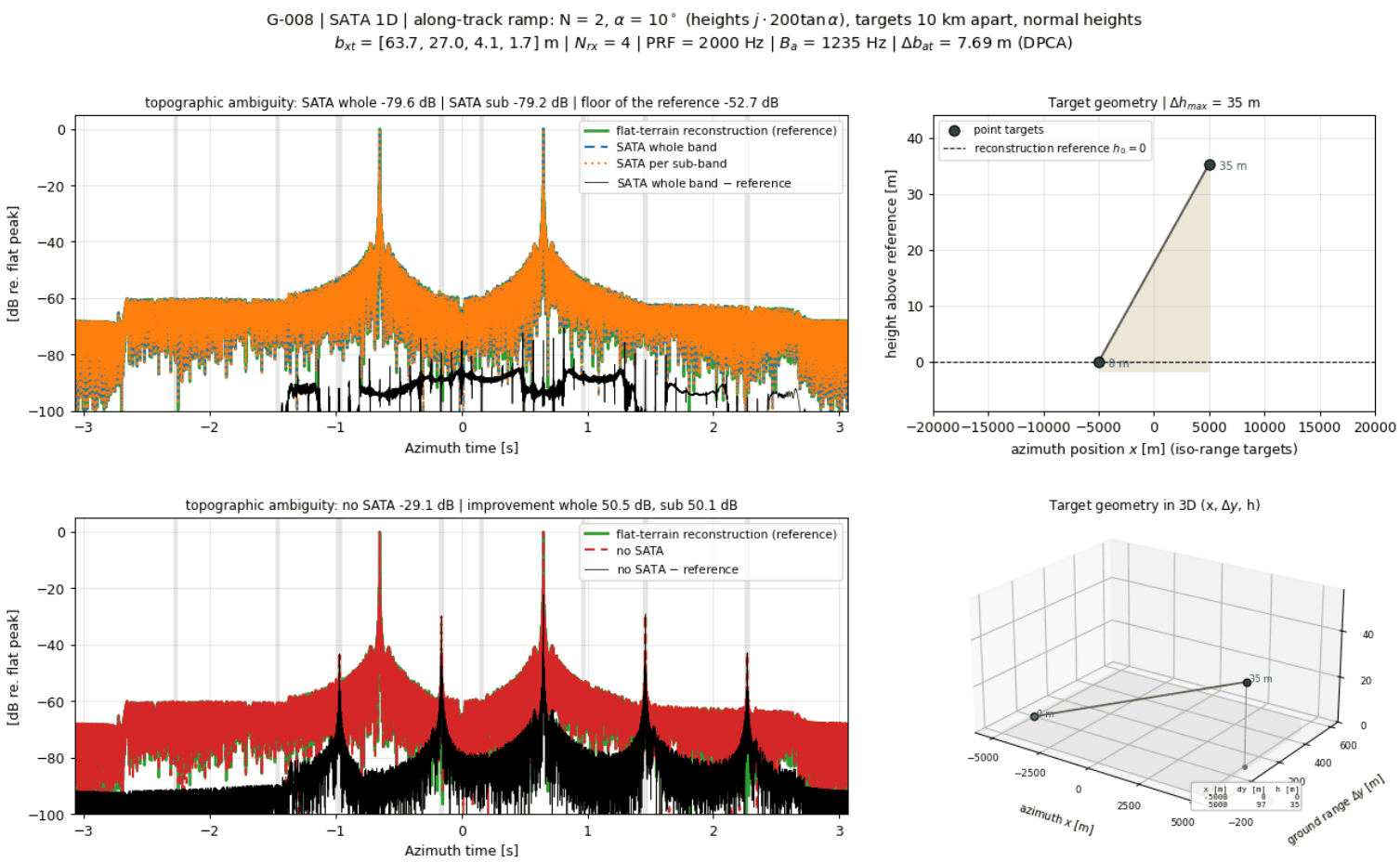


Figure 86: G-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -29.1, SATA whole band -79.6, per sub-band -79.2 dB (improvement 50.5 dB); floor -52.7 dB.

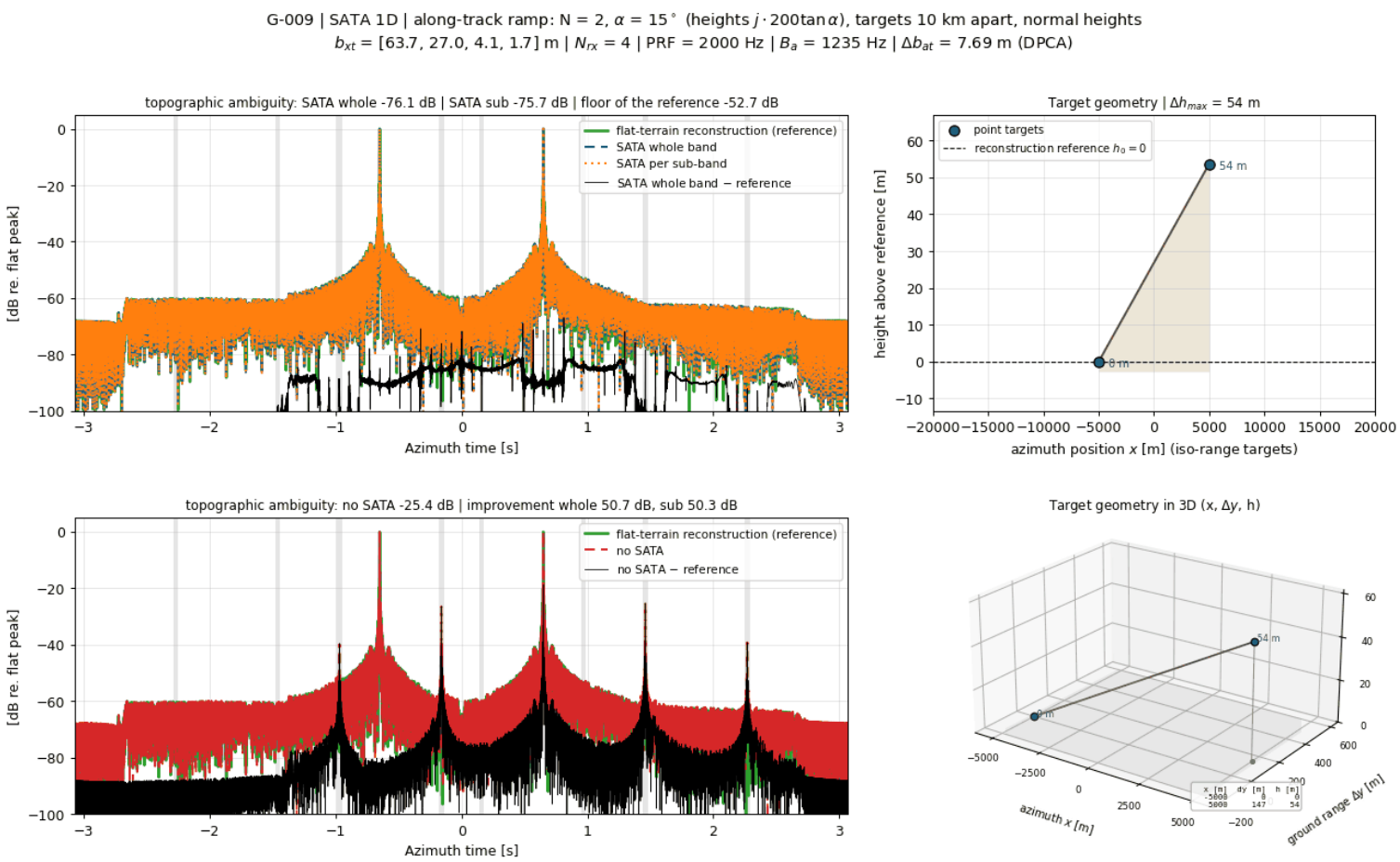


Figure 87: G-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -25.4, SATA whole band -76.1, per sub-band -75.7 dB (improvement 50.7 dB); floor -52.7 dB.

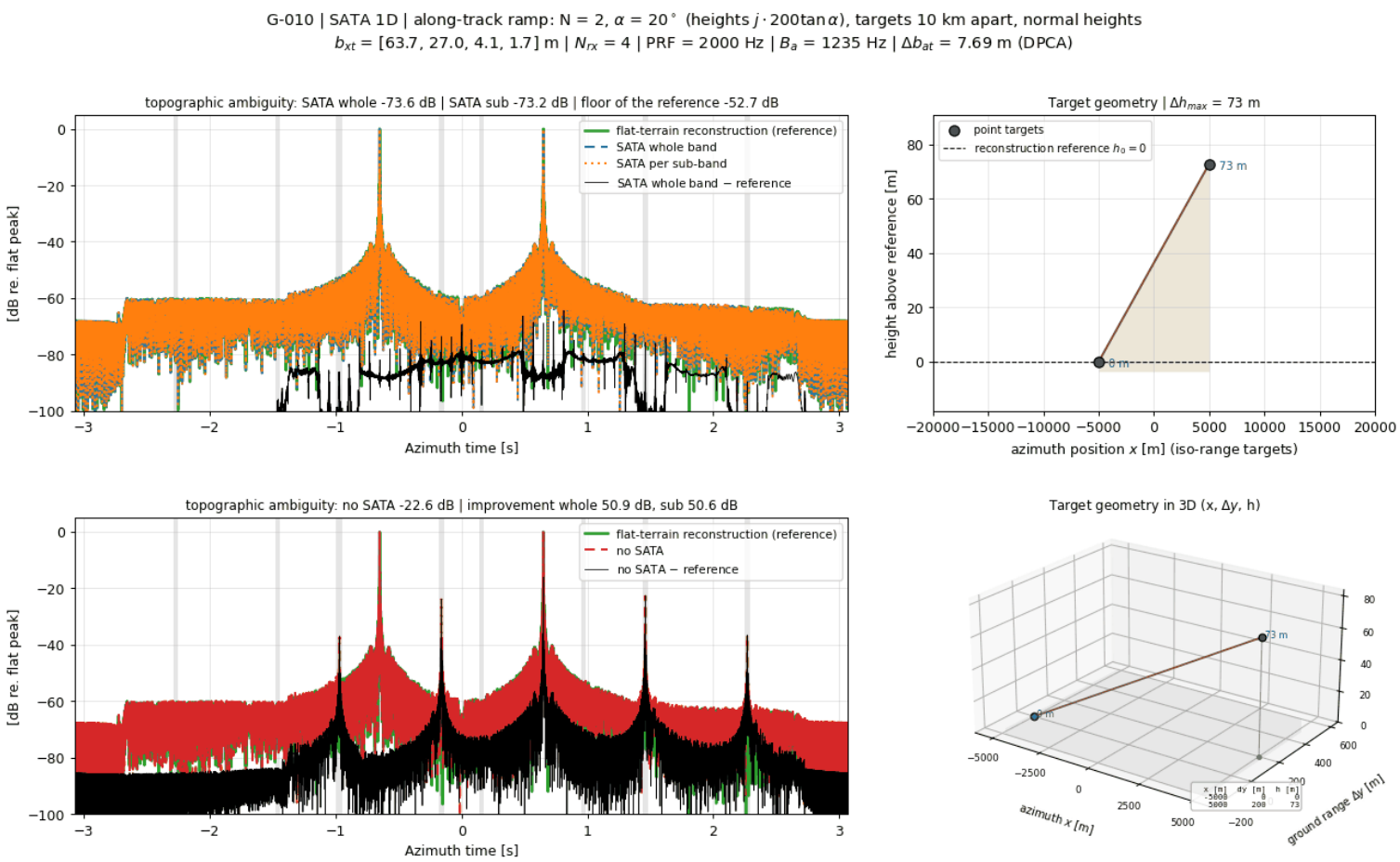


Figure 88: G-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -22.6, SATA whole band -73.6, per sub-band -73.2 dB (improvement 50.9 dB); floor -52.7 dB.

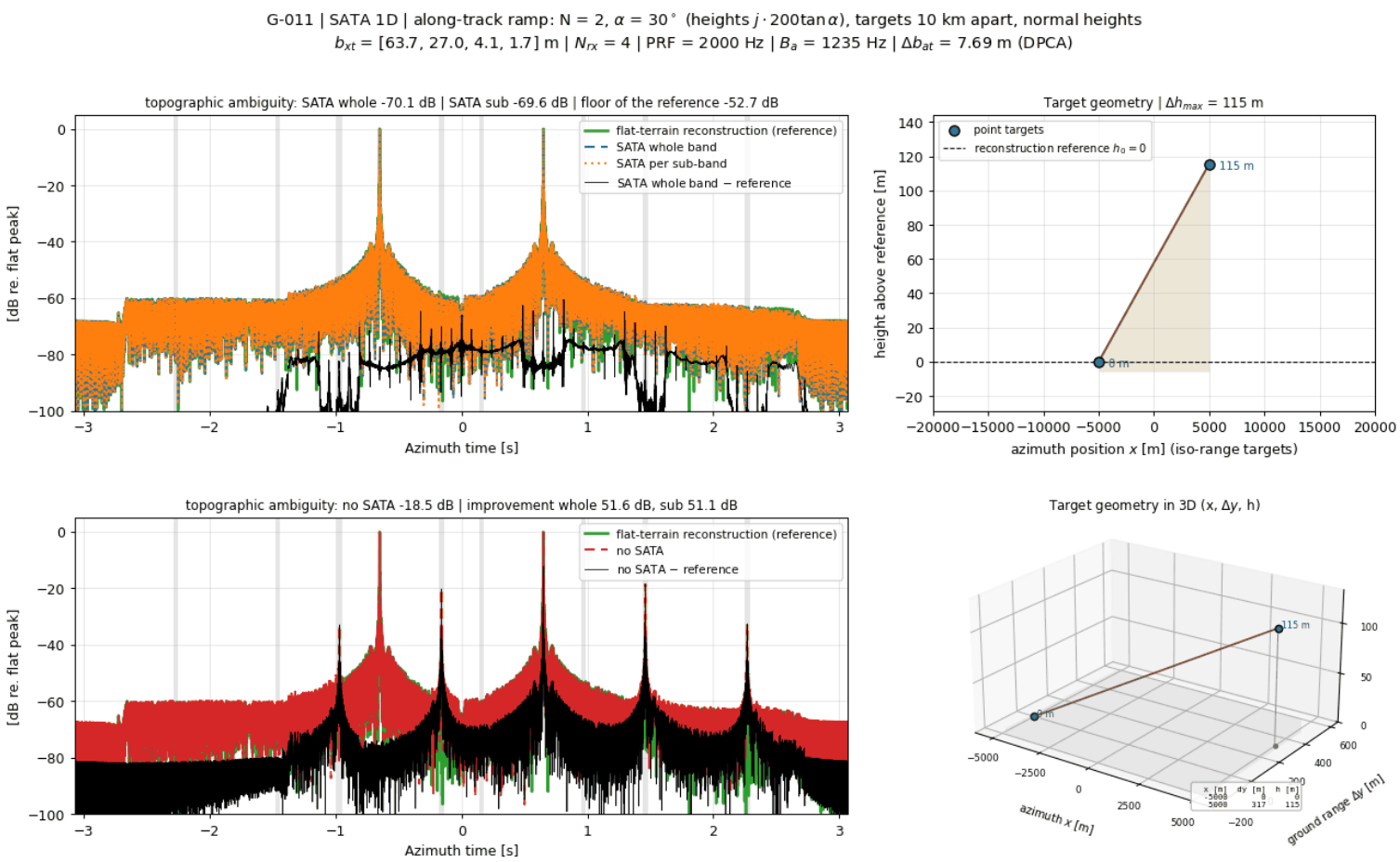


Figure 89: G-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −18.5, SATA whole band −70.1, per sub-band −69.6 dB (improvement 51.6 dB); floor −52.7 dB.

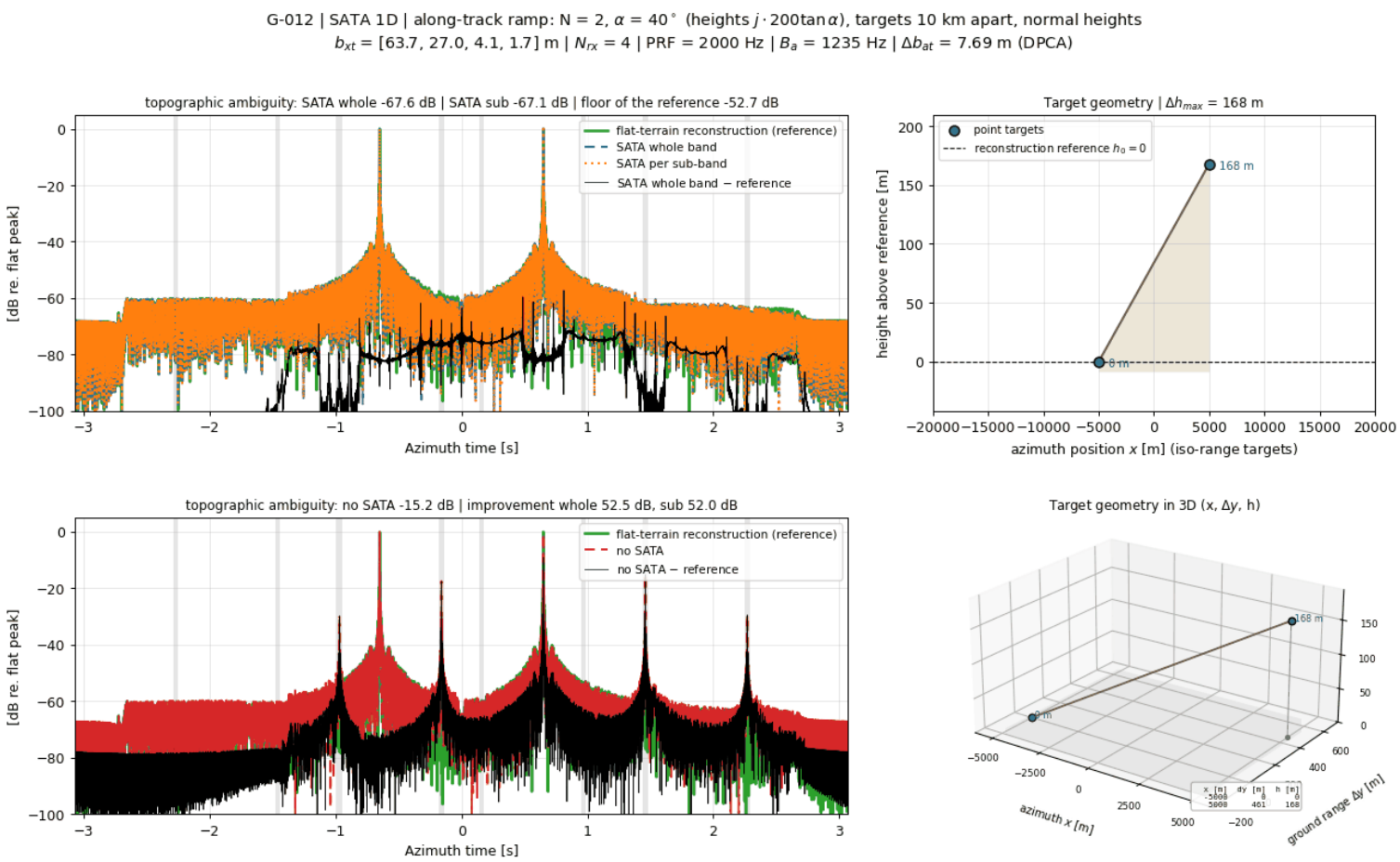


Figure 90: G-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −15.2, SATA whole band −67.6, per sub-band −67.1 dB (improvement 52.5 dB); floor −52.7 dB.

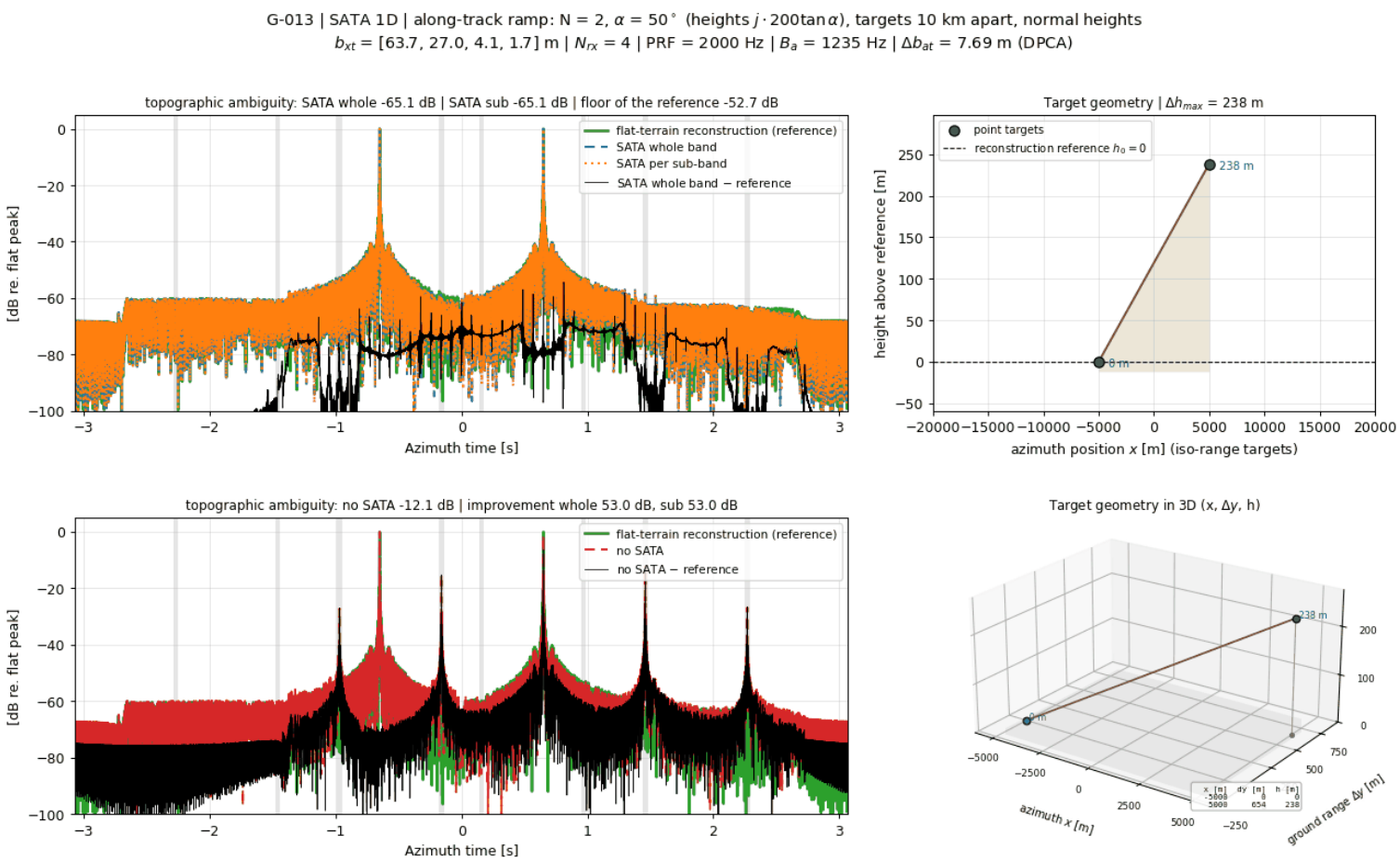


Figure 91: G-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −12.1, SATA whole band −65.1, per sub-band −65.1 dB (improvement 53.0 dB); floor −52.7 dB.

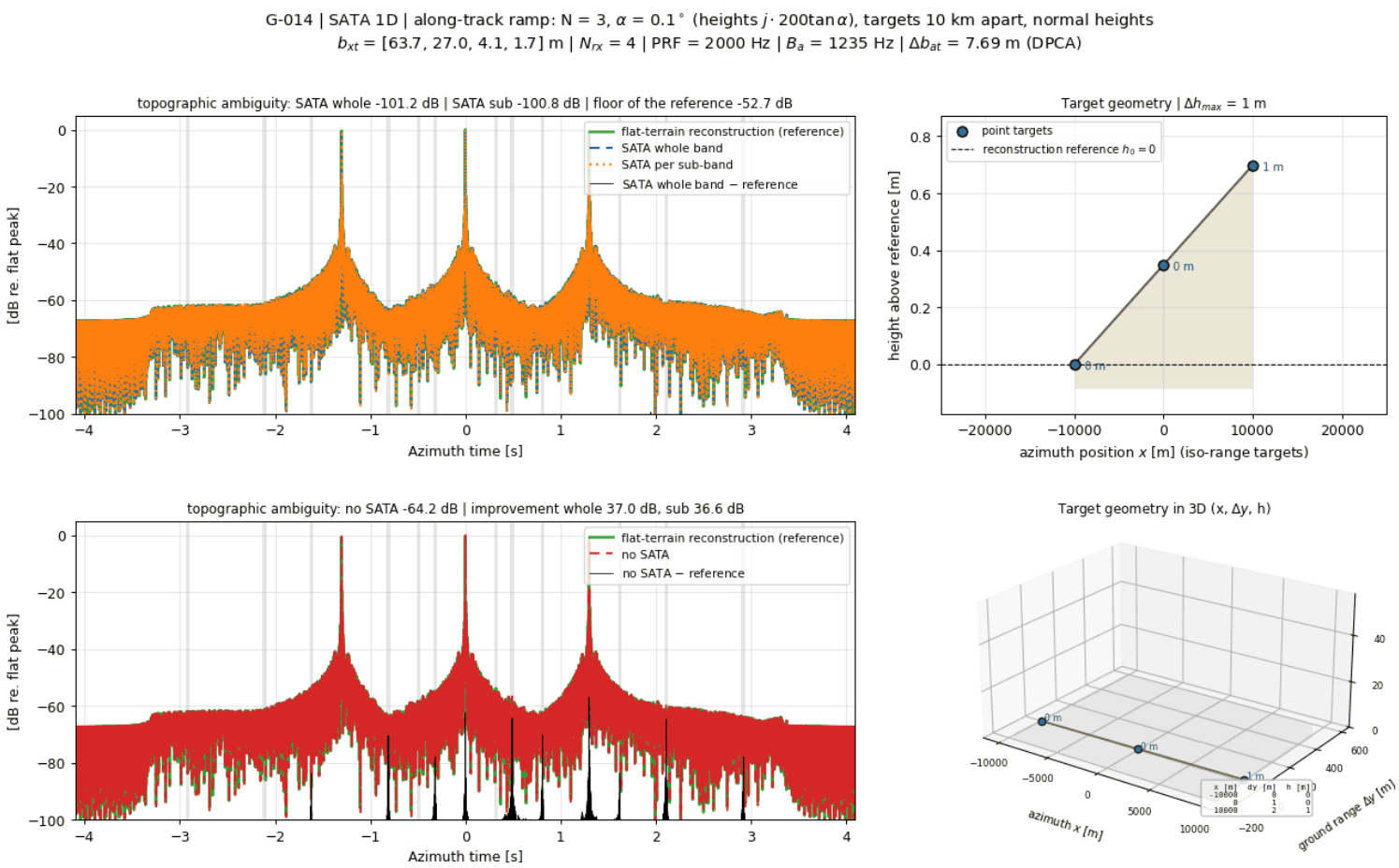


Figure 92: G-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −64.2, SATA whole band −101.2, per sub-band −100.8 dB (improvement 37.0 dB); floor −52.7 dB.

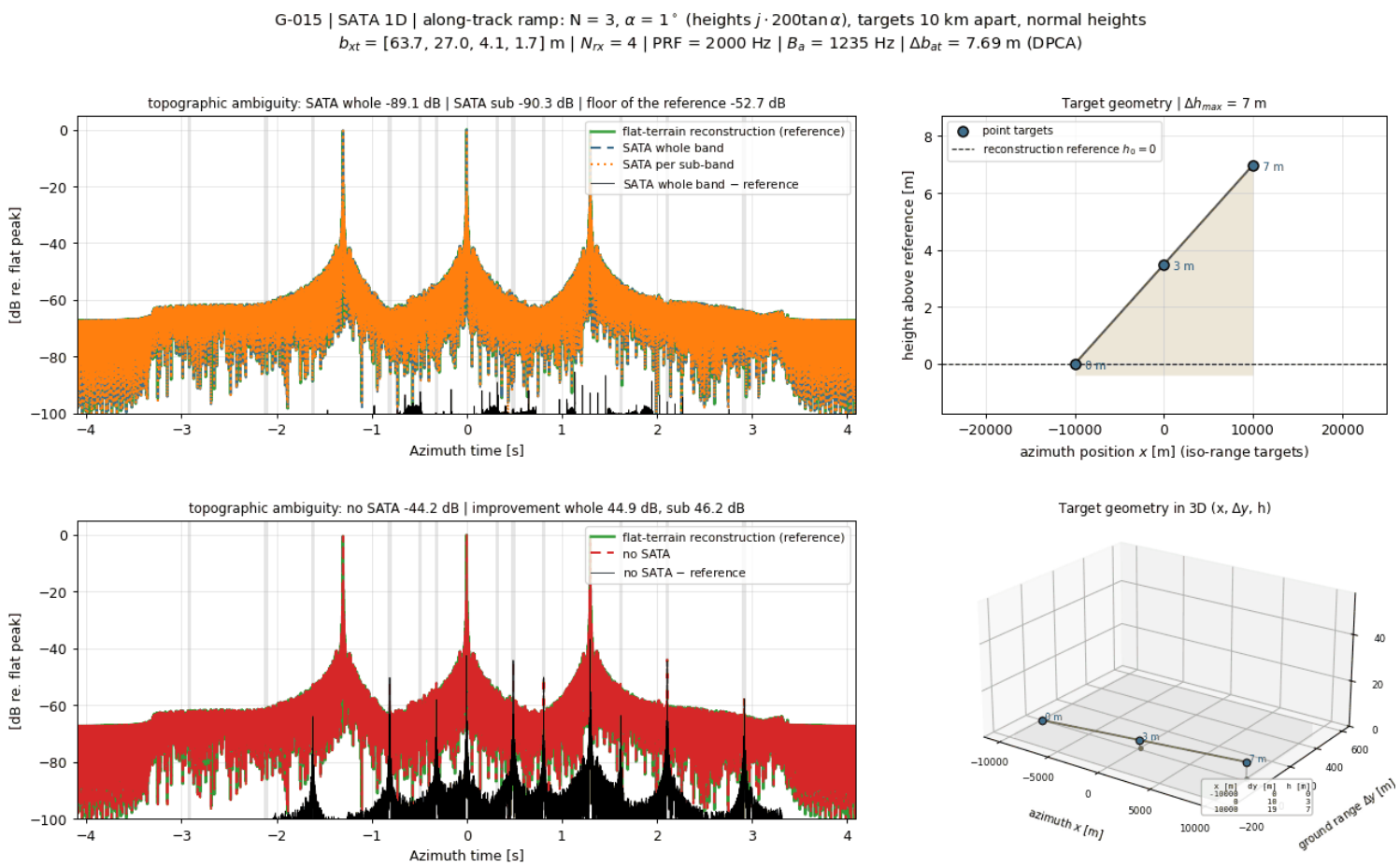


Figure 93: G-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -44.2, SATA whole band -89.1, per sub-band -90.3 dB (improvement 44.9 dB); floor -52.7 dB.

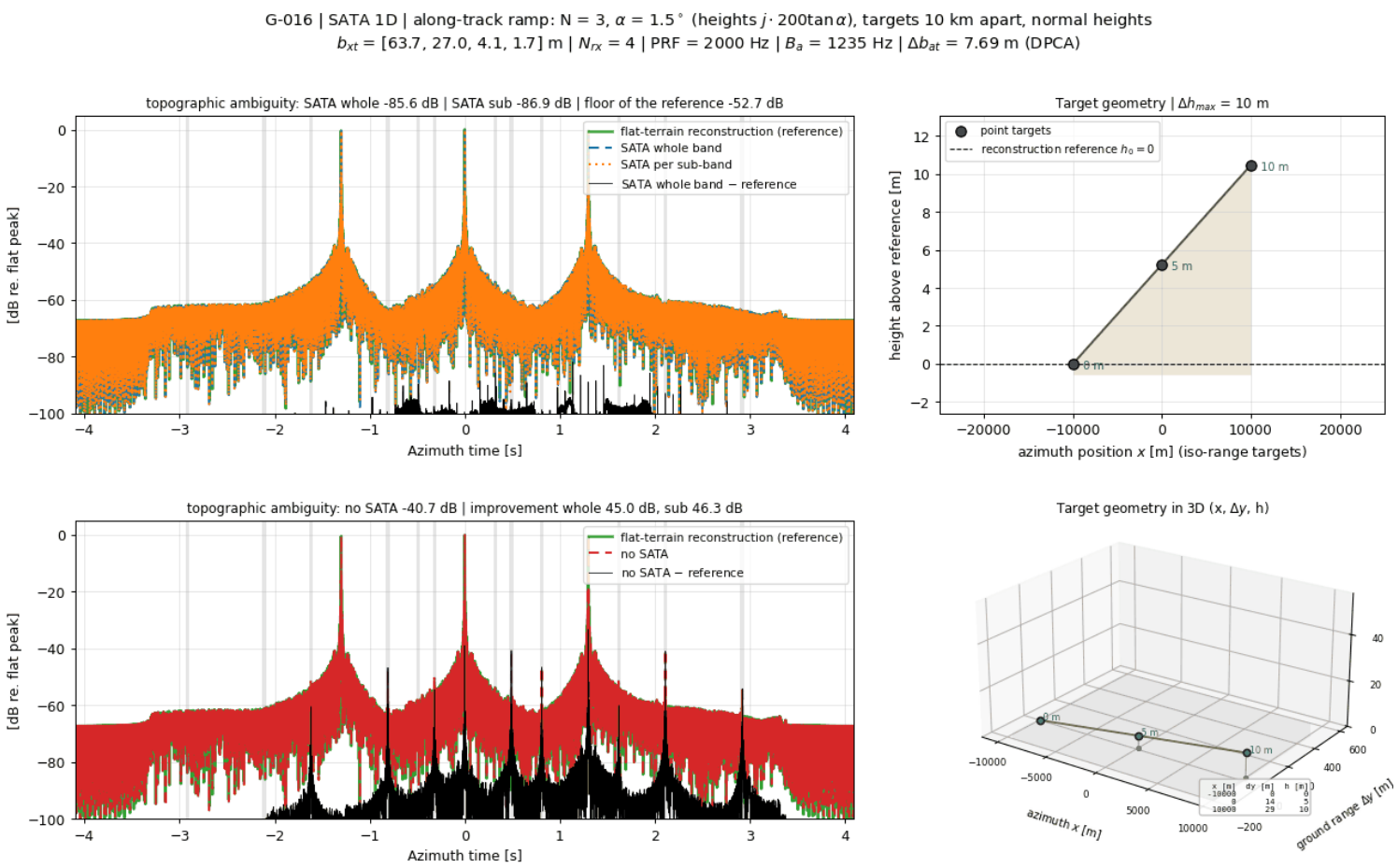


Figure 94: G-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j{\cdot}200\tan\alpha$). Topographic ambiguity: no SATA -40.7, SATA whole band -85.6, per sub-band -86.9 dB (improvement 45.0 dB); floor -52.7 dB.

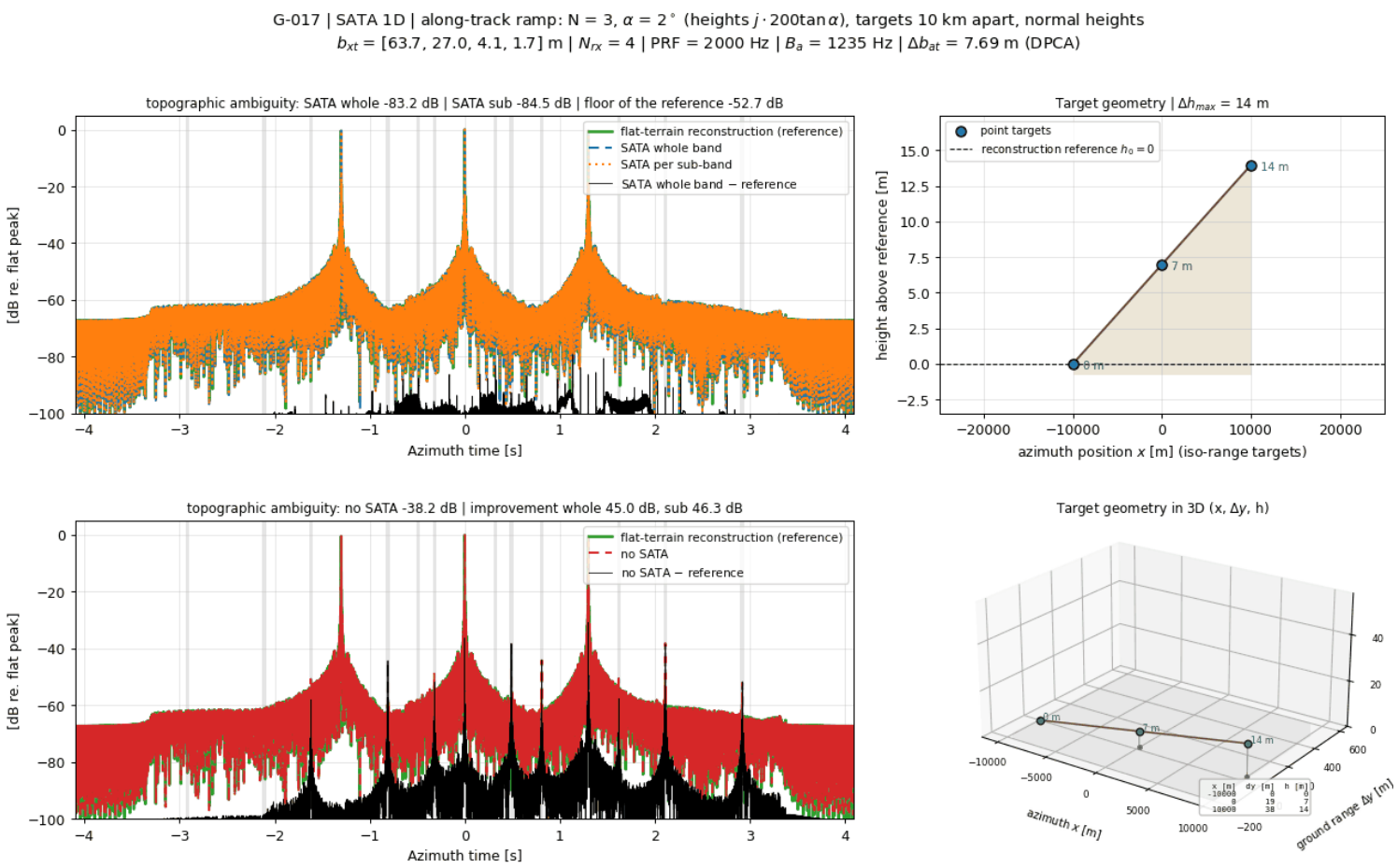


Figure 95: G-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −38.2, SATA whole band −83.2, per sub-band −84.5 dB (improvement 45.0 dB); floor −52.7 dB.

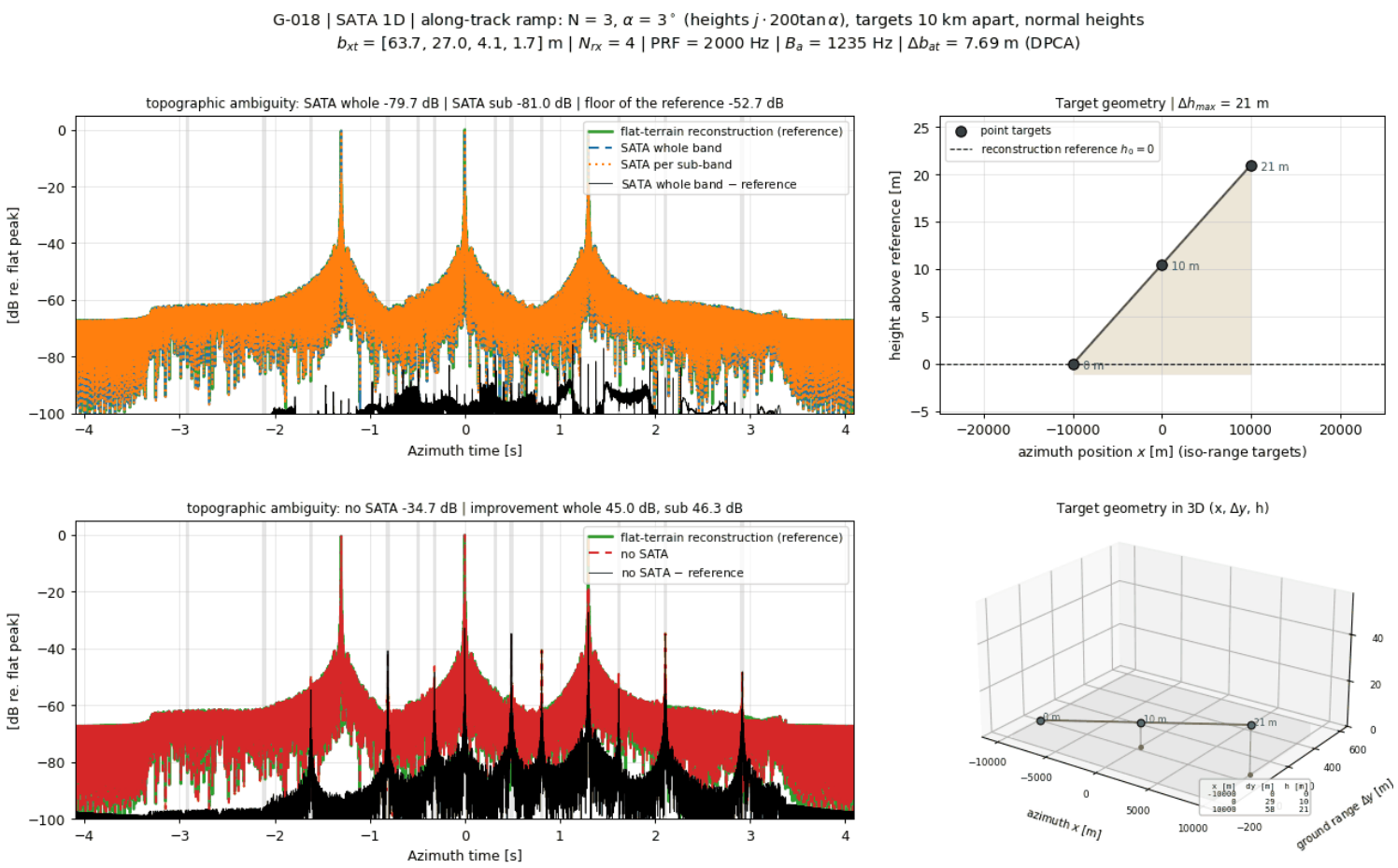


Figure 96: G-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −34.7, SATA whole band −79.7, per sub-band −81.0 dB (improvement 45.0 dB); floor −52.7 dB.

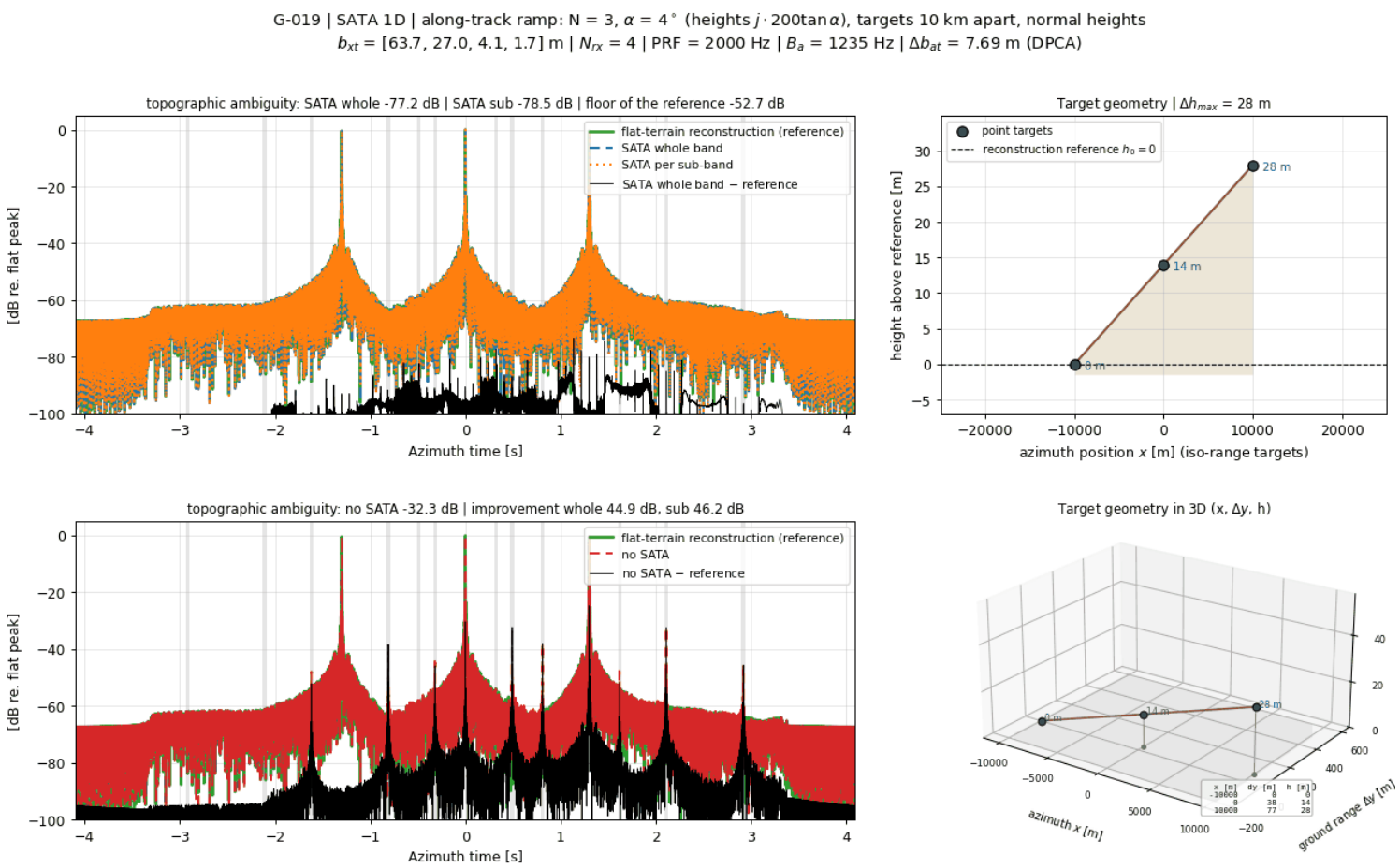


Figure 97: G-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -32.3, SATA whole band -77.2, per sub-band -78.5 dB (improvement 44.9 dB); floor -52.7 dB.

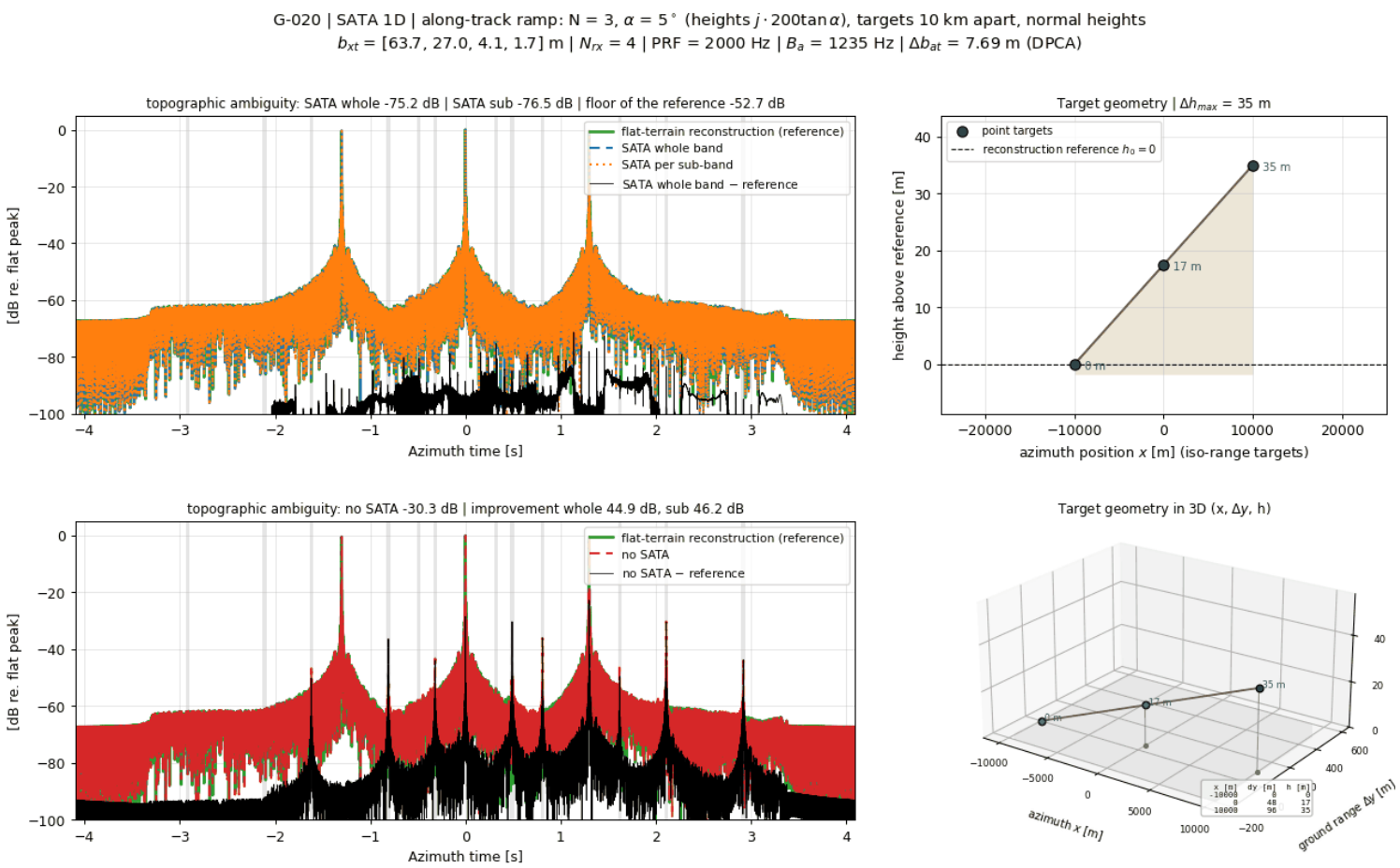


Figure 98: G-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -30.3, SATA whole band -75.2, per sub-band -76.5 dB (improvement 44.9 dB); floor -52.7 dB.

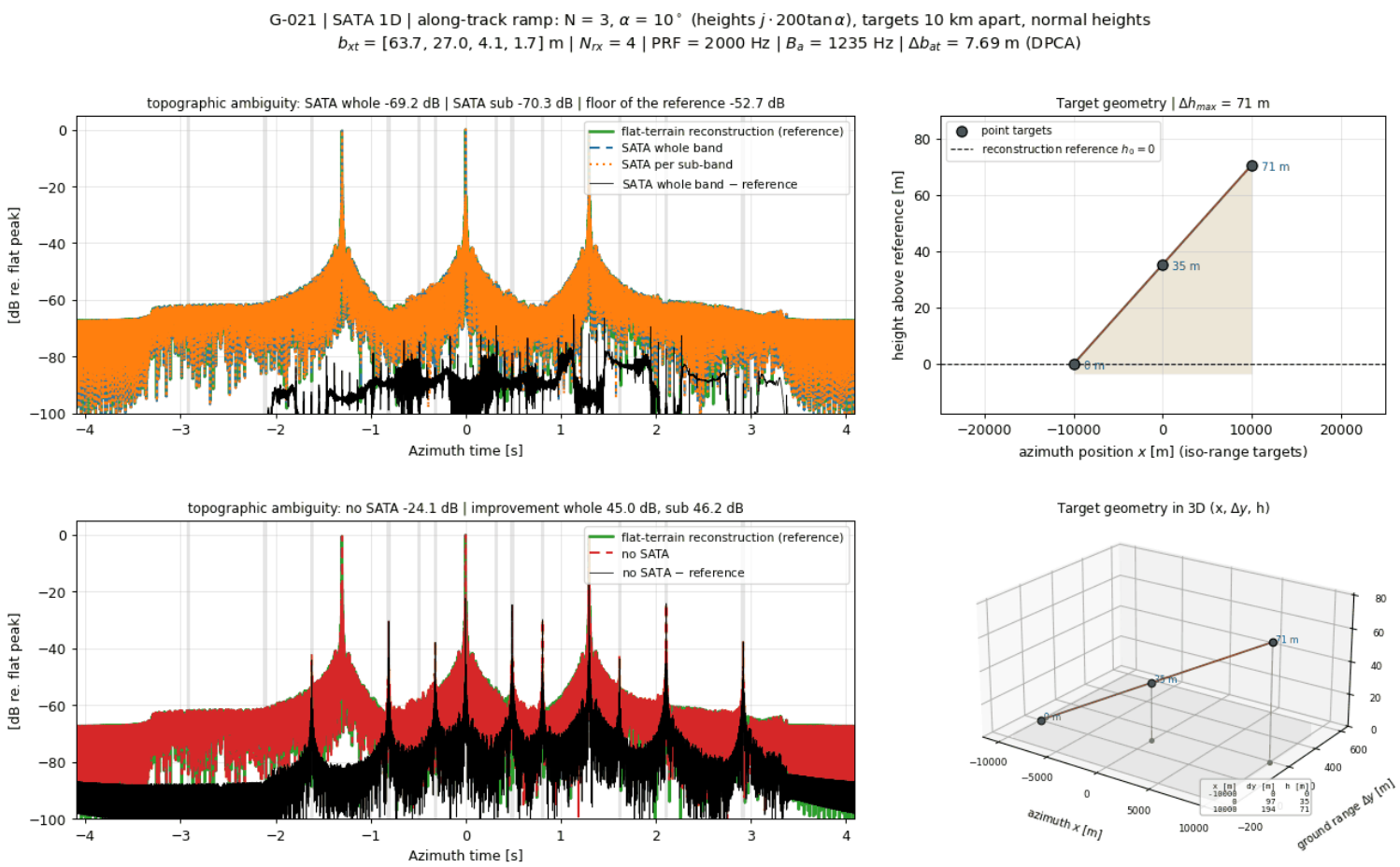


Figure 99: G-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -24.1, SATA whole band -69.2, per sub-band -70.3 dB (improvement 45.0 dB); floor -52.7 dB.

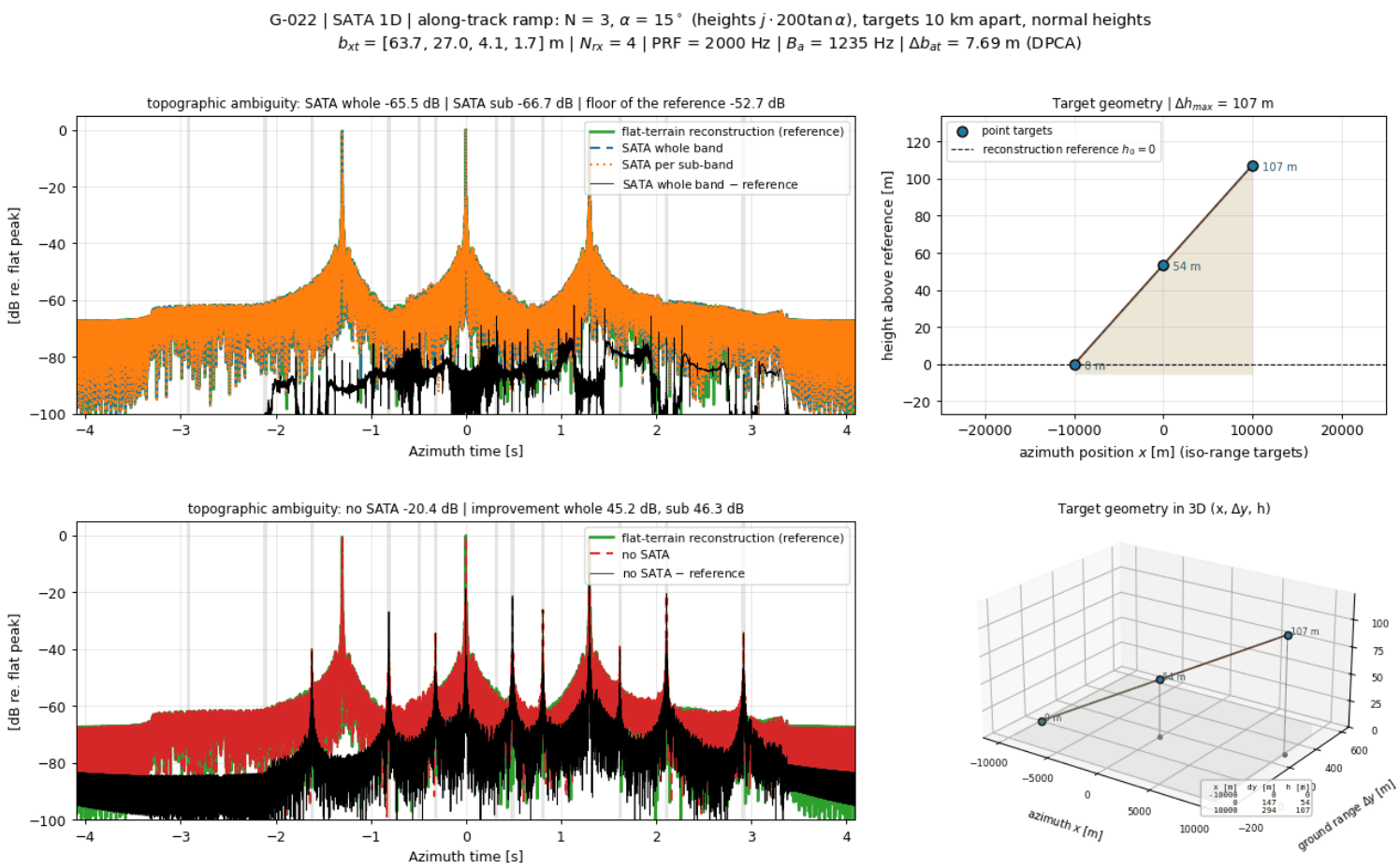


Figure 100: G-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −20.4, SATA whole band −65.5, per sub-band −66.7 dB (improvement 45.2 dB); floor −52.7 dB.

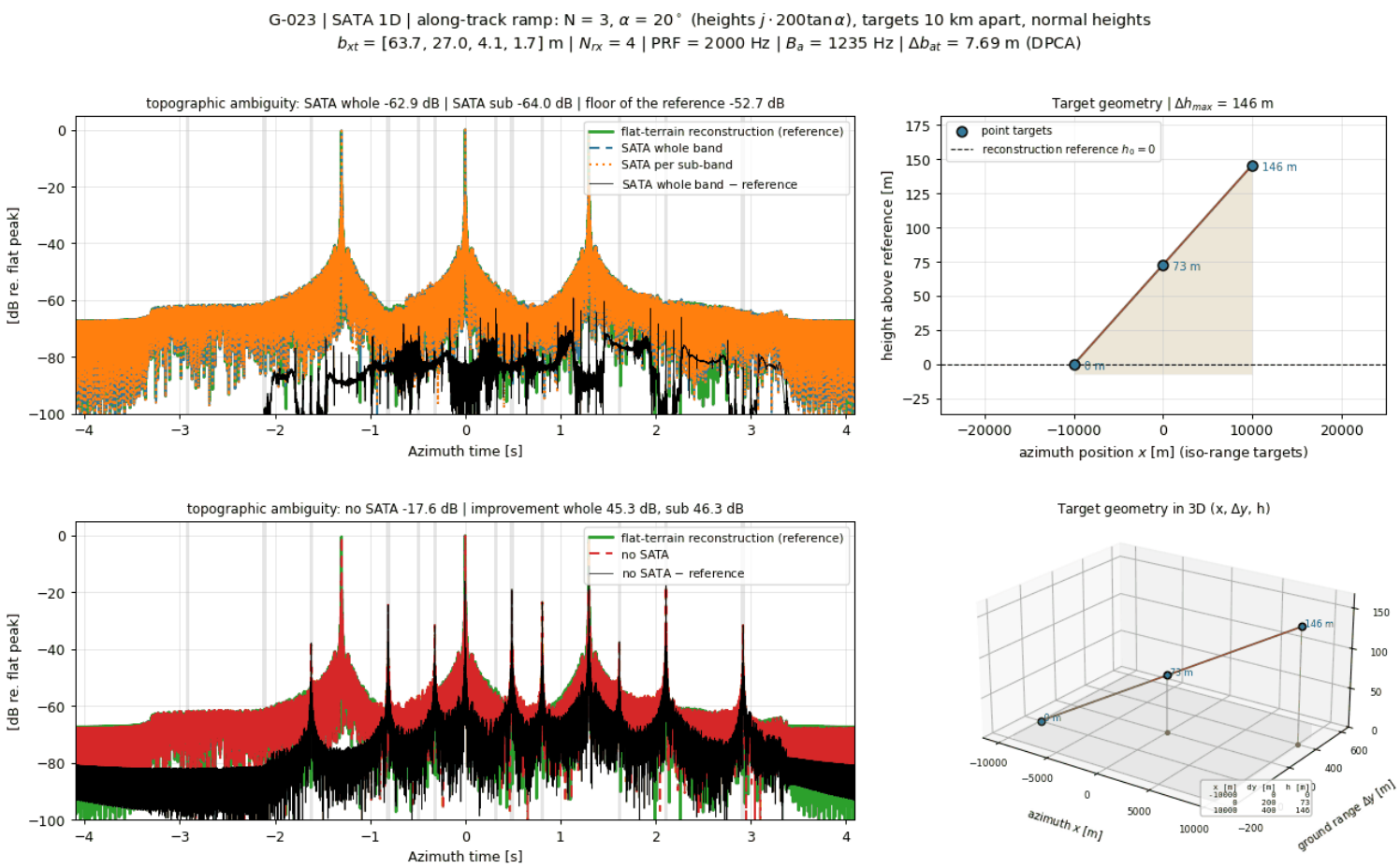


Figure 101: G-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −17.6, SATA whole band −62.9, per sub-band −64.0 dB (improvement 45.3 dB); floor −52.7 dB.

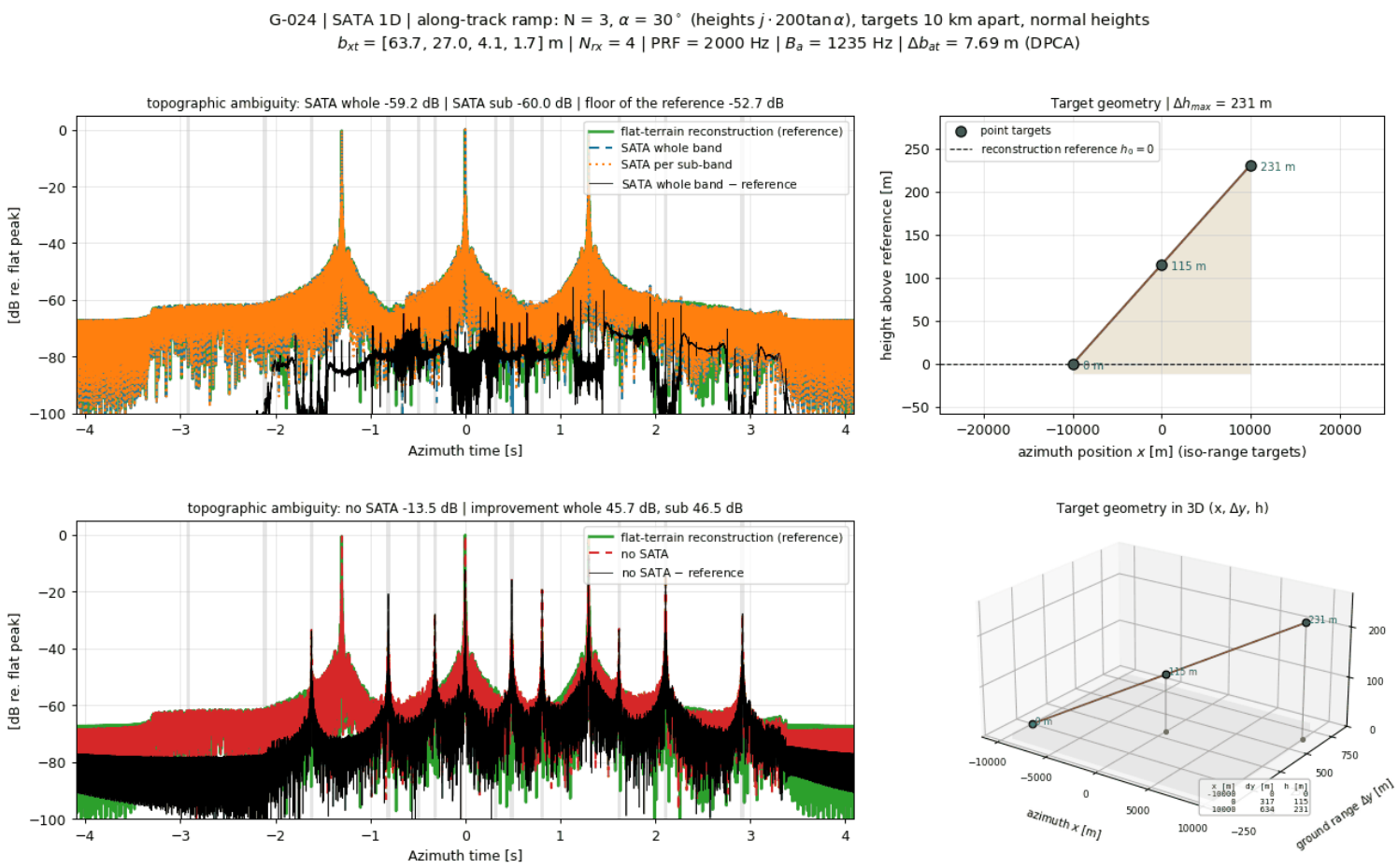


Figure 102: G-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −13.5, SATA whole band −59.2, per sub-band −60.0 dB (improvement 45.7 dB); floor −52.7 dB.

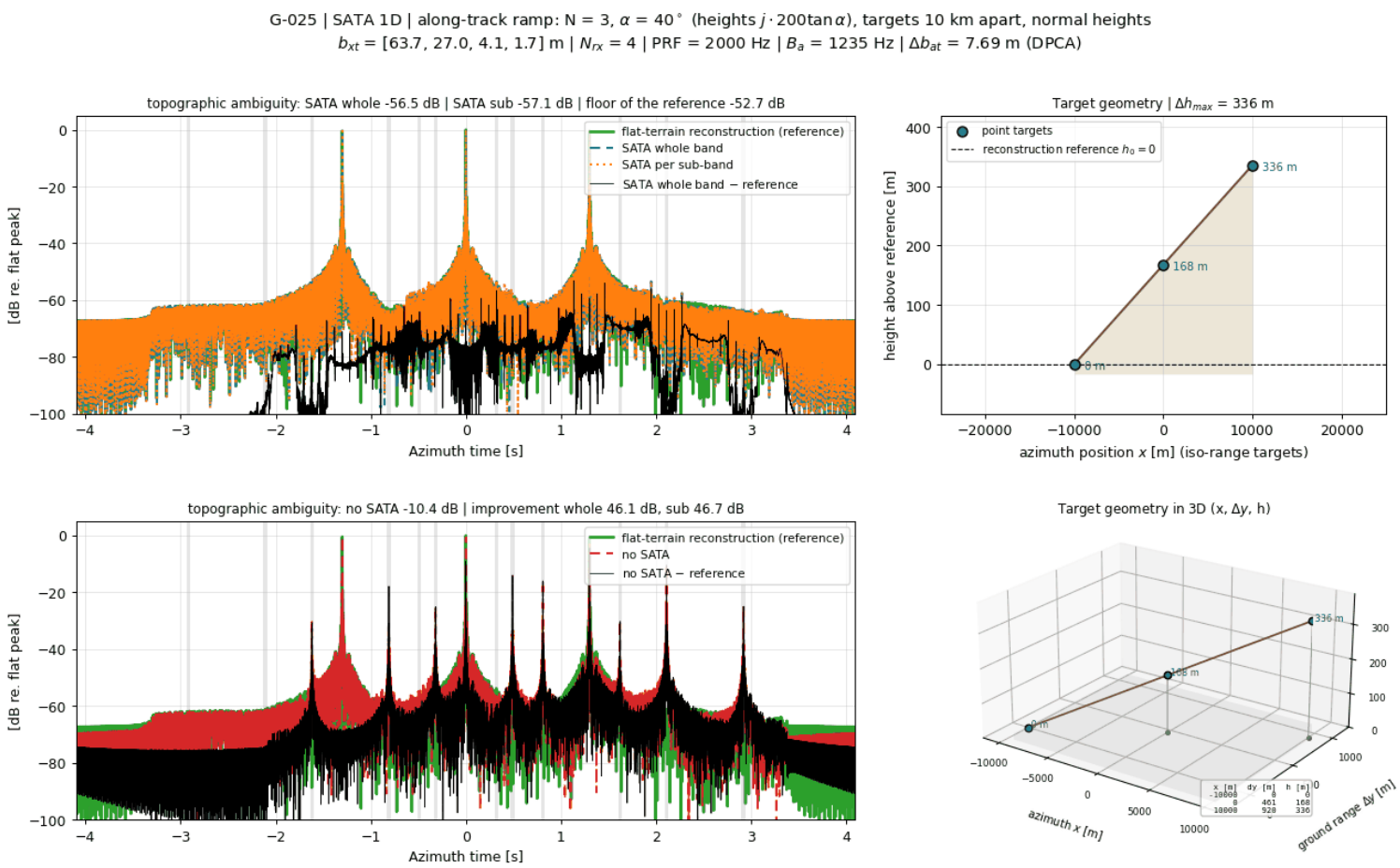


Figure 103: G-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −10.4, SATA whole band −56.5, per sub-band −57.1 dB (improvement 46.1 dB); floor −52.7 dB.

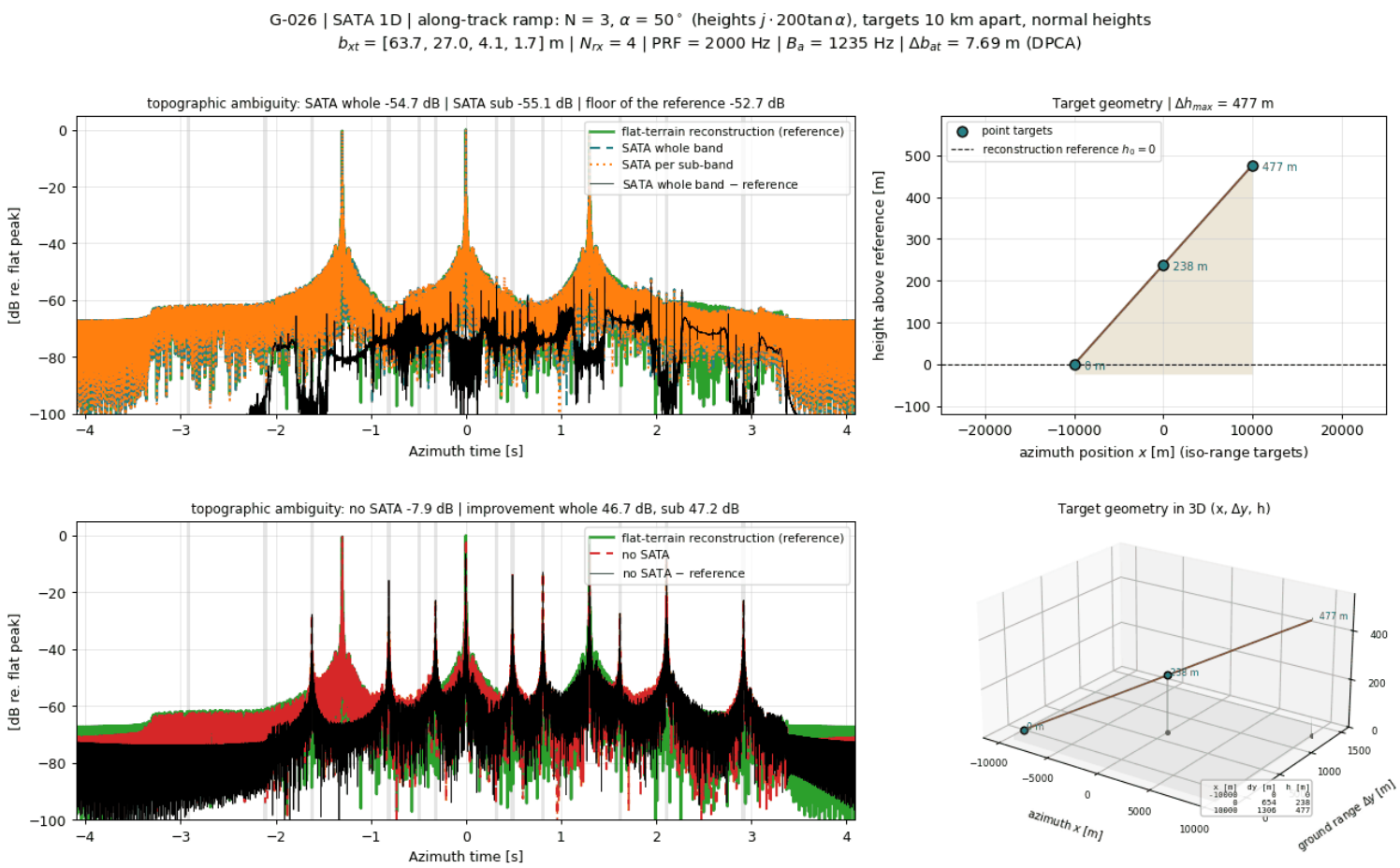


Figure 104: G-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −7.9, SATA whole band −54.7, per sub-band −55.1 dB (improvement 46.7 dB); floor −52.7 dB.

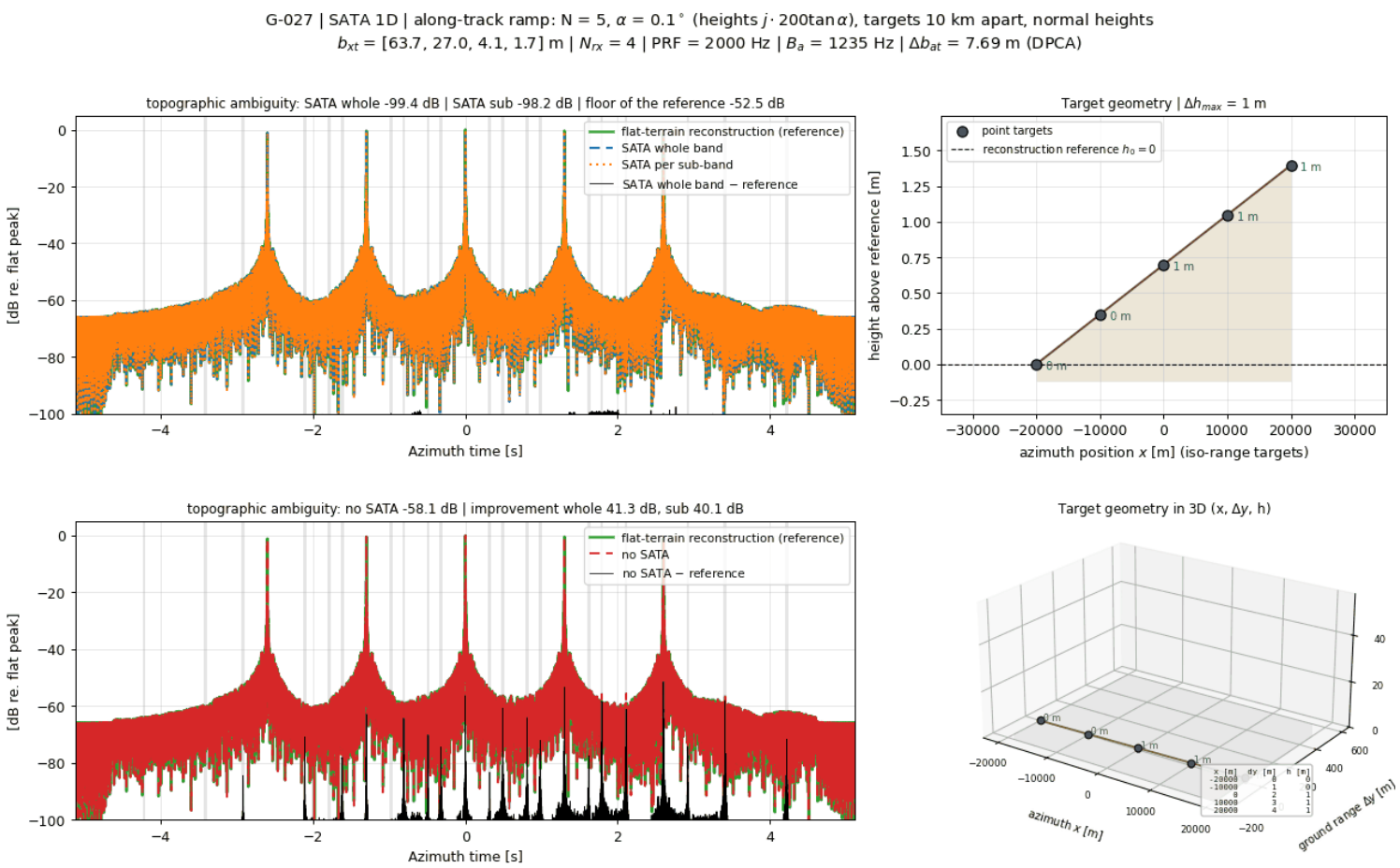


Figure 105: G-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −58.1, SATA whole band −99.4, per sub-band −98.2 dB (improvement 41.3 dB); floor −52.5 dB.

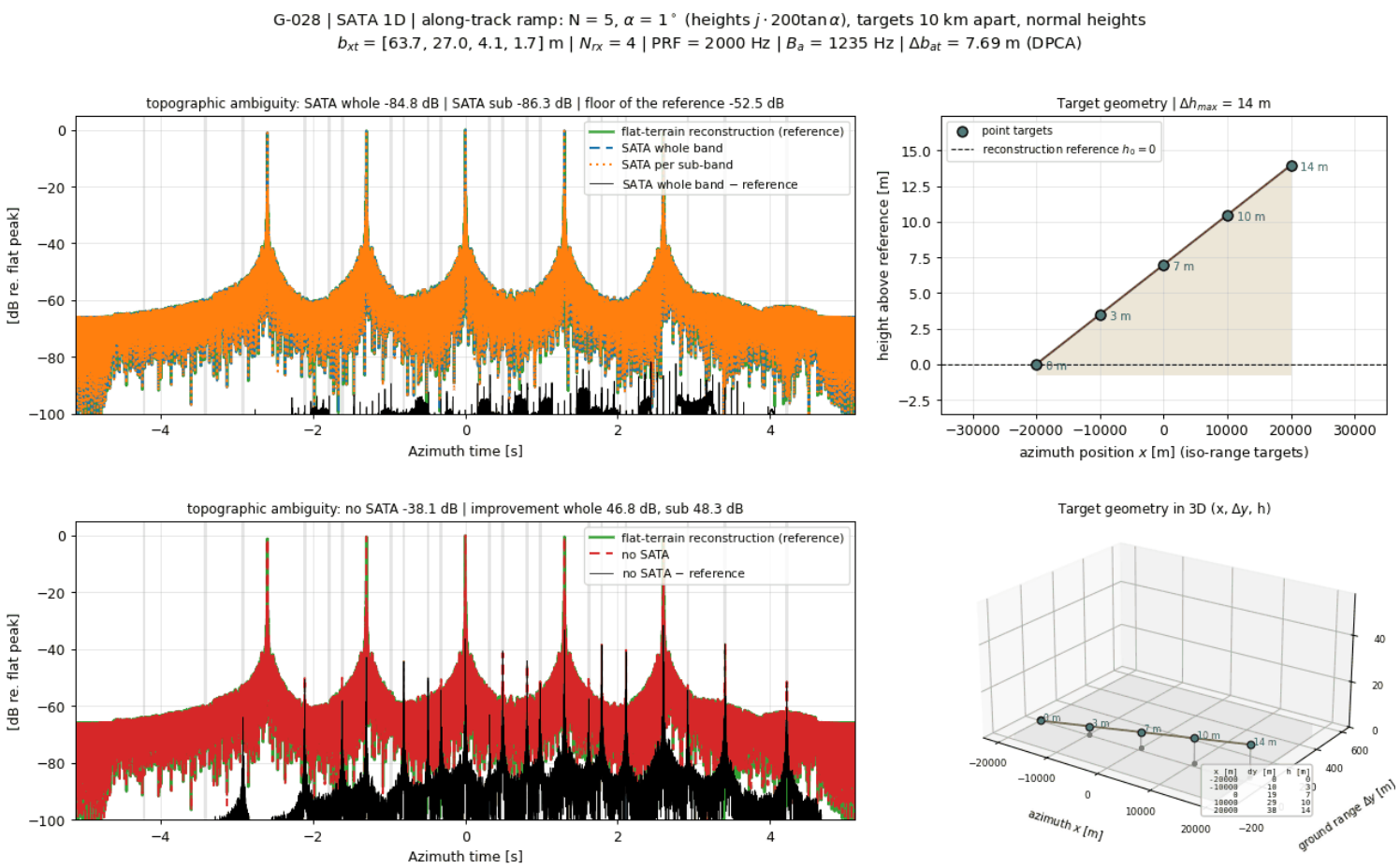


Figure 106: G-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −38.1, SATA whole band −84.8, per sub-band −86.3 dB (improvement 46.8 dB); floor −52.5 dB.

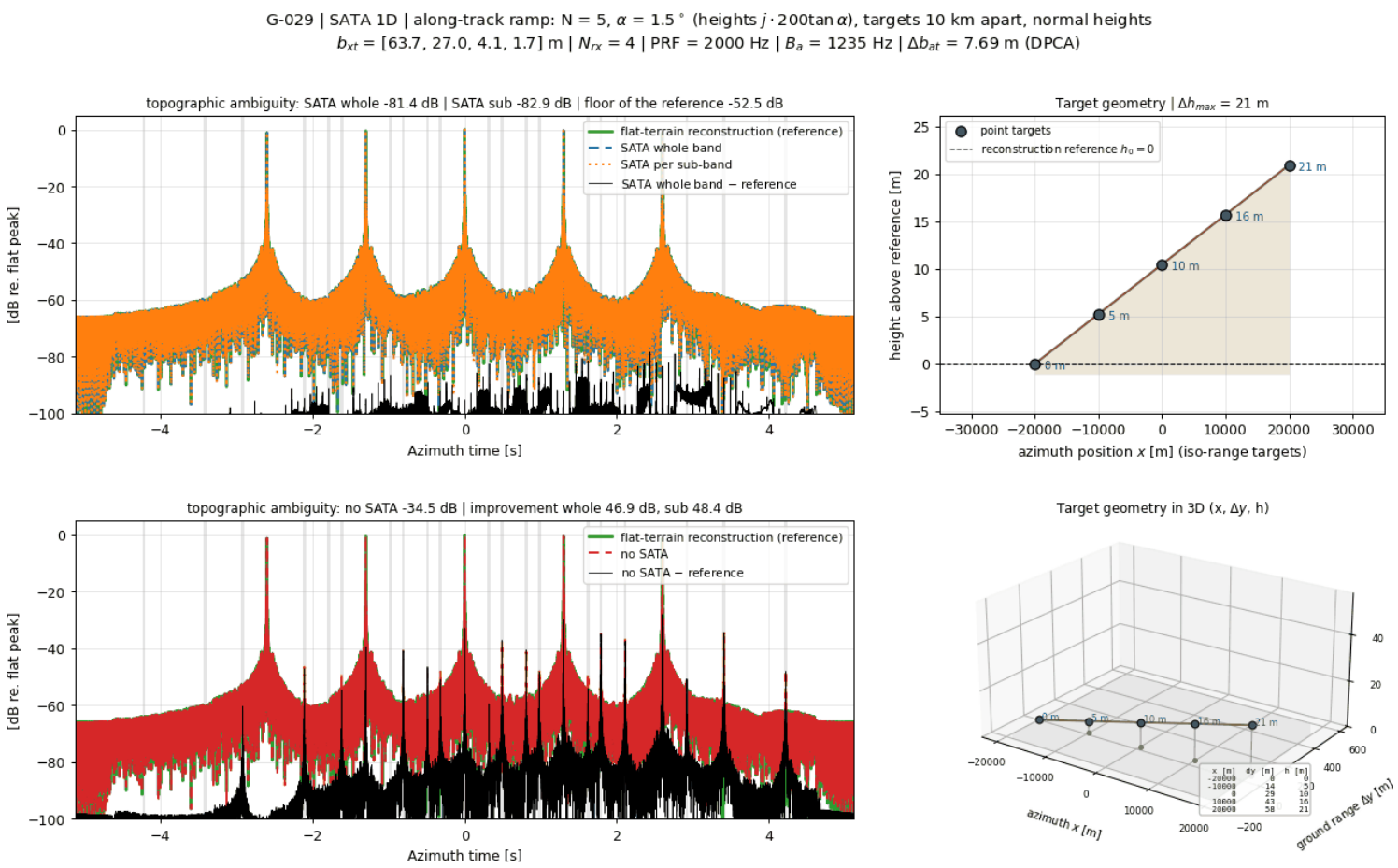


Figure 107: G-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −34.5, SATA whole band −81.4, per sub-band −82.9 dB (improvement 46.9 dB); floor −52.5 dB.

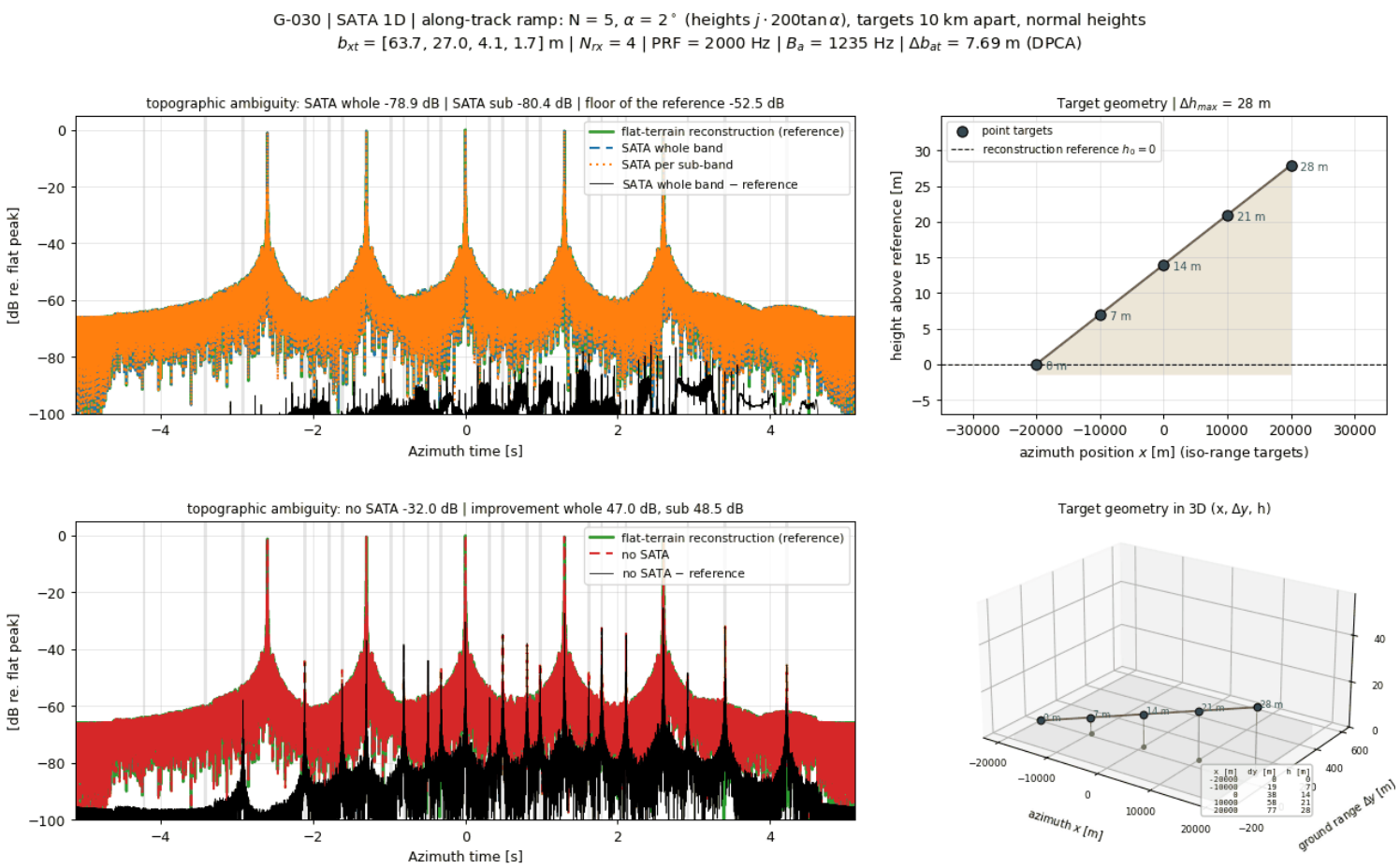


Figure 108: G-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA −32.0, SATA whole band −78.9, per sub-band −80.4 dB (improvement 47.0 dB); floor −52.5 dB.

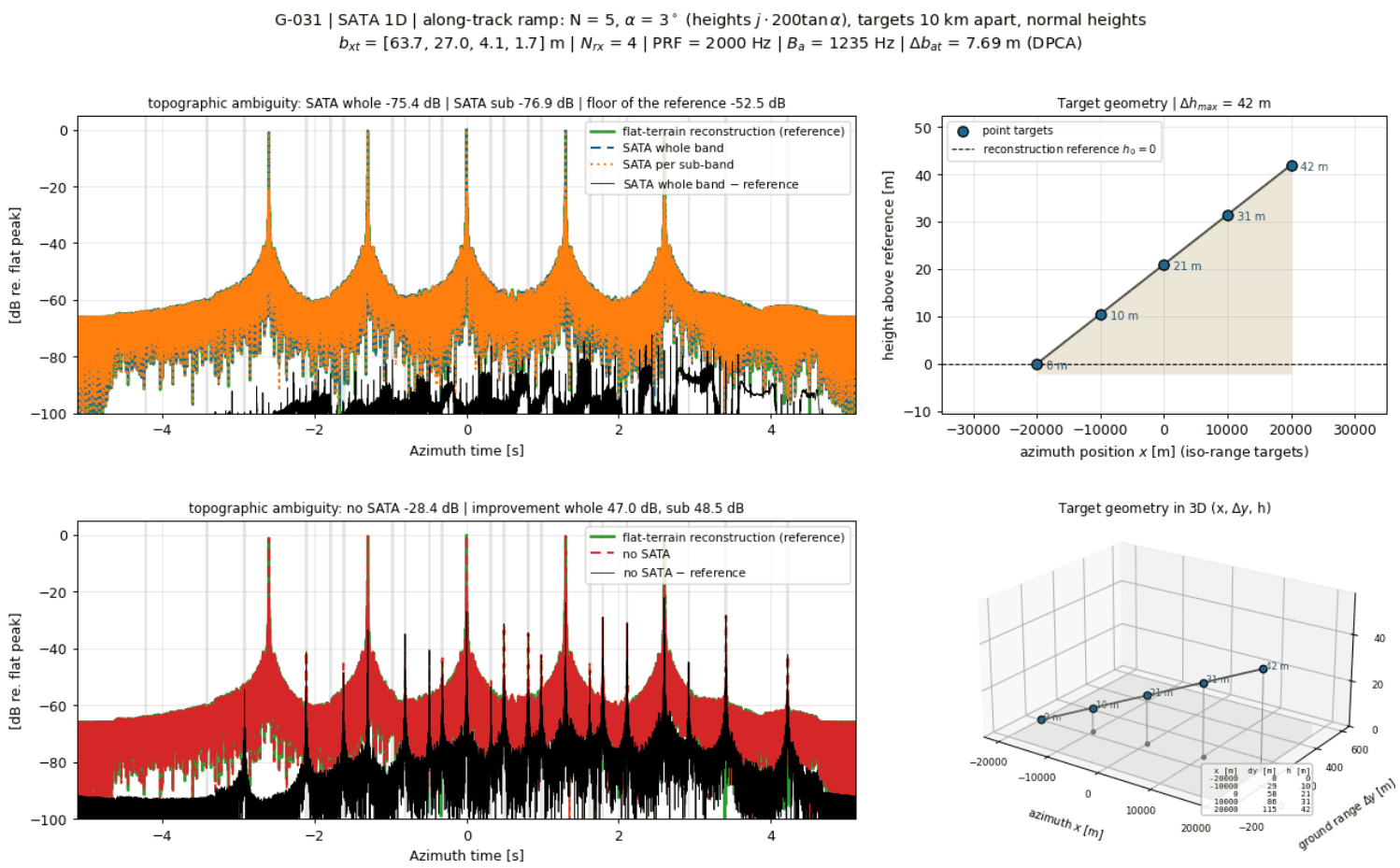


Figure 109: G-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -28.4, SATA whole band -75.4, per sub-band -76.9 dB (improvement 47.0 dB); floor -52.5 dB.

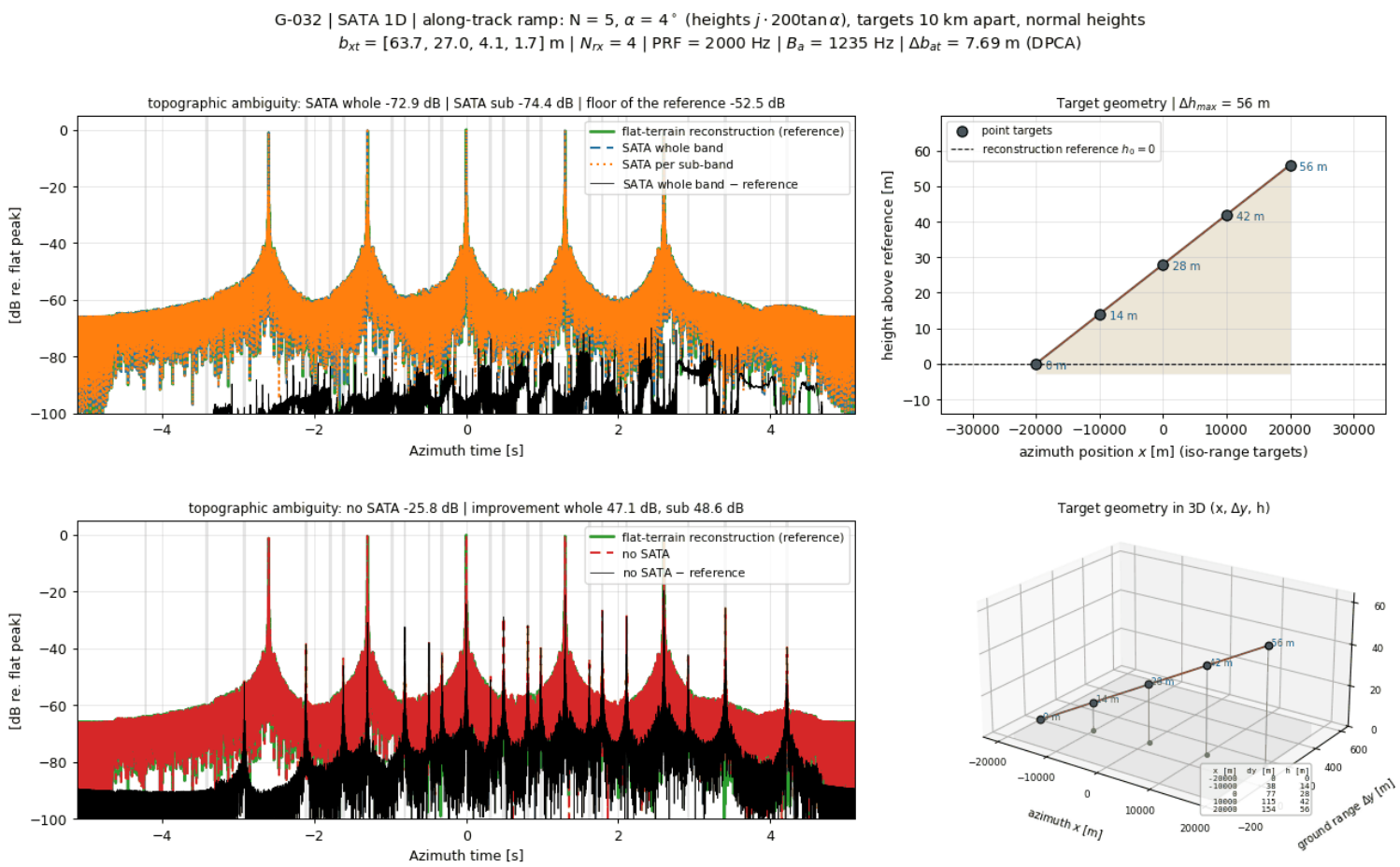


Figure 110: G-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -25.8, SATA whole band -72.9, per sub-band -74.4 dB (improvement 47.1 dB); floor -52.5 dB.

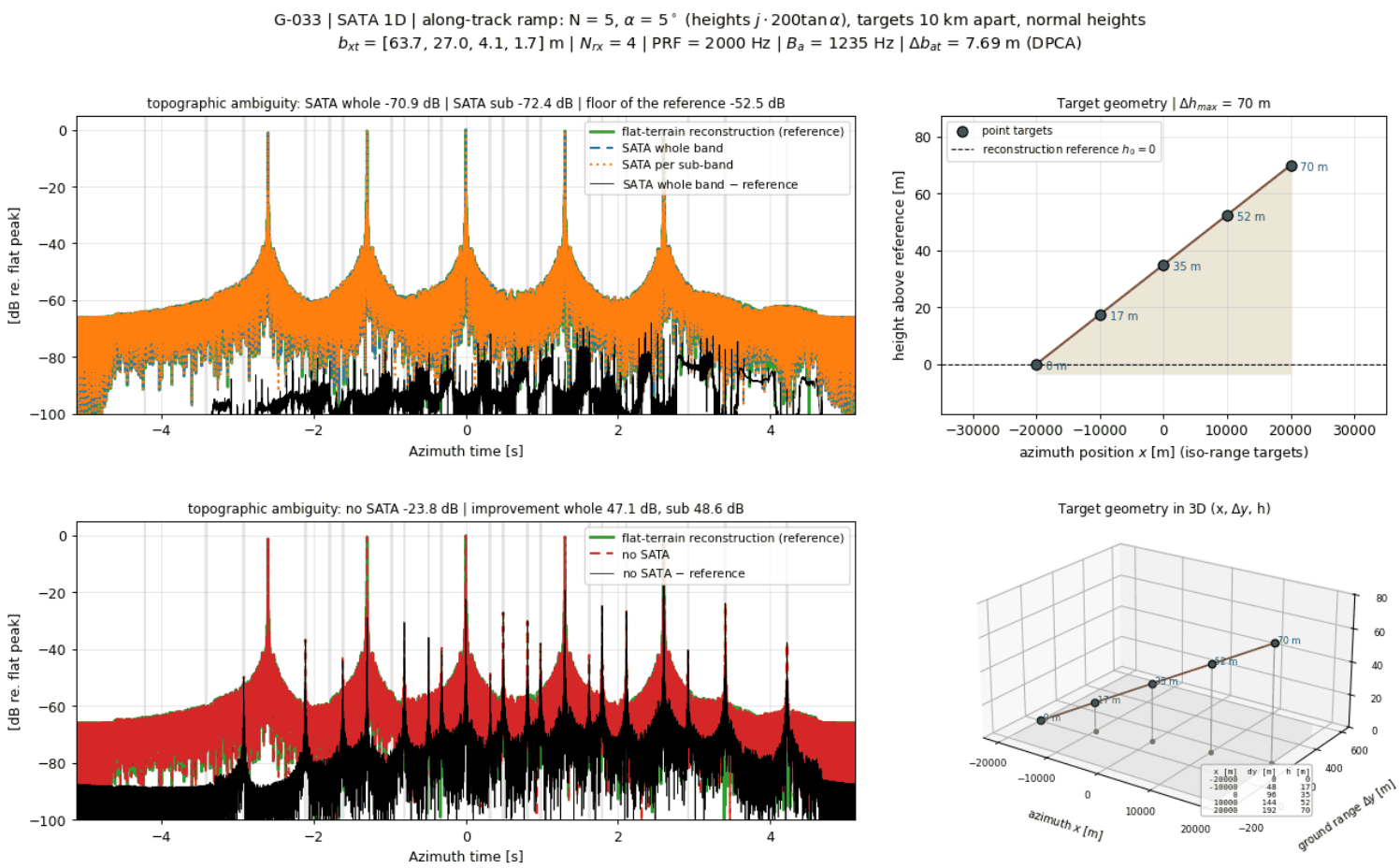


Figure 111: G-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Topographic ambiguity: no SATA -23.8, SATA whole band -70.9, per sub-band -72.4 dB (improvement 47.1 dB); floor -52.5 dB.

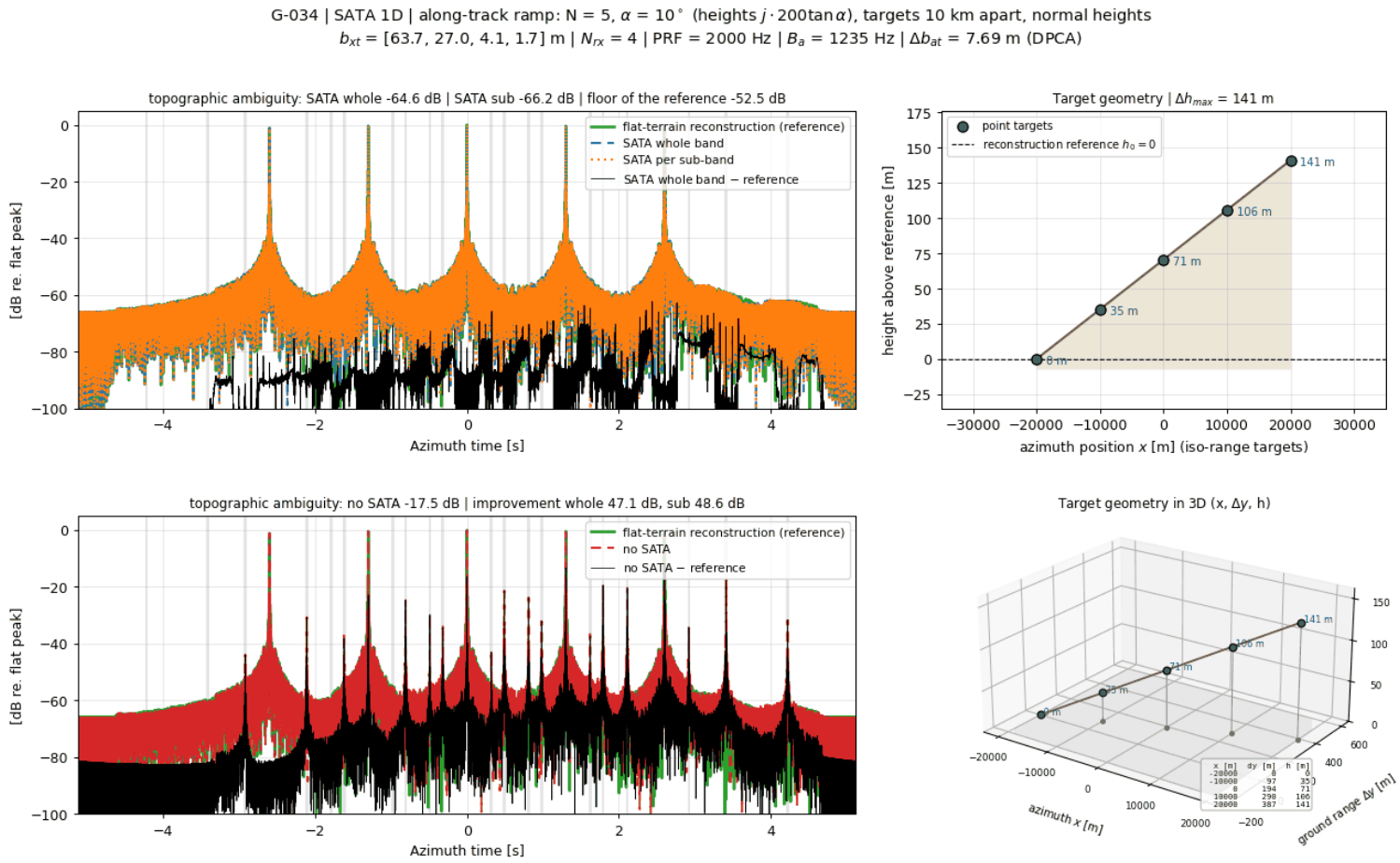


Figure 112: G-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −17.5, SATA whole band −64.6, per sub-band −66.2 dB (improvement 47.1 dB); floor −52.5 dB.

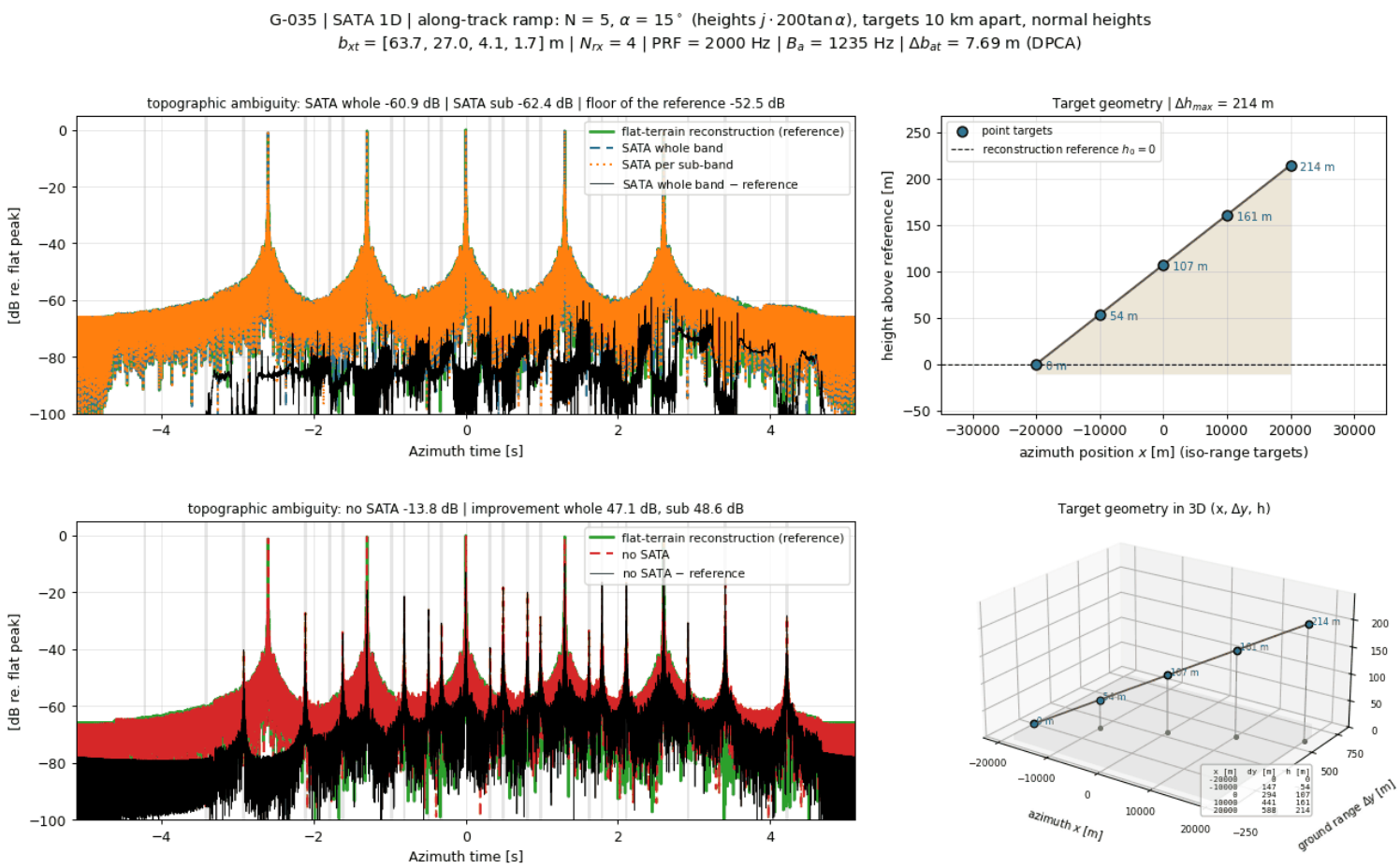


Figure 113: G-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −13.8, SATA whole band −60.9, per sub-band −62.4 dB (improvement 47.1 dB); floor −52.5 dB.

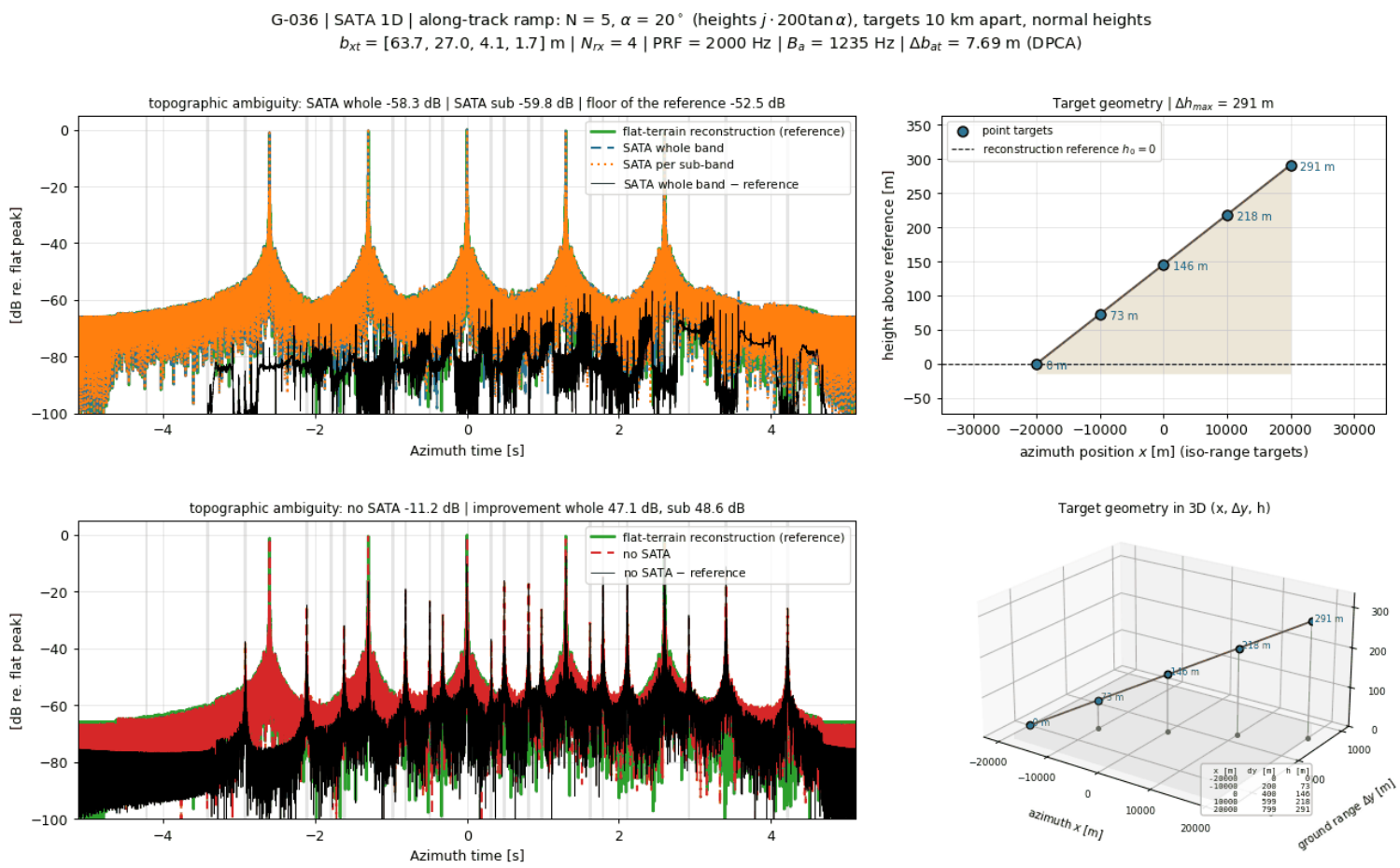


Figure 114: G-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −11.2, SATA whole band −58.3, per sub-band −59.8 dB (improvement 47.1 dB); floor −52.5 dB.

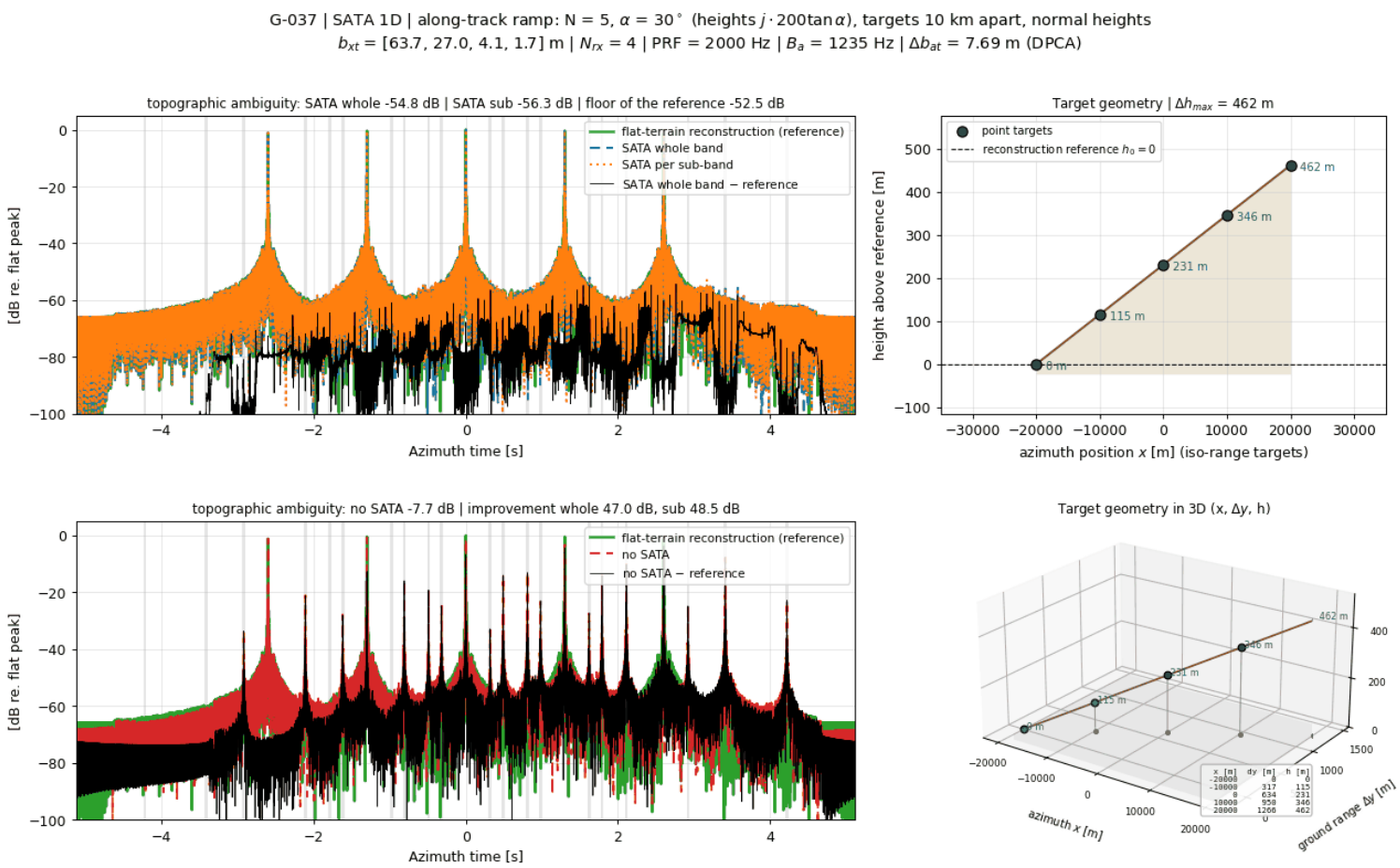


Figure 115: G-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −7.7, SATA whole band −54.8, per sub-band −56.3 dB (improvement 47.0 dB); floor −52.5 dB.

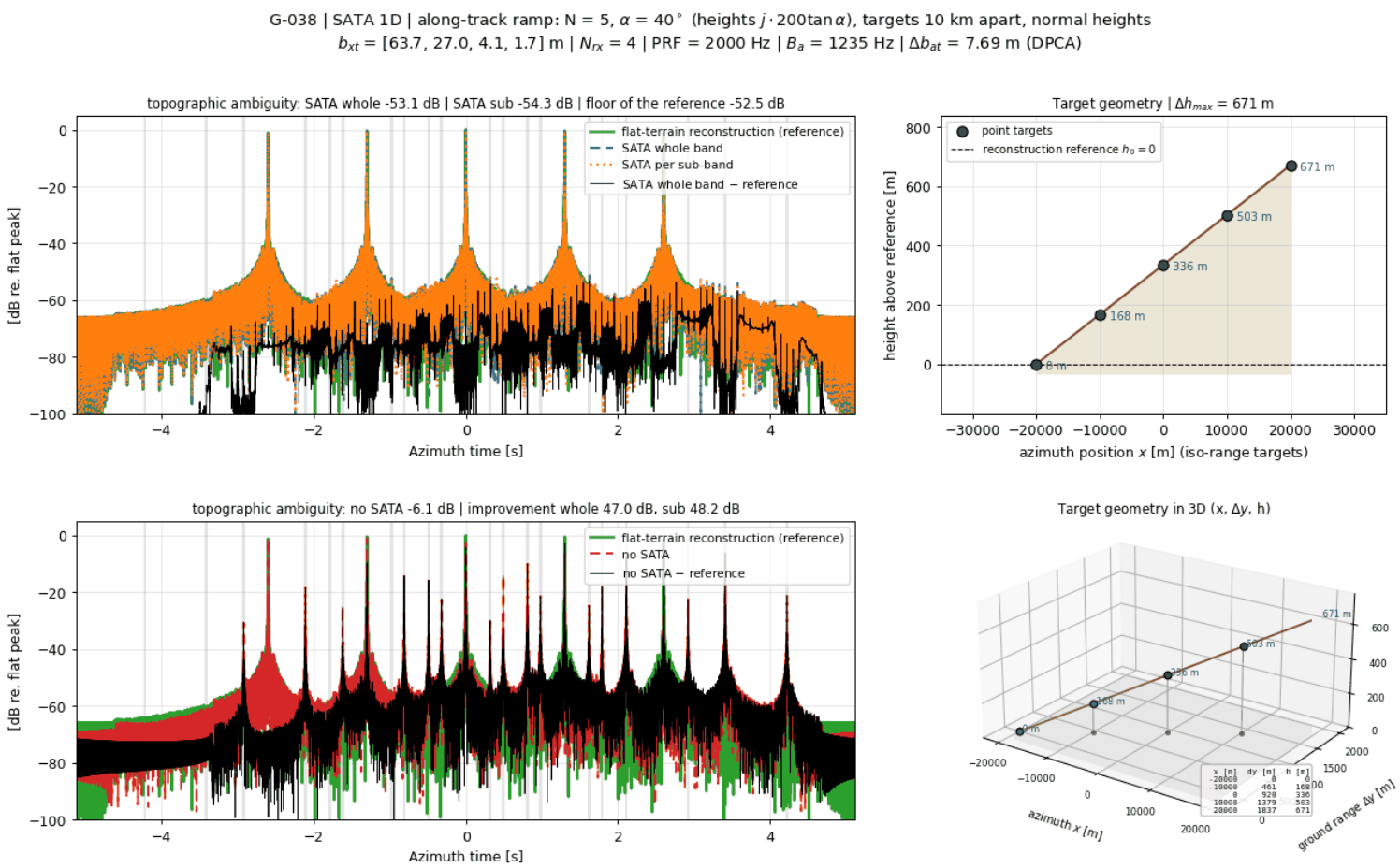


Figure 116: G-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA -6.1, SATA whole band -53.1, per sub-band -54.3 dB (improvement 47.0 dB); floor -52.5 dB.

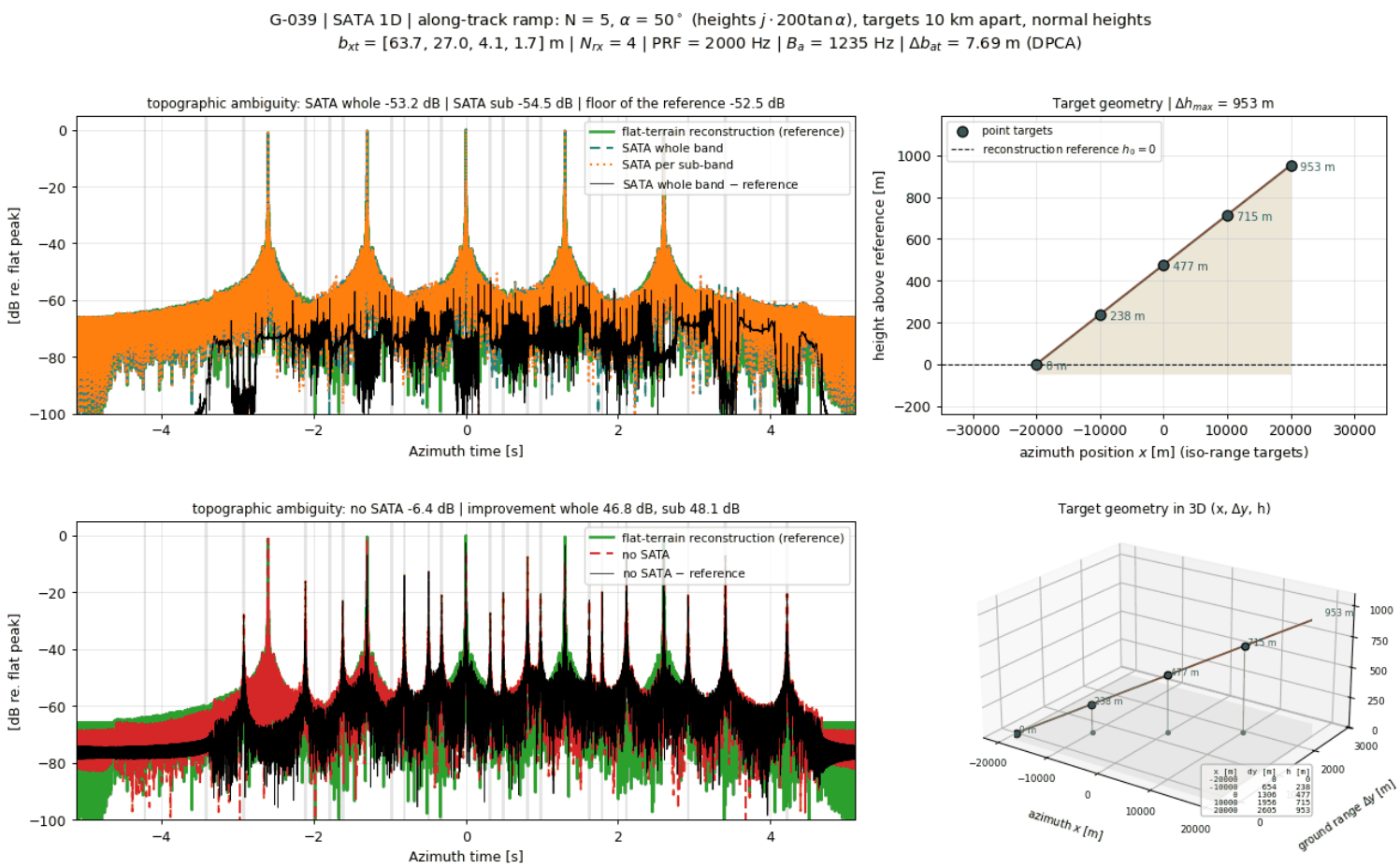


Figure 117: G-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Topographic ambiguity: no SATA −6.4, SATA whole band −53.2, per sub-band −54.5 dB (improvement 46.8 dB); floor −52.5 dB.

4 Configuration H: along-track ramps, targets 10 km apart, large heights

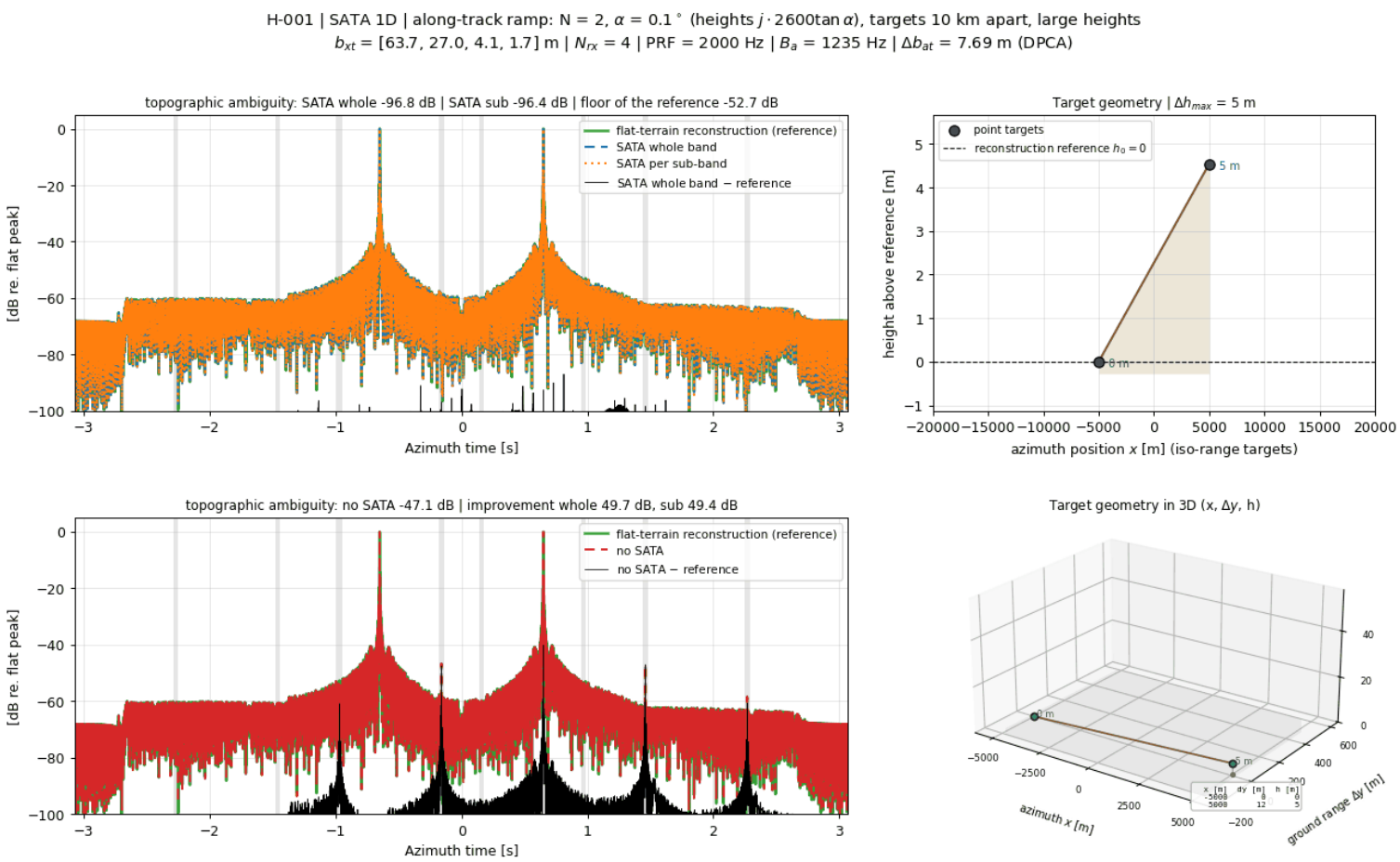


Figure 118: H-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −47.1, SATA whole band −96.8, per sub-band −96.4 dB (improvement 49.7 dB); floor −52.7 dB.

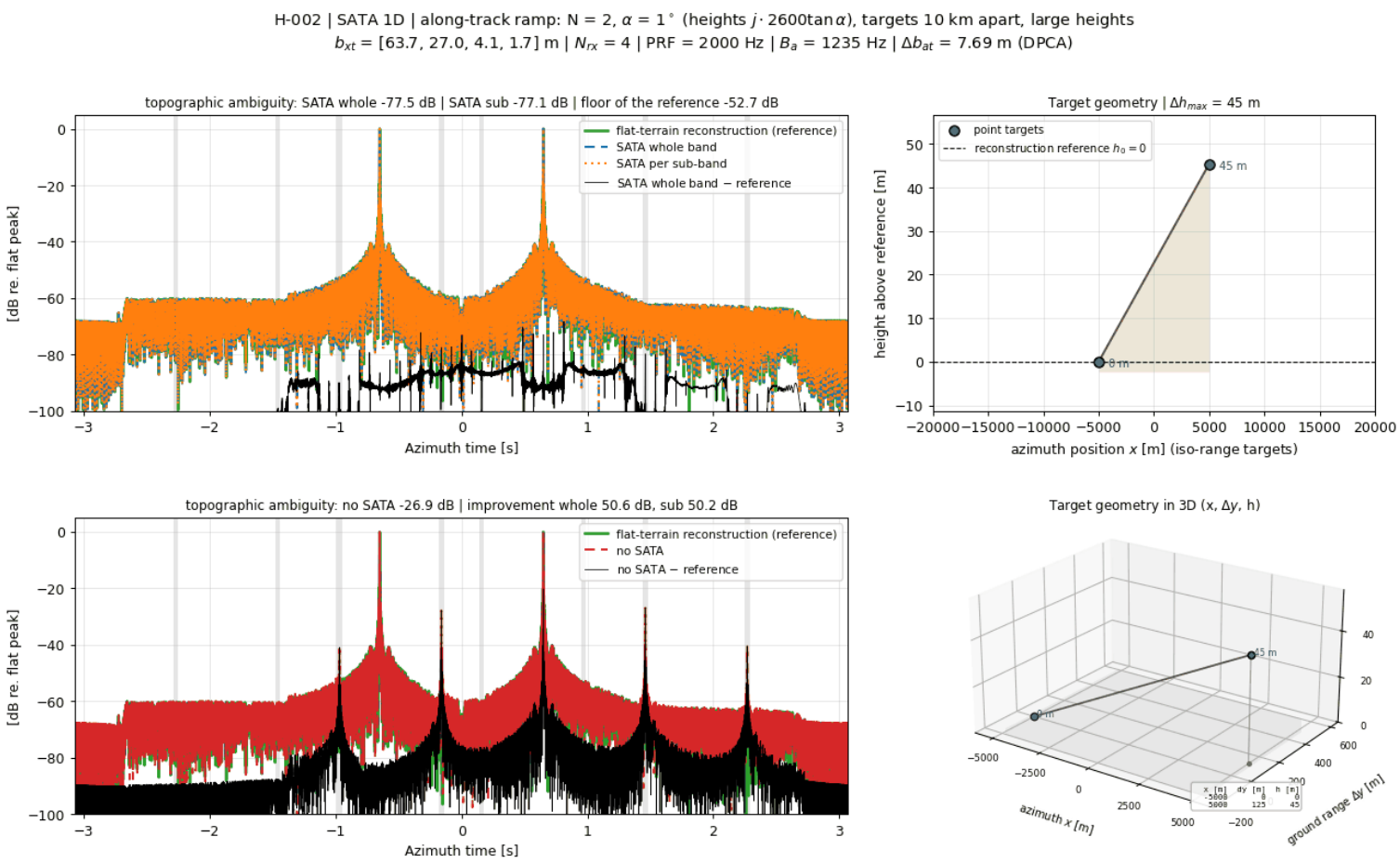


Figure 119: H-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −26.9, SATA whole band −77.5, per sub-band −77.1 dB (improvement 50.6 dB); floor −52.7 dB.

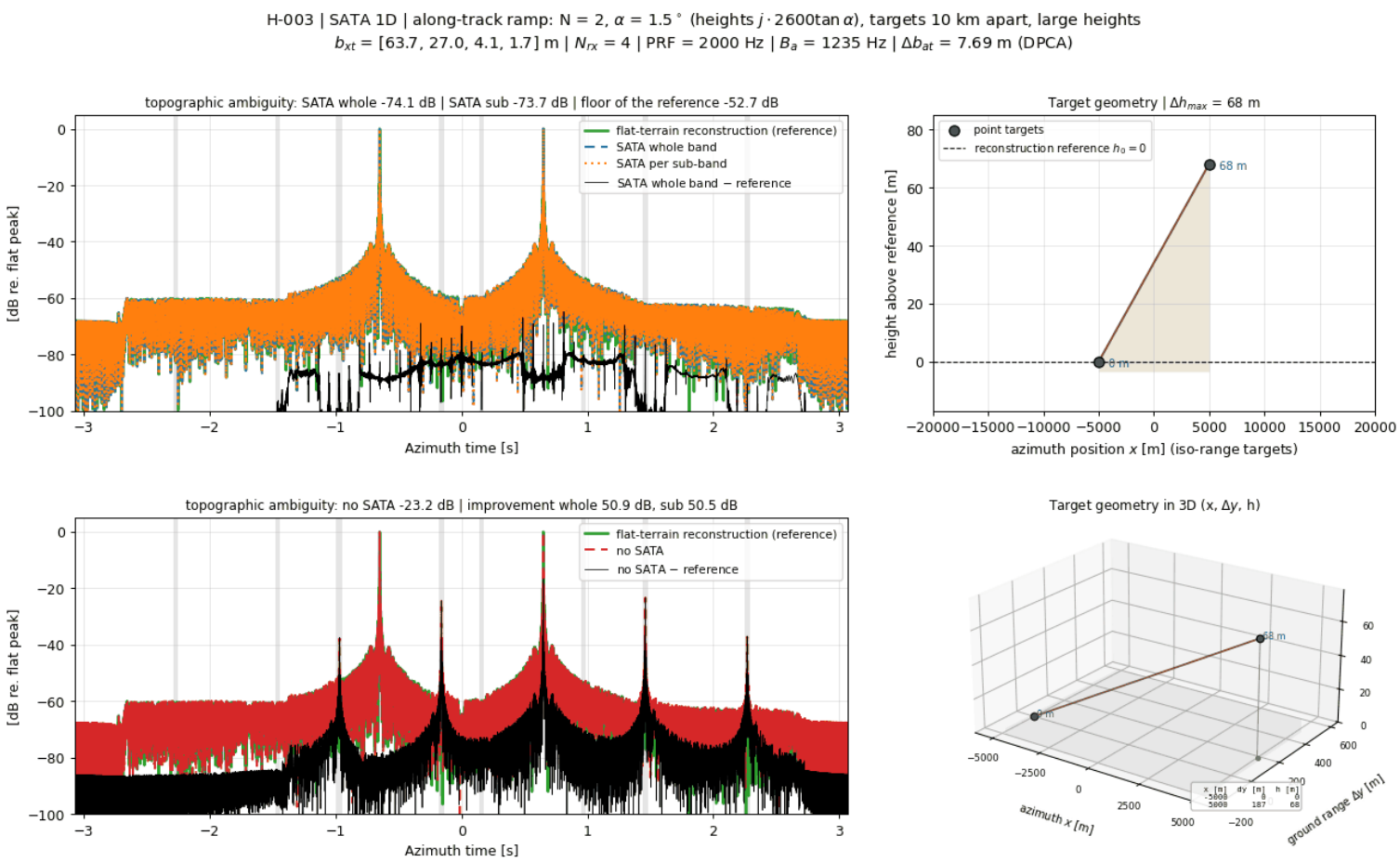


Figure 120: H-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −23.2, SATA whole band −74.1, per sub-band −73.7 dB (improvement 50.9 dB); floor −52.7 dB.

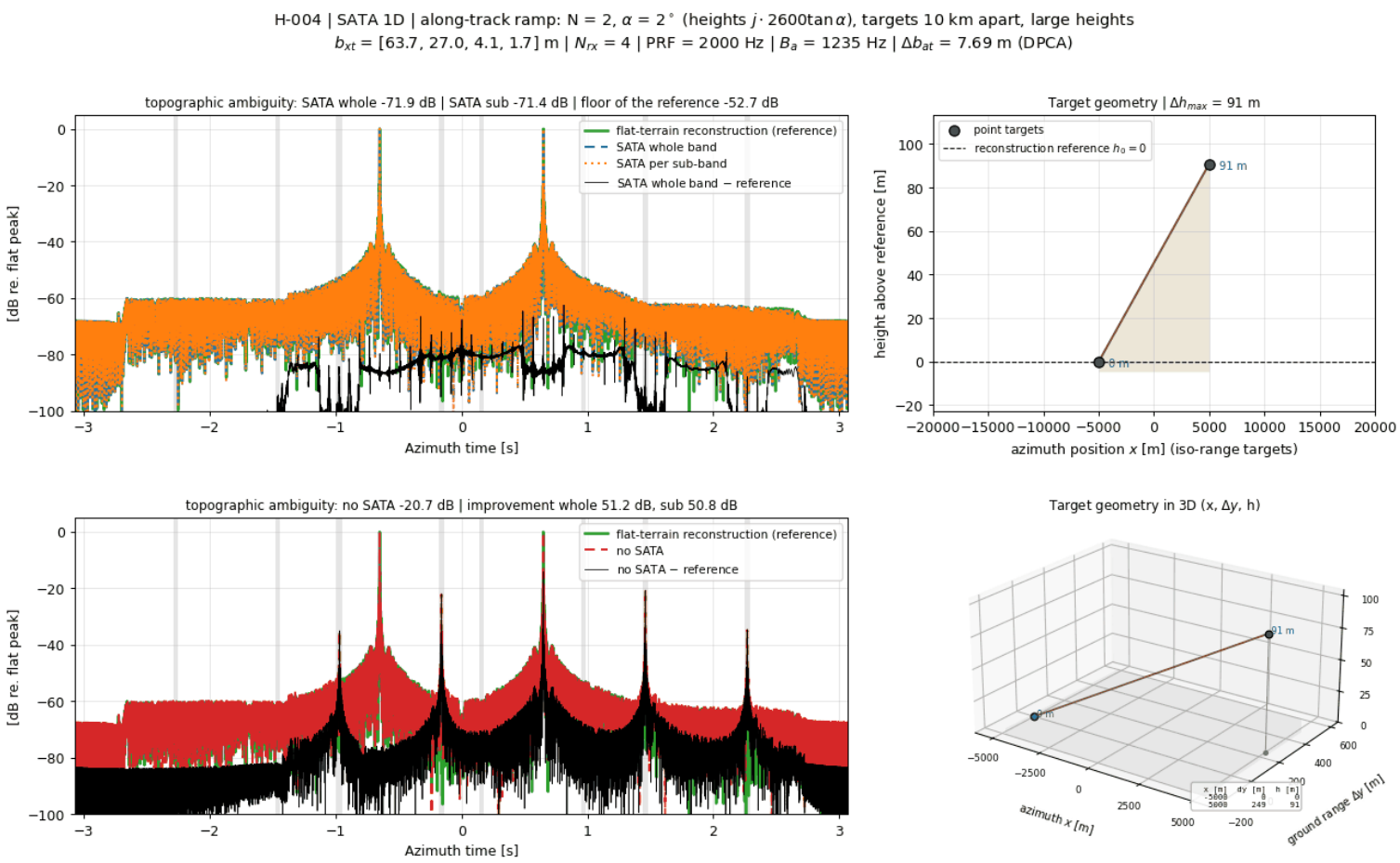


Figure 121: H-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −20.7, SATA whole band −71.9, per sub-band −71.4 dB (improvement 51.2 dB); floor −52.7 dB.

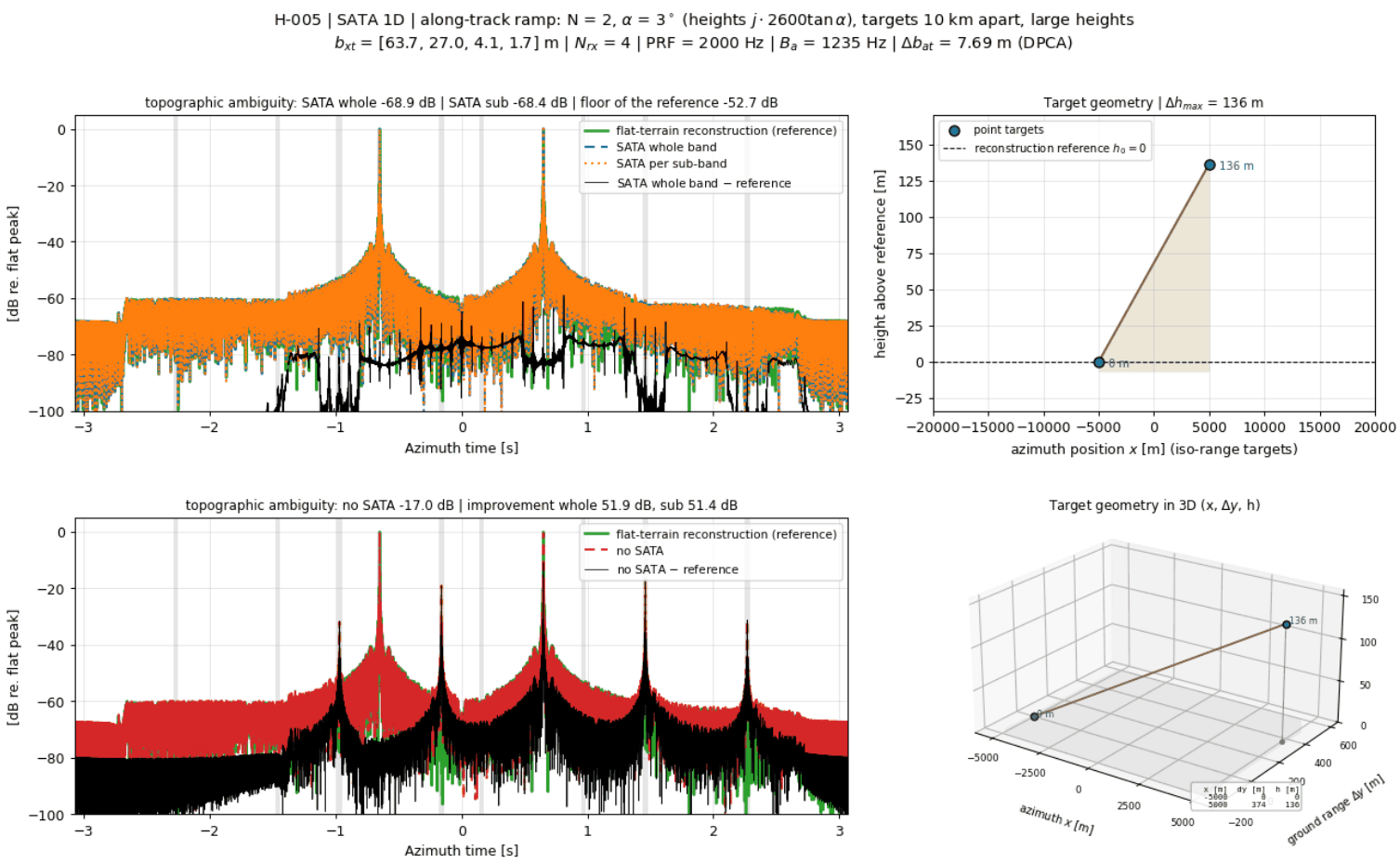


Figure 122: H-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −17.0, SATA whole band −68.9, per sub-band −68.4 dB (improvement 51.9 dB); floor −52.7 dB.

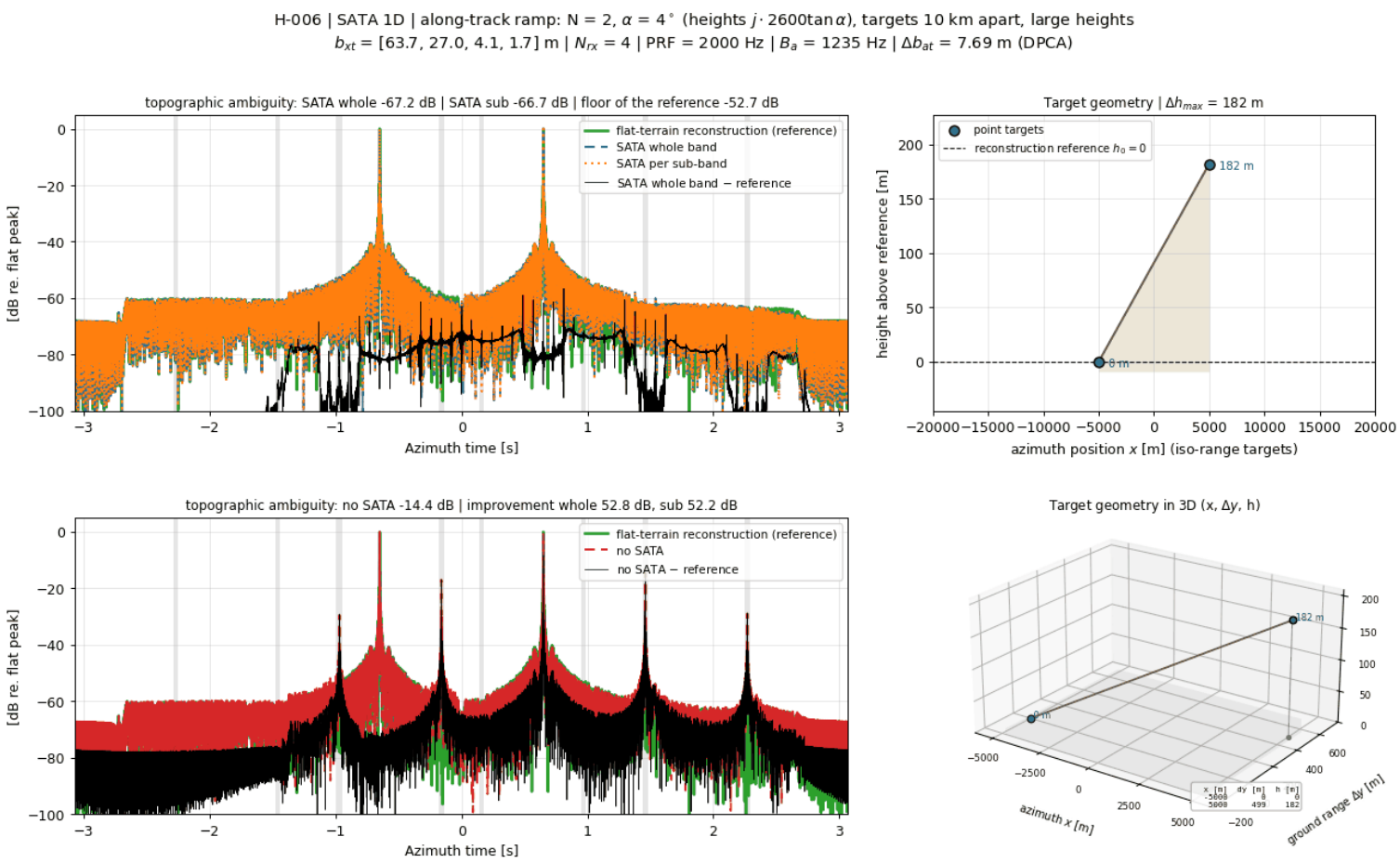


Figure 123: H-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −14.4, SATA whole band −67.2, per sub-band −66.7 dB (improvement 52.8 dB); floor −52.7 dB.

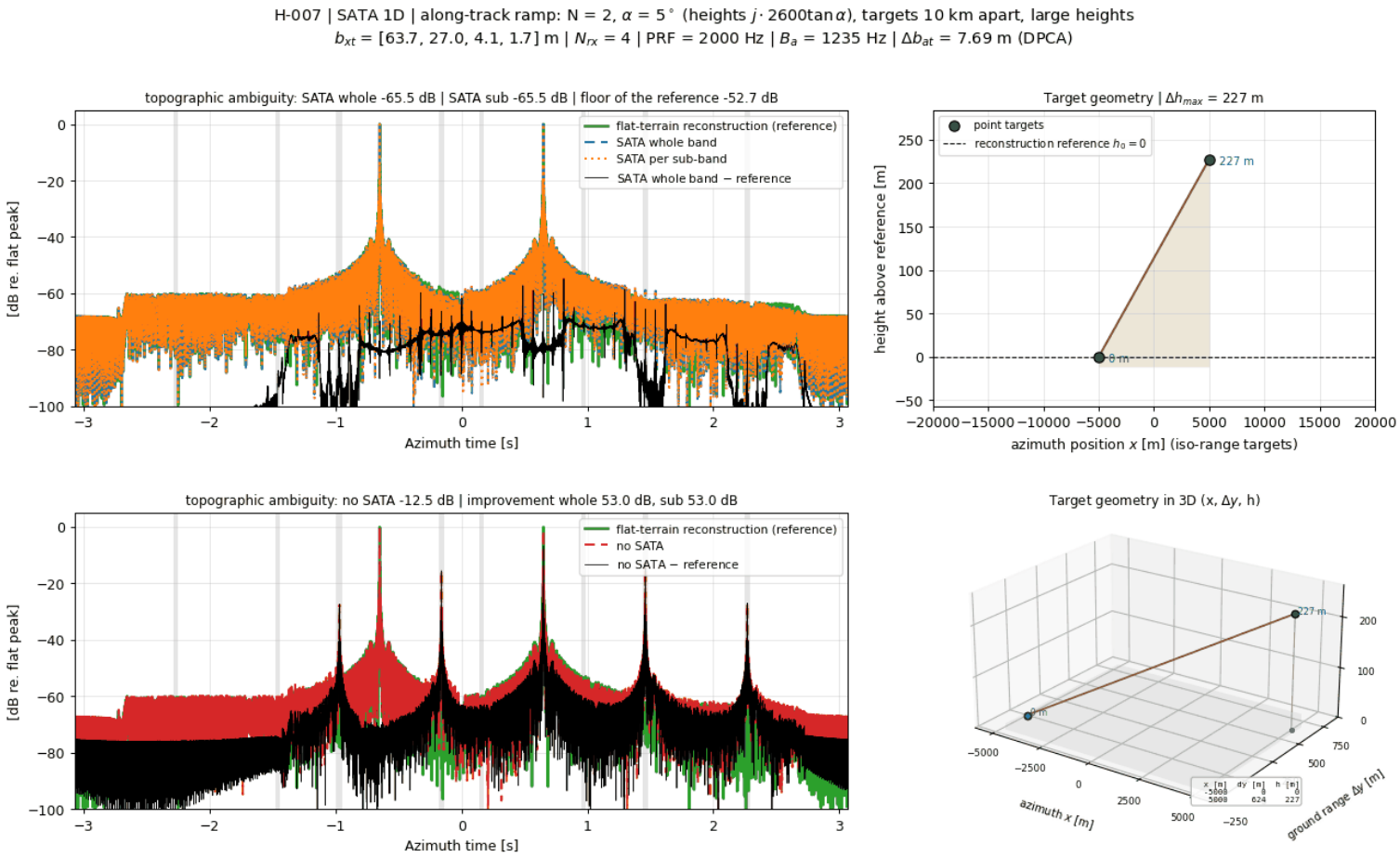


Figure 124: H-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −12.5, SATA whole band −65.5, per sub-band −65.5 dB (improvement 53.0 dB); floor −52.7 dB.

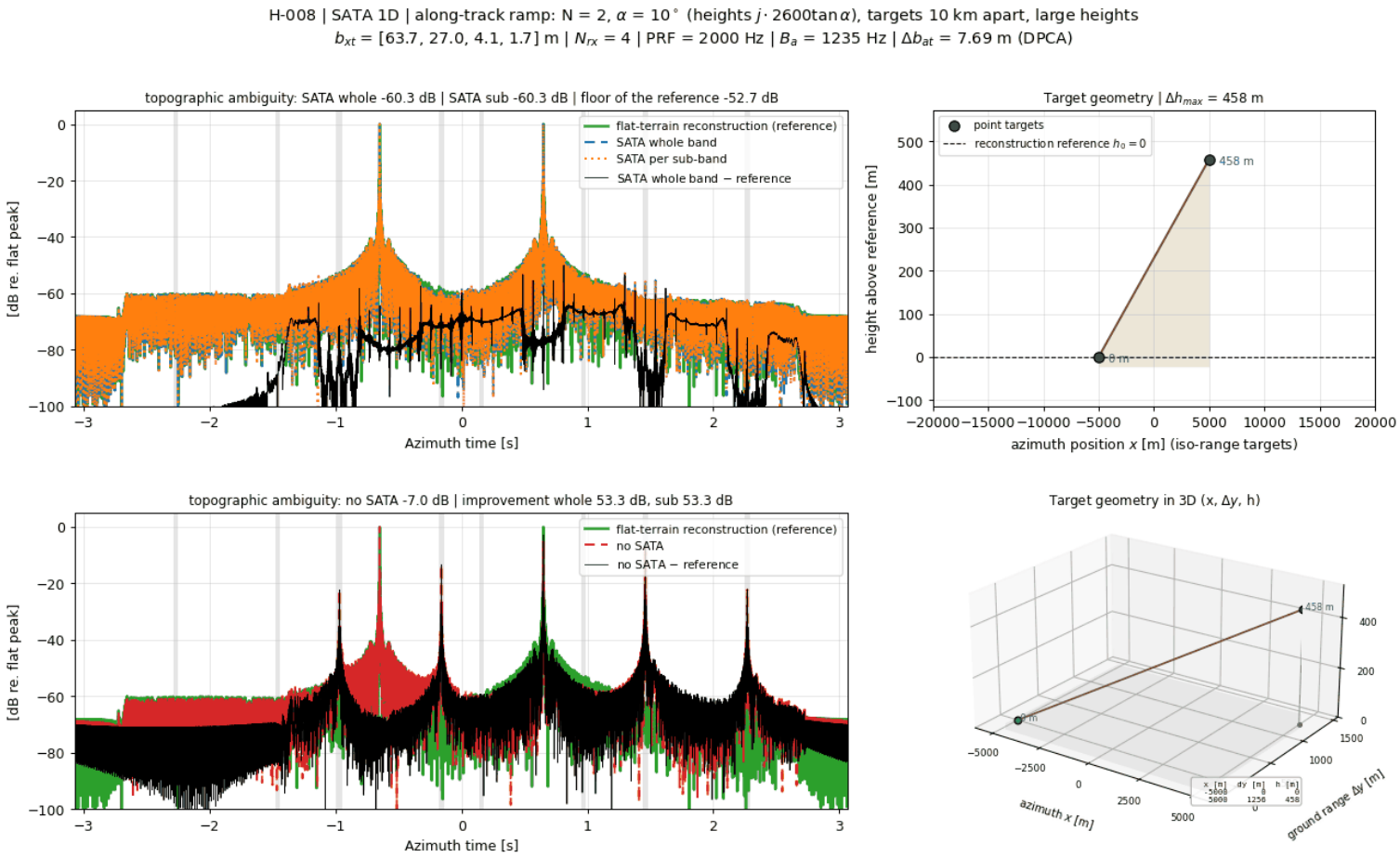


Figure 125: H-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.0, SATA whole band −60.3, per sub-band −60.3 dB (improvement 53.3 dB); floor −52.7 dB.

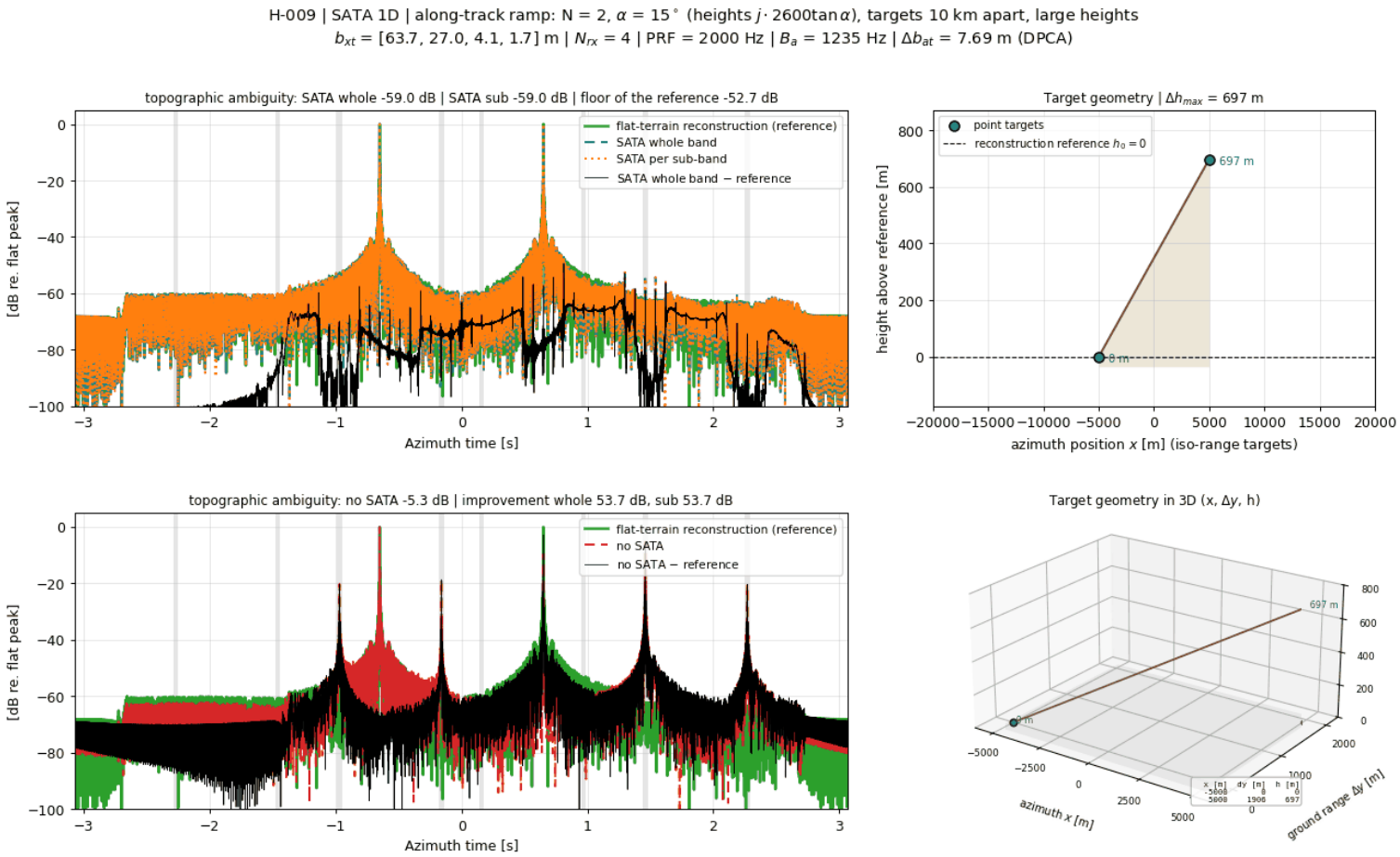


Figure 126: H-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −5.3, SATA whole band −59.0, per sub-band −59.0 dB (improvement 53.7 dB); floor −52.7 dB.

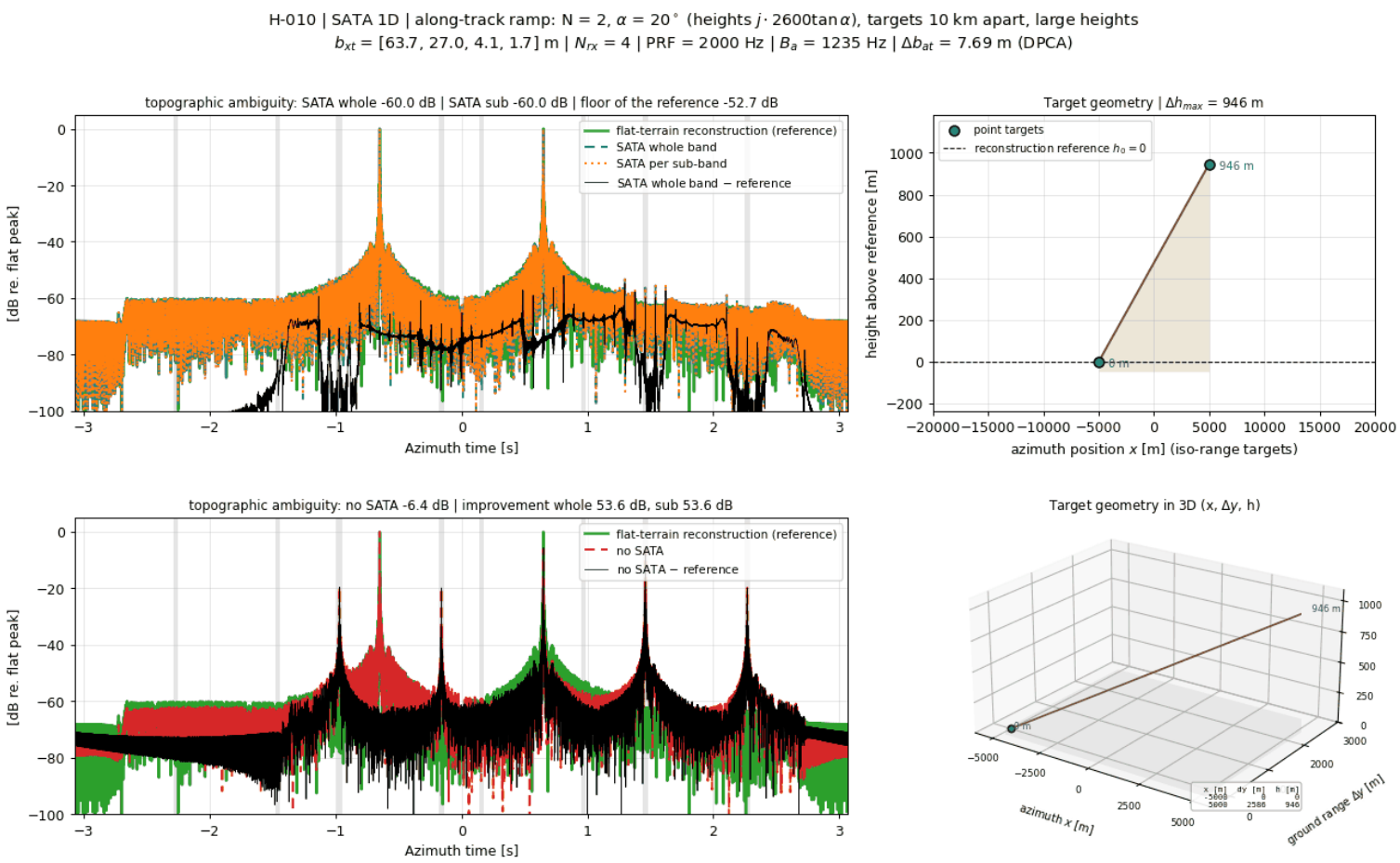


Figure 127: H-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.4, SATA whole band −60.0, per sub-band −60.0 dB (improvement 53.6 dB); floor −52.7 dB.

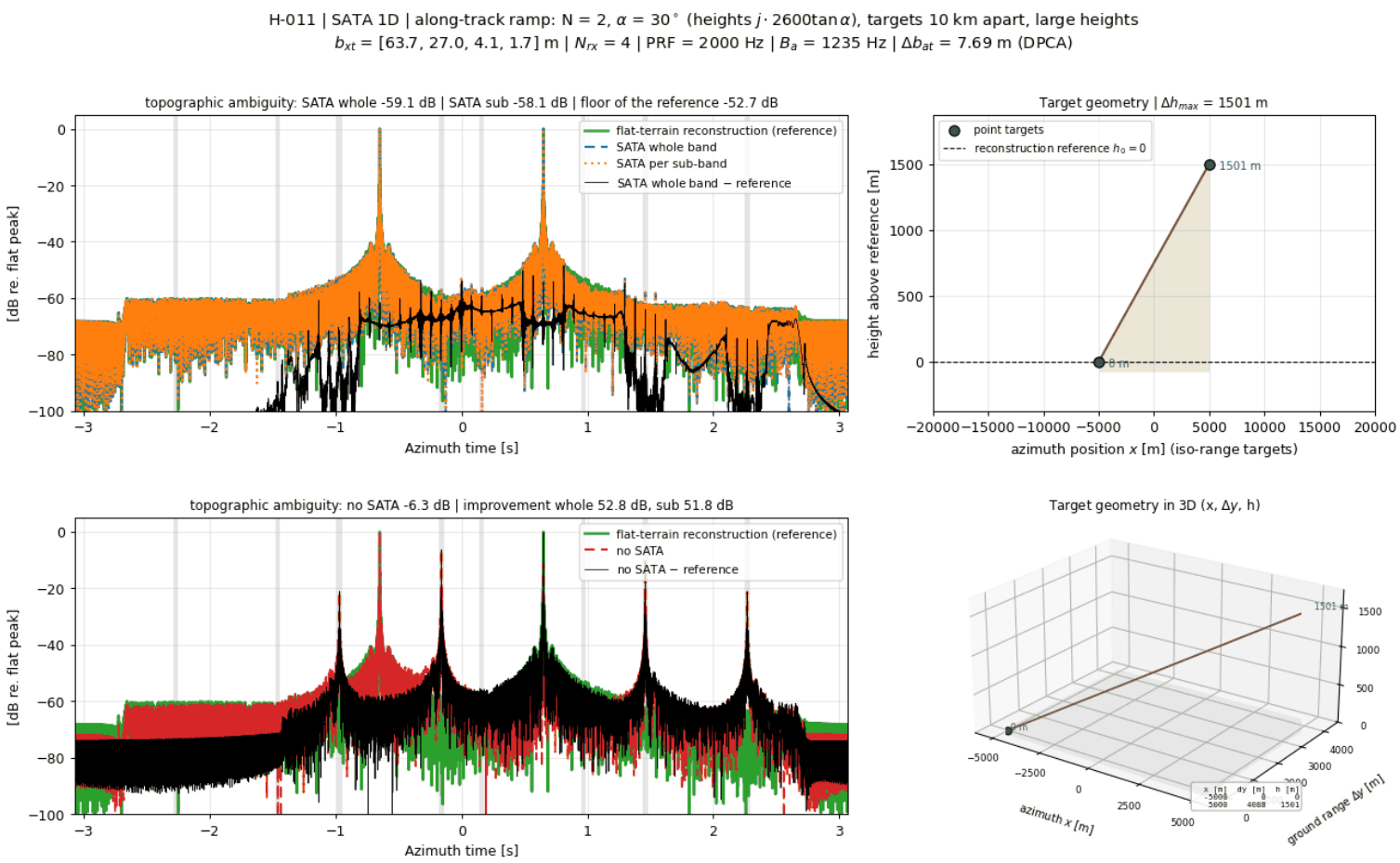


Figure 128: H-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.3, SATA whole band −59.1, per sub-band −58.1 dB (improvement 52.8 dB); floor −52.7 dB.

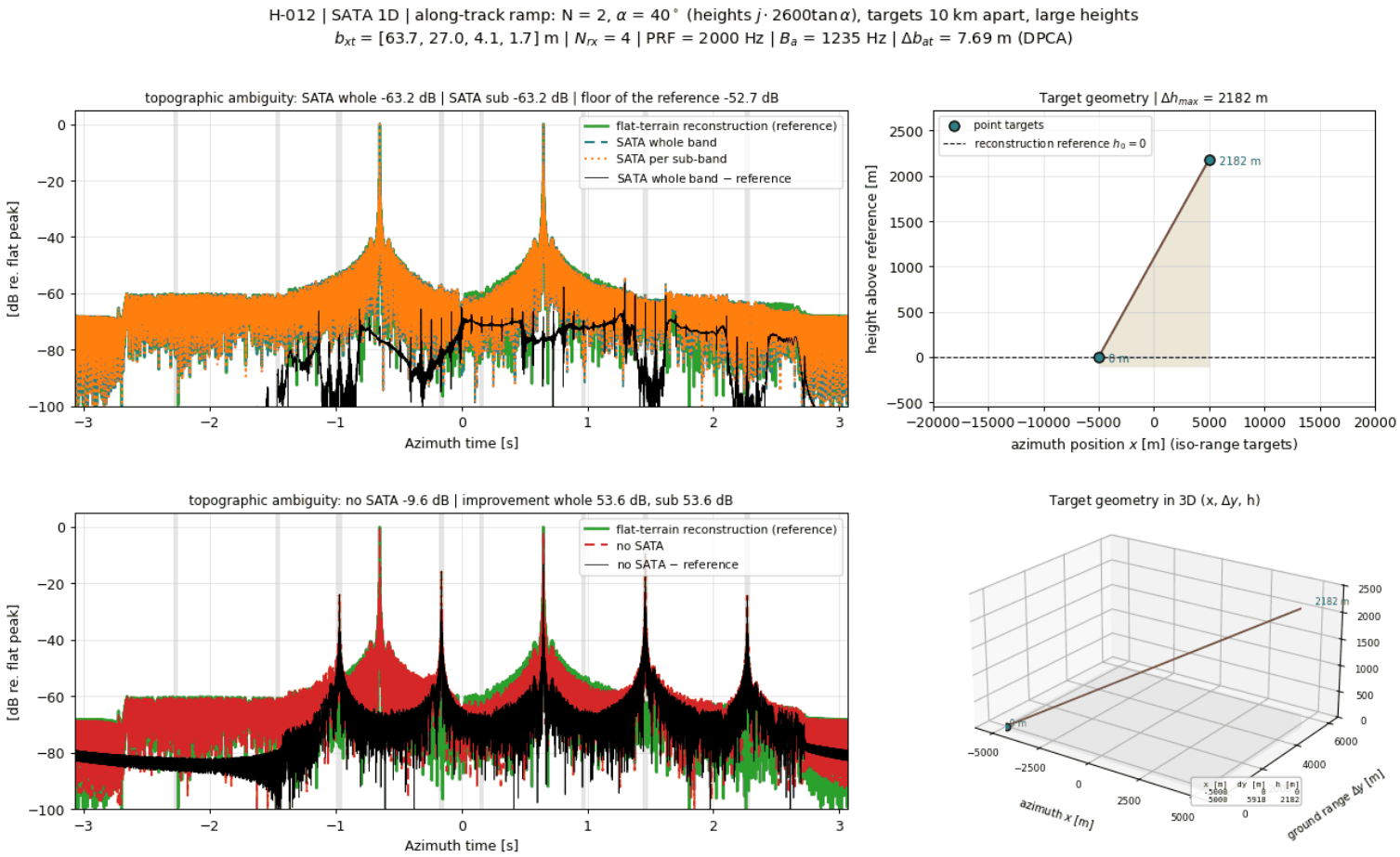


Figure 129: H-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −9.6, SATA whole band −63.2, per sub-band −63.2 dB (improvement 53.6 dB); floor −52.7 dB.

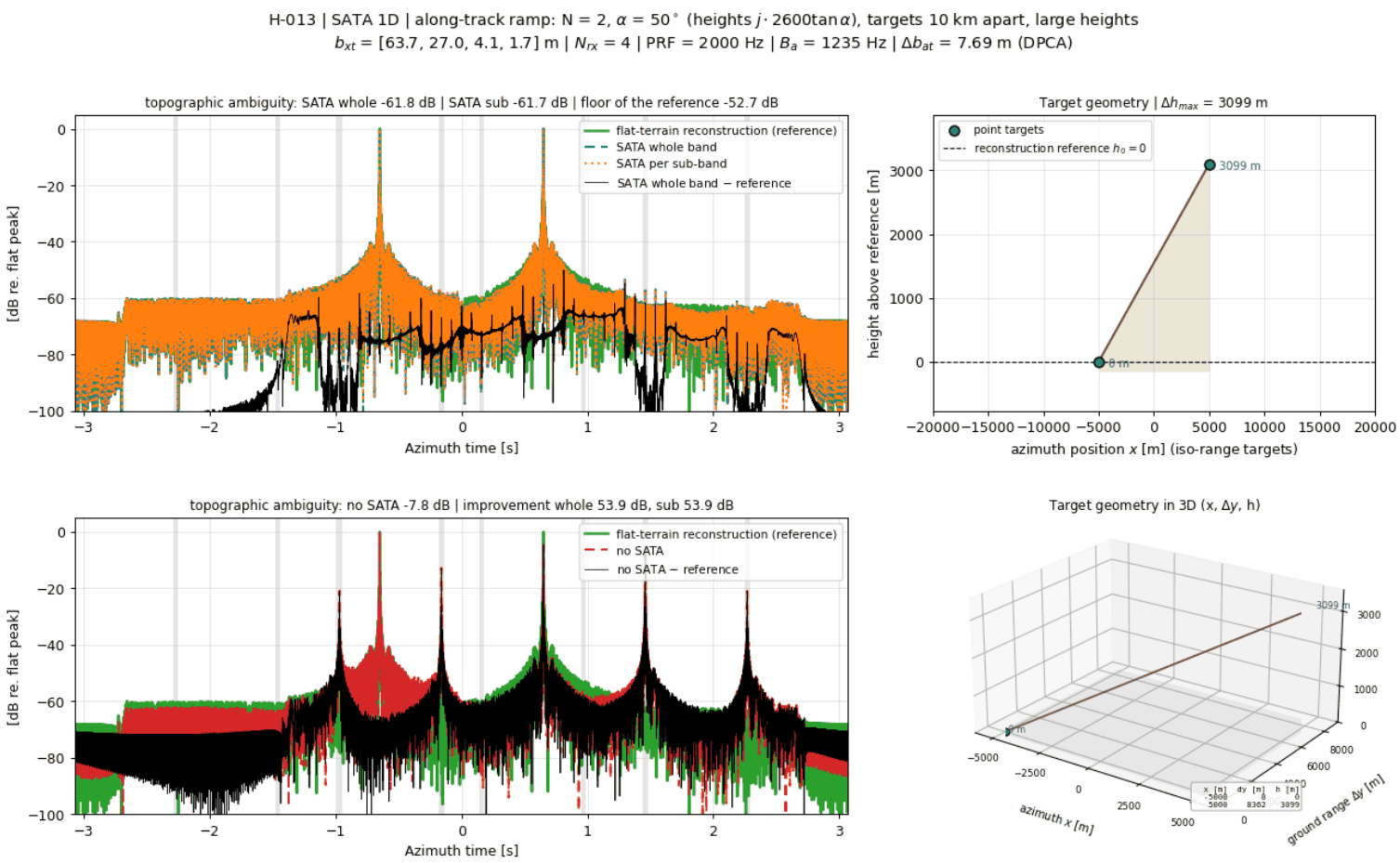


Figure 130: H-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.8, SATA whole band −61.8, per sub-band −61.7 dB (improvement 53.9 dB); floor −52.7 dB.

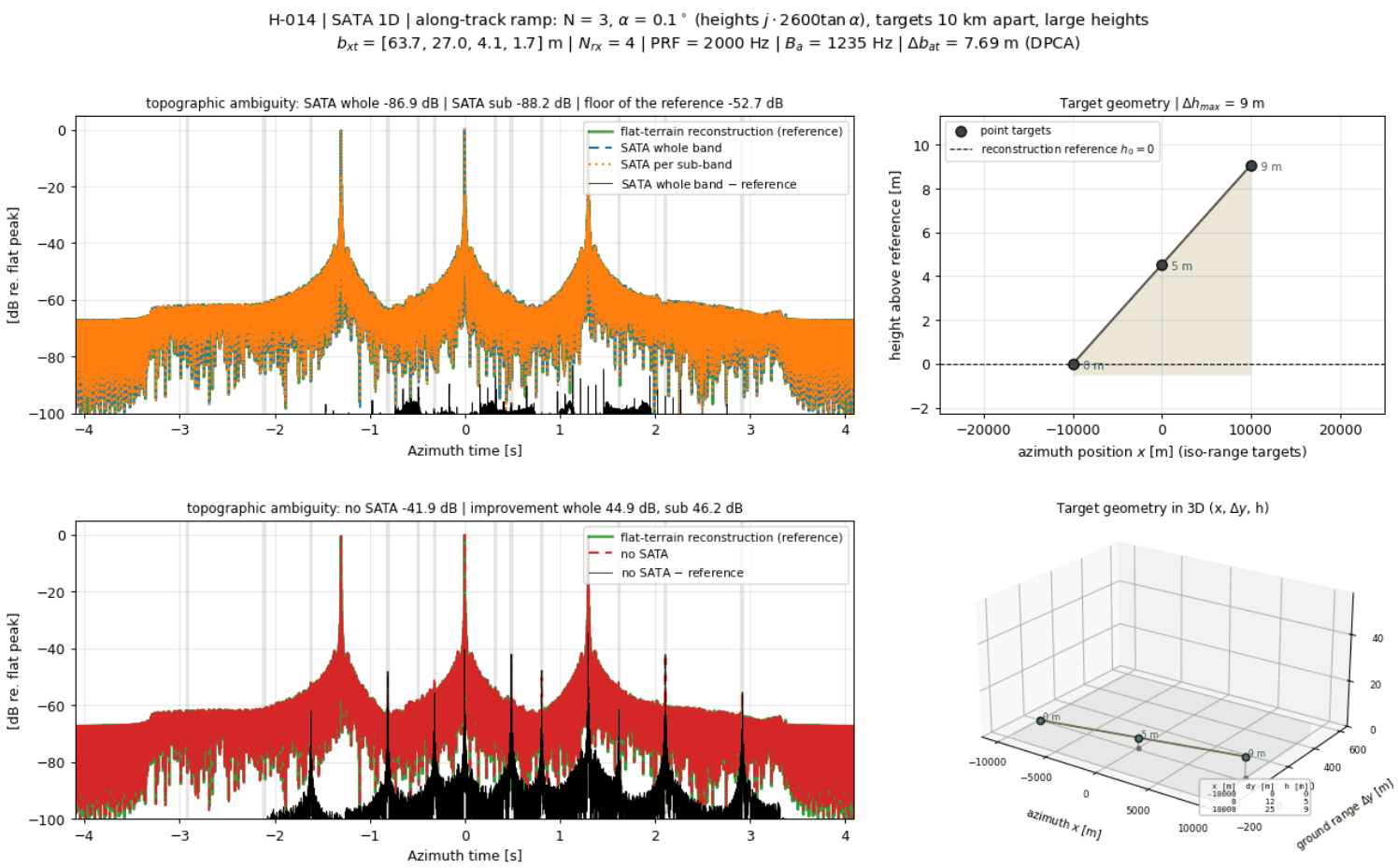


Figure 131: H-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −41.9, SATA whole band −86.9, per sub-band −88.2 dB (improvement 44.9 dB); floor −52.7 dB.

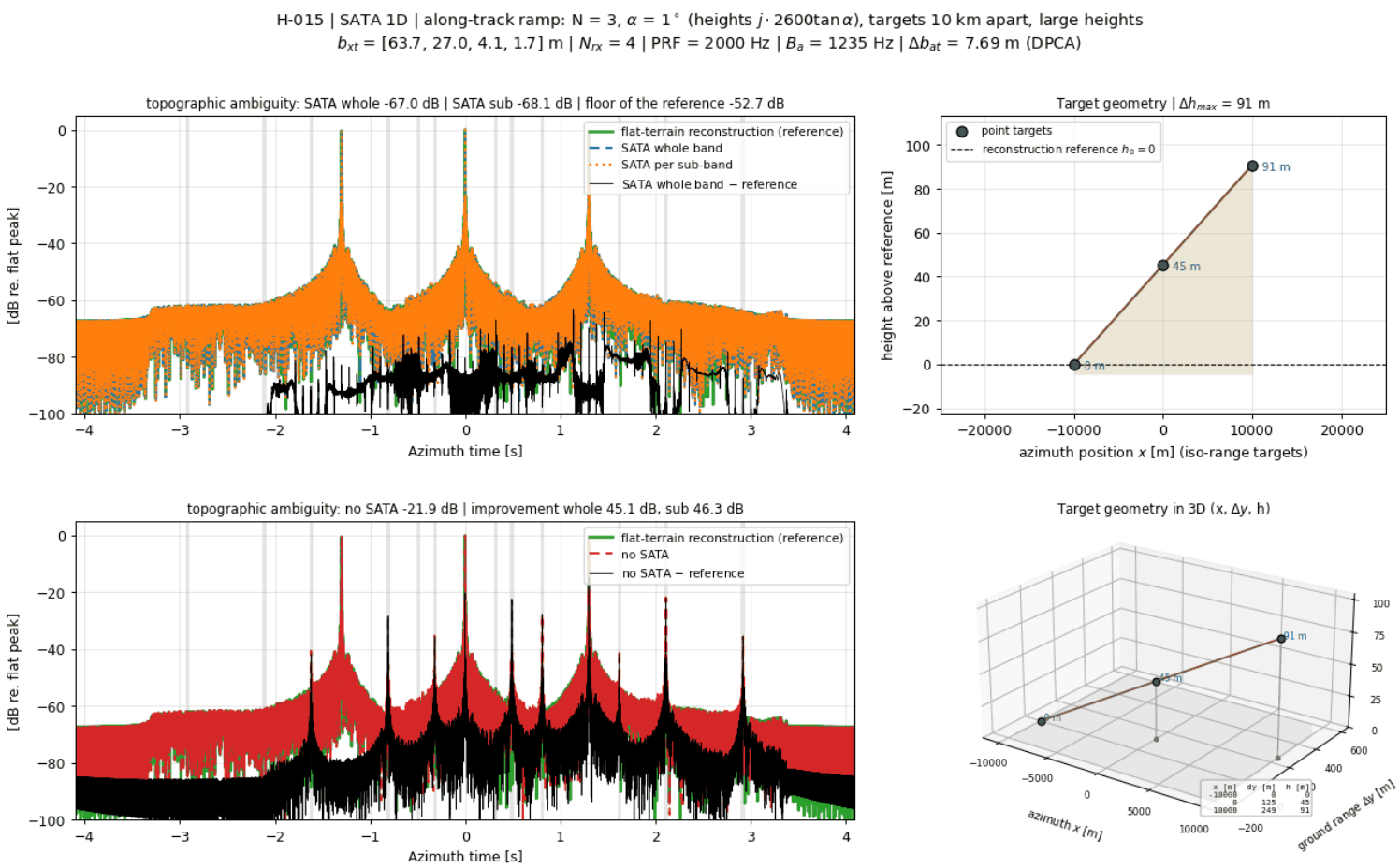


Figure 132: H-015, $N = 3$, $\alpha = 1^\circ$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −21.9, SATA whole band −67.0, per sub-band −68.1 dB (improvement 45.1 dB); floor −52.7 dB.

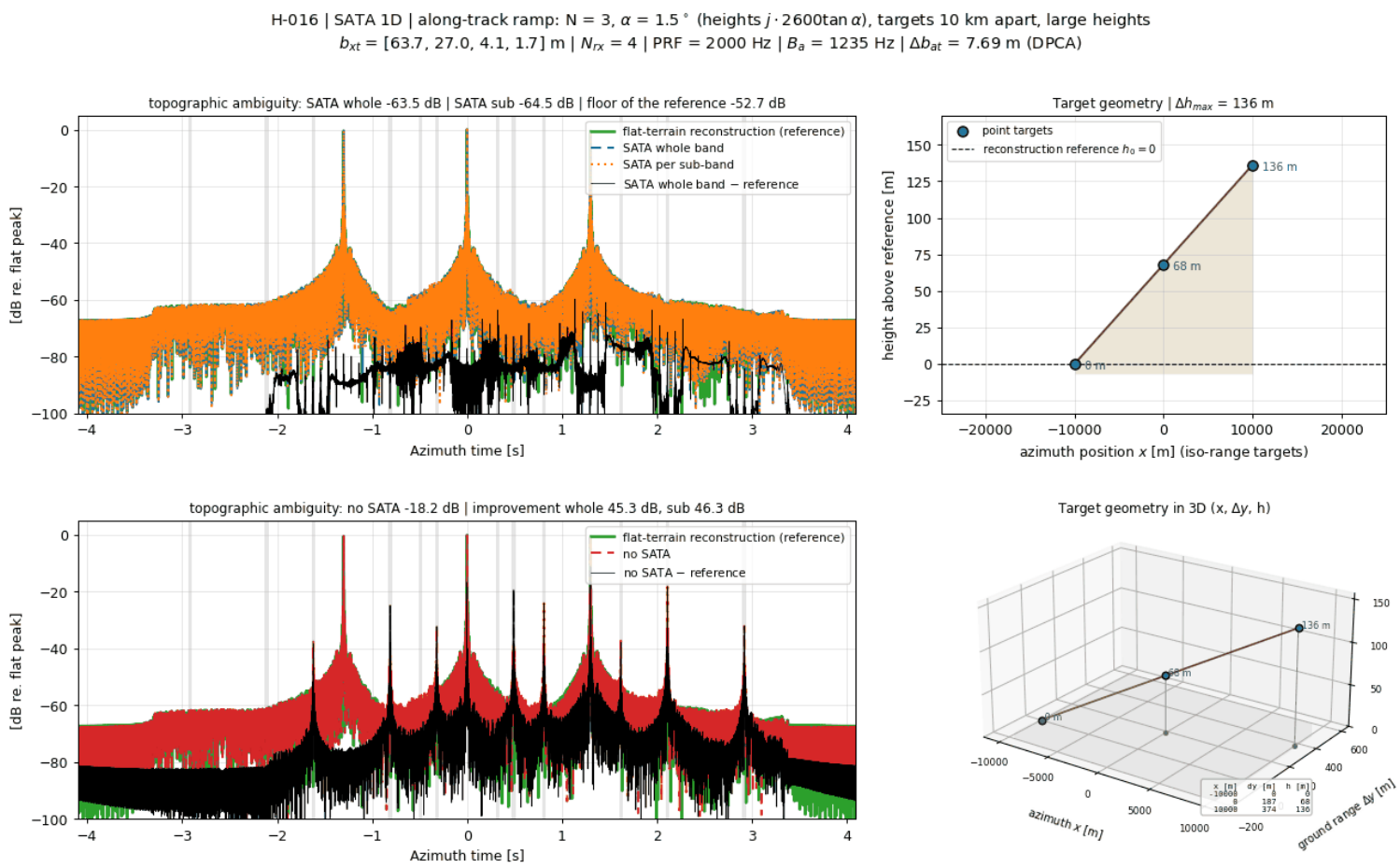


Figure 133: H-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −18.2, SATA whole band −63.5, per sub-band −64.5 dB (improvement 45.3 dB); floor −52.7 dB.

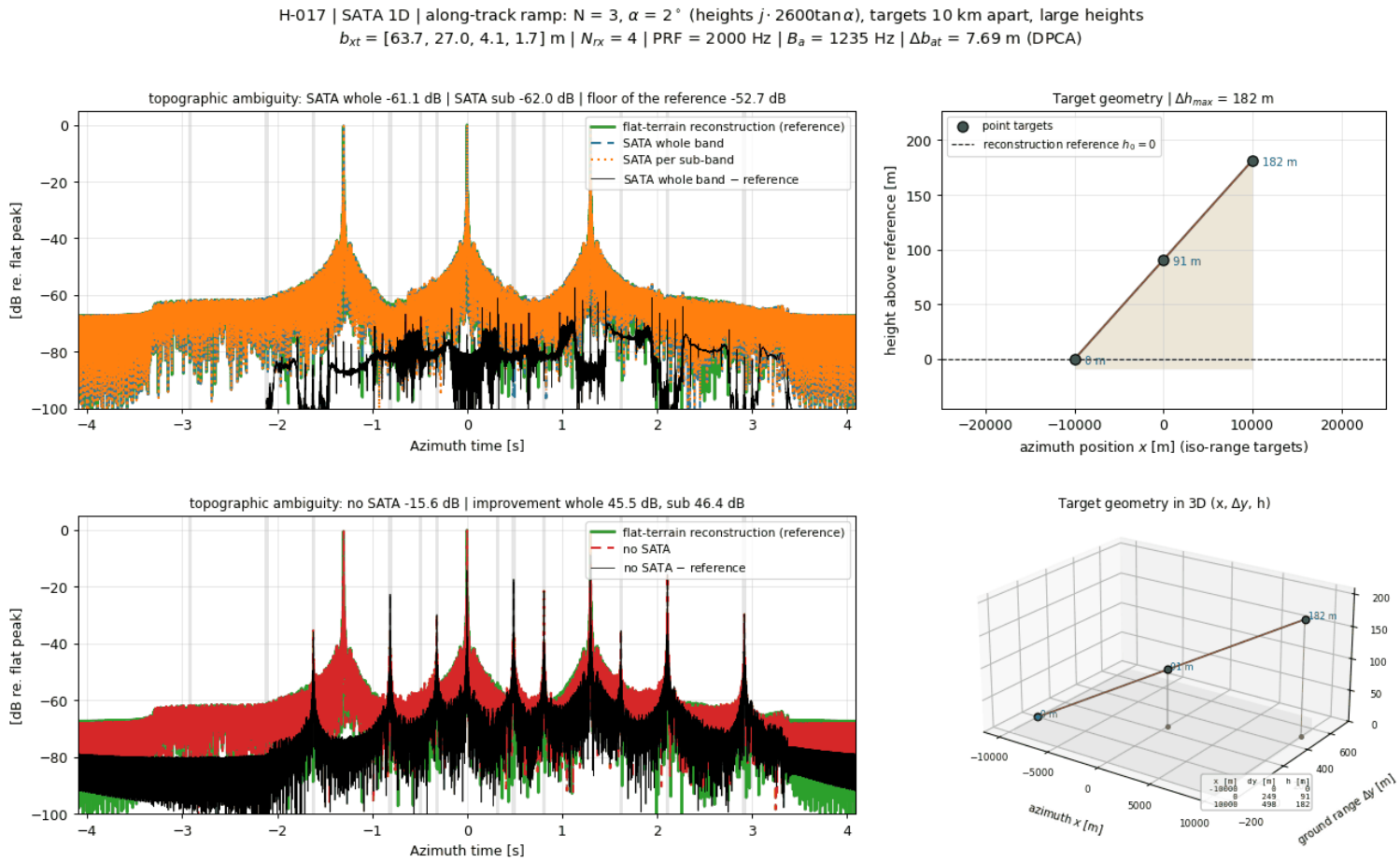


Figure 134: H-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −15.6, SATA whole band −61.1, per sub-band −62.0 dB (improvement 45.5 dB); floor −52.7 dB.

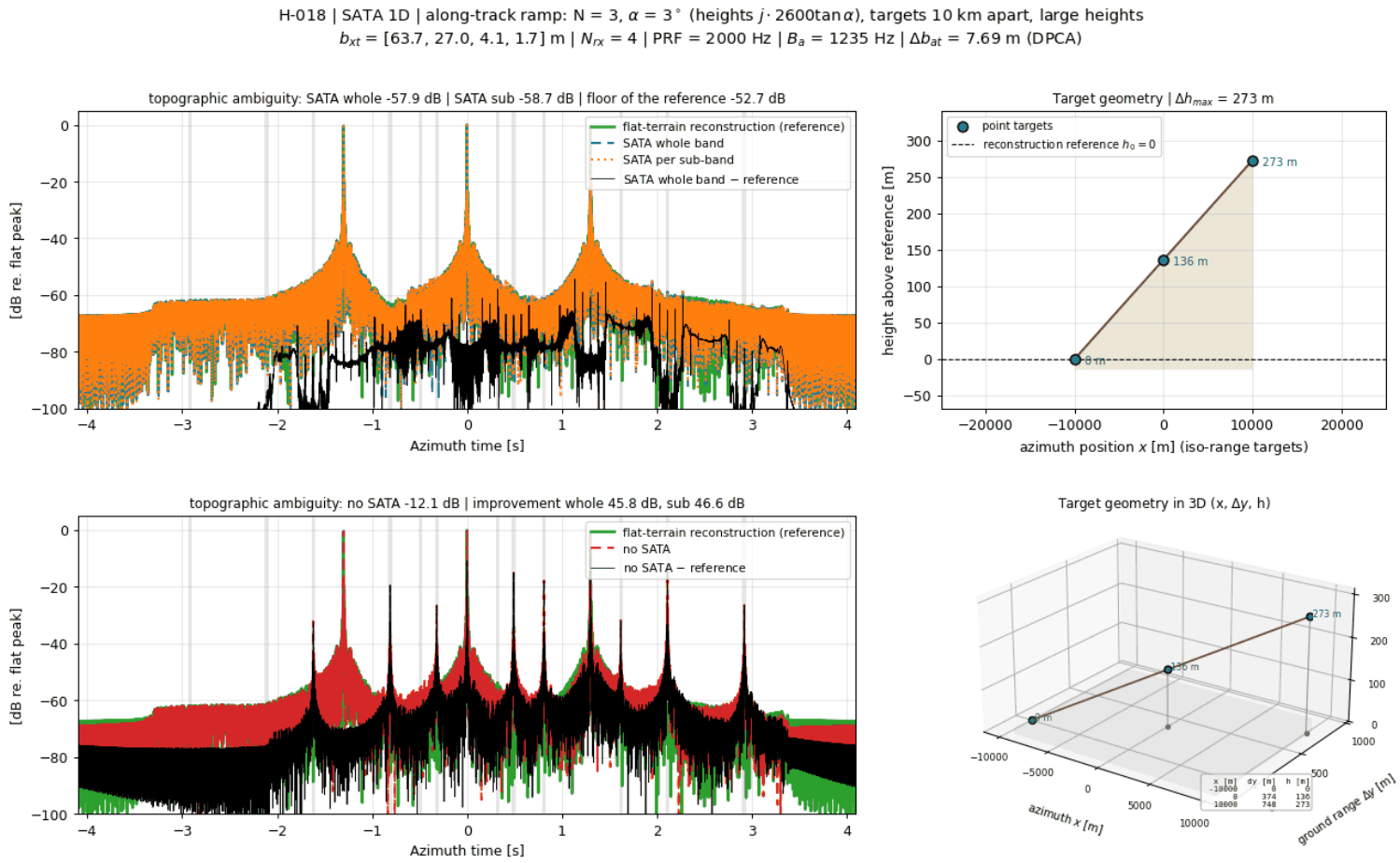


Figure 135: H-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −12.1, SATA whole band −57.9, per sub-band −58.7 dB (improvement 45.8 dB); floor −52.7 dB.

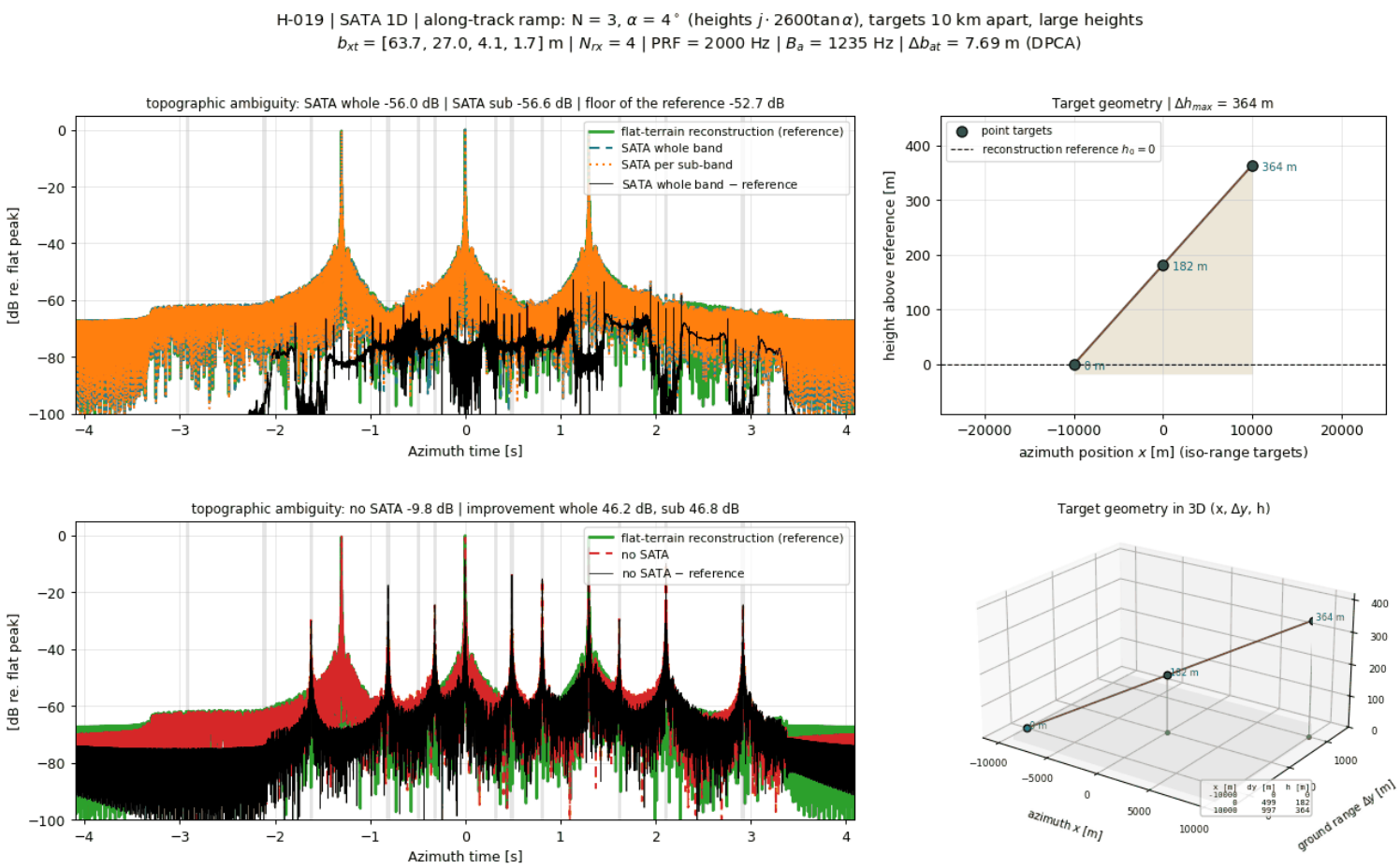


Figure 136: H-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −9.8, SATA whole band −56.0, per sub-band −56.6 dB (improvement 46.2 dB); floor −52.7 dB.

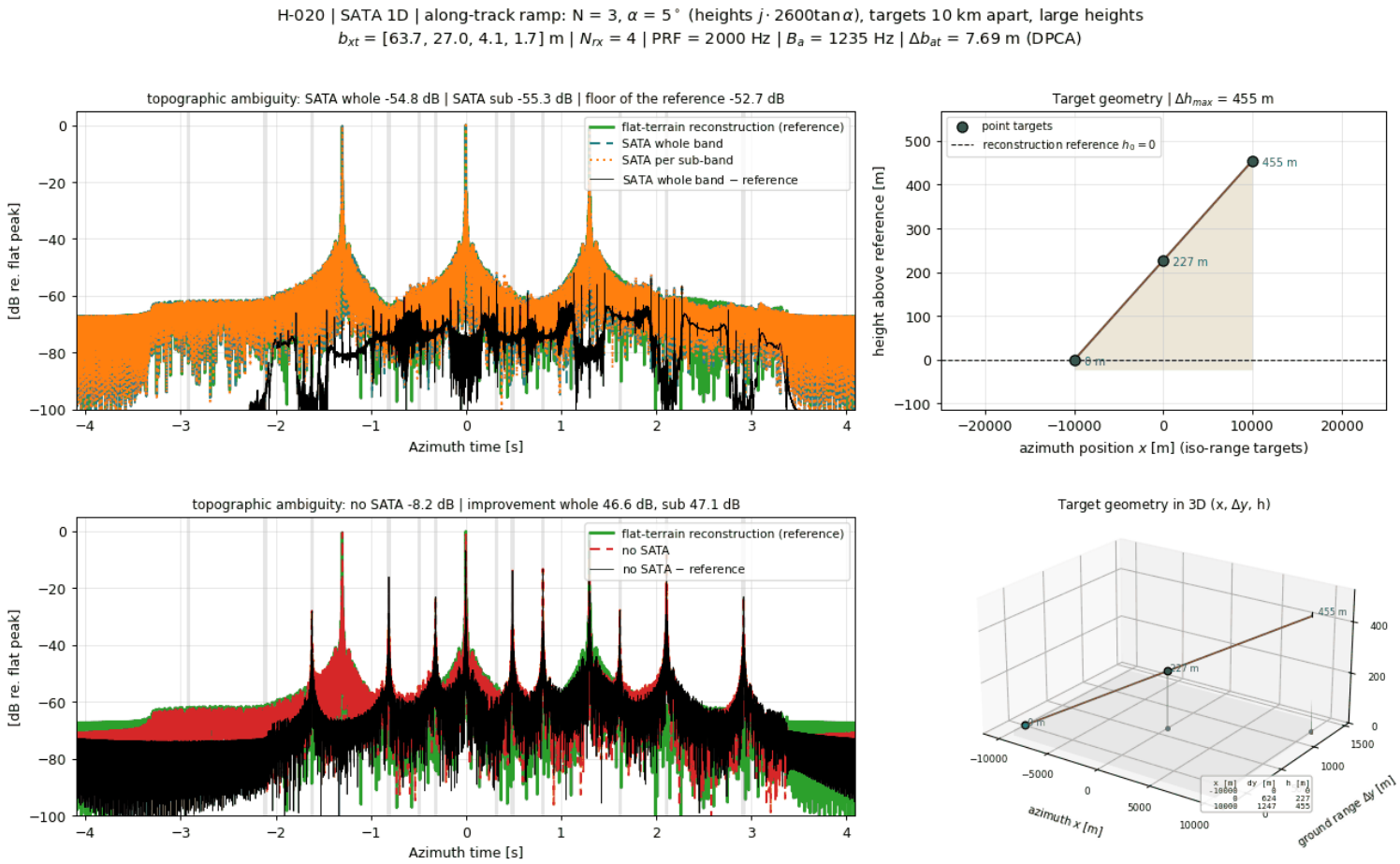


Figure 137: H-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −8.2, SATA whole band −54.8, per sub-band −55.3 dB (improvement 46.6 dB); floor −52.7 dB.

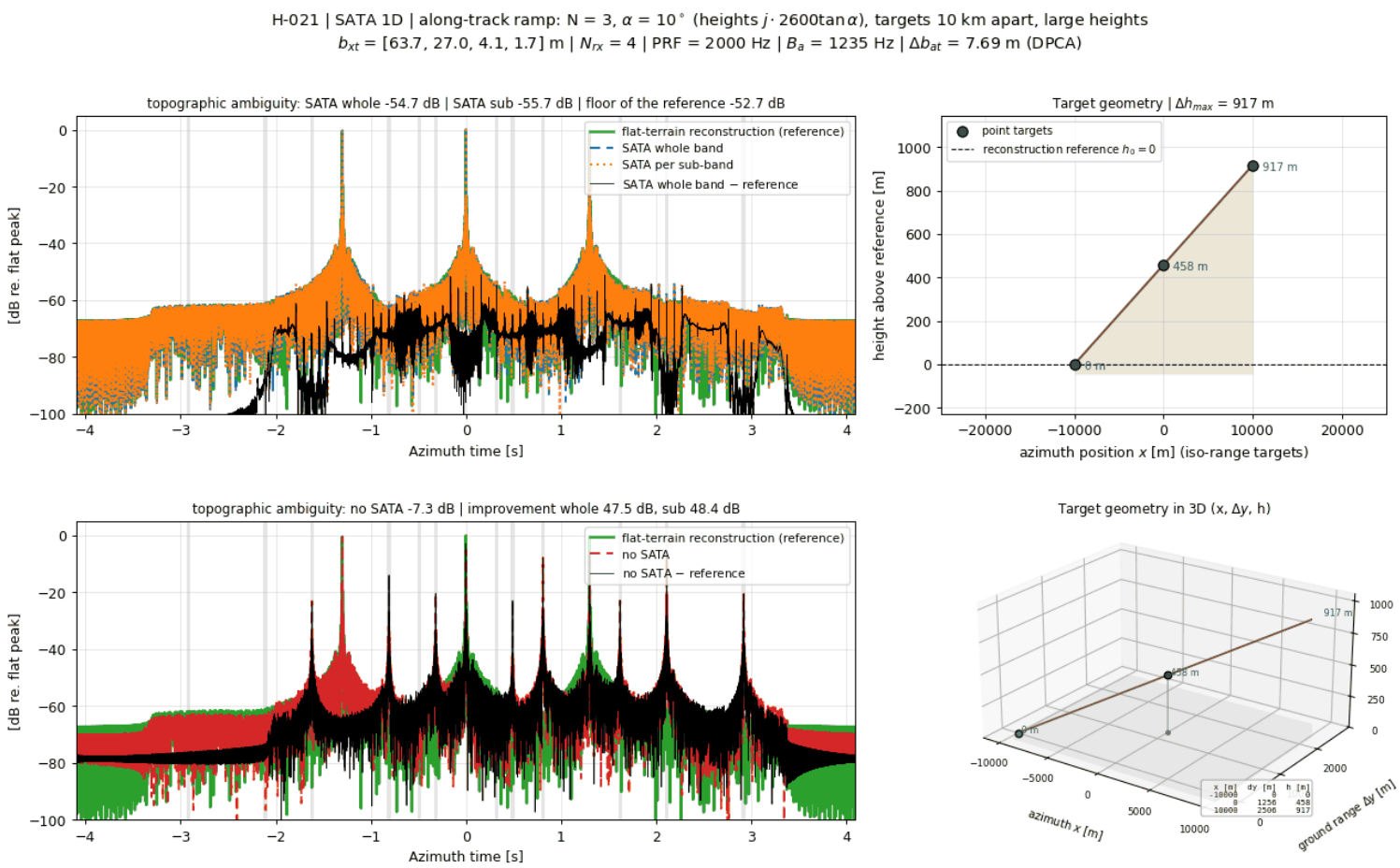


Figure 138: H-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.3, SATA whole band −54.7, per sub-band −55.7 dB (improvement 47.5 dB); floor −52.7 dB.

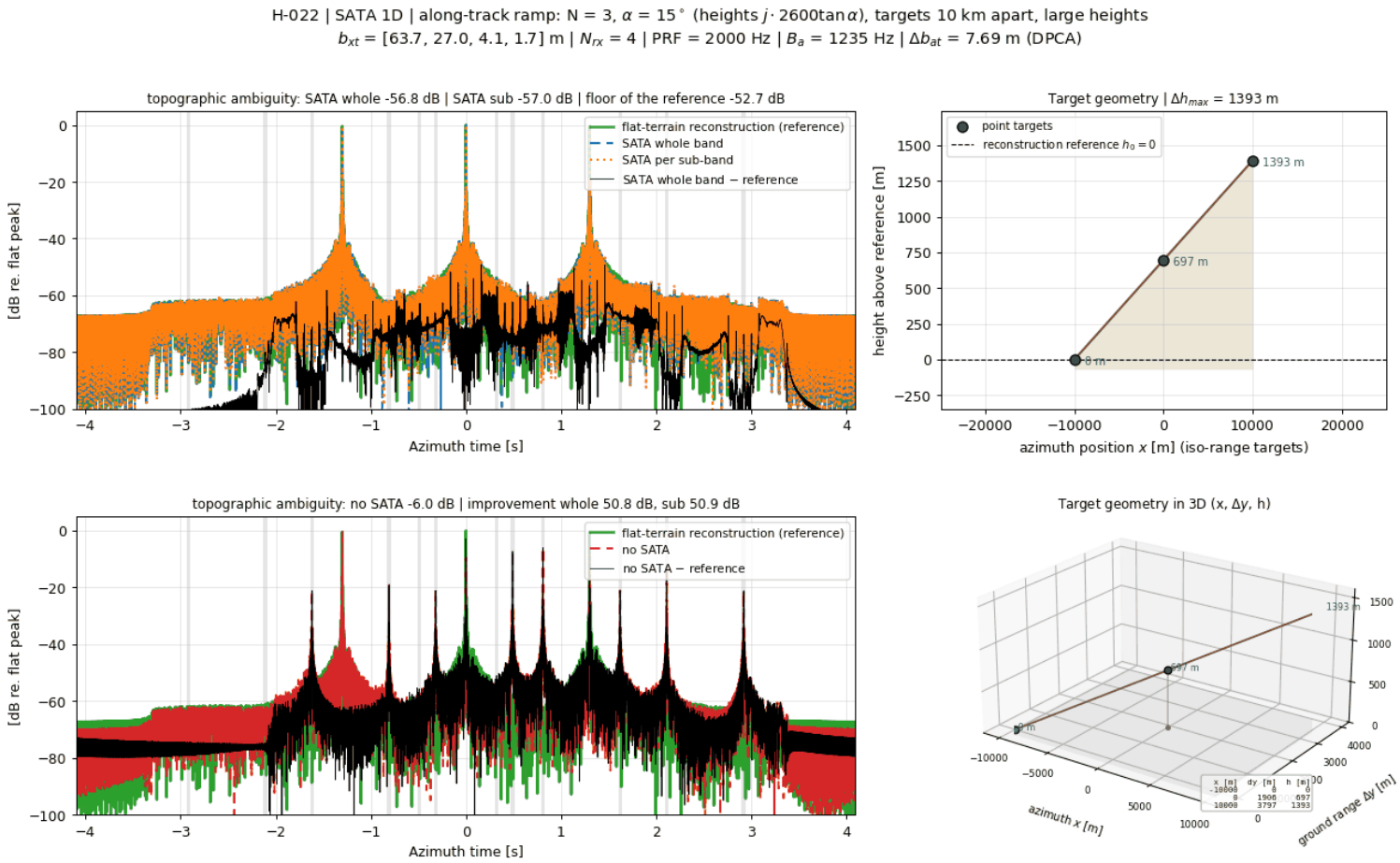


Figure 139: H-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.0, SATA whole band −56.8, per sub-band −57.0 dB (improvement 50.8 dB); floor −52.7 dB.

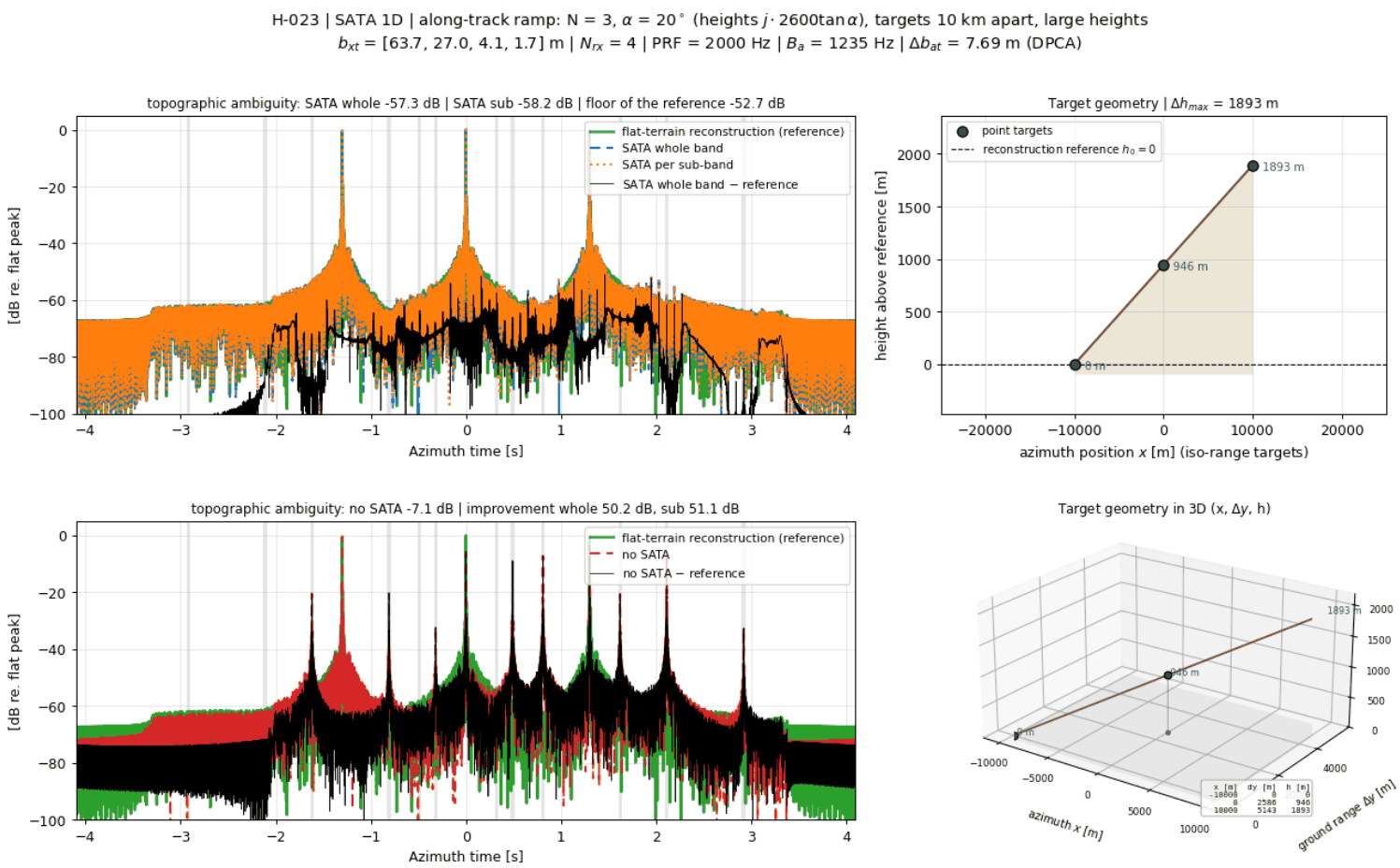


Figure 140: H-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.1, SATA whole band −57.3, per sub-band −58.2 dB (improvement 50.2 dB); floor −52.7 dB.

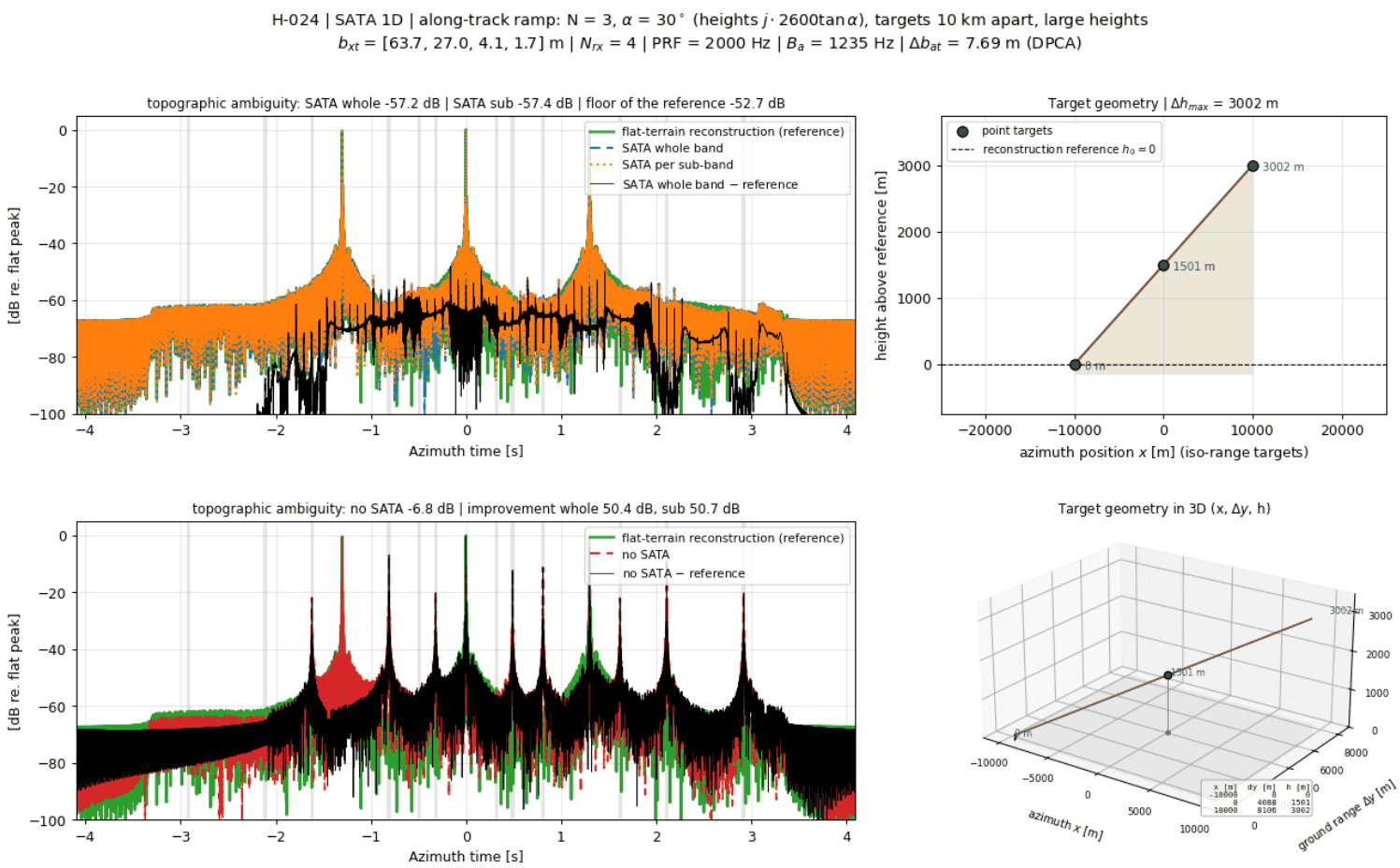


Figure 141: H-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.8, SATA whole band −57.2, per sub-band −57.4 dB (improvement 50.4 dB); floor −52.7 dB.

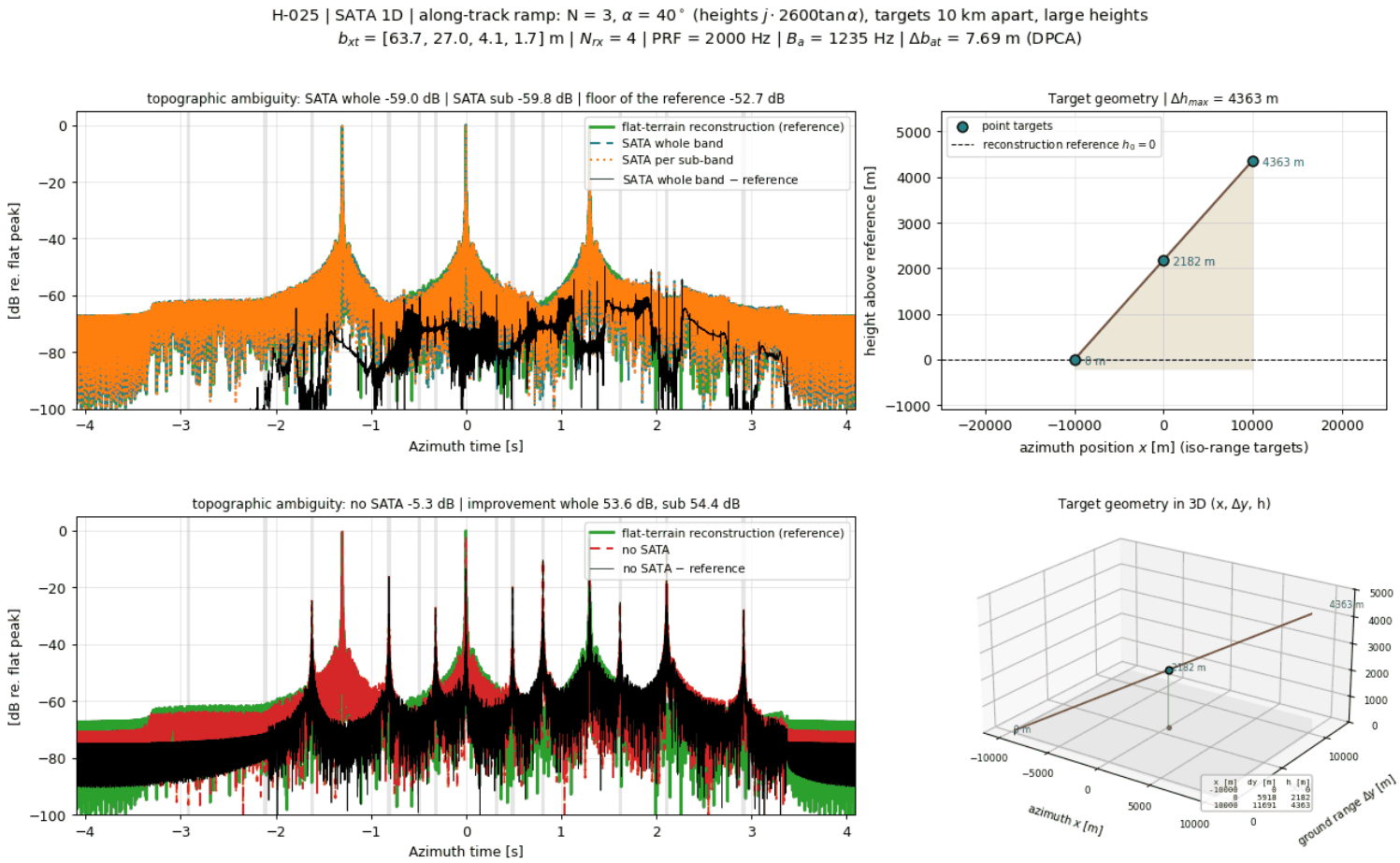


Figure 142: H-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −5.3, SATA whole band −59.0, per sub-band −59.8 dB (improvement 53.6 dB); floor −52.7 dB.

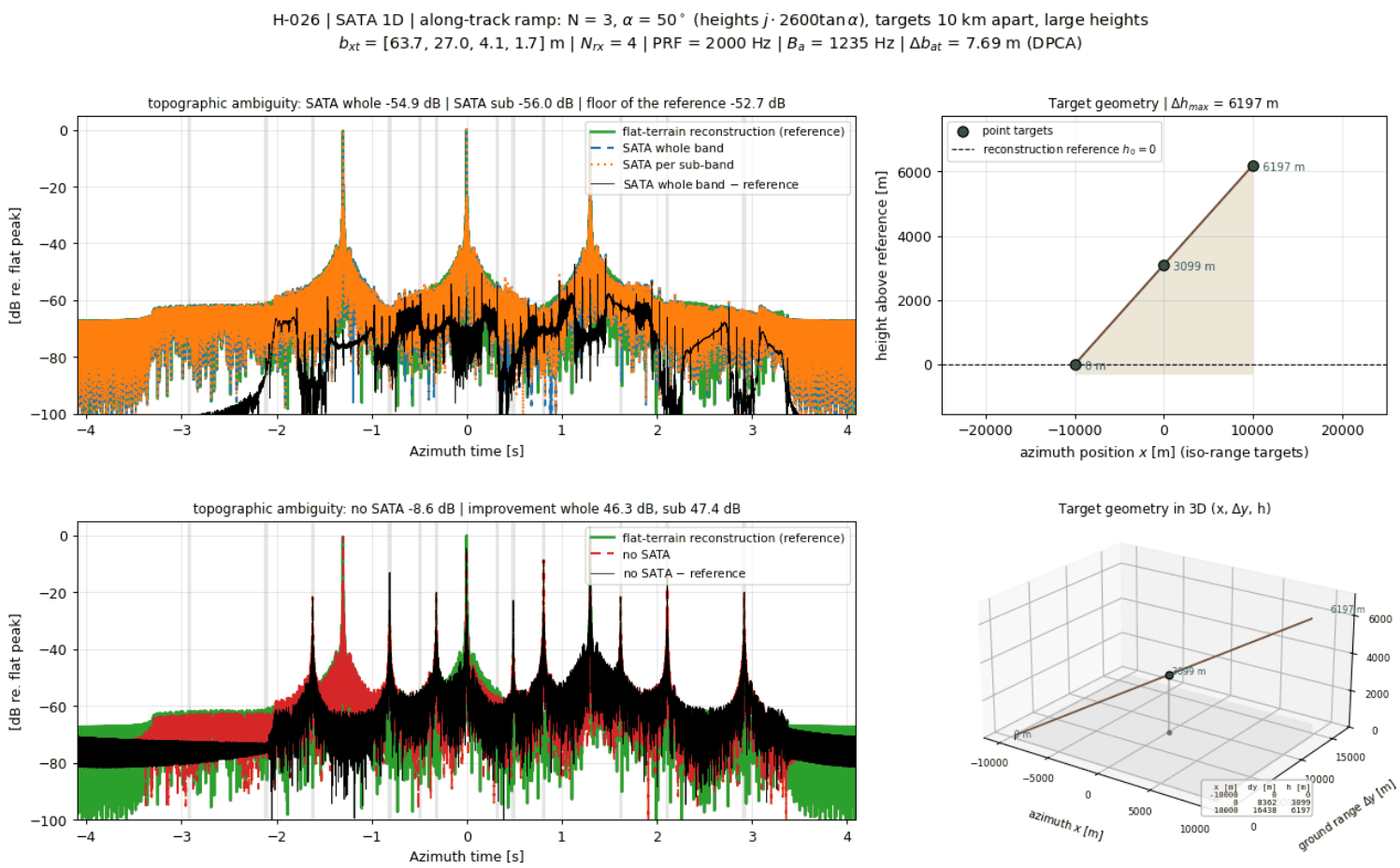


Figure 143: H-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −8.6, SATA whole band −54.9, per sub-band −56.0 dB (improvement 46.3 dB); floor −52.7 dB.

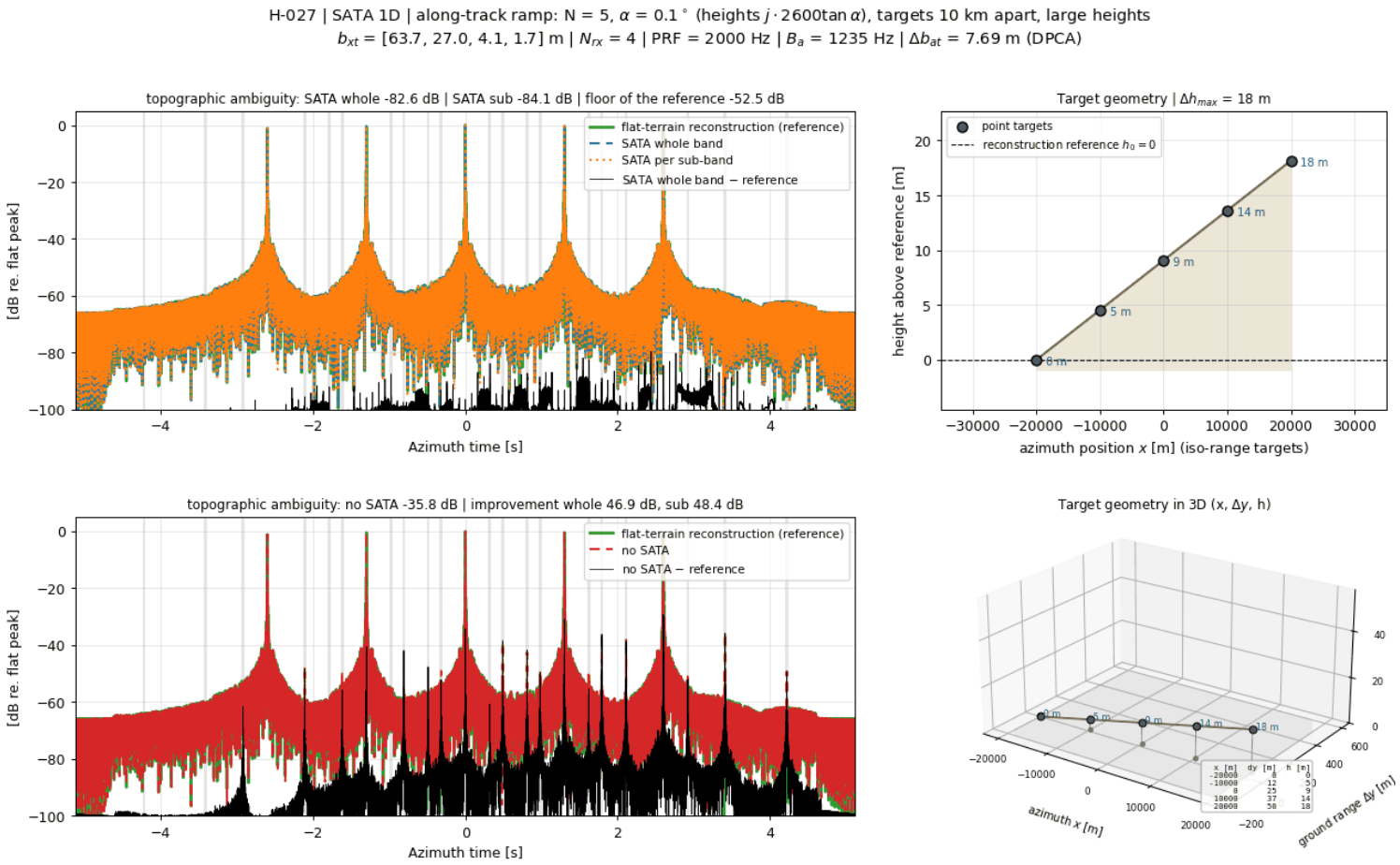


Figure 144: H-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −35.8, SATA whole band −82.6, per sub-band −84.1 dB (improvement 46.9 dB); floor −52.5 dB.

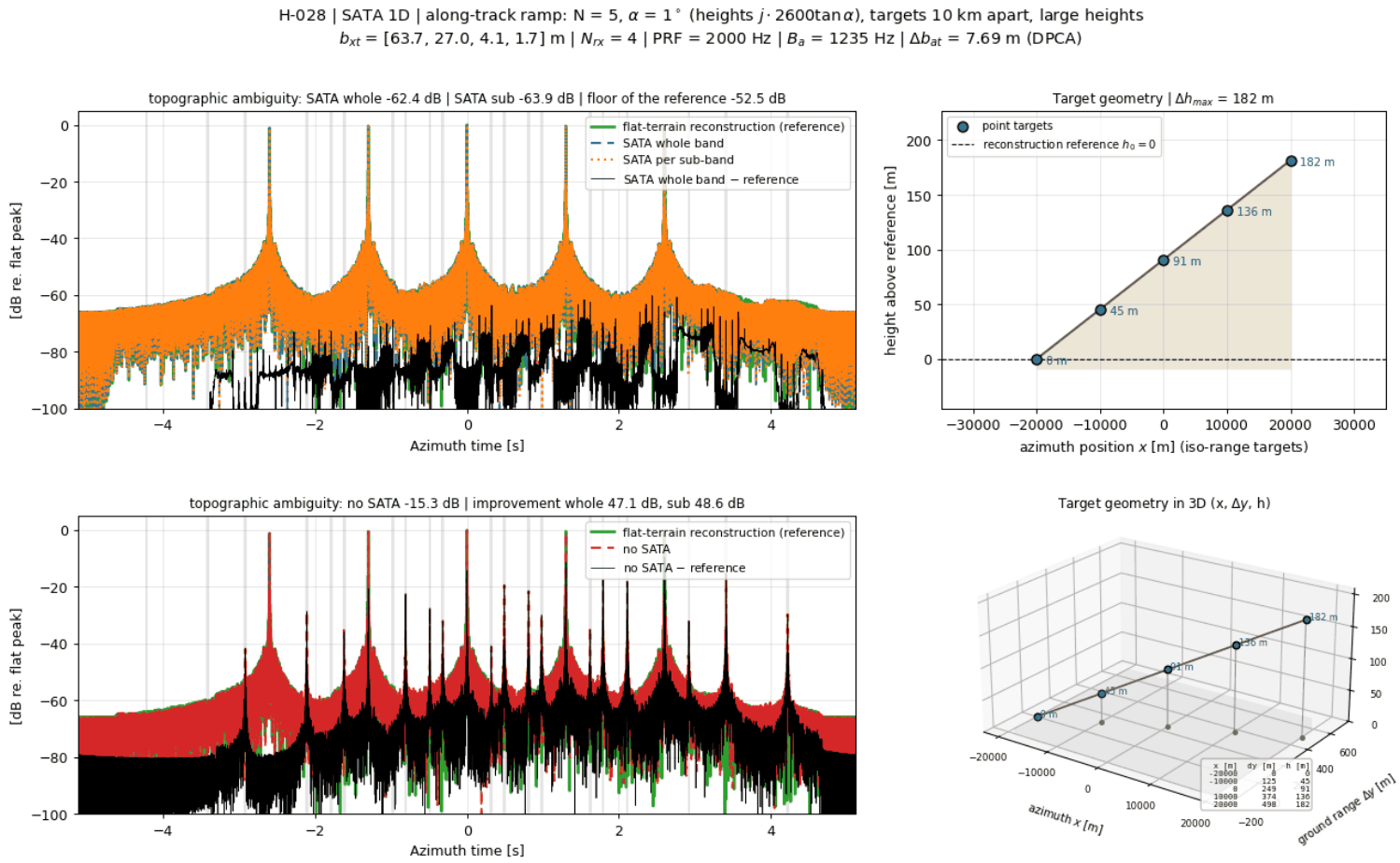


Figure 145: H-028, $N = 5$, $\alpha = 1^\circ$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −15.3, SATA whole band −62.4, per sub-band −63.9 dB (improvement 47.1 dB); floor −52.5 dB.

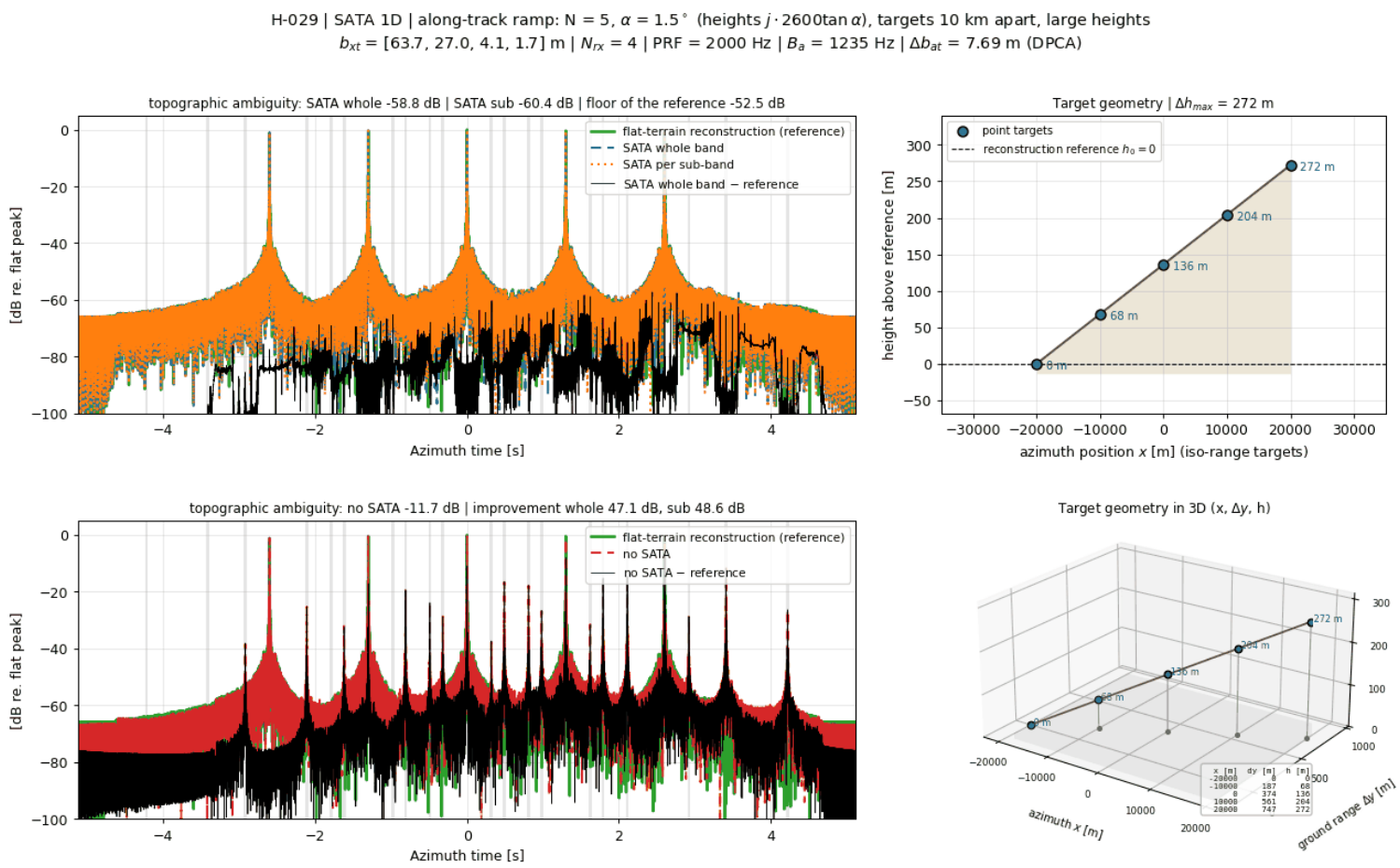


Figure 146: H-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −11.7, SATA whole band −58.8, per sub-band −60.4 dB (improvement 47.1 dB); floor −52.5 dB.

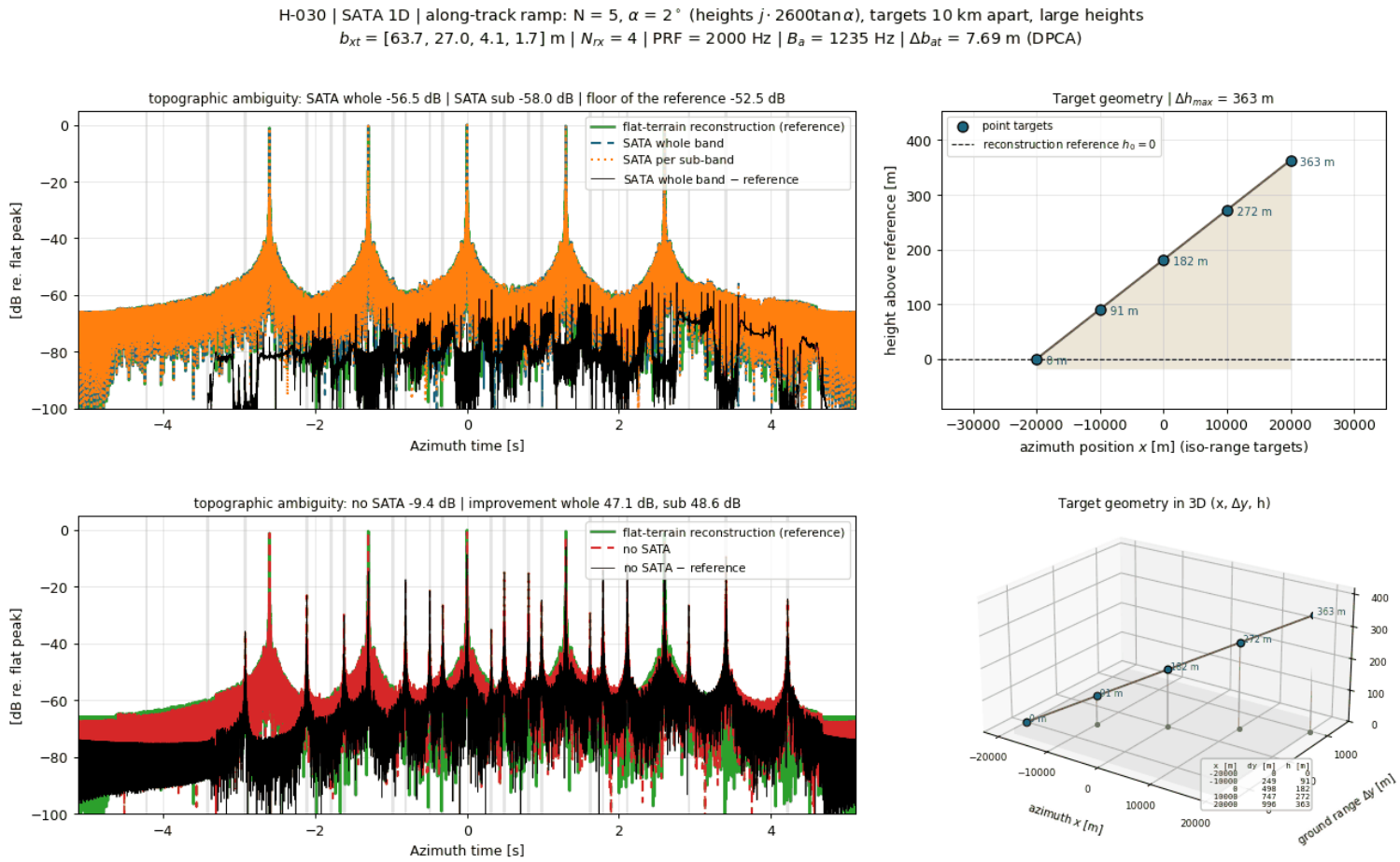


Figure 147: H-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −9.4, SATA whole band −56.5, per sub-band −58.0 dB (improvement 47.1 dB); floor −52.5 dB.

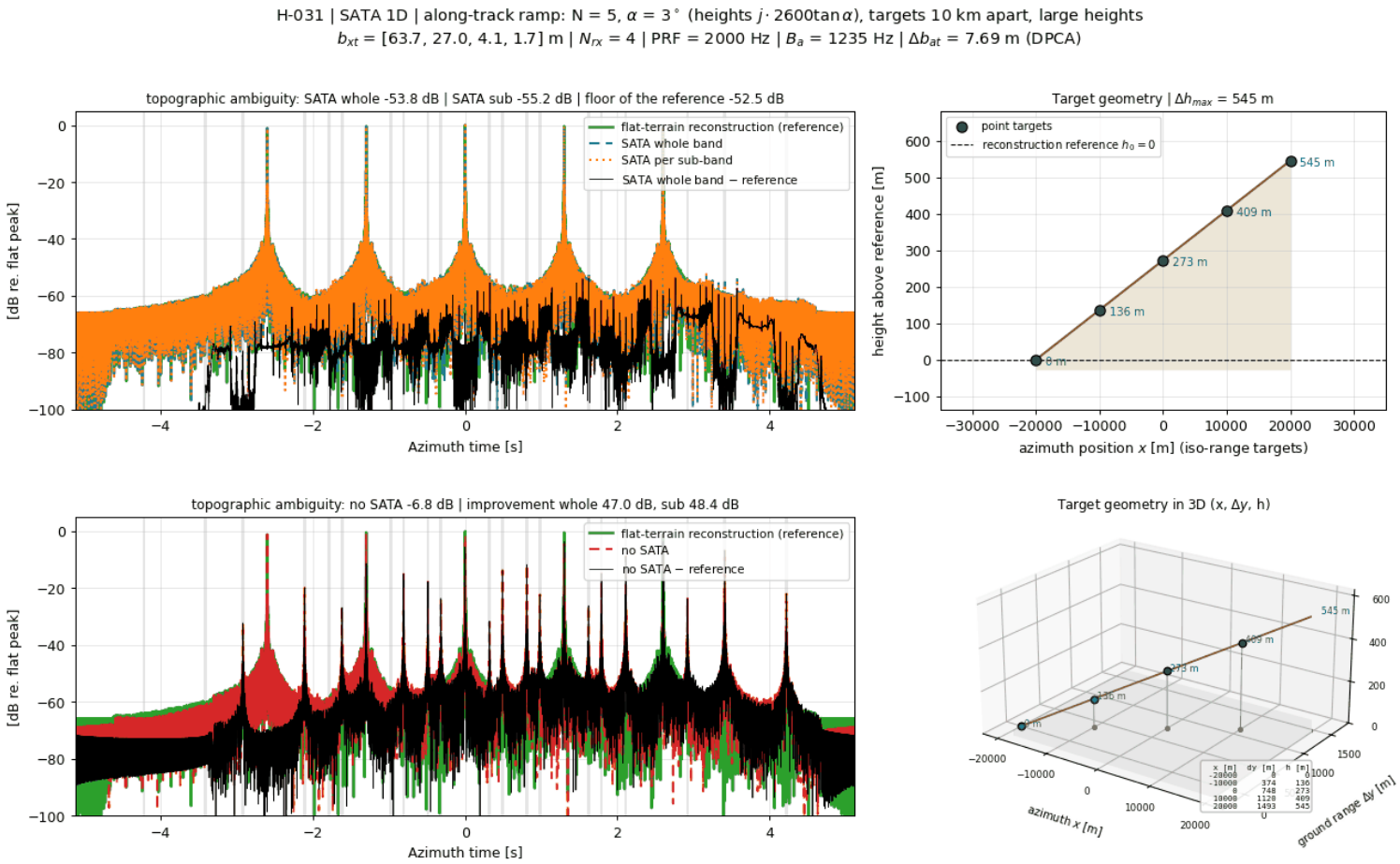


Figure 148: H-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.8, SATA whole band −53.8, per sub-band −55.2 dB (improvement 47.0 dB); floor −52.5 dB.

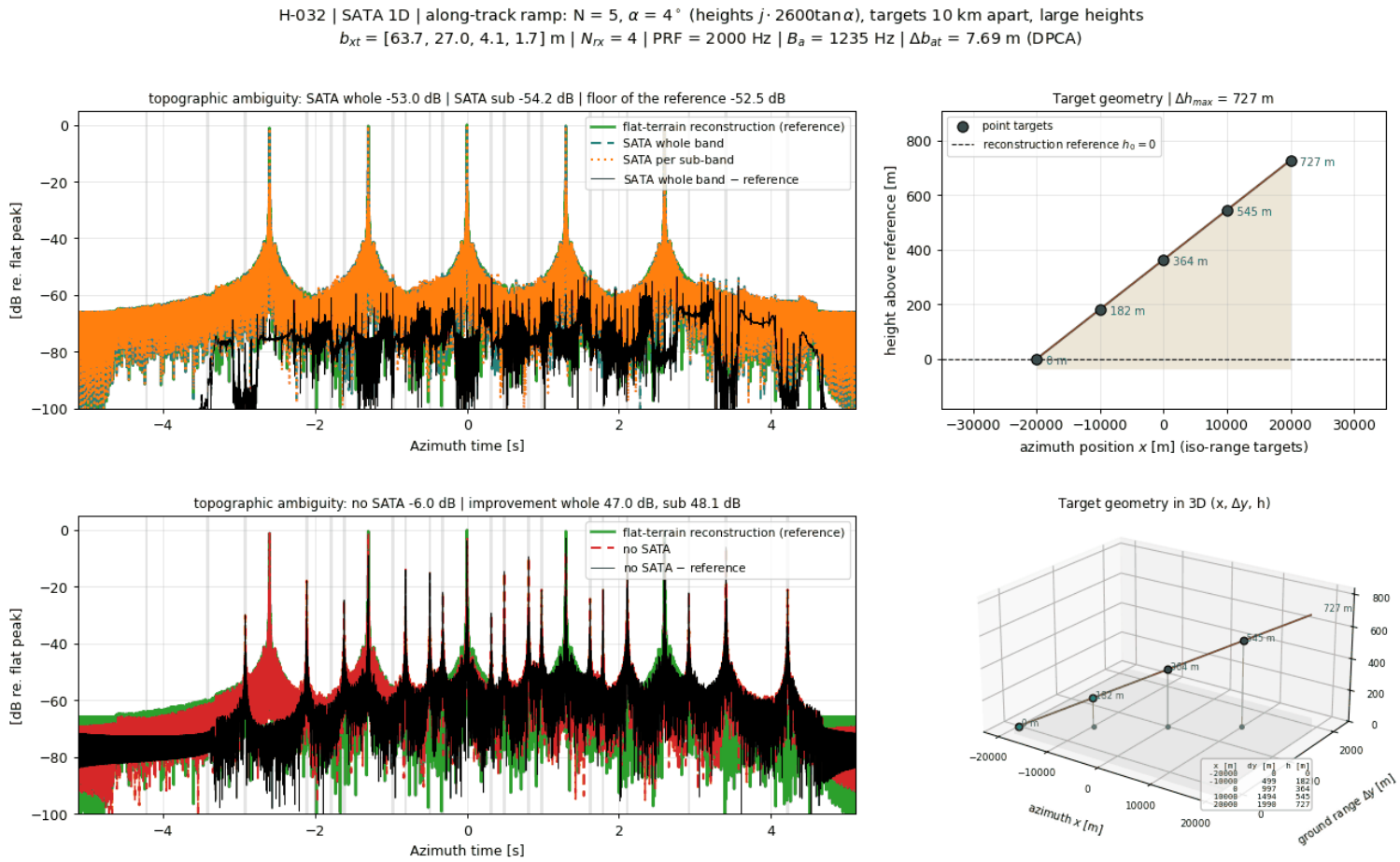


Figure 149: H-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.0, SATA whole band −53.0, per sub-band −54.2 dB (improvement 47.0 dB); floor −52.5 dB.

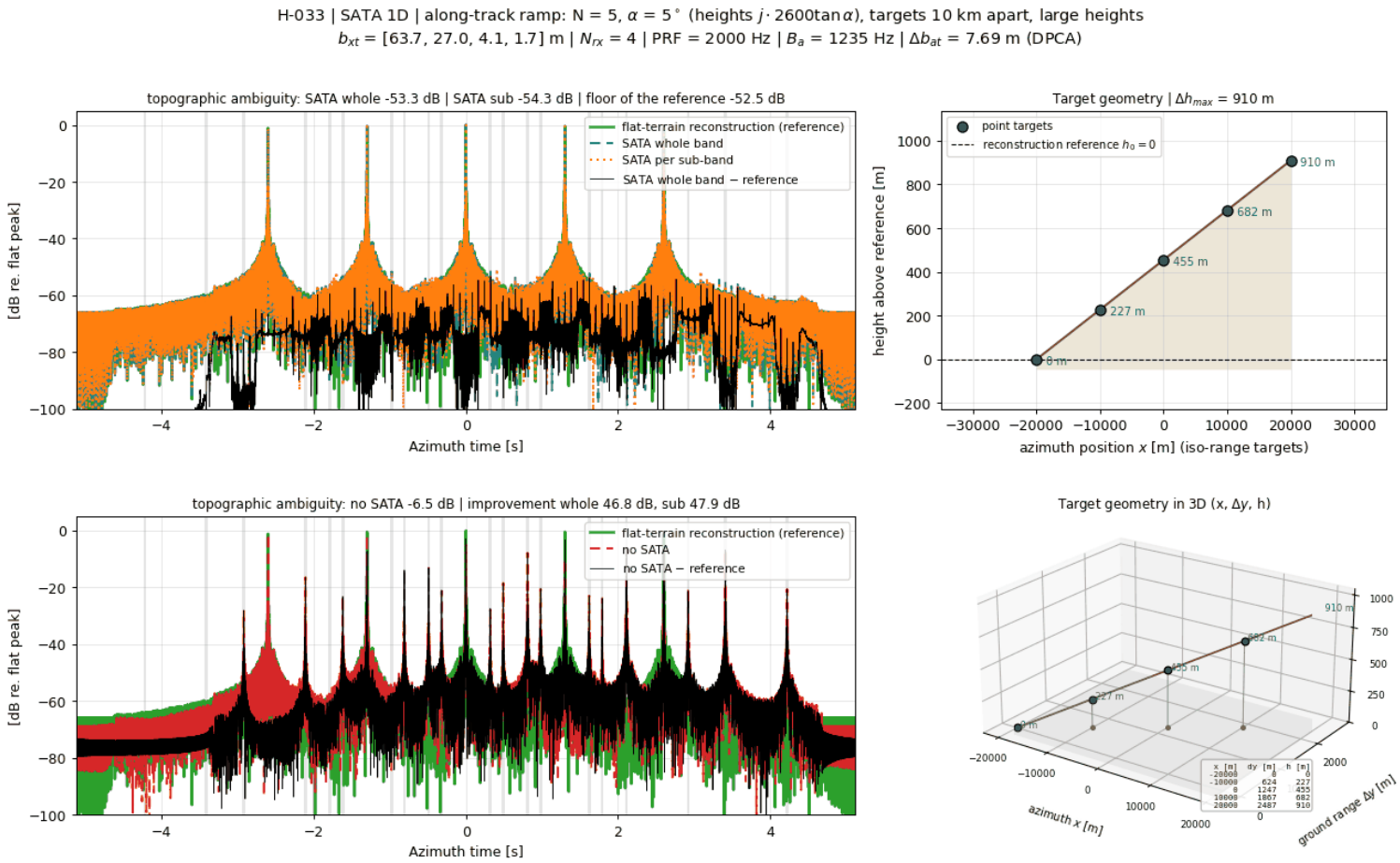


Figure 150: H-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.5, SATA whole band −53.3, per sub-band −54.3 dB (improvement 46.8 dB); floor −52.5 dB.

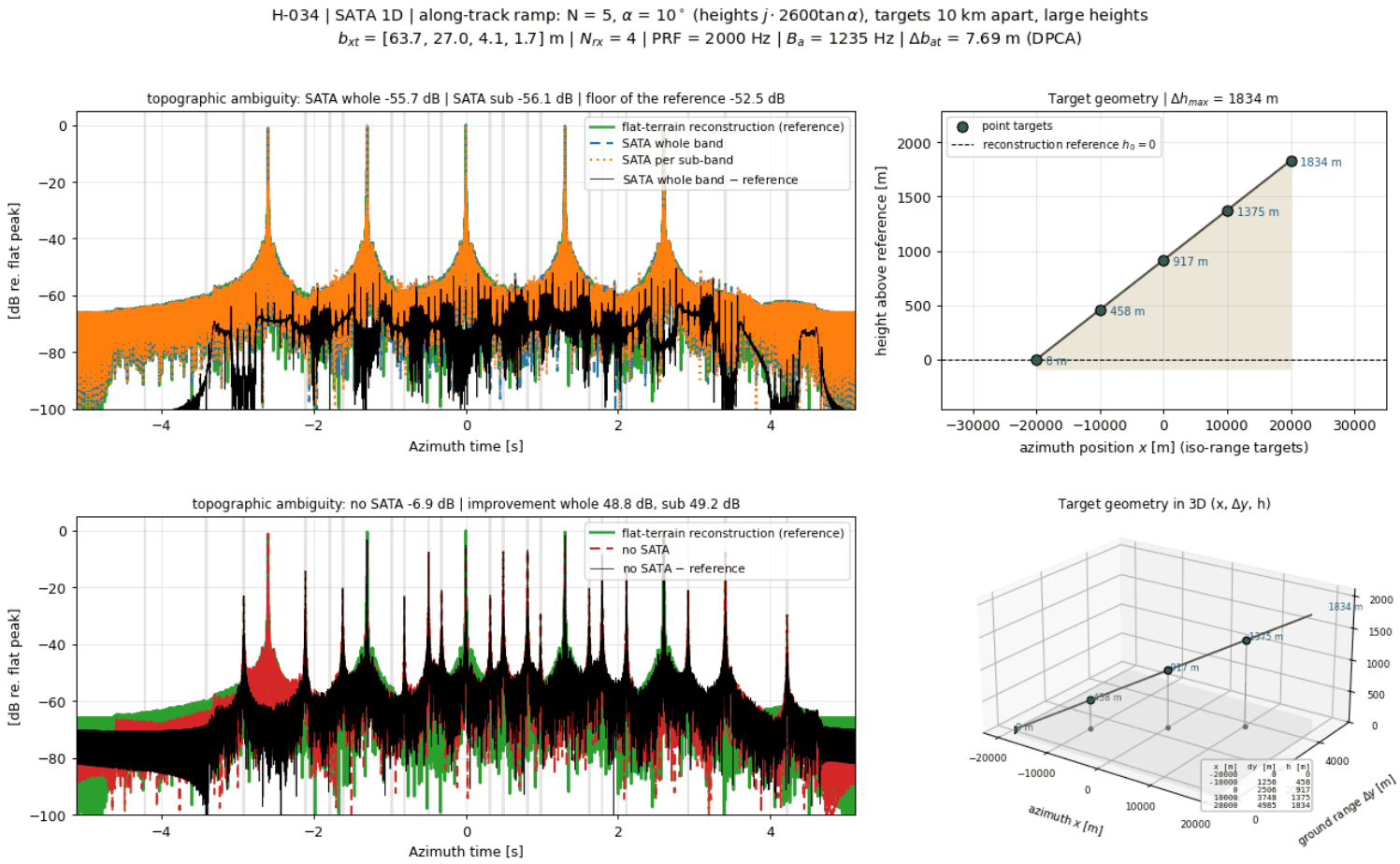


Figure 151: H-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.9, SATA whole band −55.7, per sub-band −56.1 dB (improvement 48.8 dB); floor −52.5 dB.

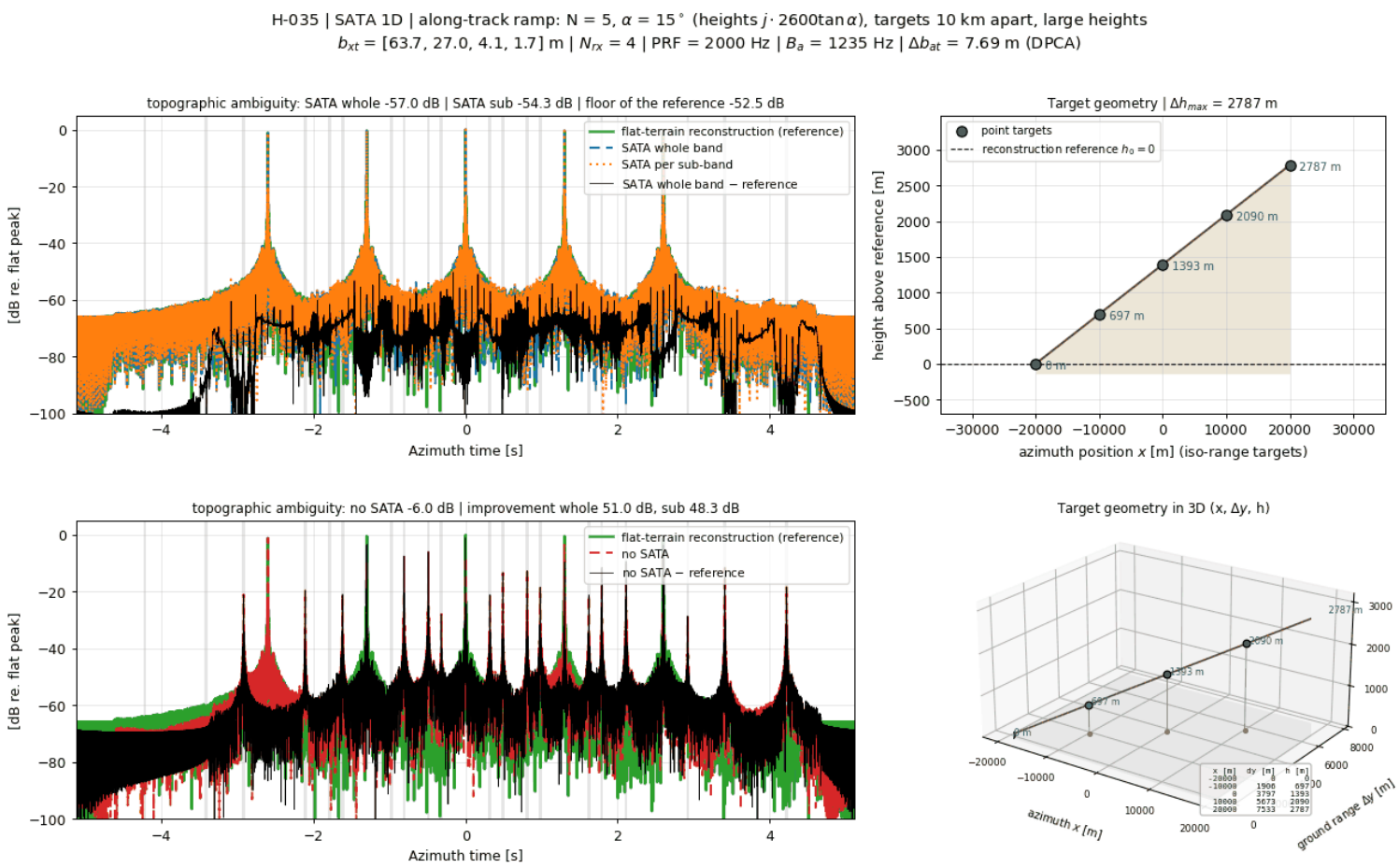


Figure 152: H-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.0, SATA whole band −57.0, per sub-band −54.3 dB (improvement 51.0 dB); floor −52.5 dB.

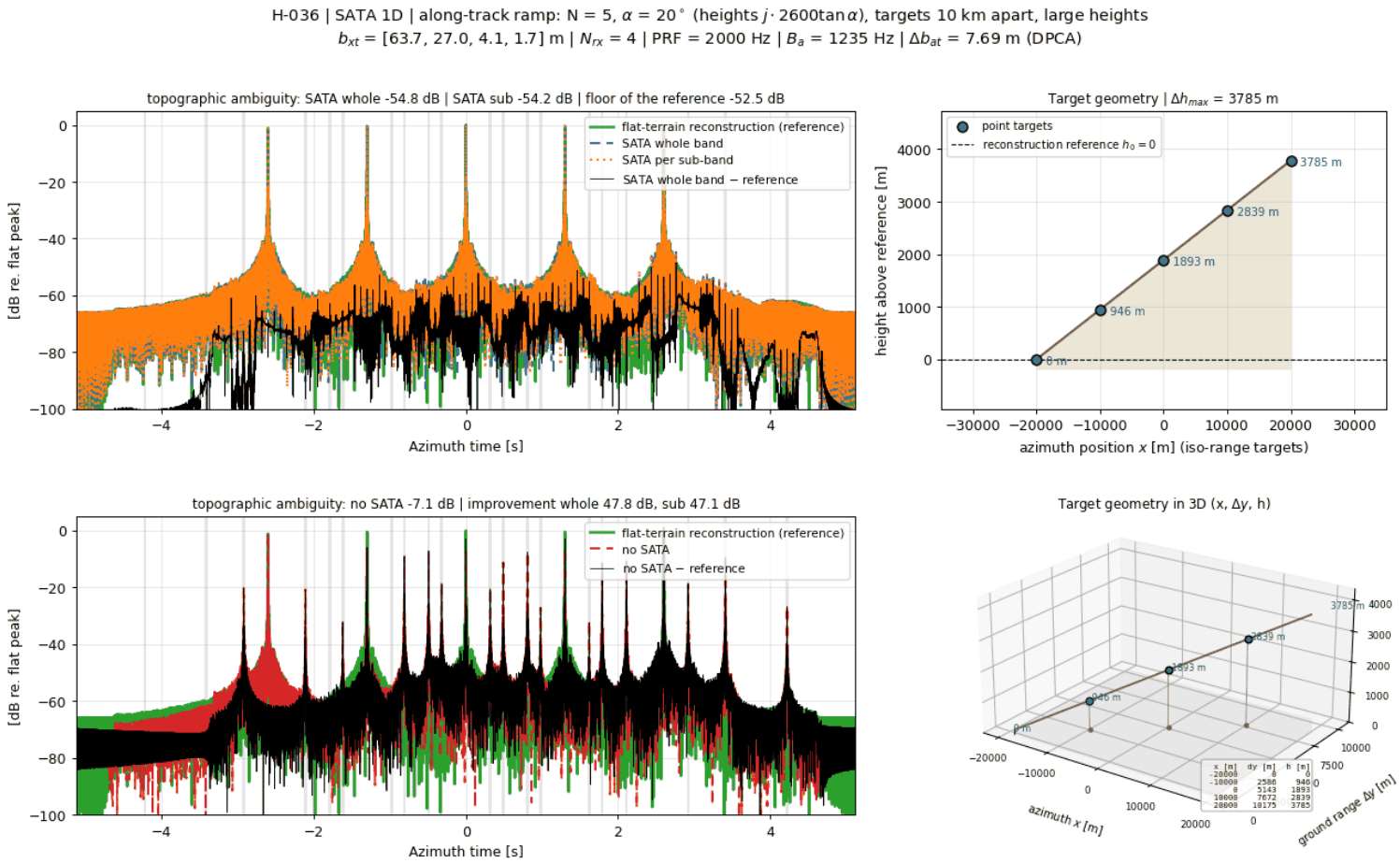


Figure 153: H-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −7.1, SATA whole band −54.8, per sub-band −54.2 dB (improvement 47.8 dB); floor −52.5 dB.

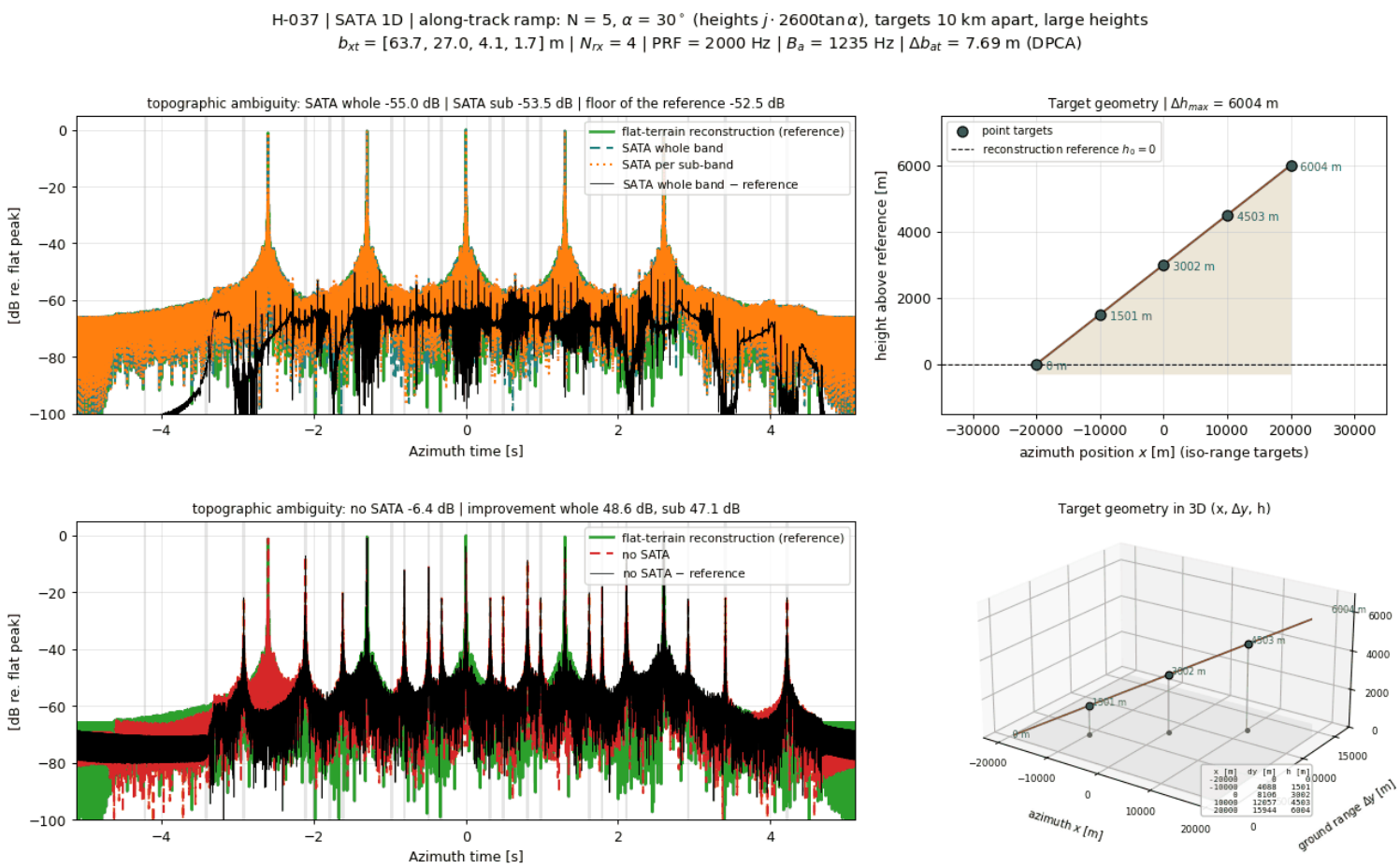


Figure 154: H-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −6.4, SATA whole band −55.0, per sub-band −53.5 dB (improvement 48.6 dB); floor −52.5 dB.

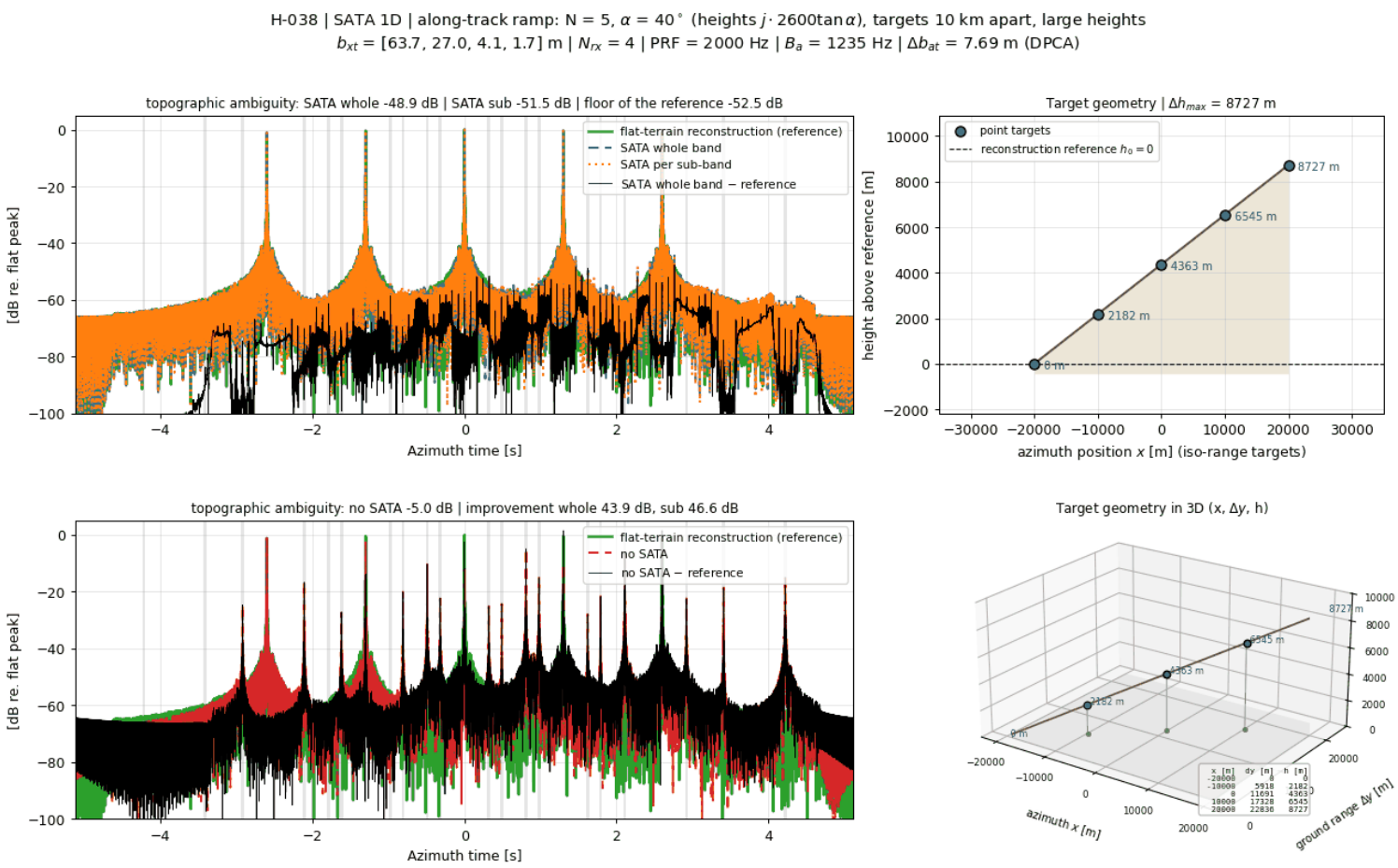


Figure 155: H-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −5.0, SATA whole band −48.9, per sub-band −51.5 dB (improvement 43.9 dB); floor −52.5 dB.

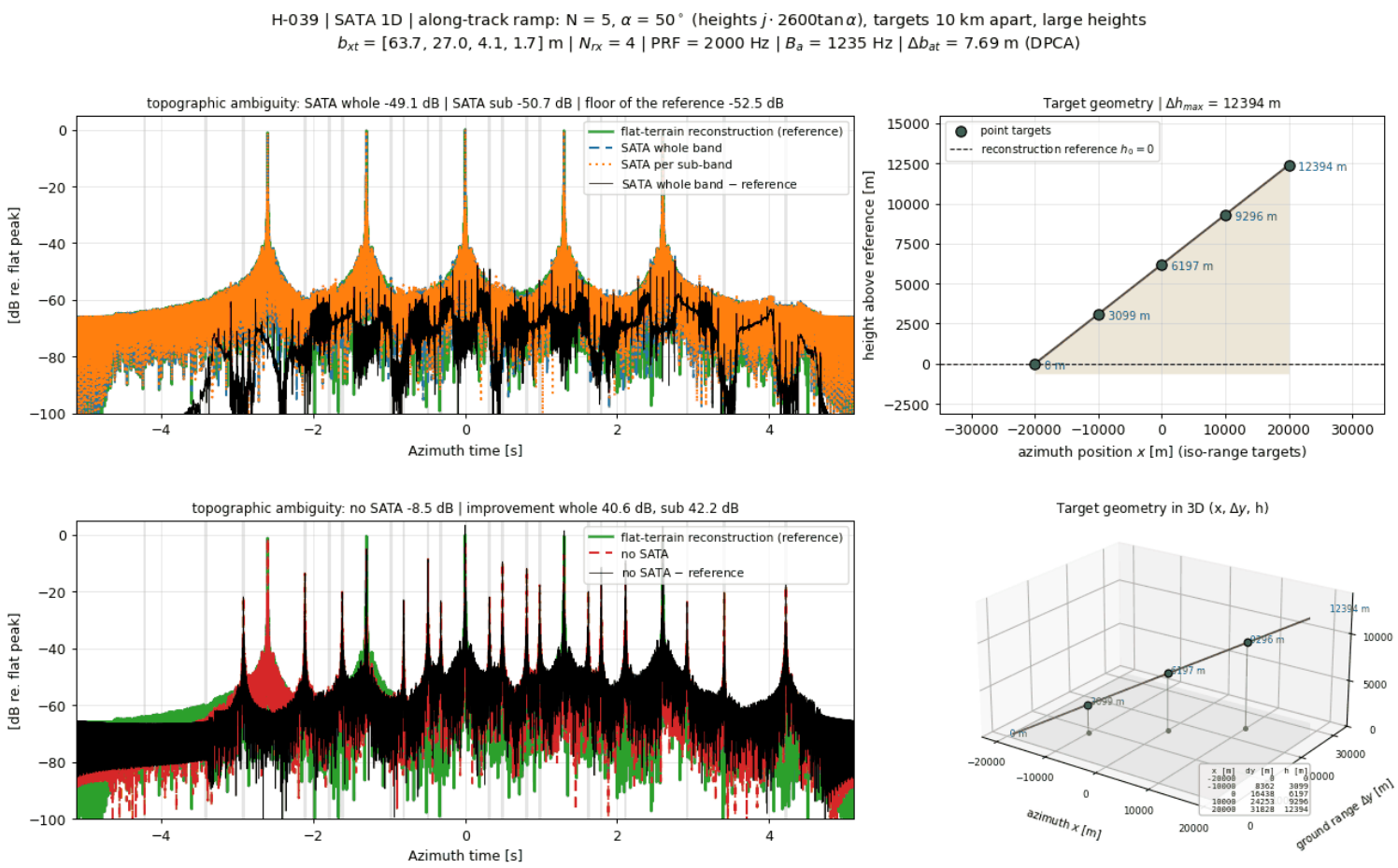


Figure 156: H-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Topographic ambiguity: no SATA −8.5, SATA whole band −49.1, per sub-band −50.7 dB (improvement 40.6 dB); floor −52.5 dB.

5 Random heights, targets 5.2 km apart

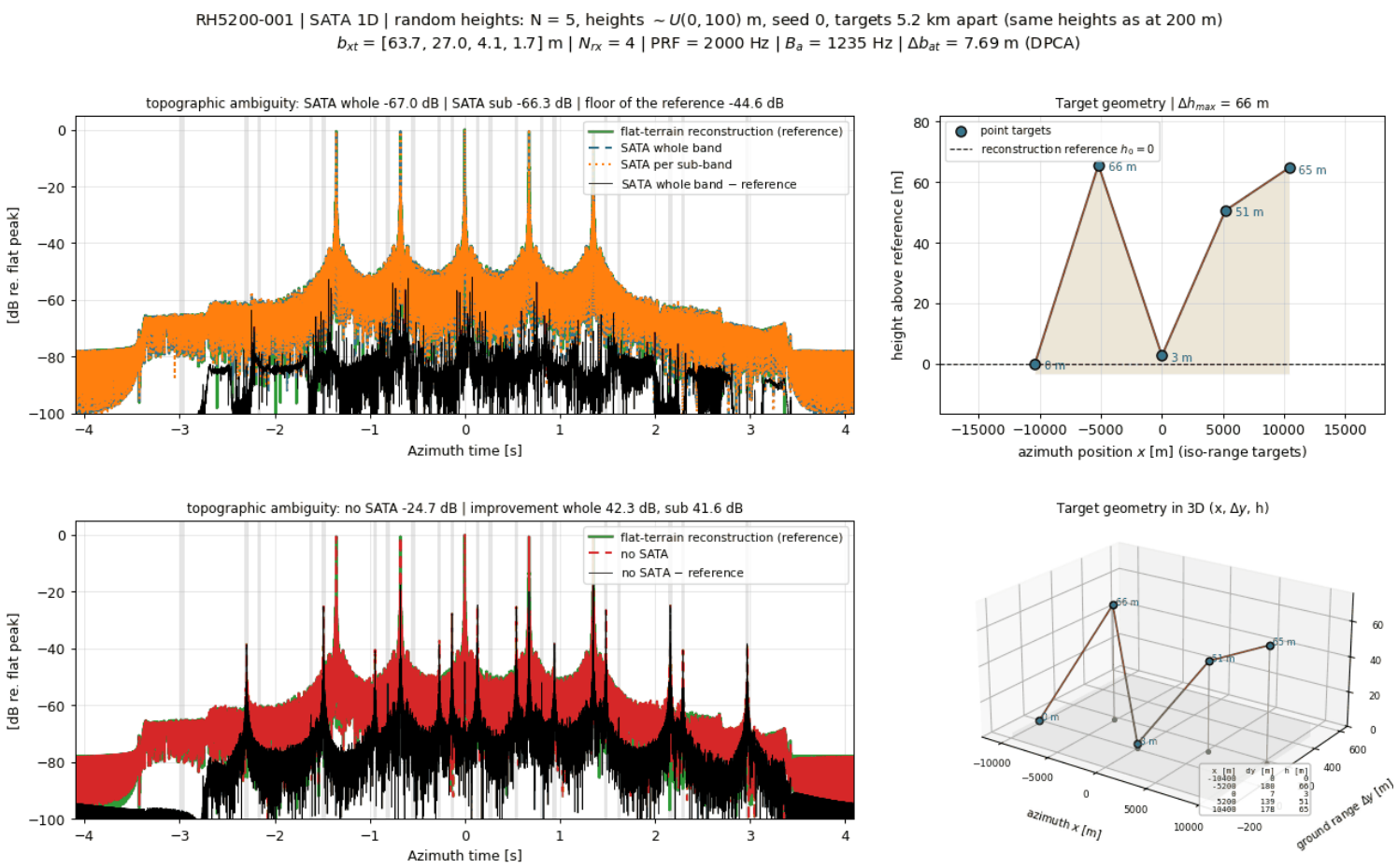


Figure 157: RH5200-001, $N = 5$, $h \sim U(0, 100)$ m, seed 0. Topographic ambiguity: no SATA −24.7, SATA whole band −67.0, per sub-band −66.3 dB (improvement 42.3 dB); floor −44.6 dB.

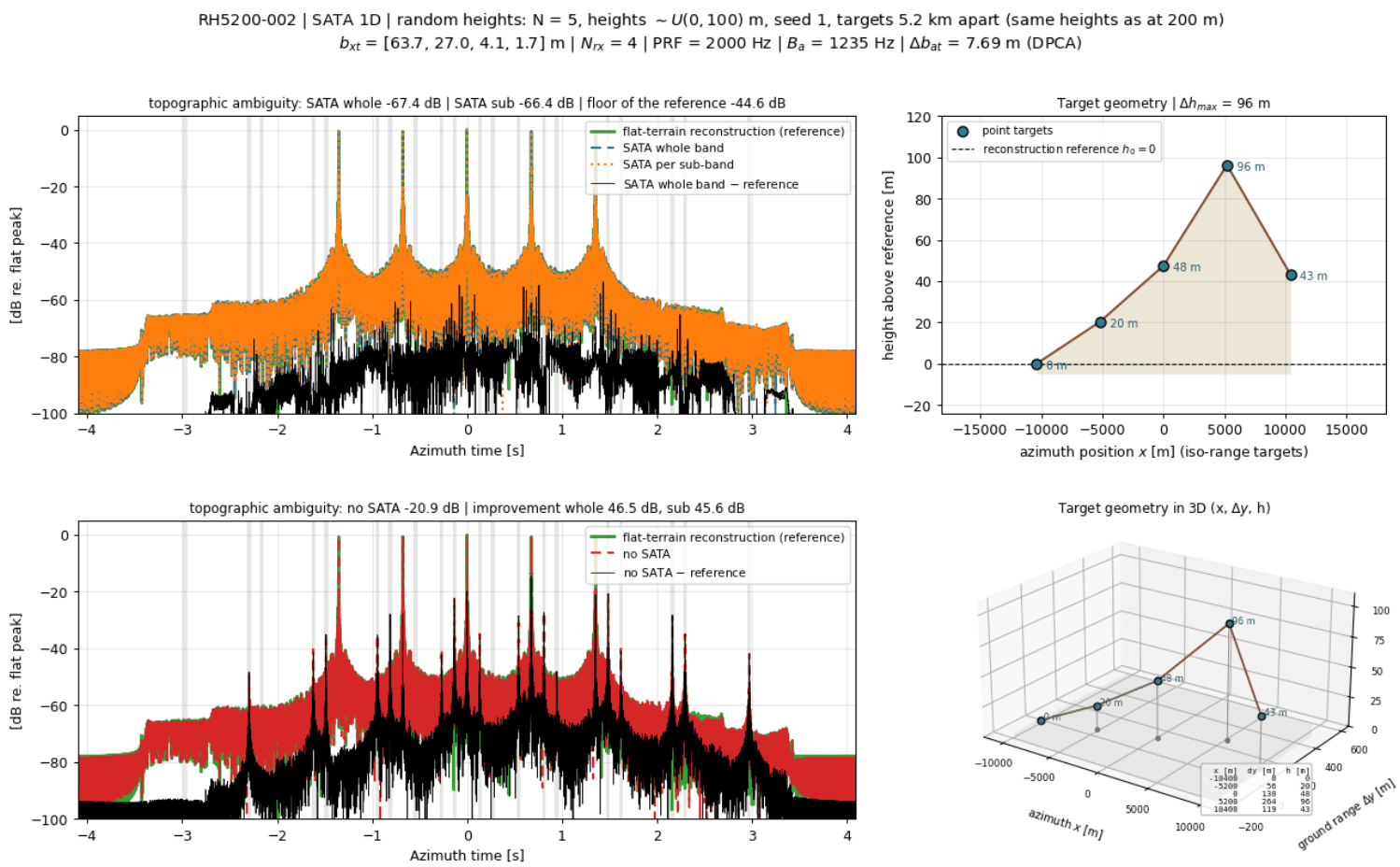


Figure 158: RH5200-002, $N = 5$, $h \sim U(0, 100)$ m, seed 1. Topographic ambiguity: no SATA −20.9, SATA whole band −67.4, per sub-band −66.4 dB (improvement 46.5 dB); floor −44.6 dB.

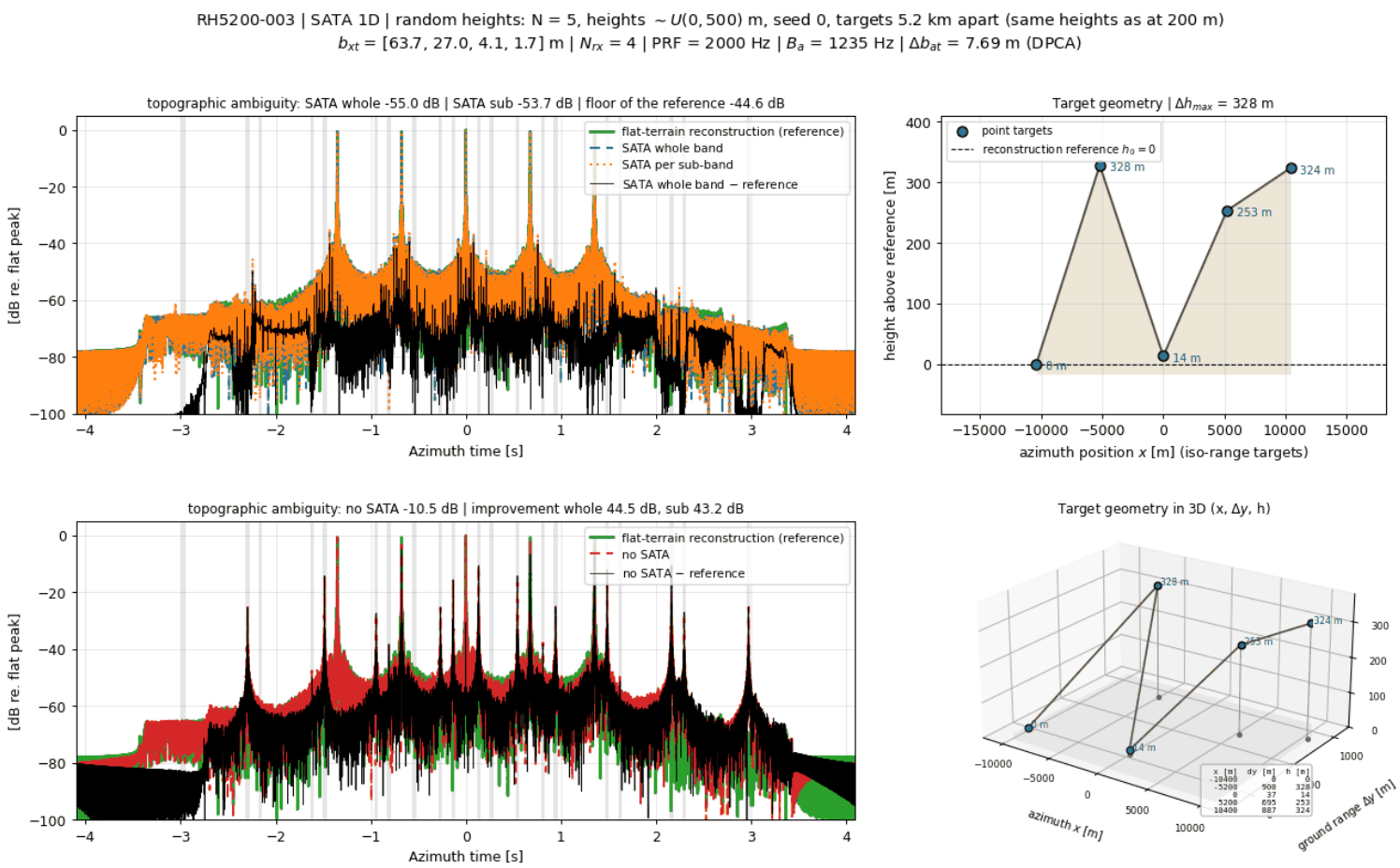


Figure 159: RH5200-003, $N = 5$, $h \sim U(0, 500)$ m, seed 0. Topographic ambiguity: no SATA −10.5, SATA whole band −55.0, per sub-band −53.7 dB (improvement 44.5 dB); floor −44.6 dB.

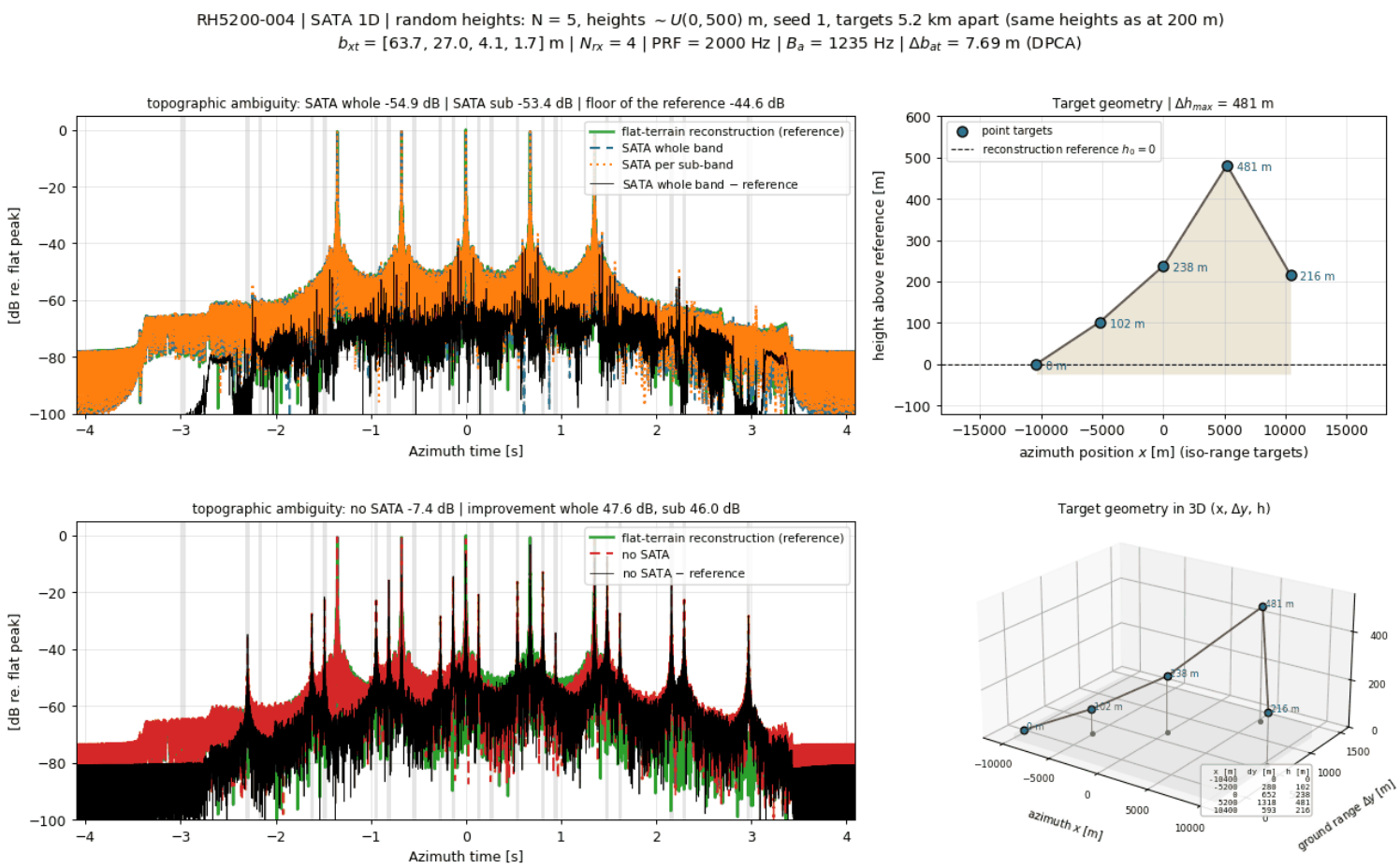


Figure 160: RH5200-004, $N = 5$, $h \sim U(0, 500)$ m, seed 1. Topographic ambiguity: no SATA −7.4, SATA whole band −54.9, per sub-band −53.4 dB (improvement 47.6 dB); floor −44.6 dB.

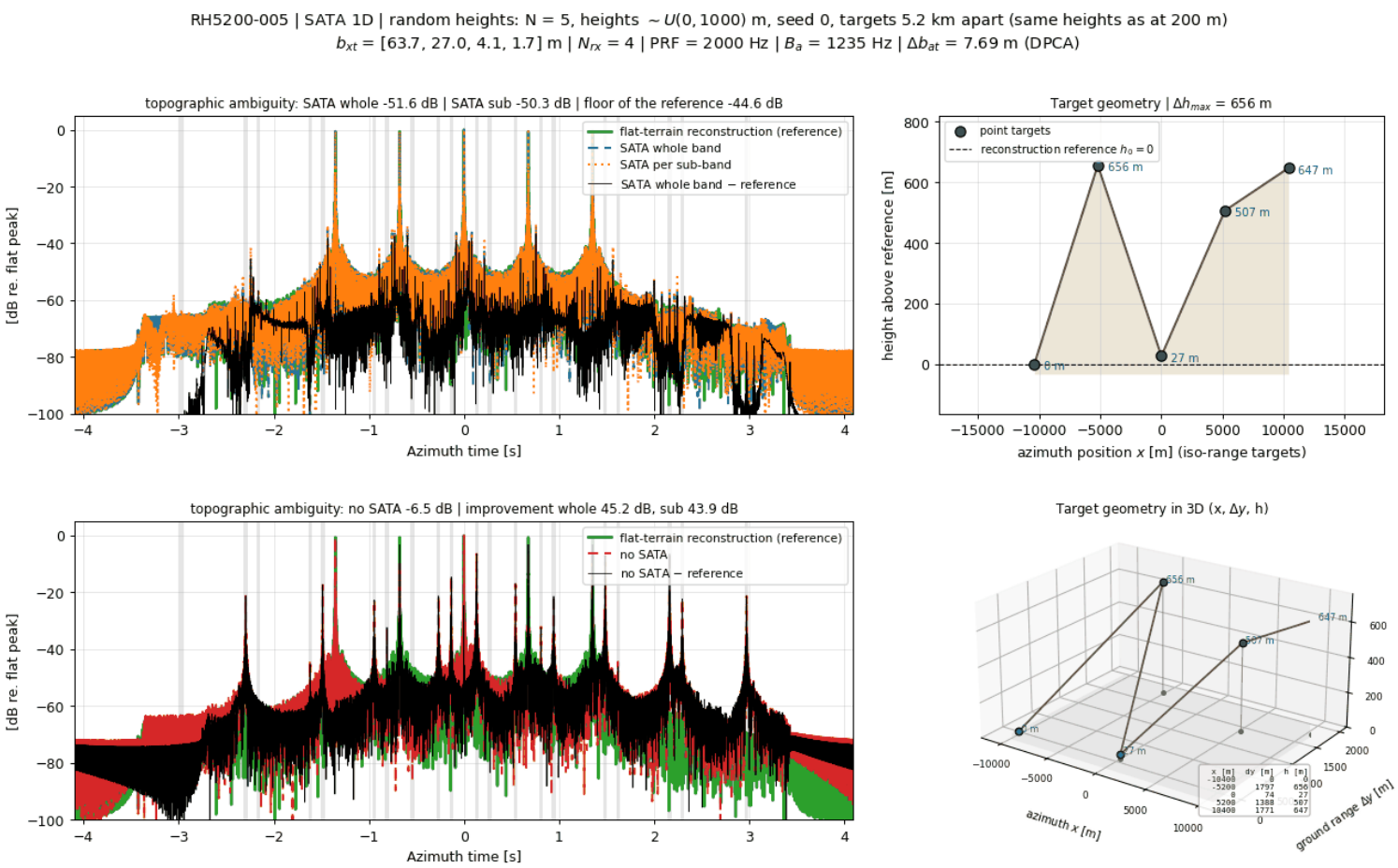


Figure 161: RH5200-005, $N = 5$, $h \sim U(0, 1000)$ m, seed 0. Topographic ambiguity: no SATA −6.5, SATA whole band −51.6, per sub-band −50.3 dB (improvement 45.2 dB); floor −44.6 dB.

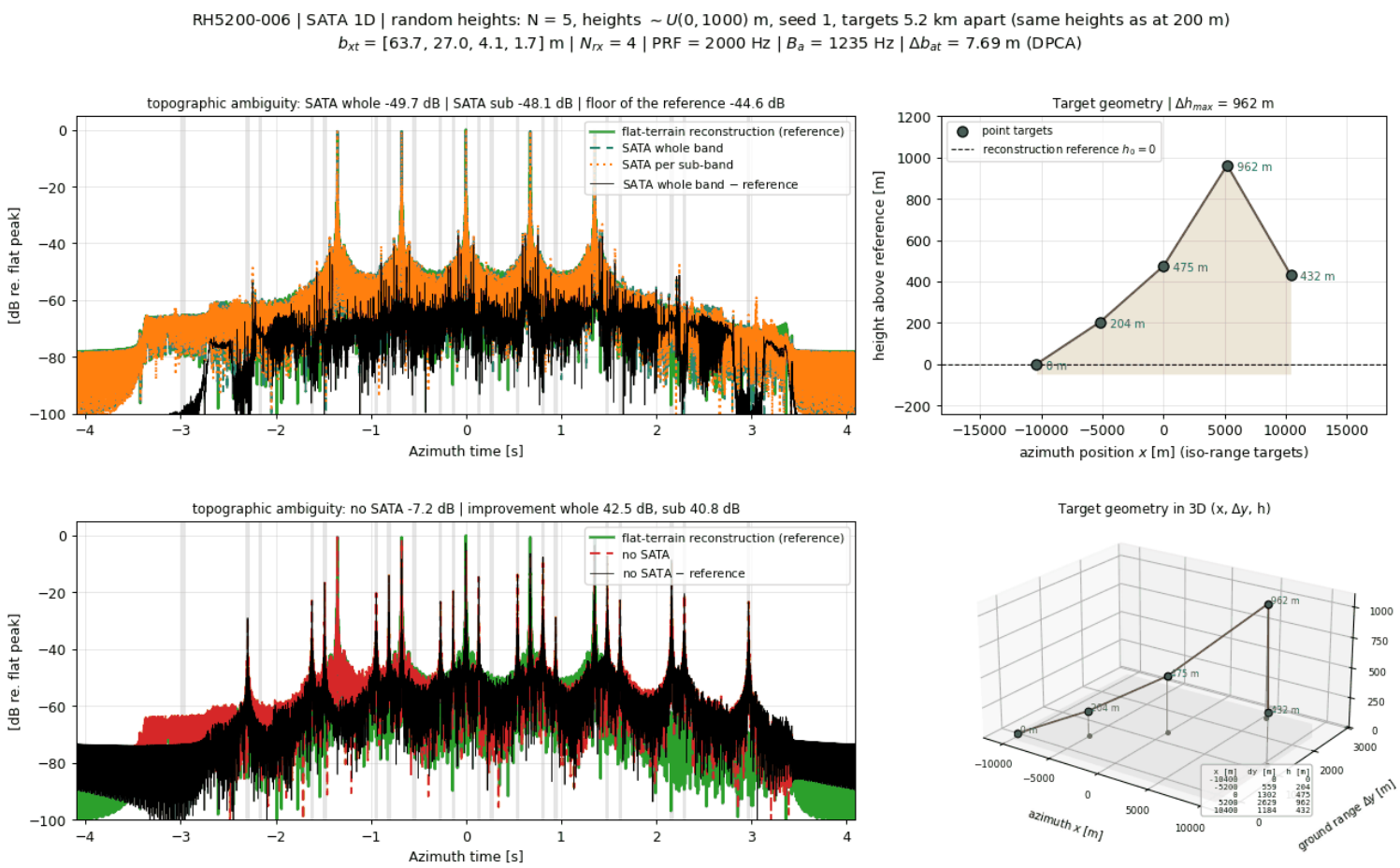


Figure 162: RH5200-006, $N = 5$, $h \sim U(0, 1000)$ m, seed 1. Topographic ambiguity: no SATA −7.2, SATA whole band −49.7, per sub-band −48.1 dB (improvement 42.5 dB); floor −44.6 dB.

6 Random heights, targets 10 km apart

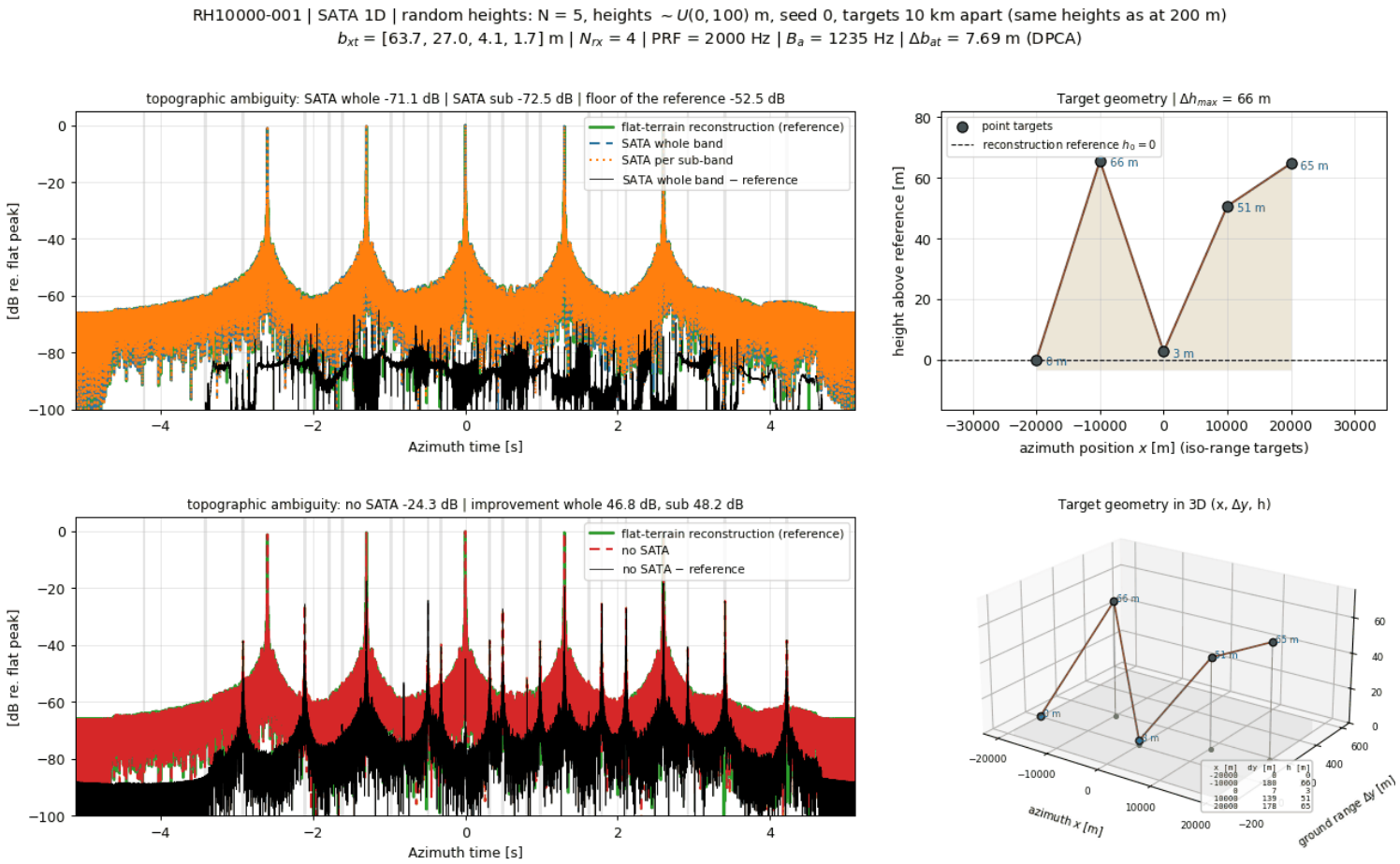


Figure 163: RH10000-001, $N = 5$, $h \sim U(0, 100)$ m, seed 0. Topographic ambiguity: no SATA −24.3, SATA whole band −71.1, per sub-band −72.5 dB (improvement 46.8 dB); floor −52.5 dB.

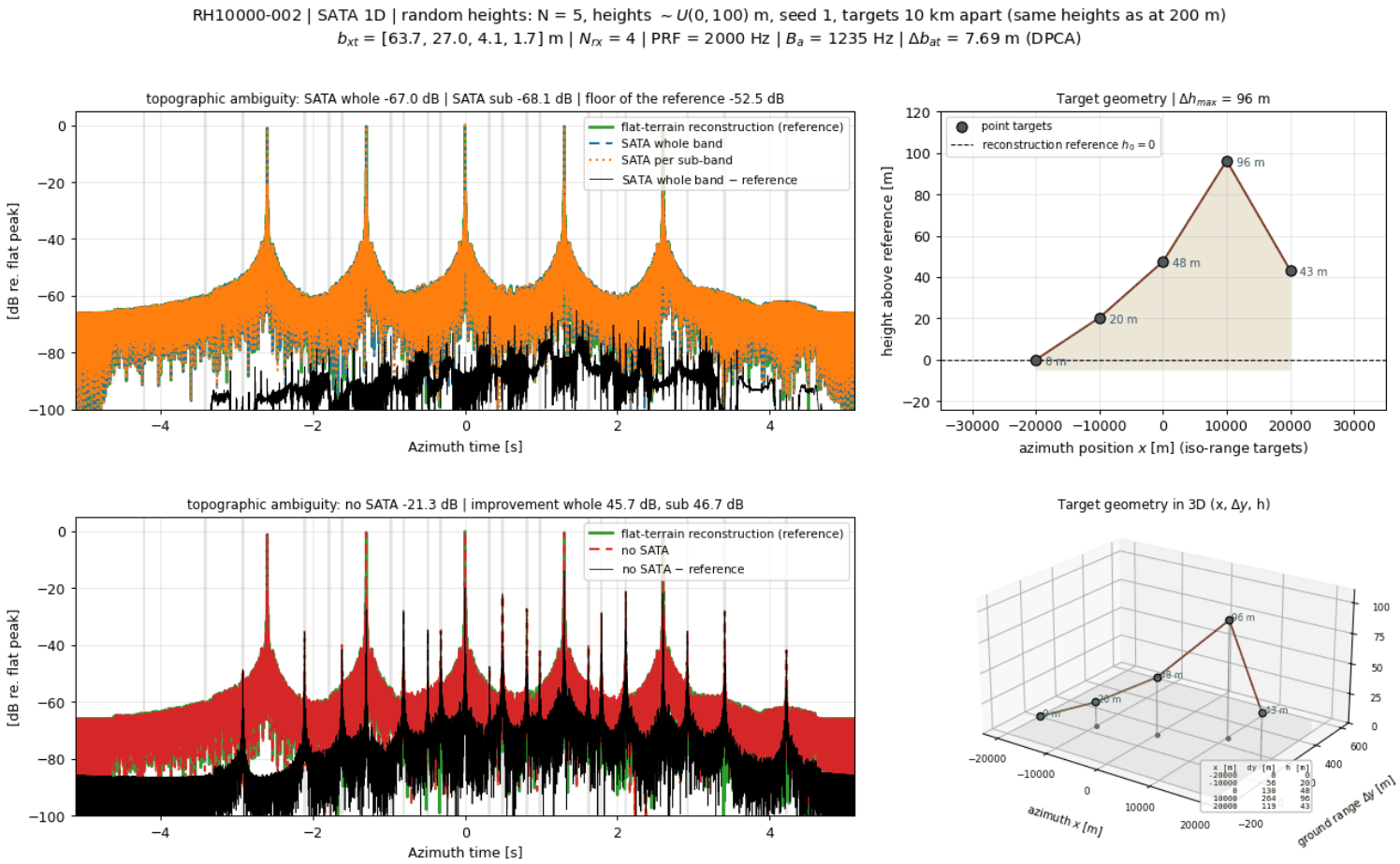


Figure 164: RH10000-002, $N = 5$, $h \sim U(0, 100)$ m, seed 1. Topographic ambiguity: no SATA −21.3, SATA whole band −67.0, per sub-band −68.1 dB (improvement 45.7 dB); floor −52.5 dB.

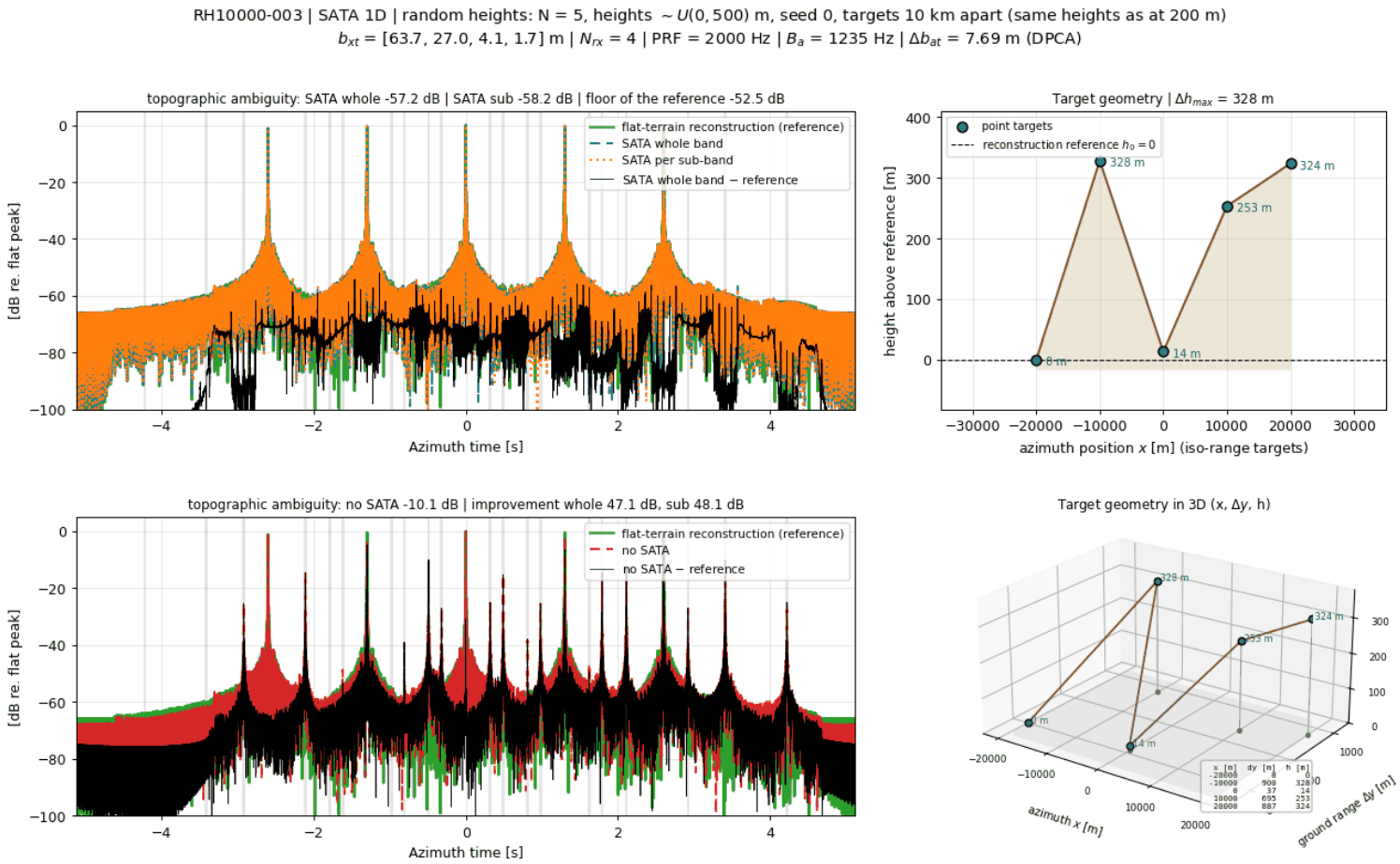


Figure 165: RH10000-003, $N = 5$, $h \sim U(0, 500)$ m, seed 0. Topographic ambiguity: no SATA −10.1, SATA whole band −57.2, per sub-band −58.2 dB (improvement 47.1 dB); floor −52.5 dB.

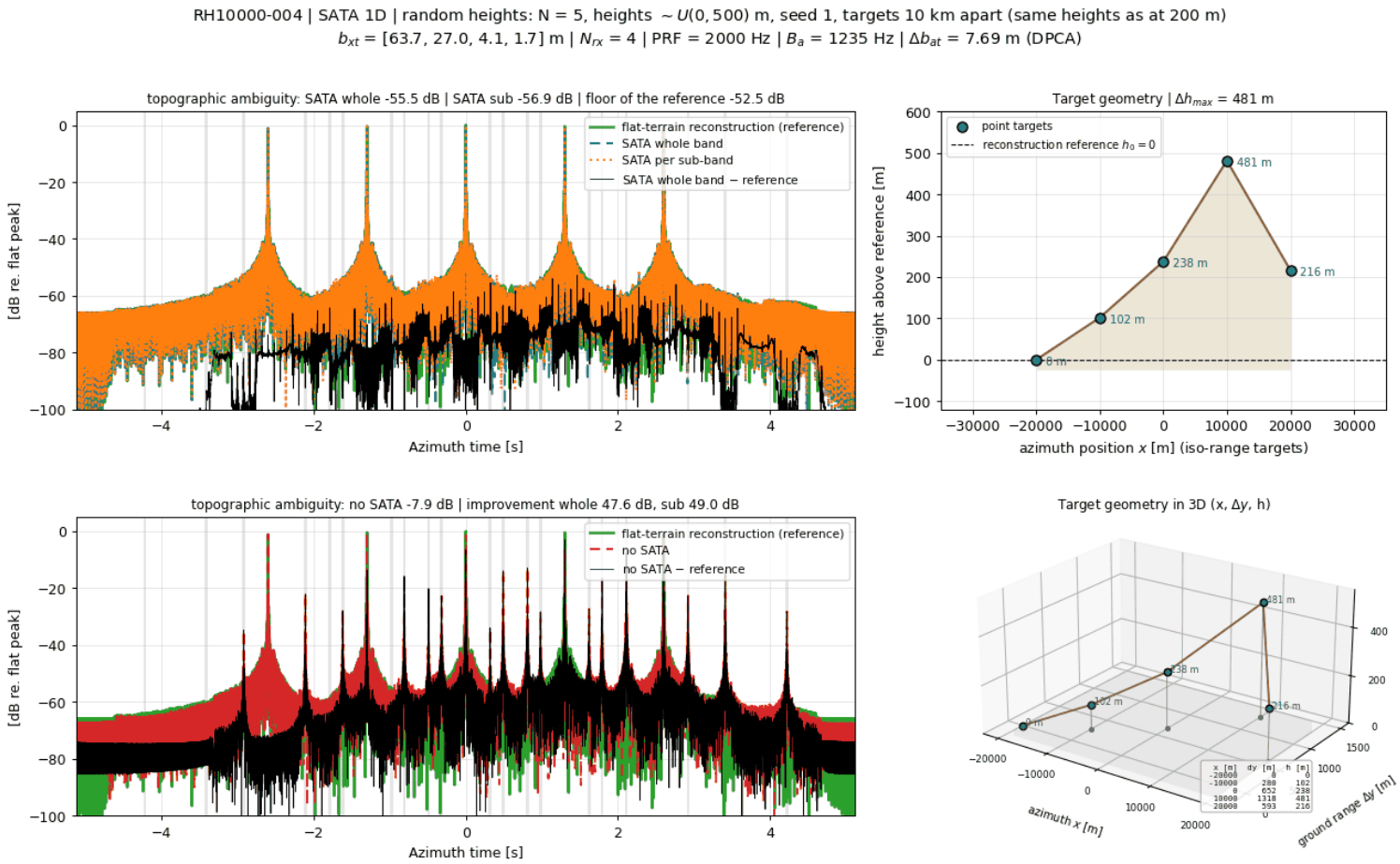


Figure 166: RH10000-004, $N = 5$, $h \sim U(0, 500)$ m, seed 1. Topographic ambiguity: no SATA −7.9, SATA whole band −55.5, per sub-band −56.9 dB (improvement 47.6 dB); floor −52.5 dB.

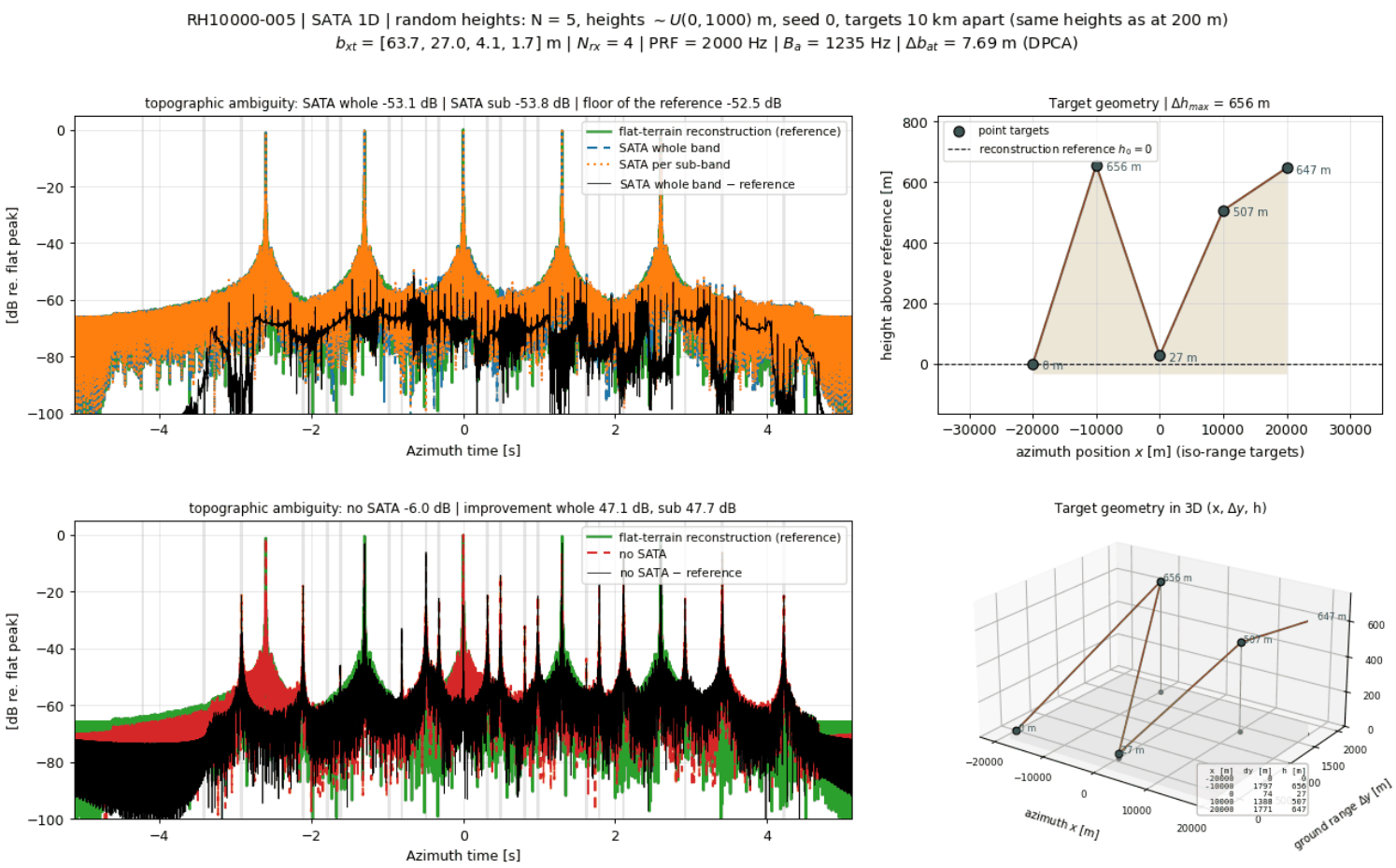


Figure 167: RH10000-005, $N = 5$, $h \sim U(0, 1000)$ m, seed 0. Topographic ambiguity: no SATA −6.0, SATA whole band −53.1, per sub-band −53.8 dB (improvement 47.1 dB); floor −52.5 dB.

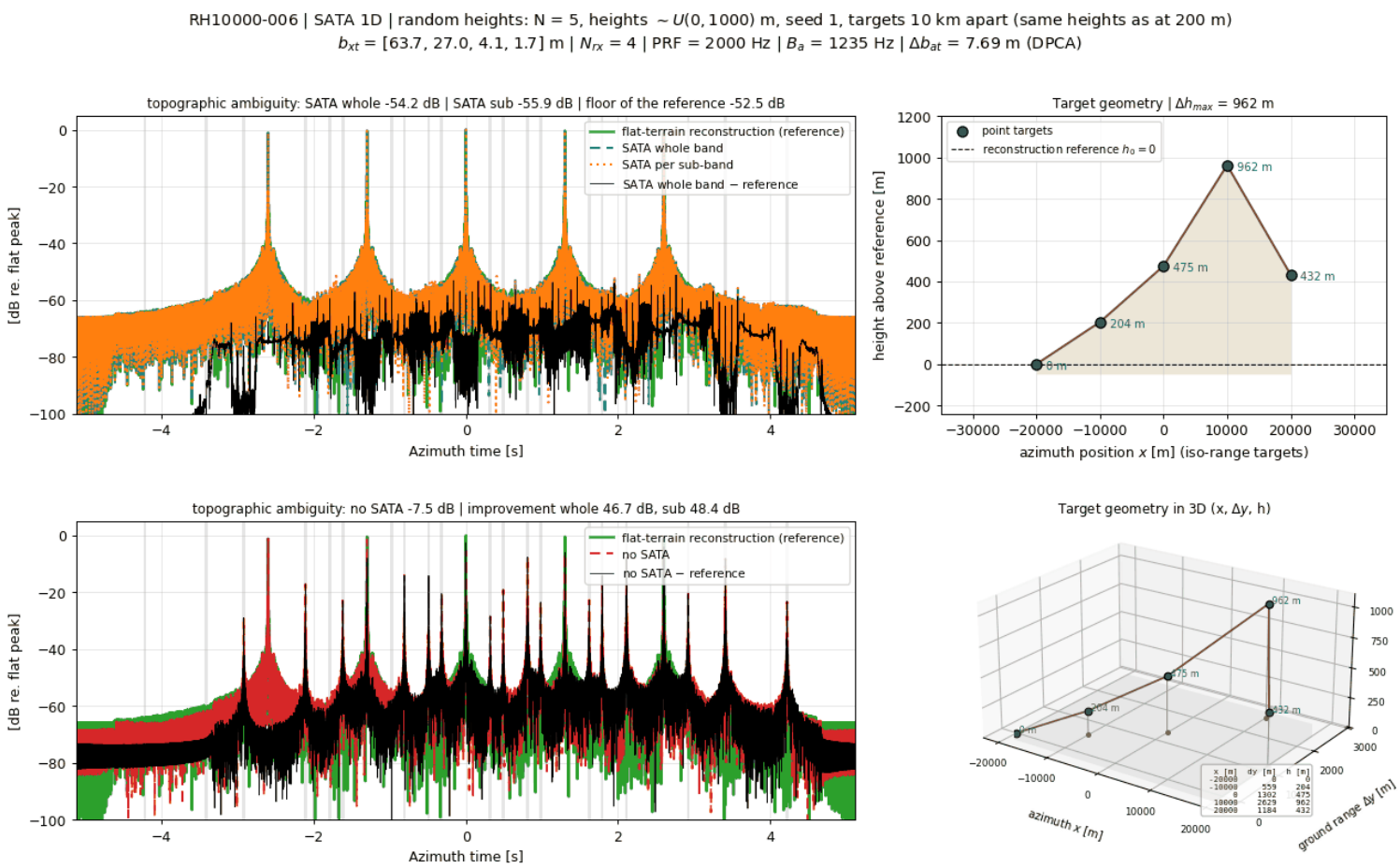


Figure 168: RH10000-006, $N = 5$, $h \sim U(0, 1000)$ m, seed 1. Topographic ambiguity: no SATA −7.5, SATA whole band −54.2, per sub-band −55.9 dB (improvement 46.7 dB); floor −52.5 dB.

7 Piecewise ramps, targets 5.2 km apart

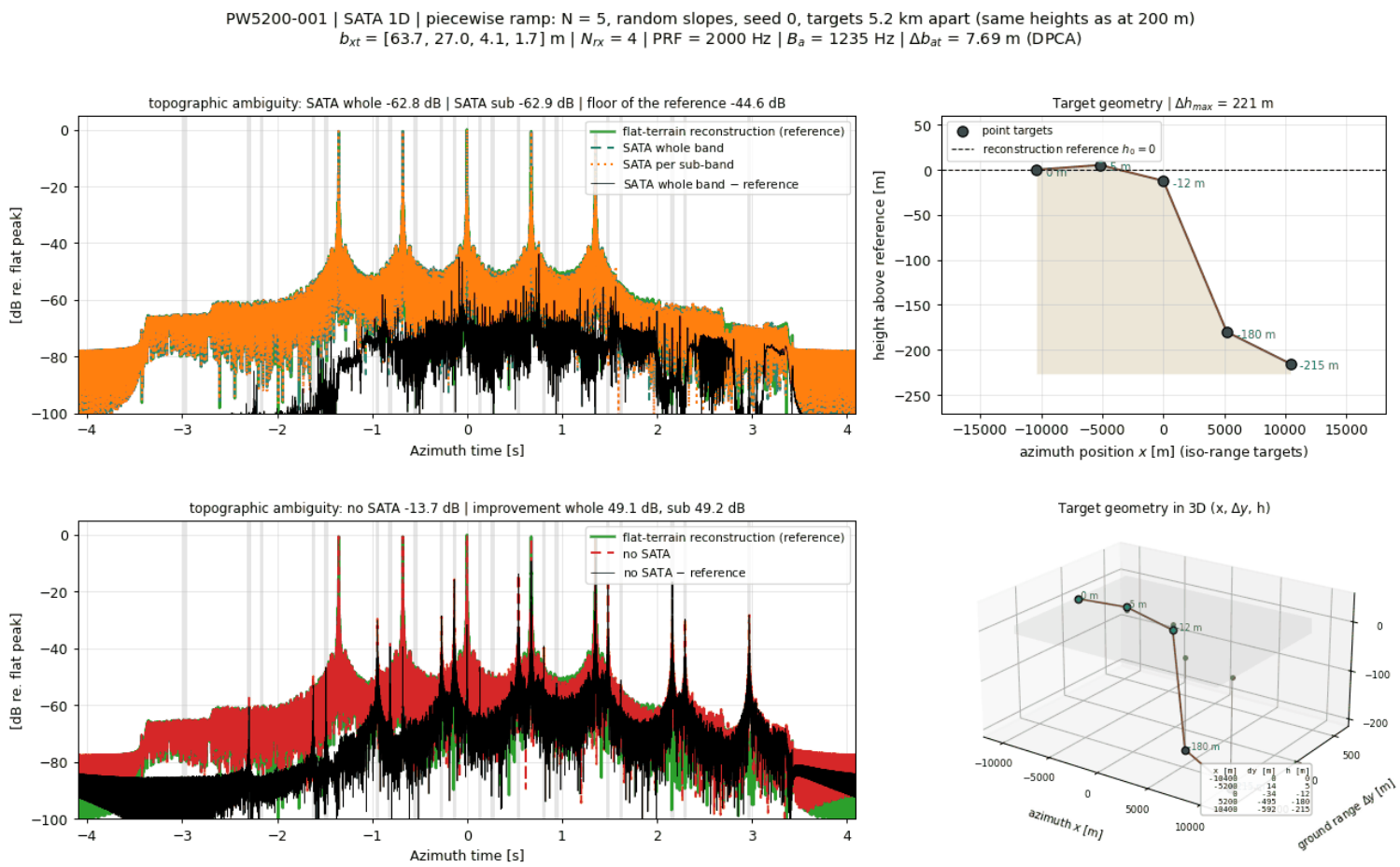


Figure 169: PW5200-001, $N = 5$, random slopes, seed 0. Topographic ambiguity: no SATA −13.7, SATA whole band −62.8, per sub-band −62.9 dB (improvement 49.1 dB); floor −44.6 dB.

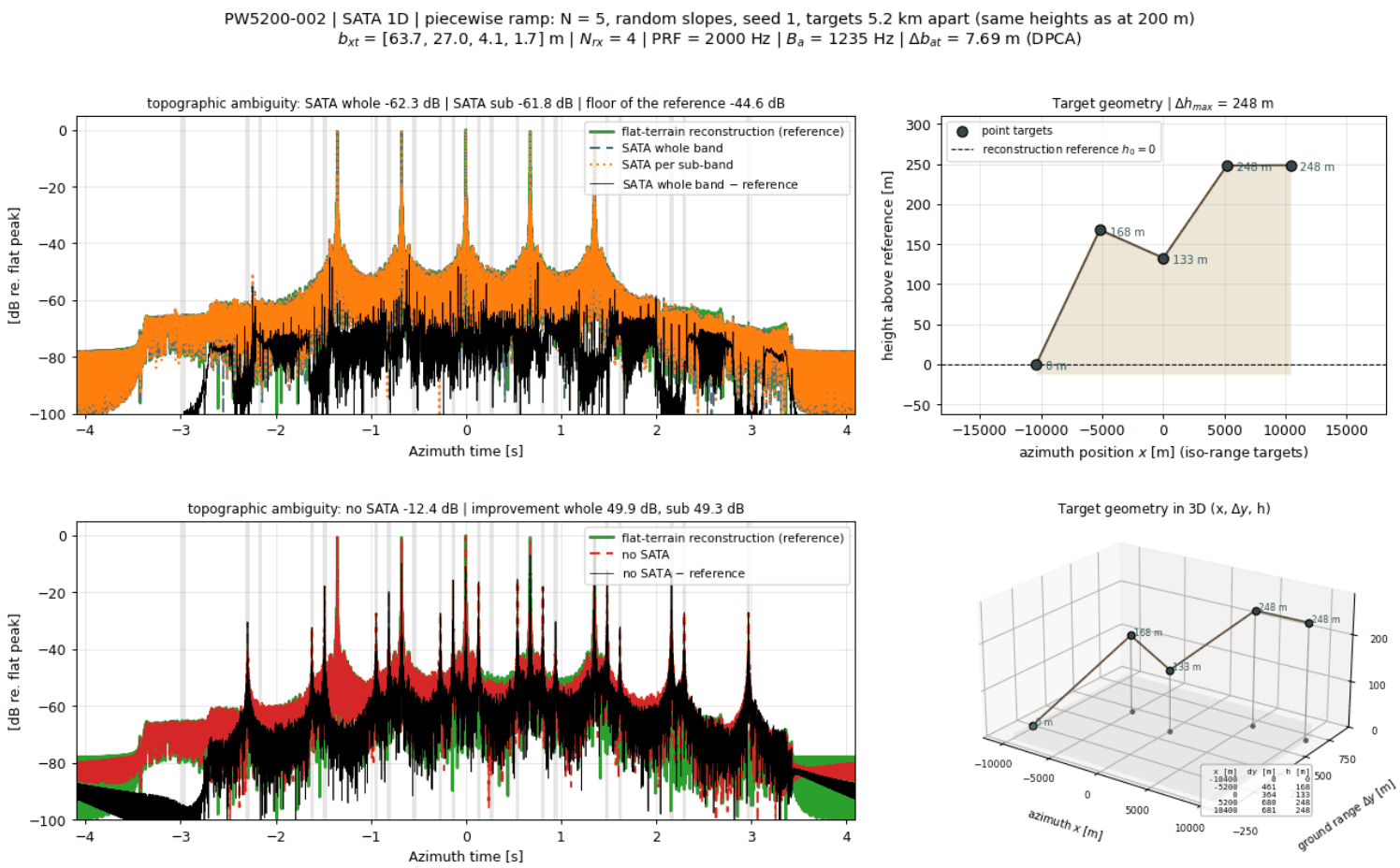


Figure 170: PW5200-002, $N = 5$, random slopes, seed 1. Topographic ambiguity: no SATA −12.4, SATA whole band −62.3, per sub-band −61.8 dB (improvement 49.9 dB); floor −44.6 dB.

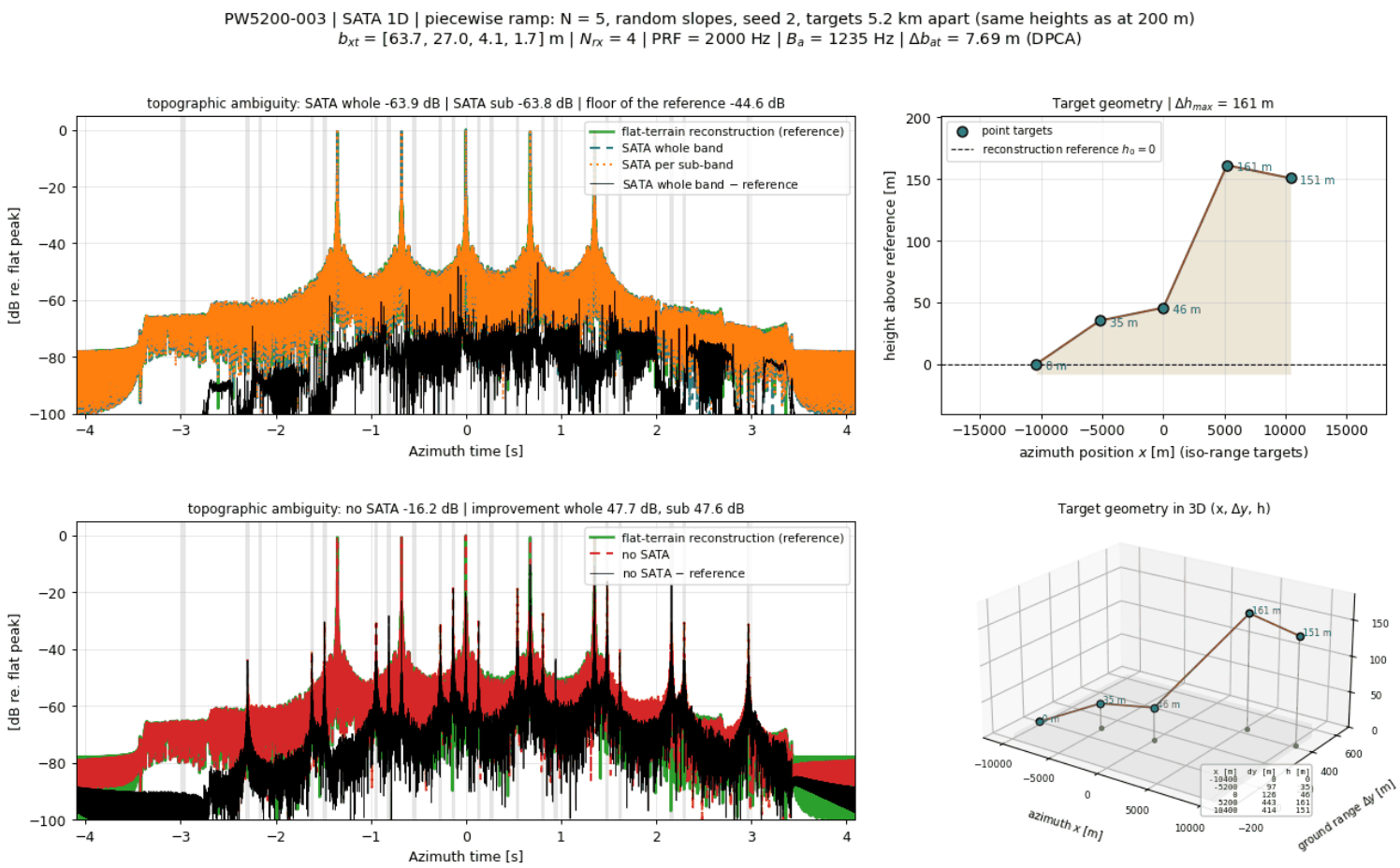


Figure 171: PW5200-003, $N = 5$, random slopes, seed 2. Topographic ambiguity: no SATA −16.2, SATA whole band −63.9, per sub-band −63.8 dB (improvement 47.7 dB); floor −44.6 dB.

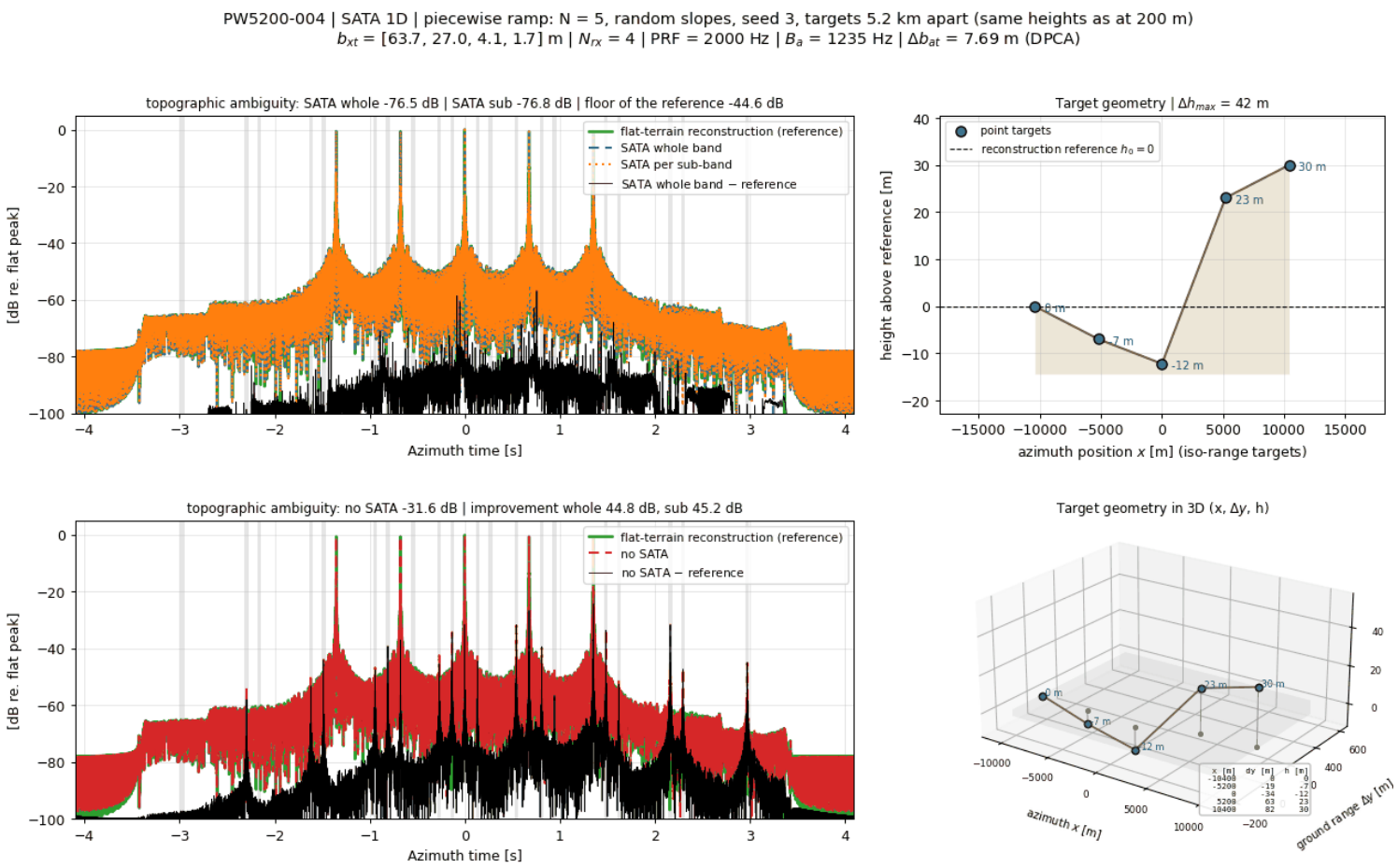


Figure 172: PW5200-004, $N = 5$, random slopes, seed 3. Topographic ambiguity: no SATA −31.6, SATA whole band −76.5, per sub-band −76.8 dB (improvement 44.8 dB); floor −44.6 dB.

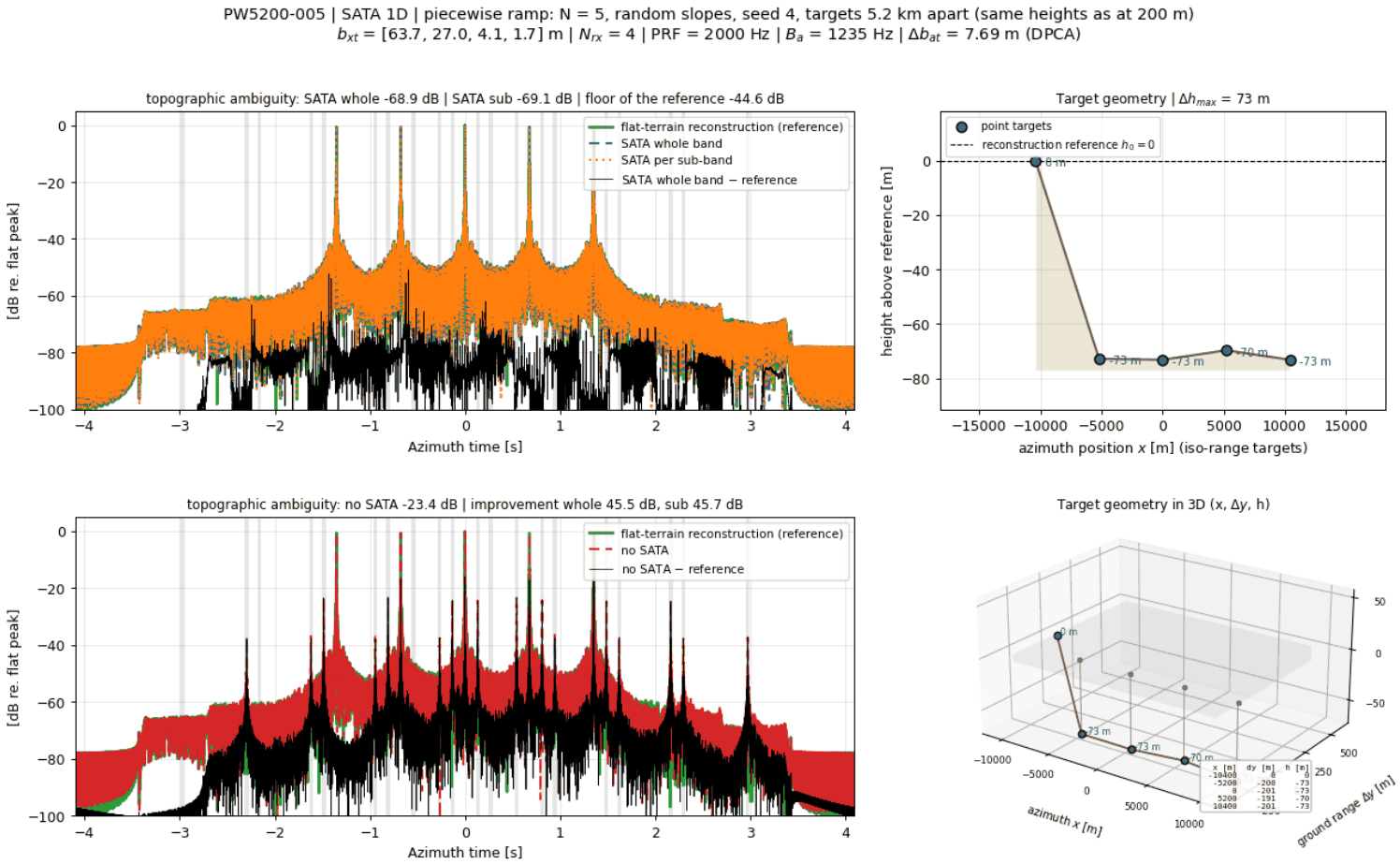


Figure 173: PW5200-005, $N = 5$, random slopes, seed 4. Topographic ambiguity: no SATA −23.4, SATA whole band −68.9, per sub-band −69.1 dB (improvement 45.5 dB); floor −44.6 dB.

8 Piecewise ramps, targets 10 km apart

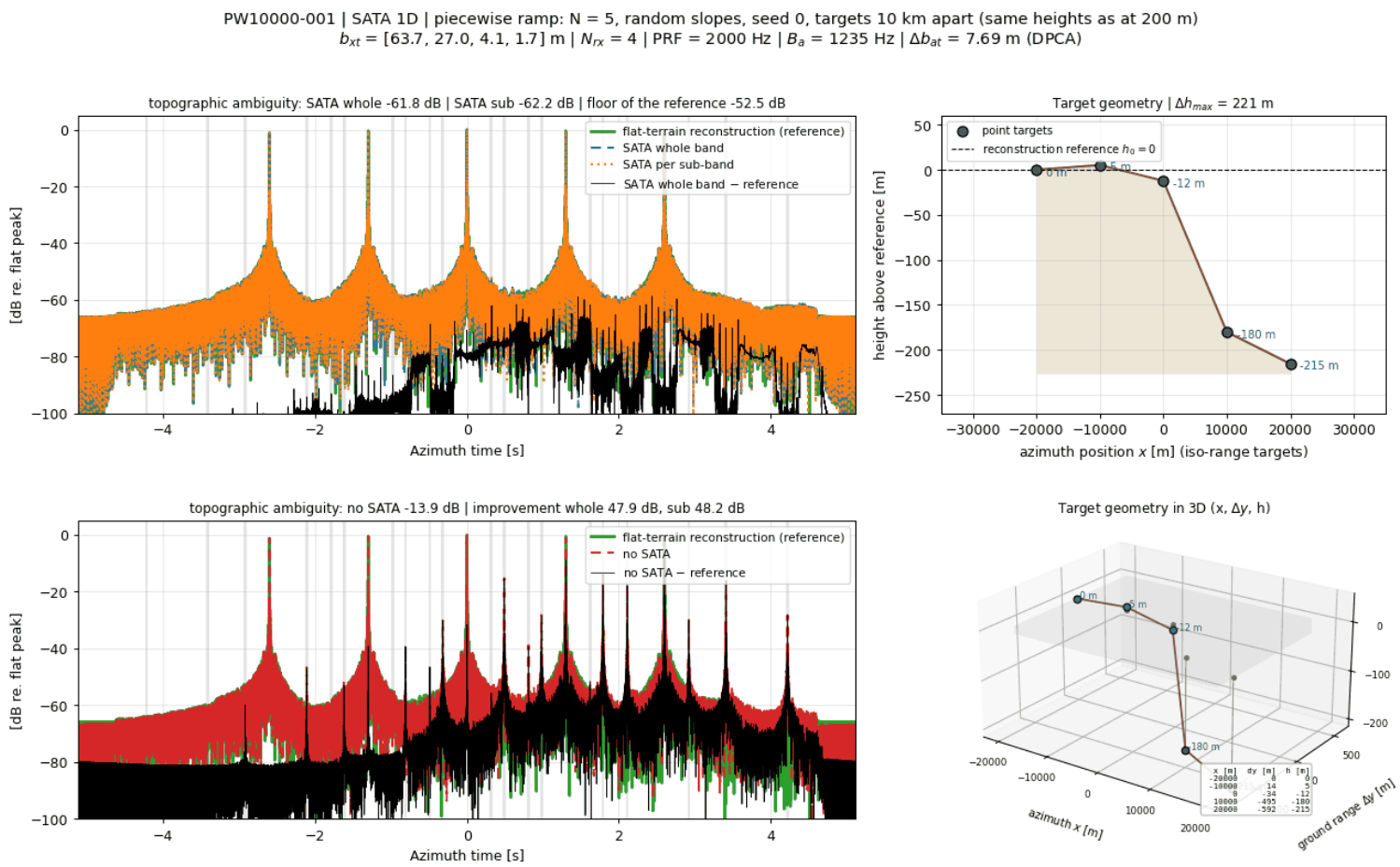


Figure 174: PW10000-001, $N = 5$, random slopes, seed 0. Topographic ambiguity: no SATA −13.9, SATA whole band −61.8, per sub-band −62.2 dB (improvement 47.9 dB); floor −52.5 dB.

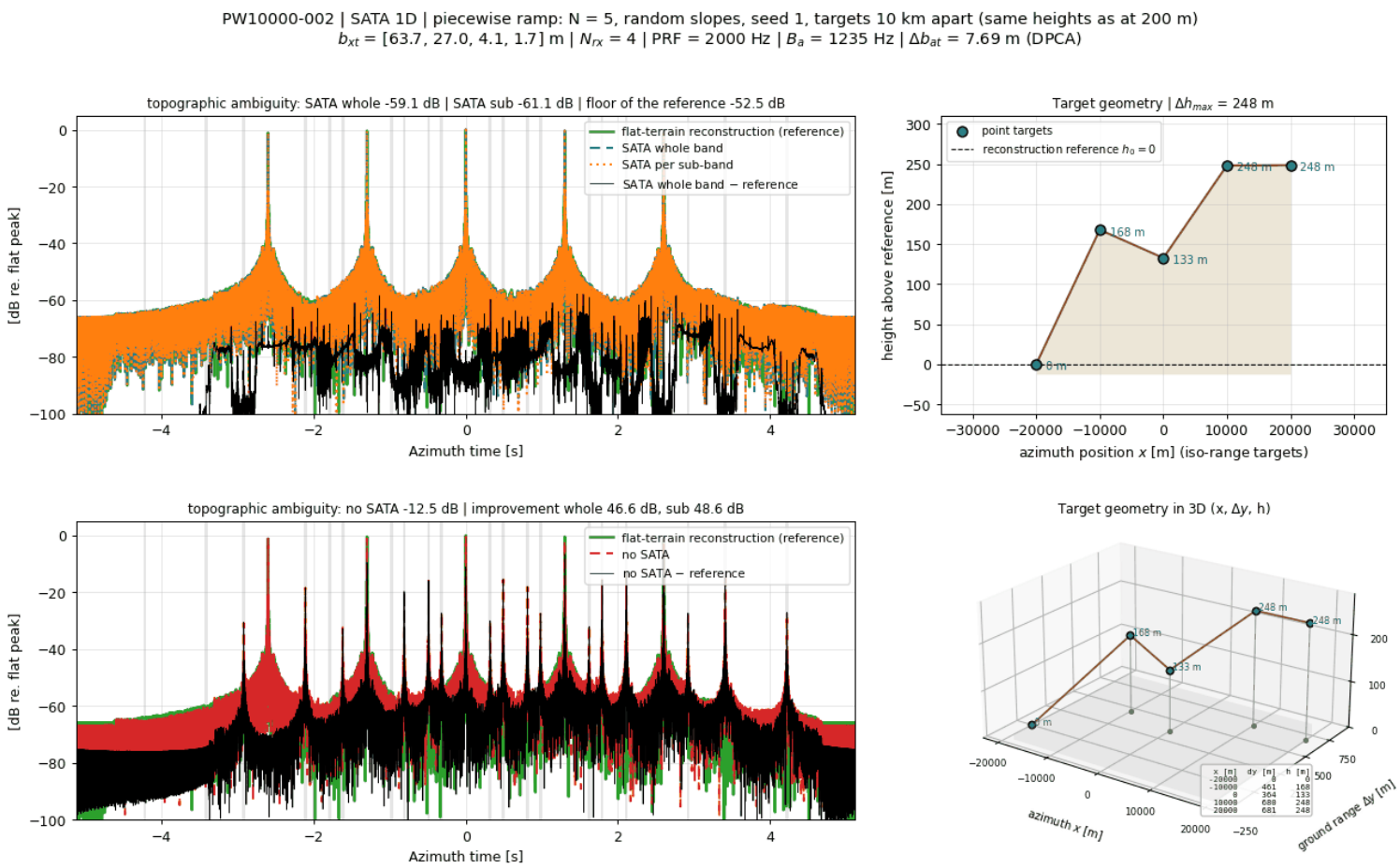


Figure 175: PW10000-002, $N = 5$, random slopes, seed 1. Topographic ambiguity: no SATA −12.5, SATA whole band −59.1, per sub-band −61.1 dB (improvement 46.6 dB); floor −52.5 dB.

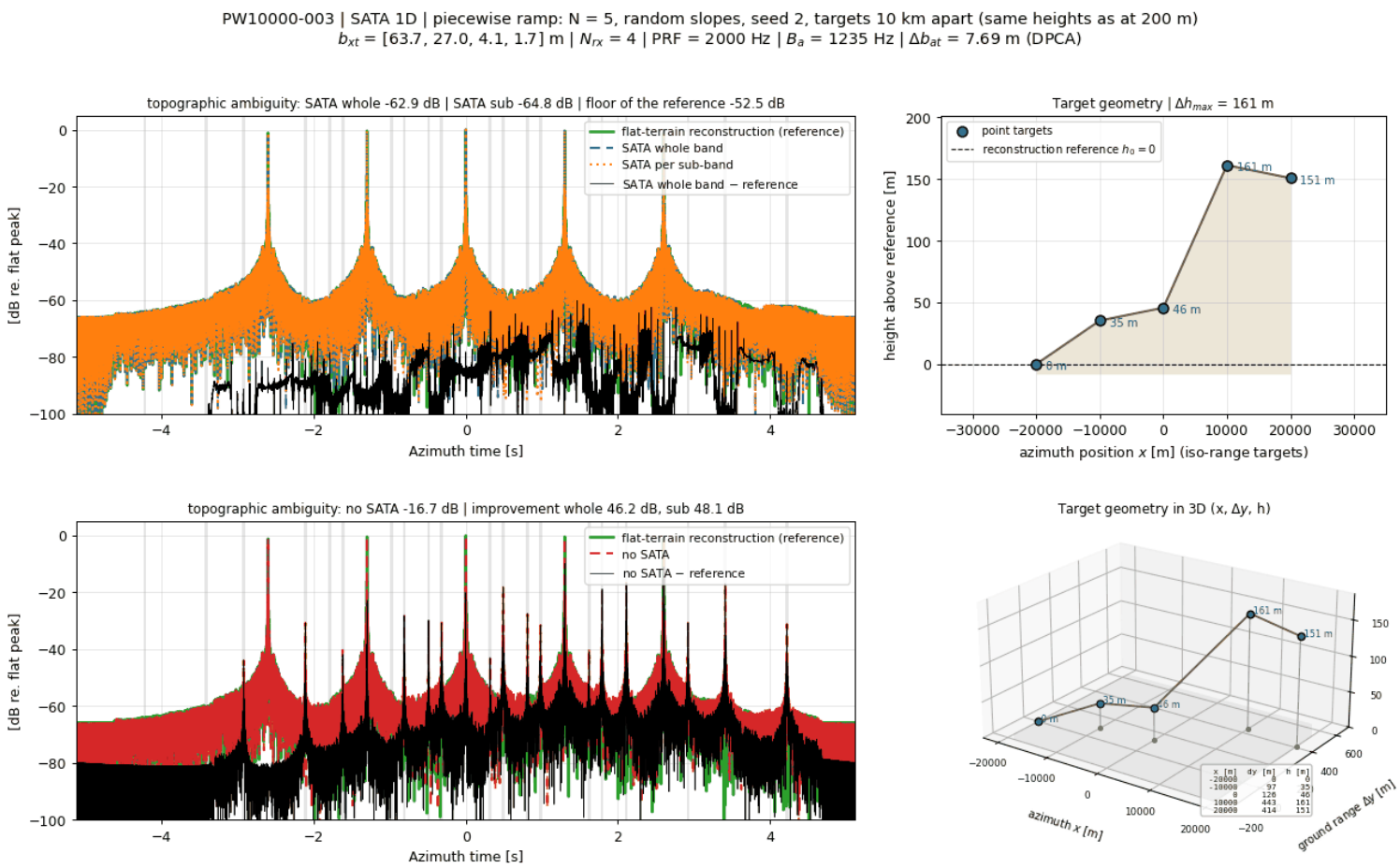


Figure 176: PW10000-003, $N = 5$, random slopes, seed 2. Topographic ambiguity: no SATA −16.7, SATA whole band −62.9, per sub-band −64.8 dB (improvement 46.2 dB); floor −52.5 dB.

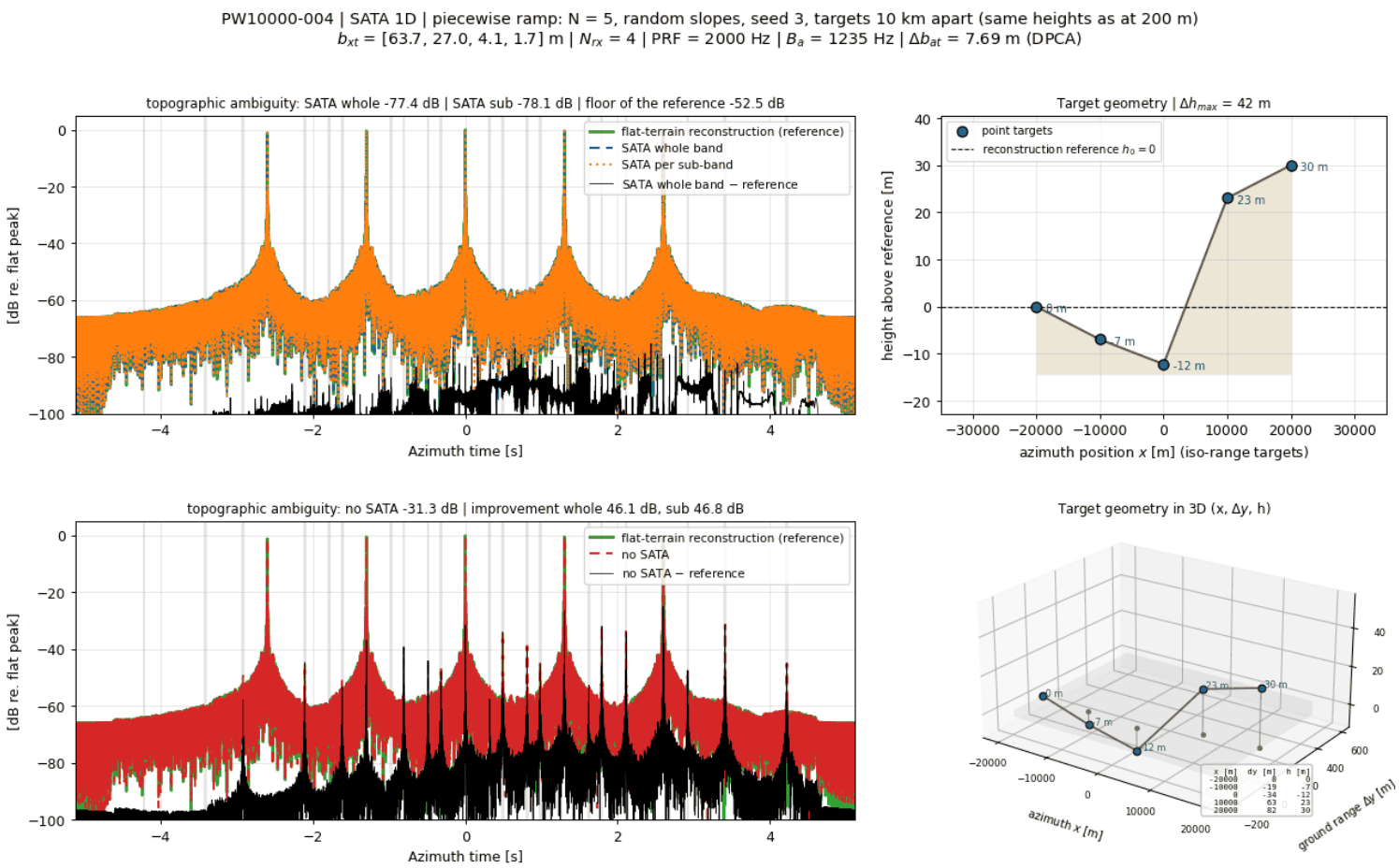


Figure 177: PW10000-004, $N = 5$, random slopes, seed 3. Topographic ambiguity: no SATA −31.3, SATA whole band −77.4, per sub-band −78.1 dB (improvement 46.1 dB); floor −52.5 dB.

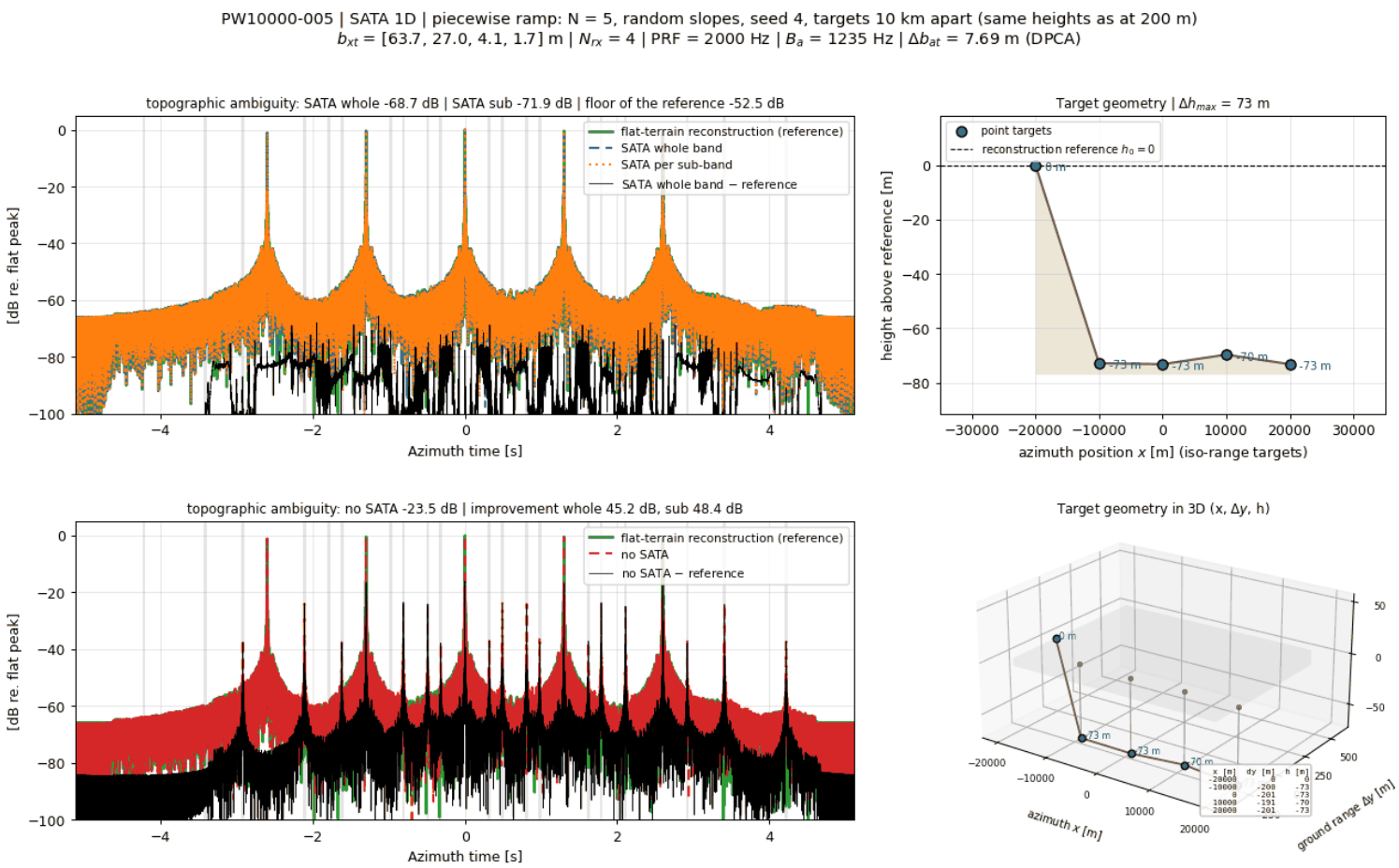


Figure 178: PW10000-005, $N = 5$, random slopes, seed 4. Topographic ambiguity: no SATA −23.5, SATA whole band −68.7, per sub-band −71.9 dB (improvement 45.2 dB); floor −52.5 dB.